

CONTRACTOR SUBMITTAL TRANSMITTAL FORM REV. A

Owner: County of Hawaii
Contractor: Nan, Inc. **Project No.:** WW-4705R
Project Name: Hilo WWTP Phase 1 **Submittal Number:**
Submittal Title: For Information Only
TO:
From: Nan Inc.

Specification No. and Subject of Submittal / Equipment Supplier	
Spec:	Paragraph:
Authored By:	Date Submitted:

Submittal Certification	
Check Either (A) or (B):	
☐ (A)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings with <u>no exceptions</u> .
☐ (B)	We have verified that the equipment or material contained in this submittal meets all the requirements specified in the project manual or shown on the contract drawings <u>except</u> for the deviations listed.
Certification Statement: By this submittal, I hereby represent that I have determined and verified all field measurements, field construction criteria, materials, dimensions, catalog numbers and similar data, and I have checked and coordinated each item with other applicable approved shop drawings and all Contract requirements.	
General Contractor's Reviewer's Signature:	
Printed Name and Title:	
In the event, Contractor believes the Submittal response does or will cause a change to the requirements of the Contract, Contractor shall immediately give written notice stating that Contractor considers the response to be a Change Order.	
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Nan, Inc

PROJECT: HILO WWTP REHABILITATION
AND REPLACEMENT PROJECT - PHASE 1

JOB NO. WW-4705R

THIS SUBMITTAL HAS BEEN CHECKED BY
THIS CONTRACTOR. IT IS CERTIFIED
CORRECT, COMPLETE, AND IN
COMPLIANCE WITH CONTRACT
DRAWINGS AND SPECIFICATIONS. ALL
AFFECTED CONTRACTORS AND
SUPPLIERS ARE AWARE OF, AND WILL
INTEGRATE THIS SUBMITTAL (UPON
APPROVAL) INTO THEIR OWN WORK.

DATE RECEIVED _____
 SPECIFICATION SECTION # _____
 SPECIFICATION _____
 PARAGRAPH _____
 DRAWING _____
 SUBCONTRACTOR _____
 SUPPLIER _____
 MANUFACTURER _____

CERTIFIED BY CQCM or Designee : _____

SECTION 11294B

HEAVY-DUTY FABRICATED STAINLESS STEEL SLIDE GATES

PART 1 GENERAL

1.01 SUMMARY

- A. Section includes: Heavy-duty fabricated stainless steel slide gates.
- B. Gate manufacturer's responsibility shall extend to sizing, selection, and furnishing of associated electric actuators as specified in Section 13447 - Electric Actuators.
- C. As specified in Section 01600 - Product Requirements.

1.02 REFERENCES

- A. American National Standards Institute (ANSI) and American Welding Society (AWS):
 - 1. ANSI/AWS D1.1 - Structural Welding Code.
 - 2. AWS D1.6 - Structural Welding Code - Stainless Steel.
- B. American Society of Mechanical Engineers (ASME):
 - 1. Section IX - Boiler and Pressure Vessel Code - Welding and Brazing qualifications.
- C. American Water Works Association (AWWA):
 - 1. C561 - Fabricated Stainless Steel Slide Gates.
- D. ASTM International (ASTM):
 - 1. A 240 - Standard Specification for Stainless Steel Plate, Sheet, and Strip.
 - 2. A276 - Standard Specification for Stainless Steel Bars and Shapes.
 - 3. B584 - Standard Specifications for Copper Alloy Sand Castings for General Application.
 - 4. D1248 - Standard Specification for Polyurethane Plastics Extrusion Materials for Wire and Cable.
 - 5. D2000 - Standard Classification for Rubber Products in Automotive Applications.
 - 6. D4020 - Standard Specification for Ultra-High Molecular-Weight Polyethylene Molding and Extrusion Materials.

1.03 DEFINITIONS

- A. Slenderness ratio (l/r): The largest ratio obtained by dividing the unsupported length of the stem by the radius of gyration of the stem cross section.
- B. Design head: Depth from surface of water to centerline of gate. Use value specified in the gate schedule.

- C. Seating head: Pressure applied to gate slide from weight of water column above gate centerline that forces gate slide into seat.
- D. Unseating head: Pressure applied to gate slide from weight of water column above gate centerline that forces gate slide away from seat.
- E. Substantially similar:
 - 1. Similar in size, design head, and service.
 - 2. Utilizes the proposed design for critical components including guides and seals.
- F. Operator: The terms "operator" and "actuator" are used interchangeably in this specification section. And the terms "motorized operator" and "electric actuator" are used interchangeably in this specification section.

1.04 DESIGN REQUIREMENTS

- A. Except as modified or supplemented as specified in this Section, all gates and operators shall conform to the requirements of AWWA C561, latest edition.
- B. Gate components:
 - 1. Frames:
 - a. Design for the design head scheduled with a minimum safety factor of 5 with regard to ultimate tensile, compressive, and shear strength.
 - b. Self-contained gates: Where operational loads are taken by the frame, not by the concrete at the base of the pedestal, design to be self-supporting so that no further reinforcing or support is required.
 - 2. Stem: Select stem diameter, stem guide quantity and stem guide spacing based on meeting all of the following criteria:
 - a. Slenderness ratio (l/r): Shall not exceed 200.
 - b. Tensile strength: Suitable to withstand the force generated by the operator with the application of a 200 pound force applied to the crank or handwheel or a 250 foot-pound torque applied to the wrench nut.
 - c. Compressive strength:
 - 1) Suitable to withstand buckling due to the force generated by the operator with the application of an 80 pound force applied to the crank or handwheel or a 100 foot-pound torque applied to the wrench nut.
 - 2) Determine buckling load using Euler Column formula in accordance with AWWA C 561, where $C = 2$.
 - d. Design force for gates equipped with power actuators:
 - 1) Electric motor operators: 1.25 times the output thrust in the stalled-motor condition.
 - e. Gates having widths that are greater than 2 times the height: Provide with 2 lifting mechanisms connected by a tandem shaft.
 - a. For gates with electric actuators, the gate and the actuator manufacturers shall coordinate the proposed gate stem configuration as described in Article 2.03 Paragraph D.7.
 - b. Minimum stem diameter: 1-1/2 inch.

3. Thrust nut: Suitable to withstand thrust developed by operator with the application of a 40 pound force on the crank or handwheel with safety factor of 5. Base design on ultimate strength of material used.
4. Yokes for self-contained gates:
 - a. Design yoke using design loading criteria for stem with safety factor of 5 based on the ultimate strength of the material used.
 - b. Maximum deflection at design load: Not to exceed 1/360th of the span or 1/4 inch, whichever is less.
5. Slide:
 - a. Deflection shall be less than or equal to 1/1000 of the span of the gate or 1/16 inch, whichever is less, when under the design head.
 - b. Design for the maximum design head specified with a minimum safety factor of 5 with regard to ultimate tensile, compressive, and shear strength.

1.05 PERFORMANCE REQUIREMENTS

- A. Leakage testing shall be conducted in accordance with AWWA C561. Maximum allowable leakage shall be 0.05 gallons per minute per foot of wetted seat perimeter under seating and unseating heads as follows. It is half the allowable limit set forth in AWWA C561.
 1. Gates with submerged inlet: At any depth of water above the centerline of the gate up to the maximum design head.
 2. Embedded gates: At any depth of water above the centerline of the gate up to the full height of the gate.

1.06 SUBMITTALS

-  A Submit as specified in Section 01330 - Submittal Procedures.
-  B Product data: As specified in Section 15050 - Common Work Results for Mechanical Equipment.
-  C Shop drawings: As specified in Section 15050 - Common Work Results for Mechanical Equipment.
 1. Layout and installation drawings for each gate size and type. Contractor shall field-verify existing dimensions prior to submittal preparation. Indicate available space in existing channels and identify adjacent gates and other equipment that may potentially cause conflict.
- D. Installation List:
 1. To verify conformance with the specified Manufacturer Qualifications.
 2. Include the following information as a minimum:
 - a. Name and location of installation.
 - b. Name and telephone number of person in direct responsible charge of the equipment.
 - c. Month and year the gate was placed in operation.
 - d. Size of gate.
 - e. Number of units installed.
 - f. Service.

3. Proposed Substitute "Or Equal" Submittal specified in Section 01330 - Submittal Procedures must also include shop drawings for the equipment installations that satisfy the requirements of Manufacturer Qualifications, Article 1.07 Paragraph B.3. of this Section. The drawings shall clearly indicate the design used for critical components as specified in Article 1.07 Paragraph B.3.b. to verify conformance with the specifications.
- E. Gate operators: As specified in Section 13447 - Electric Actuators:
1. Provide certification from both the operator manufacturer and the gate manufacturer confirming that the proposed gate stem configuration has been coordinated with motorized operator selection:
 - a. Confirm that the proposed gate stem configuration provides the most efficient combination of stem diameter, pitch, and lead.
 - b. Confirm the design will keep the operating temperature at the stem nut to a minimum during operation.
 - c. Confirm the design will meet the specified operator operating speed requirements.
 2. For coordination purposes, gate manufacturer shall submit calculations verifying the suitability of the selected operator for the application. For each gate include:
 - a. Open/close speed per Section 13447 - Electric Actuators.
 - b. The maximum torque required for operation of the gate (including breakaway from seat) with a safety factor of 1.4.
 - c. The torque supplied by the motorized operator scheduled in Section 13447 - Electric Actuators for the operating speed specified in Section 13447 - Electric Actuators.
 - d. The thrust output capacity of the motorized operator with the furnished motor.
 - e. Gate manufacturer shall provide a letter confirming that coordination with electric actuator manufacturer has been completed to assure compatibility and correct fit between gate and electric actuator.
- F. Design Calculations: As specified in Section 15050 - Common Work Results for Mechanical Equipment:
1. Submit calculations and design data substantiating conformance with the Drawings and Specifications.
 2. Gate opening and closing thrust forces that will be transmitted to the support structure with operator at extreme positions and load.
 3. Torque required to open and close the gate, including maximum torque at any point along gate travel. Indicate thrust valve and stem factor.
 4. Breakaway torque from seat. Indicate thrust valve and stem factor.
 5. Submit calculations that substantiate the slide design is in conformance with the minimum safety factors specified in the latest revision of AWWA C561. In particular, confirm this at the slide centerline and along the length of the slide, on each side, where it engages the guides.

- 6. Submit calculations that substantiate the guide design is in conformance with the minimum safety factors specified in the latest revision of AWWA C561. In particular confirm that all components, including but not limited to, the guide frame and guide frame attachment bolts are designed within the specified safety factors.
- 7. Calculations shall be stamped and signed by a licensed civil or structural engineer.
- ☒ G. Complete bill of materials.
- ☒ H. Wall thimbles design.
- ☒ I. Manufacturer's installation instructions.
- ☒ J. Vendor operation and maintenance manuals as specified in Section 01782 - Operation and Maintenance Manuals.
- K. Commissioning submittals:
 - 1. Provide Manufacturer's Certificate of Installation and Functionality Compliance as specified in Section 01756 - Commissioning.
 - 2. Detailed Performance Test Procedures: Factory and field testing.
- ☒ L. Detailed performance test procedures for factory and field-testing.
- M. Certificate of Installation and Functionality Compliance.
- N. Testing results reports and certifications.

1.07 QUALITY ASSURANCE

- ☒ A. Factory Markings:
 - 1. Mark gates according to the tag numbers when such numbers are used.
 - 2. Where thimbles, frames and other components are not interchangeable, the components shall be match marked.
- B. Manufacturer Qualifications:
 - 1. Minimum 10 years of experience in production of equipment substantially similar to the equipment specified.
 - 2. Submit evidence of satisfactory operation in at least 50 separate facilities that utilize the proposed design for critical components: guides and seals.
 - 3. At least 10 different operating installations shall satisfy all of the following requirements:
 - a. Years in service: Minimum 8 years.
 - b. Utilizes the proposed design for critical components: guides, seals and frames.
 - c. Gate Size/Design Head:
 - 1) At least five different installations utilizing gates 48 inch by 96 inch or larger for a submerged inlet with a design seating and unseating head of at least 20 feet.
 - 4. Multiple gates at a plant shall be considered as one installation toward meeting the experience requirements.

5. Fulfillment of the specified experience requirements shall be a condition of acceptance.
 6. Manufacturer's shop welds, welding procedures, and welders: Qualified and certified in accordance with the requirements of ANSI/AWS D1.1, AWS D1.6, or ASME Boiler and Pressure Vessel Code Section IX.
 7. Assembled Gates: Shop inspected, adjusted and tested before shipping.
- C. Fulfillment of the specified experience requirements shall be a condition of acceptance.

1.08 SEQUENCING AND SCHEDULING

- A. Coordinate work with restrictions as specified in Section 01140 - Work Restrictions.

1.09 WARRANTY

- A. Provide warranty as specified in Section 01783 - Warranties and Bonds.

PART 2 PRODUCTS

2.01 MANUFACTURERS

- A. Manufacturers: One of the following or equal, modified to comply with the Specifications.
-  1. Whipps, Inc., Series 900 or 950, modified to comply with the Specifications.
 2. H. Fontaine, Series 20 or 25, modified to comply with the Specifications.
- B. Inclusion of a specific manufacturer's name and model in the Specifications does not mean that the specific manufacturer's standard product will be acceptable. Specified manufacturer's or other manufacturer's standard product shall be modified as required to meet the Specifications.
-  Operator anchor bolts and wall thimbles: Provided by manufacturer of slide gates.

2.02 MATERIALS

- A. Stainless steel: ASTM A276, Type 316:
1. Components or structural shapes which are welded: ASTM A276, Type 316L.
 2. All wetted and unwetted parts including all fasteners and hardware, except as specified in this Section, shall be Type 316L stainless steel.
- B. Ultra-high molecular weight polyethylene: ASTM D1248 and D4020.
- C. Neoprene: ASTM D2000, Grade 2 BC 510.
- D. Manganese bronze: ASTM B584, UNS Number C86500 or Alloy 432.
- E. Silicon bronze: ASTM B584 UNS Number C87300.

2.03 COMPONENTS

- A. Slide:
 - 1. Type 316L stainless steel.
 - 2. Rectangular or square.
 - 3. Fabricated with a flat plate reinforced with formed plates or structural members. Horizontal structural members shall be welded to side vertical reinforcement members.
 - 4. The stem connector shall be constructed of two angles or plates. The stem connector shall be welded to the slide. A minimum of two bolts shall connect the stem to the stem connector.
 - 5. All structural components shall have a minimum material thickness of 1/4 inch.
- B. Frame:
 - 1. Construct gate frame of Type 316L stainless steel structural members or formed plate welded to form a rigid 1-piece frame. Provide one of the following designs:
 - a. Unitized Design:
 - 1) The vertical frame rails shall be formed from one plate creating a unitized frame design.
 - 2) Frame shall have wrap around gussets going around frame groove on three sides extending back to weld onto the base wall plate at each anchor bolt location. Gussets shall stabilize frame.
 - b. Bolted Design:
 - 1) Frame members shall consist of one or more bolted structural members. Seats shall be retained by through bolting.
 - 2. All structural components shall have a minimum material thickness of 1/4 inch.
 - 3. Mounting: As scheduled.
 - 4. Self-adjustable ultra-high molecular weight polyethylene seals. Seal systems that utilize manually adjustable seals are not acceptable.
 - 5. Flush bottom type unless otherwise scheduled on the Drawings or in the gate schedule. Fill blockouts with grout, flush with concrete floor.
 - a. When replacing flush bottom gates, widen existing blockouts as required to accommodate replacement gates.
 - 6. Allow replacement of top, side, and bottom seals without removing the gate frame from concrete or wall thimble.
 - 7. Machine surfaces matching with thimble. Provide seal between gate frame and thimble that will meet leakage performance requirements.
 - 8. Anchor bolt holes:
 - a. Where gates are installed in existing structures, anchor bolt hole locations shall be located so that anchor bolts will avoid existing rebars in the concrete. Shop drawings shall include Contractor verified locations of existing rebars to determine anchor bolt locations.
 - b. Where gates are installed in existing structures that have existing wall thimbles, oversize gate frame to avoid wall thimbles.
 - 9. Yoke for self-contained gates:
 - a. Type 316L stainless steel.
 - b. Extend guides and frame so that bottom of yoke is at least 12 inches above top of slide at full-open position or as needed to allow raising the slide high enough to comfortably allow replacing the gate top seal. Top of yolk shall also be at least 3 feet above the operation floor.

- c. Bolt or weld to frame.
 - d. Provide mounting plate on top of yoke to mount operator.
 - e. Design yoke to allow removal of gate slide.
- 10. Gates with enclosures:
 - a. Provide extension spool piece of Type 316 stainless steel to support the motor operator outside the enclosure.
- C. Guides and Seals:
 - 1. Type 316 stainless steel with ultra-high molecular weight polyethylene seal in contact with gate.
 - 2. Length: To completely support the slide fully in the open position.
 - 3. Anchor bolts shall not pass through the guides and seals. Seals must be removable without disassembly of the guide frame.
 - 4. Seals shall achieve the specified leakage requirements.
 - 5. Sealing and sliding surfaces shall provide a low coefficient of friction with the surface of the slide.
 - 6. Seals shall be field replaceable without removing gate from concrete or wall thimble.
 - 7. Anchor bolts shall not pass through the guides and seals. Seals must be removable without disassembly of the guide frame.
 - 8. Self-adjustable ultra-high molecular weight polyethylene seals. Adjustable seal systems and those with elastomeric seals on sides and top of gate are not acceptable, including systems such as "J-seals," "Bulb Seals," and "Crown Seals."
 - 9. Minimum seating surface width: 1 inch in contact with slide.
 - 10. Bottom seal:
 - a. Resilient neoprene or EPDM, minimum durometer of 45.
 - b. Attached to the bottom of the slide or embedded in gate frame invert.
 - 11. Side and top seals:
 - a. UHMWPE self-adjusting type seals, single extruded piece: Utilize a continuous silicon, EPDM, or nitrile compression cord to ensure contact between the seals and the slide. An O-ring is not acceptable in lieu of compression cord.
 - b. Side seals: Attach to frame using one of the following approaches:
 - 1) Face mounted gates.
 - a) One of the following:
 - (1) Side seal held in place between the front and back angles of the guide frame with Type 316 stainless steel bolts passing through the guide, seal, and frame along the length of the guide.
 - (2) Side seal held in place between front and back of a formed, 1-piece, rigid channel guide. Attach seals to frame using Type 316 stainless steel bolts.
 - b) Design and installation shall provide access to and removal of the bolt to allow removal of the side seal without removing the gate from the concrete.
 - 2) Embedded Gates:
 - a) Side seal held in place in the guide with 316 stainless steel bolt at the bottom of the guide. Arrangement with seals that are force fit or held in place with adhesives are unacceptable.

- b) Design and installation of seal shall provide access to and removal of the bolt to allow removal of the side seal without removing the gate from the concrete.
 - 3) Side seal held in place between front and back of a formed, one piece, rigid channel guide frame. Attach seals to frame using Type 316 stainless steel bolts.
 - c. Top Seal: UHMWPE self-adjusting type seal with compression cord.
- D. Stem:
 - 1. Type 316 stainless steel.
 - 2. Machine cut or rolled threads of the full depth Acme type, polished as necessary to a 16 micro inch finish or better.
 - 3. Stem couplings:
 - a. Silicon bronze or Type 316 stainless steel.
 - b. Threaded and keyed to stem or bolted to stem.
 - 4. Stem guides:
 - a. Type 316 stainless steel.
 - b. Split collar.
 - c. Adjustable in 2 directions.
 - d. Ultra-high molecular weight polyethylene bushing.
 - 5. Provide manganese bronze stop collar on stem above actuator.
 - 6. Drill and connect stem to slide structural sections with a minimum of two Type 316 stainless steel bolts for stem 1.75 inches and smaller. Large stems shall connect with a minimum of three type 316 stainless steel bolts or connect with a bolted stem coupling to a stub piece welded to at least two horizontal stiffeners.
 - 7. Coordinate the selection of the gate stem configuration with the gate operator and operating speed.
 - a. The selected gate stem configuration shall provide the most efficient combination of stem diameter/pitch/lead and keep the operating temperature at the stem nut to a minimum during operation.
 - b. For motorized applications, if the most efficient combination would result in any deviation from the gate operating rise rate specified in Section 13447 - Electric Actuators, base the bid price on the most efficient combination that meets the specified gate operating rise rate. However, submit proposed deviation along with cost deduct for evaluation and consideration by the Engineer.
- E. Stem covers:
 - 1. Fabricated from ultraviolet light resistant, clear butyrate or Lexan plastic pipe. Stem covers shall be capped on the upper end and either threaded into the top of the gate operators or held in place by stainless steel fasteners or bolt-down aluminum brackets.

2. For weir gates (gates indicated to be downward opening in the Slide Gate Schedule):
 - a. Staff gauges shall be adhesive-backed mylar, suitable for outdoor service. Staff gauges shall be calibrated in hundredths of feet and shall read out directly the weir crest elevation. The staff gauge range shall accommodate the full travel of the gate and elevations in increments of 0.25 feet and shall be indicated on each staff gauge. Staff gauges shall be supplemented with a stem-mounted pointer or indicator which shall permit direct observation of the weir gate crest elevation.
- F. AWWA Operating nut:
 1. Locate at operator level.
 2. Material: Manganese bronze.
- G. Gate operators: As indicated in the Schedule and as specified in Section 13447 - Electric Actuators.
 1. Equip manual operators and motorized operators with a AWWA 2-inch square nut to permit operation with a portable operator specified within this Specifications section.
- H. Coordination with electric actuator manufacturer (for gates with electric actuators):
 1. Gate manufacturer's responsibility shall extend to selection and sizing of the electric actuator for each gate based on:
 - a. Design seating and unseating head.
 - b. Open/close speed specified in Section 13447 - Electric Actuators.
 - c. Torque safety factor of 1.4, minimum, applied to the maximum torque requirement, including breakaway from seat.
 2. Preliminary sizes and model numbers of electric actuators for gates are estimated in Section 13447 - Electric Actuators.
 - a. Gate manufacturer shall verify, in writing, that the electric actuators are adequately sized.
 - b. In the event that a different size or model is required for any gate, notify the Engineer immediately.
 - c. Gate manufacturer shall advise Contractor of the proper selection and Contractor shall provide, at no additional cost, the proper operator.
 3. The gate supplier shall machine the stem nuts, provide proper mounting adaptation, and adaptation hardware to ensure adequate interface between the motorized operators and the slide gates.
 4. Electric actuator manufacturer shall provide template of bolt patterns to gate manufacturer to assure the correct mating of actuator to gate.
 5. Gate manufacturer shall provide a letter confirming that coordination with electric actuator manufacturer has been completed to assure compatibility and correct fit between gate and electric actuator.
- I. Pedestals and wall brackets: Type 316L stainless steel.

- J. Bolts, nuts, and fittings: Type 316 stainless steel.
- K. Anchor bolts: Type 316 stainless steel with a minimum diameter of ½ inch.

2.04 WALL THIMBLES

- A. Provide wall thimbles for gates where scheduled and as indicated on the Drawings.
- B. F-section of a depth equal to the thickness of the structure wall upon which the gate is mounted, unless otherwise specified or dimensioned on the Drawings:
 - 1. Modify F-sections where required for F-section and pipe bell ring connections in a wall.
 - a. Provide flange-by-mechanical joint or flange by push on joint wall thimble where ductile iron piping connects to the wall thimble. Ensure that joint wall thimble has sufficient embedment to resist pipe thrust.
 - b. Provide flange-by-bell joint where spigot connects to the wall thimble.
 - c. Provide flange-by-bell ring wall thimble insert where reinforced concrete piping connects to the wall thimble.
- C. Fabricated Type 316L stainless steel of sufficient section to resist permanent distortion; minimum 1/4 inch thick plate. Coordinate thickness of thimble with the plate thickness of the frame.
- D. Width of mounting flange of wall thimble: 1/2 inch wider than mounting flange of gate.
- E. Drill and tap the front flange to match the drilling on the flange back gate seat.
- F. Clearly mark top center of each thimble for installation.
- G. Provide Type 316 stainless steel studs for attaching the gate frame.
- H. Seal joint between thimble and gate watertight, in accordance with AWWA C561.
- I. To permit entrapped air to escape as the thimble is being cast in concrete, drill holes in each entrapment zone formed by ribs, flanges, and water stops.
- J. Provide annular weep ring to control seepage and resist thrust.
 - 1. Continuously weld weep ring to outside of the wall thimble. Stitch welding and sealing in between the welds is also acceptable.
 - 2. Weep ring shall be minimum 1/4 inch thick and minimum 2 inches deep.

2.05 GATE ENCLOSURES

- A. Completely enclose each self-contained gate to contain odors. Extend enclosure over floor opening. Unless specified otherwise in the gate schedule, bolt enclosure to floor with adjustable structural angle member.
- B. Supplied by the gate manufacturer.

- C. Features: As shown on the Drawings and as follows:
1. Removable without removing gate operator. Fabricated in 2 pieces with a joint in the same plane as the gate frame.
 2. Enclosure, including the mounting flange, shall not protrude more than 3 inches beyond the edge of the deck opening except where noted in the gate schedule. No additional bracing shall be allowed.
 3. Access openings: As required for maintenance.
 4. Seal opening at top of enclosure where operator mounting extension spool piece passes through the enclosure, or bolt operator on cover over head rails.
 5. Where gate extends through a concrete deck opening, bolt enclosure mounting flange to concrete on all four sides. Provide gasket between mounting flange and concrete deck.
 6. Where gate extends above an open channel, support enclosure from gate as follows:
 - a. Support enclosure from gate yoke and/or gate frame, and bolt bottom of enclosure to concrete deck with bottom flange. Provide gasket on bottom of to provide an air-tight seal between bottom of enclosure and deck.
 - b. Where enclosure sits on checker plate or where concrete deck slopes across gate, provide adjustable angle piece with slotted holes to allow vertical adjustment across bottom of enclosure. Provide gasket on bottom of adjustable angle piece to provide an air-tight seal between bottom of enclosure and checker plate.
- D. Material of construction: Type 316L Stainless steel plate. Stainless steel plate shall have a minimum thickness of 14 gauge and reinforced as necessary to reduce deflection on gates greater than 18 inches wide. Stiffeners shall be located at least 24 inches on center.

2.06 FINISHES

- A. Stainless steel:
1. Shot blast gates and wall thimbles after fabrication in accordance with ASTM A 380 to remove weld burn, slag, and splatter and to polish scratches.
 2. Clean the entire surface to produce an even color and sheen.
- B. Operators, stands, and other accessory equipment: Surface preparation, factory prime, field prime, and finish coats as specified in Section 09960 - High-Performance Coatings.

2.07 FABRICATION

- A. Shop assembly:
1. Gates shall be factory assembled, adjusted, and tested.
 2. Mount all accessories and appurtenances including, but not limited to, motor operators and limit switches so that the complete system can be tested at the factory.
 3. Electric actuators shall be shop tested by the original manufacturer prior to shipment.

2.08 SOURCE QUALITY CONTROL

- A. Factory testing:
 - 1. All gates shall be factory assembled and tested before shipment.
 - 2. Test equipment as specified in Section 15958 - Mechanical Equipment Testing.
 - a. Testing Level 1 General Equipment Performance Test.
 - 3. Operational testing:
 - a. Test functionality of all controls and limit switches.
 - b. Verify gate slide moves smoothly without binding.
 - 4. Leakage test:
 - a. Verify compliance with allowable leakage limits as specified.
 - b. Conduct water test in accordance with AWWA C561. Feeler gauge test is not acceptable.
 - 5. Test witnessing:
 - a. Witnessed: 3 gates as follows:
 - 1) One screen inlet gate (36 inches Width by 96 inches Height).
 - 2) One screen outlet gate (48 inches Width by 78 inches Height).
 - 3) One grit effluent bypass gate (36 inches Width by 36 inches Height).
 - b. Not witnessed: Remaining gates.
 - c. Provide Manufacturer's Certificate of Source Testing.

PART 3 EXECUTION

3.01 INSTALLATION

- A. Install as indicated on the Drawings and in strict conformance with the manufacturer's installation instructions, shop drawings, and recommendations.
- B. Mount thimbles and gates plumb in both vertical planes and level in horizontal plane.
- C. Coat seating surfaces between frame and wall thimble with a waterproof plastic compound or provide EPDM gasket prior to tightening frame studs.
- D. Adjust limit switches in electric actuators in accordance with manufacturer's instructions.
- E. Face mounted gates:
 - 1. Existing concrete:
 - a. Where a new gate replaces an existing gate in an existing structure, Contractor shall scan the existing concrete for rebar locations and coordinate with gate manufacturer to develop a plan for anchors (quantities and locations) needed for the new gate such that the new anchors do not cut existing rebar.
 - b. Where wall thimbles do not exist, mount gate to wall with anchors and provide a 1-inch non-shrink epoxy grout pad in accordance with manufacturer's recommendations.

- c. Where wall thimbles are existing, mount gate to existing wall thimbles or to the wall with anchors and provide a 1-inch epoxy grout pad in accordance with manufacturer's recommendations. Drill through existing wall thimbles if required to mount gate.
- 2. New concrete:
 - a. Where wall thimbles are not required in the Schedule, mount gate to wall with anchor bolts and provide a 1-inch non-shrink epoxy grout pad and in accordance with manufacturer's recommendations.
- F. Embedded gates:
 - 1. Provide blockouts in sidewalls and channel bottom for installation of gates.
 - 2. After gate placement, adjustment, and alignment in accordance with manufacturer's recommendations, grout frame with non-shrink grout.
- G. Electric Actuators:
 - 1. Orient electric actuators on gates to provide appropriate operation and maintenance access for all components of the unit.
- H. Floor mounted operators:
 - 1. In new concrete, attach floor mounted operators with anchor bolts.
 - 2. In existing concrete:
 - a. Contractor shall scan the concrete to identify location of existing reinforcing steel and coordinate anchor locations such that the new anchors do not cut existing rebar. Larger custom base plate may be required.
 - b. Core-drill to carefully remove existing anchors as needed for floor mounted operators being demolished.

3.02 FIELD QUALITY CONTROL

- A. As specified in Section 01756 - Commissioning and this Section and as follows.
- B. Testing:
 - 1. Leakage tests: Conduct in accordance with AWWA C561. Comply with specified allowable leakage limits.
 - 2. After gate installation and checking, run gates through at least 5 full cycles from the closed position to full-open position and back to the closed position.
 - 3. Verify functionality of all switches and controls.
 - 4. Provide Certificate of Installation and Functionality Compliance. Comply with Section 01756 - Commissioning.
- C. Manufacturer's field services:
 - 1. Contractor shall furnish services of a qualified manufacturer's representative experienced in installation and operation of the gates.
 - 2. Contractor shall coordinate field service work with the manufacturer's representative and City Representative prior to initiating field service work.

- D. Manufacturer's representative shall perform the services described below and as specified in Section 01756 - Commissioning. The specified durations are the minimum required time on the job site. Additional services and/or longer durations shall be provided as needed at no cost to the City to meet the required quality of work. Work to be done in a minimum of 3 trips:
1. Installation assistance: As required:
 - a. Advise/observe the Contractor on the installation of the equipment.
 - b. Check and verify that the installation of the gates is in accordance with the Drawings and manufacturer's installation instructions.
 - c. Provide additional assistance as required.
 2. Installation inspection: 1 trip; 3 workdays.
 3. Start-up/performance testing assistance: 1 trip; 3 workdays:
 - a. Prior to start-up, the equipment shall be inspected for proper alignment, operation, and satisfactory performance.
 - b. Provide additional start-up/testing assistance as required.
 - c. Provide Certificate of Installation and Functionality Compliance.
 4. Training: As defined in Operation and Maintenance Manuals and this Section:
 - a. Operations: 2-hour class, 2 sessions.
 - b. Mechanical Maintenance: 3-hour class, 1 session.
 - c. Combined Electrical and I&C Maintenance: 3-hour class, 1 session.
 5. Final acceptance checkout: 1 workday (trip may be combined with training).

3.03 SCHEDULE

- A. The Slide Gate Schedule is on the following pages. The Slide Gate Schedule is not a take-off list. Contractor shall provide additional gates per specifications and as indicated on the Drawings.

HEAVY-DUTY FABRICATED STAINLESS STEEL SLIDE GATE SCHEDULE											
Gate Tag Number	Dwg Number	Equipment Name	Opening Size W X H (inches)	Design Head, Seating and Unseating (feet) ⁽¹⁾	Type of Frame ⁽²⁾	Type of Closure ⁽³⁾	Gate Mounting ⁽⁴⁾	Type of Stem	Operating Head (feet) ⁽¹⁾⁽⁵⁾	Type of Operator ⁽⁶⁾	Remarks ⁽⁷⁾
✓ 02-GAT-1110 ✓ 02-GAT-1120	02-M-01-104	Screen Inlet Gate 1 Screen Inlet Gate 2	36x96	20	SC	FB	EMB	RS	12	MO/BS	(b)(c)
✓ 02-GAT-1130	02-M-01-104	Bypass Inlet Gate	36x72	20	SC	FB	EMB	RS	10	MOD/BS	(b)(c)
✓ 02-GAT-1115 ✓ 02-GAT-1125	02-M-01-104	Screen Outlet Gate 1 Screen Outlet Gate 2	48x78	20	SC	FB	EMB	RS	12	MO/BS	(b)(c)
✓ 02-GAT-1135	02-M-01-104	Bypass Outlet Gate	48x78	20	SC	FB	EMB	RS	12	MO/BS	(b)(c)
✓ 02-GAT-1140 ✓ 02-GAT-1141 ✓ 02-GAT-1142	02-M-01-104	Screen Channel Drain Gate 1 Screen Channel Drain Gate 2 Bypass Channel Drain Gate	6x6	20	NSC	FB	F x FLG	RS	13	CO/FS	
✓ 02-GAT-1210 ✓ 02-GAT-1220	02-M-01-107	Grit Inlet Gate 1 Grit inlet Gate 2	*45x78	20	SC	FB	EMB	RS	13	MO/BS	(b)(c)(l)
✓ 02-GAT-1215 ✓ 02-GAT-1225	02-M-01-107	Grit Outlet Gate 1 Grit Outlet Gate 2	*45x66	20	SC	FB	EMB	RS	13	MO/FS	(b)(c)(l)
✓ 03-GAT-1230	03-M-01-102	Grit Effluent Bypass Gate 1	36x36	20	SC	STD	F x FLG/ES	RS	13	MO/BS	(l)(m)
✓ 02-GAT-1240	02-M-01-107	Grit Effluent Bypass Gate 2	36x36	20	NSC	STD	F x FLG/ES	RS	13	MO/FS	(m)
✓ 03-FG-1110 ✓ 03-FG-1120 ✓ 03-FG-1130	03-M-01-103	Primary Inlet Gate 1 Primary Inlet Gate 2 Primary Inlet Gate 3	24x48	10	SC	STD	FM/ES	RS	5	MO/BS	(b)(f)
✓ 03-FG-1440 ✓ 03-FG-1445	03-M-01-104	Primary Effluent Outlet Gate 1 Primary Effluent Outlet Gate 2	48x48	20	SC	STD	WT/ES FM	RS	9	MO/BS	(b)(f)(k) (b)
✓ 03-FG-1495	03-M-01-104	Primary Effluent Overflow Isolation Gate	24x48	20	SC	STD	FM	RS	9	MO/BS	(b)
✓ 06-FG-1410 ✓ 06-FG-1420 ✓ 06-FG-1430	06-M-01-101	Secondary Clarifier Inlet Gate 1 Secondary Clarifier Inlet Gate 2 Secondary Clarifier Inlet Gate 3	30x72	20	SC	STD	FM/ES	RS	7.5	MO/BS	(f)
✓ 06-FG-1515 ✓ 06-FG-1525 ✓ 06-FG-1535	06-M-01-101	Secondary Clarifier Scum Well Gate 1 Secondary Clarifier Scum Well Gate 2 Secondary Clarifier Scum Well Gate 3	12" DIA	22	FL	STD	F x FLG FLG ^{AD6}	RS	20	CO	(f)(n) ^{AD6}
✓ 06-FG-1610 ✓ 06-FG-1620 ✓ 06-FG-1630	06-M-01-101	Secondary Clarifier RSS Well Gate 1 Secondary Clarifier RSS Well Gate 2 Secondary Clarifier RSS Well Gate 3	12" DIA	12	FL	STD	F x FLG FLG ^{AD6}	RS	10	CO	(f)(n) ^{AD6}
✓ 09-FG-1100	09-M-01-101	Chlorine Contact Tank Inlet Gate	16" DIA	10	SC	STD	FM/ES	RS	7	CO/BS	(f) ^{AD4}
Notes: (1) Gate design head and operating head applied at centerline of gate. (2) Frame: SC = Self-Contained; NSC = Non-Self Contained; F = Flatback; FL = Flange back. (3) Closure: DO = Downward Opening; FB = Flush Bottom; STD = Standard; FM = Mount on Existing Floor. For FB closure, see Typical Details P718, P720, S900/TYP for installation details. (4) Mounting: FM = Face Mounted; ES = Existing Structure; SM = Surface Mount on Channel Wall; EMB = Embedded in Channel Wall; SP = Spigot back; Wall Thimble: FWT = "F" Wall Thimble. F x FLG = F Thimble with Flange. (5) Head to be used to size gate operator.											

HEAVY-DUTY FABRICATED STAINLESS STEEL SLIDE GATE SCHEDULE											
Gate Tag Number	Dwg Number	Equipment Name	Opening Size W X H (inches)	Design Head, Seating and Unseating (feet) ⁽¹⁾	Type of Frame ⁽²⁾	Type of Closure ⁽³⁾	Gate Mounting ⁽⁴⁾	Type of Stem	Operating Head (feet) ⁽¹⁾⁽⁵⁾	Type of Operator ⁽⁶⁾	Remarks ⁽⁷⁾
(6) Operator: CO = Hand crank operator with 2-inch AWWA nut for portable operator; HW = Handwheel; HC = Hand crank; MO = Motor Operator; MOD = Modulating Motor Operator; HO = Hydraulic Operator; MHO = Manual Hydraulic Operator (Hand Pump); BS = Bench Stand; FS = Floor Stand; IFS = Interconnect Floor Stand; PS = Pedestal Support on Cantilevered Wall Bracket. (7) Top seals are required unless indicated otherwise. Remarks: (a) Provide yoke-mounted pedestal (torque tube) to transfer thrust from operator to gate frame. (b) Provide SST gate enclosure per the Specifications. (c) Top seals are not required. (d) Provide 3'-10" long "F" wall thimble, 5/8 inch minimum thickness. Include inlet vane in "F" wall thimble as indicated on Drawings. (e) Locate stem on backside (i.e., downstream) of slide. (f) Located at existing structure. Field verify opening size dimensions. (g) One side of frame is face mounted and one side of frame is surface mounted in existing channel. (h) Opening size width dimension is channel width. Gate opening width will be less due to side mounted frame. (i) Create blockouts in existing structure per S900/TYP. (j) Confirm side frame width to provide sufficient edge distance and embedment for anchor bolts to accommodate angled channel wall as shown on the Drawings. (k) Mount gate to existing wall thimble or to the wall with anchors and provide a 1 inch epoxy grout pad in accordance with manufacturer's recommendations. Drill through existing wall thimbles if required to mount gate. (l) Contractor shall confirm grit inlet channel size and grit inlet gate width (*) based on grit basin manufacturer selected. (m) Provide F x FLG thimble minimum 1/2-inch wall thickness. Extend thimble past wall. Refer to drawings for required length. Top seals are not required. ^{AD6} (m)(n)Field verify pipe size, bolt size, and bolt pattern prior to submitting shop drawings. ^{AD6}											

END OF SECTION

AD6

Addendum No. 6 - July 2024

AD4

Addendum No. 4 - June 2024

Whipps, inc.

5/21/25

P.O. Box 1058 • 370 South Athol Rd.
Athol, Massachusetts 01331
Phone: (978) 249-7924
Fax: (978) 249-3072

Nan, Inc.
636 Laumaka St.
Honolulu, HI 96819

Attn: Jason Tagawa

Subject: Hilo, HI WWTF Phase 1
Your P.O. No.: 24077-00044
Our Ref. # 32752

Mr. Tagawa:

Please find enclosed the following shop drawings and submittal information for the above-mentioned project for your review and approval.

Certificate of Unit Responsibility

A-327-520	Stainless Steel Slide Gate Material & Engineering Standards
A-327-521	Gate Design Calculations (3 Pages)
A-327-522	Gate Design Calculations (3 Pages)
A-327-523	Gate Design Calculations (3 Pages)
A-327-524	Gate Design Calculations (3 Pages)
A-327-525	Gate Design Calculations (3 Pages)
A-327-526	Gate Design Calculations (3 Pages)
A-327-527	Gate Design Calculations (3 Pages)
A-327-528	Gate Design Calculations (3 Pages)
A-327-529	Gate Design Calculations (3 Pages)
A-327-5201	Gate Design Calculations (3 Pages)
A-327-5202	Gate Design Calculations (3 Pages)
A-327-5203	Gate Design Calculations (3 Pages)
A-327-5204	Gate Design Calculations (2 Pages)
A-327-5205	Gate Design Calculations (3 Pages)
A-327-5206	Gate Design Calculations (3 Pages)
A-327-5207	Gate Design Calculations (3 Pages)
BC-921-2609	Series 921 S.S. Slide Gate Install. 36" x 96"
BC-921-2609-S2	Series 921 Slide Gate Sections
BC-921-2609-S3	Adapter Plate Detail
BC-921-2609-S4	Odor Control Covers
BC-921-2610	Series 921 S.S. Slide Gate Install. 36" x 72"
BC-921-2610-S2	Series 921 Slide Gate Sections
BC-921-2610-S3	Adapter Plate Detail
BC-921-2610-S4	Odor Control Covers
BC-921-2611	Series 921 S.S. Slide Gate Install. 48" x 78"
BC-921-2611-S2	Series 921 Slide Gate Sections
BC-921-2611-S3	Adapter Plate Detail
BC-921-2611-S4	Odor Control Covers
BC-921-2611-S5	Stem Guide Detail
BC-921-2612	Series 921 S.S. Slide Gate Install. 48" x 78"
BC-921-2612-S2	Series 921 Slide Gate Sections
BC-921-2612-S3	Adapter Plate Detail
BC-921-2612-S4	Odor Control Covers
BC-921-2612-S5	Stem Guide Detail
BC-921-2613	Series 921 S.S. Slide Gate Install. 45" x 78"
BC-921-2613-S2	Series 921 Slide Gate Sections
BC-921-2613-S3	Adapter Plate Detail
BC-921-2613-S4	Odor Control Covers

(over)

BC-921-2614	Series 921 S.S. Slide Gate Install. 45" x 66"
BC-921-2614-S2	Series 921 Slide Gate Sections
BC-921-2614-S3	Adapter Plate Detail
BC-921-2614-S4	Odor Control Covers
BC-921-2614-S5	Stem Guide Detail
BC-923-3269	Series 923 S.S. Slide Gate Install. 24" x 48"
BC-923-3269-S2	Series 923 Slide Gate Sections
BC-923-3269-S3	Adapter Plate Detail
BC-923-3269-S4	Odor Control Covers
BC-924-3115	Series 924 S.S. Slide Gate Install. 24" x 48"
BC-924-3115-S2	Series 924 Slide Gate Sections
BC-924-3115-S3	Adapter Plate Detail
BC-924-3116	Series 924 S.S. Slide Gate Install. 48" x 48"
BC-924-3116-S2	Series 924 Slide Gate Sections
BC-924-3116-S3	Adapter Plate Detail
BC-924-3116-S4	Odor Control Covers
BC-924-3116-S5	48" x 48" S.S. Wall Thimble Detail
BC-924-3117	Series 924 S.S. Slide Gate Install. 24" x 48"
BC-924-3117-S2	Series 924 Slide Gate Sections
BC-924-3117-S3	Adapter Plate Detail
BC-924-3117-S4	Odor Control Covers
BC-924-3117-S5	24" x 48" S.S. Wall Thimble Detail
BC-924-3118	Series 924 S.S. Slide Gate Install. 30" x 72"
BC-924-3118-S2	Series 924 Slide Gate Sections
BC-924-3118-S3	Adapter Plate Detail
BC-924-3119	Series 924 S.S. Slide Gate Install. 16" Dia.
BC-924-3119-S2	Series 924 Slide Gate Sections
BC-924-3119-S3	Series 924 Flange-Mounted Frame Detail
BC-925-3656	Series 925 S.S. Slide Gate Install. 6" Dia.
BC-925-3656-S2	Series 925 Slide Gate Sections
BC-925-3656-S3	Series 924 Flange-Mounted Frame Detail
BC-925-3657	Series 925 S.S. Slide Gate Install. 36" x 36"
BC-925-3657-S2	Series 925 Slide Gate Sections
BC-925-3657-S3	36" x 36" Wall Thimble Detail
BC-925-3658	Series 925 S.S. Slide Gate Install. 12" Dia.
BC-925-3658-S2	Series 925 Slide Gate Sections
BC-925-3658-S3	Series 924 Flange-Mounted Frame Detail
BC-925-3658-S4	Operator Support Bracket Detail
BC-925-3659	Series 925 S.S. Slide Gate Install. 12" Dia.
BC-925-3659-S2	Series 925 Slide Gate Sections
BC-925-3659-S3	Series 924 Flange-Mounted Frame Detail
ST-100-068-A	Fabricated Floor-Mounted Pedestal
ST-100-082	Fabricated Floor-Mounted Pedestal
ST-100-050	Stem Guide Detail
ST-200-013	Stem Guide Detail
ST-100-071	Adapter Plate Detail
AC-200-490	Non-Rising Stem Thrust Nut
ST-100-2012	T-Wrench Detail
AC-102-016-S	S.S. Hand Crank Assembly
BC-101-164	Type 101 Manual Operator for Non-Rising Stem
BC-102-263A	Type 102 Manual Operator
Adhesive Anchors Recommendations Sheet	
Whipps O & M Installation Instructions (14 Pages)	
Procedure 043-01	Shop Leakage Test (4 Pages)
Procedure 043-03	Shop Performance/Operational Test (4 Pages)
Field Leakage Test Procedure (2 Pages)	

Please note that the above-listed proposed equipment is submitted as our best interpretation of the latest versions of available contract documents. **It is Whipps, Inc. policy that all equipment is fabricated per approved submittal drawings only. Verification of all elevations and dimensions relating to any existing and/or proposed structures is the responsibility of the reviewing engineer and the contractor.** We ask that this careful verification be provided to ensure that the equipment submitted herein is properly fabricated to suit the desired locations.

Please also find enclosed the following electric actuator submittal information for the above-mentioned project for your review and approval.

Flowserve Customer Data Sheets (30 Pages)

03-653-0001	Limitorque MX-05 Actuator Dimensional Dwg.
03-653-0003	Limitorque MX-10 Actuator Dimensional Dwg.
03-653-0006	Limitorque MX-20 Actuator Dimensional Dwg.
03-654-0094	Limitorque MX/V1 Actuator Dimensional Dwg.
03-654-0096	Limitorque MX/V3 Actuator Dimensional Dwg.
03-615-0002-1	MX-05 Base Details
03-615-0005-1	MX-10 Base Details
03-615-0008-1	MX-05 Base Details
WD-MXb-041100-1101	Wiring Diagram
WD-MXb-141110-1101	Wiring Diagram

Please carefully review the information submitted herein and verify its compatibility with the job specification, as well as with any existing and or proposed onsite wiring requirements.

Please contact us at your earliest convenience should you have any questions or concerns during your review of this material.

Sincerely,
Whipps, Inc.



Steve Barnes

Cc: Promark Corp.

Whipps, inc.

P.O. Box 1058 • 370 South Athol Rd.
Athol, Massachusetts 01331
Phone: (978) 249-7924
Fax: (978) 249-3072

5/15/25

Nan, Inc.
636 Laumaka St.
Honolulu, HI 96819

Subject:	Hilo WWTF Phase 1
Your P.O. No.:	24077-00044
Our Ref. #	32752

Certificate of Unit Responsibility

Please accept this as certification that the slide gate installations submitted herein, proposed for service in the above-mentioned project shall be supplied by Whipps, Inc. All components of the proposed equipment shall be wholly designed and fabricated by Whipps and designed to function as a complete unit as shown, with use of the proposed Limitorque electric actuators. Flowserve (Limitorque parent company) was provided all operating requirements and stem sizes of the gates, which they utilized to propose appropriate model actuators with adequate capacities to safely operate the gates.



Steve Barnes
Project Manager



Date

ENGINEERING STANDARDS	Whipps, inc. 370 South Athol Rd, Athol MA 01331	DATE 5/14/25	DRAWN BY KS																																		
STAINLESS STEEL SLIDE GATE		SS.O. NUMBER 327-520																																			
		DRAWING NUMBER A-327-520	REV A																																		
GATE DESIGN TO AWWA STANDARD C561-21 WELDING TO AWS D1.6																																					
SLIDE																																					
• MIN THICKNESS 1/4" ALL PARTS • MAX BENDING STRESS 15,000 PSI • MAX BENDING DEFLECTION < 1/720 OF SPAN																																					
GATE FRAME																																					
• MIN THICKNESS 1/4" ALL PARTS																																					
STEM																																					
• MAX UNSUPPORTED STEM LENGTH <=200 L/r BASED ON PITCH DIAMETER																																					
HOIST																																					
• MAX 40# PULL @ SPECIFIED OPERATING HEAD																																					
MATERIAL STANDARDS																																					
<table><tr><td>SLIDE PLATE</td><td>ASTM A240 STAINLESS STEEL TYPE 316L</td></tr><tr><td>SLIDE STIFFENERS</td><td>ASTM A240 STAINLESS STEEL TYPE 316L</td></tr><tr><td>SLIDE STEM CONNECTOR</td><td>ASTM A240 STAINLESS STEEL TYPE 316L</td></tr><tr><td>FRAME GUIDE</td><td>ASTM A240 STAINLESS STEEL TYPE 316L</td></tr><tr><td>FRAME INVERT</td><td>ASTM A240 STAINLESS STEEL TYPE 316L</td></tr><tr><td>FRAME SEALS</td><td>ASTM D4020 UHMWPE</td></tr><tr><td>INVERT SEAL</td><td>ASTM D2000 55-60 NEOPRENE</td></tr><tr><td>STEM</td><td>ASTM A276 STAINLESS STEEL TYPE 316</td></tr><tr><td>STEM COVER</td><td>ASTM D3935 POLYCARBONATE W/ CAP & VENTS</td></tr><tr><td>STOP COLLAR</td><td>ALUMINUM 6061-T6</td></tr><tr><td>PEDESTAL</td><td>ASTM A240 STAINLESS STEEL TYPE 316L</td></tr><tr><td>THRUST NUT</td><td>MANGANESE BRONZE</td></tr><tr><td>STOP COLLAR</td><td>MANGANESE BRONZE</td></tr><tr><td>ADAPTER PLATE</td><td>ALUMINUM 6061-T6</td></tr><tr><td>STEM GUIDE</td><td>ASTM D4020 UHMWPE</td></tr><tr><td>STEM GUIDE BRACKET</td><td>ASTM A240 STAINLESS STEEL TYPE 316L</td></tr><tr><td>FASTENERS/ANCHORS</td><td>ASTM A276 STAINLESS STEEL TYPE 316</td></tr></table>				SLIDE PLATE	ASTM A240 STAINLESS STEEL TYPE 316L	SLIDE STIFFENERS	ASTM A240 STAINLESS STEEL TYPE 316L	SLIDE STEM CONNECTOR	ASTM A240 STAINLESS STEEL TYPE 316L	FRAME GUIDE	ASTM A240 STAINLESS STEEL TYPE 316L	FRAME INVERT	ASTM A240 STAINLESS STEEL TYPE 316L	FRAME SEALS	ASTM D4020 UHMWPE	INVERT SEAL	ASTM D2000 55-60 NEOPRENE	STEM	ASTM A276 STAINLESS STEEL TYPE 316	STEM COVER	ASTM D3935 POLYCARBONATE W/ CAP & VENTS	STOP COLLAR	ALUMINUM 6061-T6	PEDESTAL	ASTM A240 STAINLESS STEEL TYPE 316L	THRUST NUT	MANGANESE BRONZE	STOP COLLAR	MANGANESE BRONZE	ADAPTER PLATE	ALUMINUM 6061-T6	STEM GUIDE	ASTM D4020 UHMWPE	STEM GUIDE BRACKET	ASTM A240 STAINLESS STEEL TYPE 316L	FASTENERS/ANCHORS	ASTM A276 STAINLESS STEEL TYPE 316
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WHIPPS INC MANUFACTURES ALL STAINLESS STEEL FORMED SECTIONS FROM ASTM A240 PLATE. ASTM A276 RELATES TO STAINLESS STEEL BAR (OPERATING STEM) & TUBE SECTIONS ONLY																																					

CALCULATION SHEET

Sheet No. A-327-521
Project: Hilo WWTF Phase 1
Whipps #: 32752
Gate Ref: 02-GAT-1110 & 1120

Prepared By: S. Barnes
5/15/2025

Calculations

1. Allowable Leakage Rates
2. Slide design
3. Stem Design
4. Yoke Design
5. Loads transferred to Structure

1. Leakage Rates

Gate Aperture Width = 36.0 in
Gate Aperture Depth = 96.0 in
Seal Periphery = 228.0 in
Allowable Seating Leakage rate = 0.05 gal/min/ft of seal periphery
Allowable Seating Leakage = 0.95 gal/min

Allowable Unseating Seating Leakage rate = 0.1 gal/min/ft of seal periphery
Allowable Unseating Seating Leakage = 0.95 gal/min

2. Slide Design

Design Head on gate Invert = H = 24 ft. = 288 in
Aperture Width = L = 36 in
Density of Water = ρ = 62 lb/cu.ft = 0.0361 lb/cu.in
Slide section depth (Beam Width) = d = 16 in
Slide section Moment of Inertia = I = 4.1 in⁴
Slide – furthest fiber distance = y = 2.2 in
Material Modulus = E = 2.80E+07 psi (Stainless Steel Grade 316)

Assumed Slide section is a simply supported beam under uniform load (Roark's p.104 – 2.e)

$$W_a = \text{Load per unit Length} = \left(H - \frac{d}{2}\right) \cdot \rho \cdot d = 164.2 \text{ lb/in}$$

$$\text{Deflection} = \delta = \frac{5 \cdot W_a \cdot l^4}{384 \cdot E \cdot I} = 0.032 \text{ in} \quad \text{Acceptable, Less than 0.036 in}$$

$$\text{Deflection Ratio} = l / \delta = 1: 1142$$

$$\text{Bending Stress} = \sigma = \frac{W_a \cdot l^2 \cdot y}{8 \cdot I} = 14,337 \text{ psi}$$

Acceptable

Less than 15000 psi

R.F of 5.2 On Ultimate Stress

R.F of 2.4 On Yield Stress

Note: RF refers to Reserve factor (or safety factor)



$$\text{Vertical Shear through Stiffener section} = V = \frac{\rho \cdot H \cdot d \cdot l}{2}$$

$$\text{Area of slide plate section} = A' = 4.06 \text{ sq.in}$$

$$\begin{aligned} \text{Distance from slide plate neutral axis to combined section neutral axis} &= z' = 0.432 \text{ in} \\ \text{Effective width of Weld} &= b = 0.2192 \text{ in} \quad (\text{Based on a } 0.31" \text{ weld}) \end{aligned}$$

$$\text{Shear Flow} = \tau = \frac{V \cdot A' \cdot z'}{I} = 1310.2 \text{ lb/in}$$

$$\text{Weld Shear stress} = \tau \text{ based on skip weld, 2 on 6} = 8,967 \text{ psi} \quad \begin{array}{l} \text{Acceptable} \\ \text{Less than 10000 psi} \end{array}$$

3. Stem Design

$$\text{Stem Type} = 2.25" \text{ Rolled 3tpi}$$

$$\text{Effective Stem Diameter} = D = 2.088 \text{ in} \quad (\text{minor diameter, per AWWA C561})$$

$$\text{Stem Radius of Gyration} = r = 0.522 \text{ in}$$

$$\text{Stem Moment of Inertia} = I = 0.932 \text{ in}^4$$

$$\text{Stem Modulus Of Elasticity} = E = 2.80\text{E}+07 \text{ psi}$$

$$\text{Stem Max Free Length} = L = 96 \text{ in}$$

$$L/r \text{ ratio} = 184 \quad \text{Acceptable not greater than 200}$$

$$\text{Weight of Slide \& Stem} = W = 580 \text{ lbs.}$$

$$\text{Area of Gate} = A = 24.00 \text{ sq.ft}$$

$$\text{Operating Head to gate c/l} = H_c = 12.00 \text{ ft.}$$

$$\text{Seal Friction Coefficient} = \mu = 0.2$$

$$\text{Slide Lifting Force} = \text{SLF} = (62.4 \times \mu \times A \times H_c) = 3,594 \text{ lbs.}$$

$$\text{Seal Drag (10 lb/in of slide height)} = 960 \text{ lbs.}$$

$$\text{Design Lifting Force} = (\text{Greater of Seal Drag or SLF}) + W = F = 4,174 \text{ lbs.}$$

$$\text{Electric Operator Run motor Torque} = \text{Tr} = 427 \text{ ft.-lbs.}$$

$$\text{Stem Factor} = \text{Sf} = 0.027$$

$$\text{Thrust @ Run Torque} = \text{Tr/Sf} = 15,574 \text{ lbs.} \quad \text{Acceptable, greater than F}$$

$$\text{Electric Operator Stalled motor Torque} = T = 643 \text{ ft.lb}$$

$$\text{Stem factor} = \text{Sf} = 0.027$$

$$\text{Design Torque} = \text{Td} = 1.25 \times T = 803.8 \text{ ft.lb}$$

$$\text{Thrust @ Design Torque} = \text{Th} = \text{Td} / \text{Sf} = 29,315 \text{ lbs.}$$

$$\text{Direct stress at Design Thrust} = \sigma = \frac{\text{Th} \cdot 4}{\pi \cdot D^2} = 8,565 \text{ psi}$$

$$\text{Shear Stress at Design Thrust} = \tau = \frac{192 \cdot \text{Td}}{\pi \cdot D^3} = 5,400 \text{ psi}$$

$$\text{Combined stress} = \sigma / 2 + \sqrt{(\sigma / 2)^2 + \tau^2} = 11,175 \text{ psi} \quad \text{Acceptable, < Yield Stress (35000 psi)}$$

$$\text{Stem Buckling Load (C=2)} = \frac{2 \cdot \pi^2 \cdot E \cdot I}{l^2} = 55,901 \text{ lbs.} \quad \text{Acceptable, Greater than Th}$$

4. Yoke Design

Assumed Yoke acts as a simply supported beam under central Point Load (Roark's p.101 – 1.e)

Yoke Type = Double 6 in x 3 in Channel sections

Yoke Length = L = Aperture Width = 36 in

Yoke section Moment of Inertia = I = 31.74 in⁴

Material Modulus = E = 2.80E+07 psi (Stainless Steel Grade 316)

Yoke – neutral axis to furthest fiber = y = 3.00 in

At Design Lifting Force (F from section 3)

$$\text{Deflection} = \delta_{Lf} = \frac{F \bullet L^3}{48 \bullet E \bullet I} = 0.005 \text{ in}$$

Deflection Ratio = l / δ = **1: 7887** **Acceptable, Less than 1:360 (AWWA C.561)**

$$\text{Max bending moment at design} = M_{Lf} = \frac{F \bullet L}{4} = 37,568 \text{ in lbs.}$$

$$\text{Max Bending Stress} = \sigma_{Lf} = \frac{M \bullet y}{I} = 3,550 \text{ psi} \quad \textbf{Acceptable Less than 15000 psi}$$

At Electric Operator Stalled motor Torque (T from section 3)

$$\text{Max bending moment} = M_{St} = \frac{T \bullet L}{Sf \bullet 4} = 211,070 \text{ in lbs.}$$

$$\text{Max Bending Stress} = \sigma_{St} = \frac{M \bullet y}{I} = 19,947 \text{ psi}$$

Reserve Factor On Yield Stress = 1.755 **Acceptable, Greater than 1.5 (AWWA C.561)**

$$\text{Deflection (at Stall)} = \frac{T \bullet L^3}{Sf \bullet 48 \bullet E \bullet I} = 0.026 \text{ in} \quad \textbf{Acceptable Less than 0.1 in}$$

5. Loads transferred to Structure

Self contained gates do not transmit Operating loads to the mounting structure.

Loads transferred are those due to the Operating Head (Hc from section 3) acting on the gate area

$$\text{Water load} = (62.4 \times A \times Hc) = 17,971 \text{ lbs.}$$

CALCULATION SHEET

Calc Self Cont STD Electric
D. Postema February 8, 2018
Master Doc.: F030-20 Rev.1]

Sheet No. A-327-522
Project: Hilo WWTF Phase 1
Whipps #: 32752
Gate Ref: 02-GAT-1130

Prepared By: S. Barnes
5/15/2025

Calculations

1. Allowable Leakage Rates
2. Slide design
3. Stem Design
4. Yoke Design
5. Loads transferred to Structure

1. Leakage Rates

Gate Aperture Width = 36.0 in
Gate Aperture Depth = 72.0 in
Seal Periphery = 180.0 in
Allowable Seating Leakage rate = 0.05 gal/min/ft of seal periphery
Allowable Seating Leakage = 0.75 gal/min

Allowable Unseating Seating Leakage rate = 0.1 gal/min/ft of seal periphery
Allowable Unseating Seating Leakage = 0.75 gal/min

2. Slide Design

Design Head on gate Invert = H = 23 ft. = 276 in
Aperture Width = L = 36 in
Density of Water = ρ = 62 lb/cu.ft = 0.0361 lb/cu.in
Slide section depth (Beam Width) = d = 12 in
Slide section Moment of Inertia = I = 3.8 in⁴
Slide – furthest fiber distance = y = 2.1 in
Material Modulus = E = 2.80E+07 psi (Stainless Steel Grade 316)

Assumed Slide section is a simply supported beam under uniform load (Roark's p.104 – 2.e)

$$W_a = \text{Load per unit Length} = \left(H - \frac{d}{2}\right) \cdot \rho \cdot d = 117.0 \text{ lb/in}$$

$$\text{Deflection} = \delta = \frac{5 \cdot W_a \cdot l^4}{384 \cdot E \cdot I} = 0.024 \text{ in} \quad \text{Acceptable, Less than 0.036 in}$$

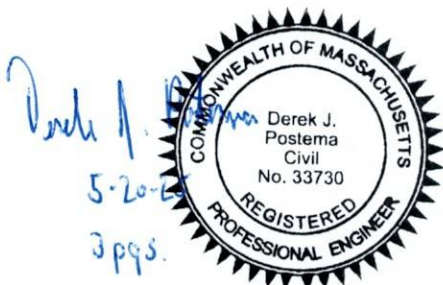
$$\text{Deflection Ratio} = l / \delta = 1: 1504$$

$$\text{Bending Stress} = \sigma = \frac{W_a \cdot l^2 \cdot y}{8 \cdot I} = 10,349 \text{ psi}$$

Acceptable
Less than 15000 psi

R.F of 7.2 On Ultimate Stress
R.F of 3.4 On Yield Stress

Note: RF refers to Reserve factor (or safety factor)



$$\text{Vertical Shear through Stiffener section} = V = \frac{\rho \cdot H \cdot d \cdot l}{2}$$

$$\text{Area of slide plate section} = A' = 3 \text{ sq.in}$$

$$\begin{aligned} \text{Distance from slide plate neutral axis to combined section neutral axis} &= z' = 0.541 \text{ in} \\ \text{Effective width of Weld} &= b = 0.1768 \text{ in} \quad (\text{Based on a 0.25" weld}) \end{aligned}$$

$$\text{Shear Flow} = \tau = \frac{V \cdot A' \cdot z'}{I} = 915.5 \text{ lb/in}$$

$$\text{Weld Shear stress} = \tau \text{ based on skip weld, 2 on 6} = 7,770 \text{ psi} \quad \begin{matrix} \text{Acceptable} \\ \text{Less than 10000 psi} \end{matrix}$$

3. Stem Design

$$\text{Stem Type} = 1.75" \text{ Rolled 4tpi}$$

$$\text{Effective Stem Diameter} = D = 1.625 \text{ in} \quad (\text{minor diameter, per AWWA C561})$$

$$\text{Stem Radius of Gyration} = r = 0.406 \text{ in}$$

$$\text{Stem Moment of Inertia} = I = 0.342 \text{ in}^4$$

$$\text{Stem Modulus Of Elasticity} = E = 2.80\text{E}+07 \text{ psi}$$

$$\text{Stem Max Free Length} = L = 72 \text{ in}$$

$$L/r \text{ ratio} = 177 \quad \text{Acceptable not greater than 200}$$

$$\text{Weight of Slide \& Stem} = W = 380 \text{ lbs.}$$

$$\text{Area of Gate} = A = 18.00 \text{ sq.ft}$$

$$\text{Operating Head to gate c/l} = H_c = 10.00 \text{ ft.}$$

$$\text{Seal Friction Coefficient} = \mu = 0.2$$

$$\text{Slide Lifting Force} = \text{SLF} = (62.4 \times \mu \times A \times H_c) = 2,246 \text{ lbs.}$$

$$\text{Seal Drag (10 lb/in of slide height)} = 720 \text{ lbs.}$$

$$\text{Design Lifting Force} = (\text{Greater of Seal Drag or SLF}) + W = F = 2,626 \text{ lbs.}$$

$$\text{Electric Operator Run motor Torque} = \text{Tr} = 157 \text{ ft.-lbs.}$$

$$\text{Stem Factor} = \text{Sf} = 0.021$$

$$\text{Thrust @ Run Torque} = \text{Tr/Sf} = 7,452 \text{ lbs.} \quad \text{Acceptable, greater than F}$$

$$\text{Electric Operator Stalled motor Torque} = T = 338 \text{ ft.lb}$$

$$\text{Stem factor} = \text{Sf} = 0.021$$

$$\text{Design Torque} = T_d = 1.25 \times T = 422.5 \text{ ft.lb}$$

$$\text{Thrust @ Design Torque} = T_h = T_d / \text{Sf} = 20,055 \text{ lbs.}$$

$$\text{Direct stress at Design Thrust} = \sigma = \frac{T_h \cdot 4}{\pi \cdot D^2} = 9,670 \text{ psi}$$

$$\text{Shear Stress at Design Thrust} = \tau = \frac{192 \cdot T_d}{\pi \cdot D^3} = 6,018 \text{ psi}$$

$$\text{Combined stress} = \sigma / 2 + \sqrt{(\sigma / 2)^2 + \tau^2} = 12,554 \text{ psi} \quad \text{Acceptable, < Yield Stress (35000 psi)}$$

$$\text{Stem Buckling Load (C=2)} = \frac{2 \cdot \pi^2 \cdot E \cdot I}{l^2} = 36,493 \text{ lbs.} \quad \text{Acceptable, Greater than Th}$$

4. Yoke Design

Assumed Yoke acts as a simply supported beam under central Point Load (Roark's p.101 – 1.e)

Yoke Type = Double 6 in x 3 in Channel sections

Yoke Length = L = Aperture Width = 36 in

Yoke section Moment of Inertia = I = 31.74 in⁴

Material Modulus = E = 2.80E+07 psi (Stainless Steel Grade 316)

Yoke – neutral axis to furthest fiber = y = 3.00 in

At Design Lifting Force (F from section 3)

$$\text{Deflection} = \delta_{Lf} = \frac{F \bullet L^3}{48 \bullet E \bullet I} = 0.003 \text{ in}$$

Deflection Ratio = l / δ = **1: 12534** **Acceptable, Less than 1:360 (AWWA C.561)**

$$\text{Max bending moment at design} = M_{Lf} = \frac{F \bullet L}{4} = 23,638 \text{ in lbs.}$$

$$\text{Max Bending Stress} = \sigma_{Lf} = \frac{M \bullet y}{I} = 2,234 \text{ psi} \quad \textbf{Acceptable Less than 15000 psi}$$

At Electric Operator Stalled motor Torque (T from section 3)

$$\text{Max bending moment} = M_{St} = \frac{T \bullet L}{Sf \bullet 4} = 144,393 \text{ in lbs.}$$

$$\text{Max Bending Stress} = \sigma_{St} = \frac{M \bullet y}{I} = 13,646 \text{ psi}$$

Reserve Factor On Yield Stress = 2.565 **Acceptable, Greater than 1.5 (AWWA C.561)**

$$\text{Deflection (at Stall)} = \frac{T \bullet L^3}{Sf \bullet 48 \bullet E \bullet I} = 0.018 \text{ in} \quad \textbf{Acceptable Less than 0.1 in}$$

5. Loads transferred to Structure

Self contained gates do not transmit Operating loads to the mounting structure.

Loads transferred are those due to the Operating Head (Hc from section 3) acting on the gate area

$$\text{Water load} = (62.4 \times A \times Hc) = 11,232 \text{ lbs.}$$

CALCULATION SHEET

Calc Self Cont STD Electric
D. Postema February 8, 2018
Master Doc.: F030-20 Rev.1]

Sheet No. A-327-523
Project: Hilo WWTF Phase 1
Whipps #: 32752
Gate Ref: 02-GAT-1115 & 1125

Prepared By: S. Barnes
5/15/2025

Calculations

1. Allowable Leakage Rates
2. Slide design
3. Stem Design
4. Yoke Design
5. Loads transferred to Structure

1. Leakage Rates

Gate Aperture Width = 48.0 in
Gate Aperture Depth = 78.0 in
Seal Periphery = 204.0 in
Allowable Seating Leakage rate = 0.05 gal/min/ft of seal periphery
Allowable Seating Leakage = 0.85 gal/min

Allowable Unseating Seating Leakage rate = 0.05 gal/min/ft of seal periphery
Allowable Unseating Seating Leakage = 0.85 gal/min

2. Slide Design

Design Head on gate Invert = H = 23.3 ft. = 279 in
Aperture Width = L = 48 in
Density of Water = ρ = 62.4 lb/cu.ft = 0.0361 lb/cu.in
Slide section depth (Beam Width) = d = 13 in
Slide section Moment of Inertia = I = 10.9 in⁴
Slide – furthest fiber distance = y = 3.22 in
Material Modulus = E = 2.80E+07 psi (Stainless Steel Grade 316)

Assumed Slide section is a simply supported beam under uniform load (Roark's p.104 – 2.e)

$$W_a = \text{Load per unit Length} = \left(H - \frac{d}{2}\right) \cdot \rho \cdot d = 127.9 \text{ lb/in}$$

$$\text{Deflection} = \delta = \frac{5 \cdot W_a \cdot l^4}{384 \cdot E \cdot I} = 0.029 \text{ in} \quad \text{Acceptable, Less than 0.048 in}$$

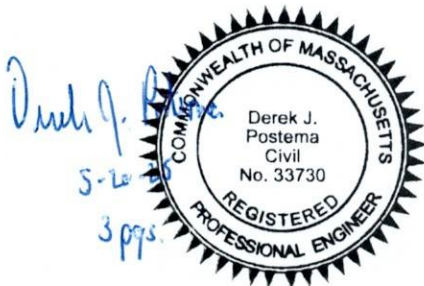
$$\text{Deflection Ratio} = l / \delta = 1:1656$$

$$\text{Bending Stress} = \sigma = \frac{W_a \cdot l^2 \cdot y}{8 \cdot I} = 10,875 \text{ psi}$$

Acceptable
Less than 15000 psi

R.F of 6.9 On Ultimate Stress
R.F of 3.2 On Yield Stress

Note: RF refers to Reserve factor (or safety factor)



$$\text{Vertical Shear through Stiffener section} = V = \frac{\rho \bullet H \bullet d \bullet l}{2}$$

$$\text{Area of slide plate section} = A' = 3.25 \text{ sq.in}$$

$$\begin{aligned} \text{Distance from slide plate neutral axis to combined section neutral axis} &= z' = 0.909 \text{ in} \\ \text{Effective width of Weld} &= b = 0.1768 \text{ in} \quad (\text{Based on a 0.25" weld}) \end{aligned}$$

$$\text{Shear Flow} = \tau = \frac{V \bullet A' \bullet z'}{I} = 852.4 \text{ lb/in}$$

$$\text{Weld Shear stress} = \tau \text{ based on skip weld, 2 on 6} = 7,234 \text{ psi} \quad \begin{matrix} \text{Acceptable} \\ \text{Less than 10000 psi} \end{matrix}$$

(both sides)

3. Stem Design

$$\begin{aligned} \text{Stem Type} &= 1.75" \text{ Rolled 4tpi} \\ \text{Effective Stem Diameter} &= D = 1.625 \text{ in} \quad (\text{minor diameter, per AWWA C561}) \\ \text{Stem Radius of Gyration} &= r = 0.406 \text{ in} \\ \text{Stem Moment of Inertia} &= I = 0.342 \text{ in}^4 \\ \text{Stem Modulus Of Elasticity} &= E = 2.80\text{E}+07 \text{ psi} \\ \text{Stem Max Free Length} &= L = 80 \text{ in} \end{aligned}$$

$$L/r \text{ ratio} = 197 \quad \text{Acceptable not greater than 200}$$

$$\begin{aligned} \text{Weight of Slide \& Stem} &= W = 530 \text{ lbs.} \\ \text{Area of Gate} &= A = 26.00 \text{ sq.ft} \\ \text{Operating Head to gate c/l} &= H_c = 12.00 \text{ ft.} \\ \text{Seal Friction Coefficient} &= \mu = 0.2 \\ \text{Slide Lifting Force} &= \text{SLF} = (62.4 \times \mu \times A \times H_c) = 3,894 \text{ lbs.} \\ \text{Seal Drag (10 lb/in of slide height)} &= 780 \text{ lbs.} \end{aligned}$$

$$\text{Design Lifting Force} = (\text{Greater of Seal Drag or SLF}) + W = F = 4,424 \text{ lbs.}$$

$$\begin{aligned} \text{Electric Operator Run motor Torque} &= \text{Tr} = 293 \text{ ft.-lbs.} \\ \text{Stem Factor} &= \text{Sf} = 0.021 \\ \text{Thrust @ Run Torque} &= \text{Tr/Sf} = 13,908 \text{ lbs.} \quad \text{Acceptable, greater than F} \end{aligned}$$

$$\begin{aligned} \text{Electric Operator Stalled motor Torque} &= \text{T} = 441 \text{ ft.lb} \\ \text{Stem factor} &= \text{Sf} = 0.021 \\ \text{Design Torque} &= \text{Td} = 1.25 \times \text{T} = 551.3 \text{ ft.lb} \\ \text{Thrust @ Design Torque} &= \text{Th} = \text{Td} / \text{Sf} = 26,166 \text{ lbs.} \end{aligned}$$

$$\text{Direct stress at Design Thrust} = \sigma = \frac{\text{Th} \bullet 4}{\pi \bullet D^2} = 12,616 \text{ psi}$$

$$\text{Shear Stress at Design Thrust} = \tau = \frac{192 \bullet \text{Td}}{\pi \bullet D^3} = 7,851 \text{ psi}$$

$$\text{Combined stress} = \sigma / 2 + \sqrt{(\sigma / 2)^2 + \tau^2} = 16,380 \text{ psi} \quad \text{Acceptable, < Yield Stress (35000 psi)}$$

$$\text{Stem Buckling Load (C=2)} = \frac{2 \bullet \pi^2 \bullet E \bullet I}{l^2} = 29,559 \text{ lbs.} \quad \text{Acceptable, Greater than Th}$$

4. Yoke Design

Assumed Yoke acts as a simply supported beam under central Point Load (Roark's p.101 – 1.e)
Yoke Type = Double 7 in x 3 in Channel sections

Yoke Length = L = Aperture Width = 48 in
Yoke section Moment of Inertia = I = 45.63 in⁴
Material Modulus = E = 2.80E+07 psi (Stainless Steel Grade 316)
Yoke – neutral axis to furthest fiber = y = 3.50 in

At Design Lifting Force (F from section 3)

$$\text{Deflection} = \delta_{Lf} = \frac{F \cdot L^3}{48 \cdot E \cdot I} = 0.008 \text{ in}$$

Deflection Ratio = l / δ = **1: 6017** **Acceptable, Less than 1:360 (AWWA C.561)**

$$\text{Max bending moment at design} = M_{Lf} = \frac{F \cdot L}{4} = 53,085 \text{ in lbs.}$$

$$\text{Max Bending Stress} = \sigma_{Lf} = \frac{M \cdot y}{I} = 4,072 \text{ psi} \quad \textbf{Acceptable}$$

Less than 15000 psi

At Electric Operator Stalled motor Torque (T from section 3)

$$\text{Max bending moment} = M_{St} = \frac{T \cdot L}{Sf \cdot 4} = 251,192 \text{ in lbs.}$$

$$\text{Max Bending Stress} = \sigma_{St} = \frac{M \cdot y}{I} = 19,267 \text{ psi}$$

Reserve Factor On Yield Stress = 1.82 **Acceptable, Greater than 1.5 (AWWA C.561)**

$$\text{Deflection (at Stall)} = \frac{T \cdot L^3}{Sf \cdot 48 \cdot E \cdot I} = 0.038 \text{ in} \quad \textbf{Acceptable}$$

Less than 0.13 in

5. Loads transferred to Structure

Self contained gates do not transmit Operating loads to the mounting structure.
Loads transferred are those due to the Operating Head (Hc from section 3) acting on the gate area

$$\text{Water load} = (62.4 \times A \times Hc) = 19,469 \text{ lbs.}$$

CALCULATION SHEET

Calc Self Cont STD Electric
D. Postema February 8, 2018
Master Doc.: F030-20 Rev.1]

Sheet No. A-327-524
Project: Hilo WWTF Phase 1
Whipps #: 32752
Gate Ref: 02-GAT-1135

Prepared By: S. Barnes
5/15/2025

Calculations

1. Allowable Leakage Rates
2. Slide design
3. Stem Design
4. Yoke Design
5. Loads transferred to Structure

1. Leakage Rates

Gate Aperture Width = 48.0 in
Gate Aperture Depth = 78.0 in
Seal Periphery = 204.0 in
Allowable Seating Leakage rate = 0.05 gal/min/ft of seal periphery
Allowable Seating Leakage = 0.85 gal/min

Allowable Unseating Seating Leakage rate = 0.1 gal/min/ft of seal periphery
Allowable Unseating Seating Leakage = 0.85 gal/min

2. Slide Design

Design Head on gate Invert = H = 23 ft. = 279 in
Aperture Width = L = 48 in
Density of Water = ρ = 62 lb/cu.ft = 0.0361 lb/cu.in
Slide section depth (Beam Width) = d = 13 in
Slide section Moment of Inertia = I = 11 in⁴
Slide – furthest fiber distance = y = 3.2 in
Material Modulus = E = 2.80E+07 psi (Stainless Steel Grade 316)

Assumed Slide section is a simply supported beam under uniform load (Roark's p.104 – 2.e)

$$W_a = \text{Load per unit Length} = \left(H - \frac{d}{2}\right) \cdot \rho \cdot d = 127.9 \text{ lb/in}$$

$$\text{Deflection} = \delta = \frac{5 \cdot W_a \cdot l^4}{384 \cdot E \cdot I} = 0.029 \text{ in} \quad \text{Acceptable, Less than 0.048 in}$$

$$\text{Deflection Ratio} = l / \delta = 1: 1656$$

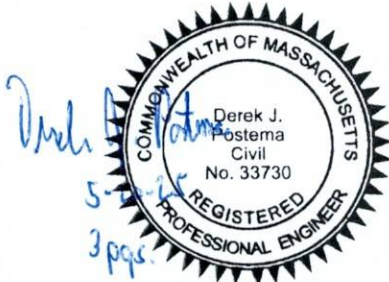
$$\text{Bending Stress} = \sigma = \frac{W_a \cdot l^2 \cdot y}{8 \cdot I} = 10,875 \text{ psi}$$

Acceptable
Less than 15000 psi

R.F of 6.9 On Ultimate Stress

R.F of 3.2 On Yield Stress

Note: RF refers to Reserve factor (or safety factor)



$$\text{Vertical Shear through Stiffener section} = V = \frac{\rho \cdot H \cdot d \cdot l}{2}$$

$$\text{Area of slide plate section} = A' = 3.25 \text{ sq.in}$$

$$\begin{aligned} \text{Distance from slide plate neutral axis to combined section neutral axis} &= z' = 0.909 \text{ in} \\ \text{Effective width of Weld} &= b = 0.1768 \text{ in} \quad (\text{Based on a 0.25" weld}) \end{aligned}$$

$$\text{Shear Flow} = \tau = \frac{V \cdot A' \cdot z'}{I} = 852.4 \text{ lb/in}$$

$$\text{Weld Shear stress} = \tau \text{ based on skip weld, 2 on 6} = 7,234 \text{ psi} \quad \begin{matrix} \text{Acceptable} \\ \text{Less than 10000 psi} \end{matrix}$$

3. Stem Design

$$\text{Stem Type} = 1.75" \text{ Rolled 4tpi}$$

$$\text{Effective Stem Diameter} = D = 1.625 \text{ in} \quad (\text{minor diameter, per AWWA C561})$$

$$\text{Stem Radius of Gyration} = r = 0.406 \text{ in}$$

$$\text{Stem Moment of Inertia} = I = 0.342 \text{ in}^4$$

$$\text{Stem Modulus Of Elasticity} = E = 2.80\text{E}+07 \text{ psi}$$

$$\text{Stem Max Free Length} = L = 80 \text{ in}$$

$$L/r \text{ ratio} = 197 \quad \text{Acceptable not greater than 200}$$

$$\text{Weight of Slide \& Stem} = W = 530 \text{ lbs.}$$

$$\text{Area of Gate} = A = 26.00 \text{ sq.ft}$$

$$\text{Operating Head to gate c/l} = H_c = 12.00 \text{ ft.}$$

$$\text{Seal Friction Coefficient} = \mu = 0.2$$

$$\text{Slide Lifting Force} = \text{SLF} = (62.4 \times \mu \times A \times H_c) = 3,894 \text{ lbs.}$$

$$\text{Seal Drag (10 lb/in of slide height)} = 780 \text{ lbs.}$$

$$\text{Design Lifting Force} = (\text{Greater of Seal Drag or SLF}) + W = F = 4,424 \text{ lbs.}$$

$$\text{Electric Operator Run motor Torque} = \text{Tr} = 293 \text{ ft.-lbs.}$$

$$\text{Stem Factor} = \text{Sf} = 0.021$$

$$\text{Thrust @ Run Torque} = \text{Tr/Sf} = 13,908 \text{ lbs.} \quad \text{Acceptable, greater than F}$$

$$\text{Electric Operator Stalled motor Torque} = T = 441 \text{ ft.lb}$$

$$\text{Stem factor} = \text{Sf} = 0.021$$

$$\text{Design Torque} = \text{Td} = 1.25 \times T = 551.3 \text{ ft.lb}$$

$$\text{Thrust @ Design Torque} = \text{Th} = \text{Td} / \text{Sf} = 26,166 \text{ lbs.}$$

$$\text{Direct stress at Design Thrust} = \sigma = \frac{\text{Th} \cdot 4}{\pi \cdot D^2} = 12,616 \text{ psi}$$

$$\text{Shear Stress at Design Thrust} = \tau = \frac{192 \cdot \text{Td}}{\pi \cdot D^3} = 7,851 \text{ psi}$$

$$\text{Combined stress} = \sigma / 2 + \sqrt{(\sigma / 2)^2 + \tau^2} = 16,380 \text{ psi} \quad \text{Acceptable, < Yield Stress (35000 psi)}$$

$$\text{Stem Buckling Load (C=2)} = \frac{2 \cdot \pi^2 \cdot E \cdot I}{l^2} = 29,559 \text{ lbs.} \quad \text{Acceptable, Greater than Th}$$

4. Yoke Design

Assumed Yoke acts as a simply supported beam under central Point Load (Roark's p.101 – 1.e)

Yoke Type = Double 7 in x 3 in Channel sections

Yoke Length = L = Aperture Width = 48 in

Yoke section Moment of Inertia = I = 45.63 in⁴

Material Modulus = E = 2.80E+07 psi (Stainless Steel Grade 316)

Yoke – neutral axis to furthest fiber = y = 3.50 in

At Design Lifting Force (F from section 3)

$$\text{Deflection} = \delta_{Lf} = \frac{F \bullet L^3}{48 \bullet E \bullet I} = 0.008 \text{ in}$$

Deflection Ratio = l / δ = **1: 6017** **Acceptable, Less than 1:360 (AWWA C.561)**

$$\text{Max bending moment at design} = M_{Lf} = \frac{F \bullet L}{4} = 53,085 \text{ in lbs.}$$

$$\text{Max Bending Stress} = \sigma_{Lf} = \frac{M \bullet y}{I} = 4,072 \text{ psi} \quad \textbf{Acceptable}$$

Less than 15000 psi

At Electric Operator Stalled motor Torque (T from section 3)

$$\text{Max bending moment} = M_{St} = \frac{T \bullet L}{Sf \bullet 4} = 251,192 \text{ in lbs.}$$

$$\text{Max Bending Stress} = \sigma_{St} = \frac{M \bullet y}{I} = 19,267 \text{ psi}$$

Reserve Factor On Yield Stress = 1.817 **Acceptable, Greater than 1.5 (AWWA C.561)**

$$\text{Deflection (at Stall)} = \frac{T \bullet L^3}{Sf \bullet 48 \bullet E \bullet I} = 0.038 \text{ in} \quad \textbf{Acceptable}$$

Less than 0.13 in

5. Loads transferred to Structure

Self contained gates do not transmit Operating loads to the mounting structure.

Loads transferred are those due to the Operating Head (Hc from section 3) acting on the gate area

$$\text{Water load} = (62.4 \times A \times Hc) = 19,469 \text{ lbs.}$$

CALCULATION SHEET

Calc Self Cont STD Electric
D. Postema February 8, 2018
Master Doc.: F030-20 Rev.1]

Sheet No. A-327-525
Project: Hilo WWTF Phase 1
Whipps #: 32752
Gate Ref: 02-GAT-1210 & 1220

Prepared By: S. Barnes
5/15/2025

Calculations

1. Allowable Leakage Rates
2. Slide design
3. Stem Design
4. Yoke Design
5. Loads transferred to Structure

1. Leakage Rates

Gate Aperture Width = 45.0 in
Gate Aperture Depth = 78.0 in
Seal Periphery = 201.0 in
Allowable Seating Leakage rate = 0.05 gal/min/ft of seal periphery
Allowable Seating Leakage = 0.84 gal/min

Allowable Unseating Seating Leakage rate = 0.05 gal/min/ft of seal periphery
Allowable Unseating Seating Leakage = 0.84 gal/min

2. Slide Design

Design Head on gate Invert = H = 23.3 ft. = 279 in
Aperture Width = L = 45 in
Density of Water = ρ = 62.4 lb/cu.ft = 0.0361 lb/cu.in
Slide section depth (Beam Width) = d = 13 in
Slide section Moment of Inertia = I = 10.9 in⁴
Slide – furthest fiber distance = y = 3.22 in
Material Modulus = E = 2.80E+07 psi (Stainless Steel Grade 316)

Assumed Slide section is a simply supported beam under uniform load (Roark's p.104 – 2.e)

$$W_a = \text{Load per unit Length} = \left(H - \frac{d}{2}\right) \cdot \rho \cdot d = 127.9 \text{ lb/in}$$

$$\text{Deflection} = \delta = \frac{5 \cdot W_a \cdot l^4}{384 \cdot E \cdot I} = 0.022 \text{ in} \quad \text{Acceptable, Less than 0.045 in}$$

$$\text{Deflection Ratio} = l / \delta = 1:2010$$

$$\text{Bending Stress} = \sigma = \frac{W_a \cdot l^2 \cdot y}{8 \cdot I} = 9,558 \text{ psi}$$

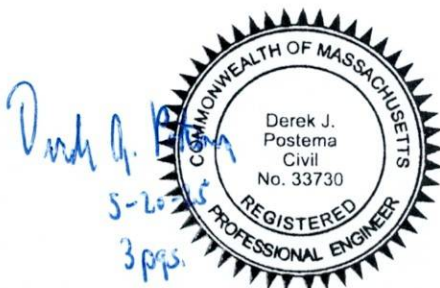
Acceptable

Less than 15000 psi

R.F of 7.8 On Ultimate Stress

R.F of 3.7 On Yield Stress

Note: RF refers to Reserve factor (or safety factor)



$$\text{Vertical Shear through Stiffener section} = V = \frac{\rho \bullet H \bullet d \bullet l}{2}$$

$$\text{Area of slide plate section} = A' = 3.25 \text{ sq.in}$$

$$\begin{aligned} \text{Distance from slide plate neutral axis to combined section neutral axis} &= z' = 0.909 \text{ in} \\ \text{Effective width of Weld} &= b = 0.1768 \text{ in} \quad (\text{Based on a 0.25" weld}) \end{aligned}$$

$$\text{Shear Flow} = \tau = \frac{V \bullet A' \bullet z'}{I} = 799.2 \text{ lb/in}$$

$$\text{Weld Shear stress} = \tau \text{ based on skip weld, 2 on 6} = 6,782 \text{ psi} \quad \begin{array}{l} \text{Acceptable} \\ \text{Less than 10000 psi} \end{array}$$

(both sides)

3. Stem Design

$$\begin{aligned} \text{Stem Type} &= 1.75" \text{ Rolled 4tpi} \\ \text{Effective Stem Diameter} &= D = 1.625 \text{ in} \quad (\text{minor diameter, per AWWA C561}) \\ \text{Stem Radius of Gyration} &= r = 0.406 \text{ in} \\ \text{Stem Moment of Inertia} &= I = 0.342 \text{ in}^4 \\ \text{Stem Modulus Of Elasticity} &= E = 2.80\text{E}+07 \text{ psi} \\ \text{Stem Max Free Length} &= L = 78 \text{ in} \end{aligned}$$

$$\text{L/r ratio} = 192 \quad \text{Acceptable not greater than 200}$$

$$\begin{aligned} \text{Weight of Slide \& Stem} &= W = 515 \text{ lbs.} \\ \text{Area of Gate} &= A = 24.38 \text{ sq.ft} \\ \text{Operating Head to gate c/l} &= H_c = 13.01 \text{ ft.} \\ \text{Seal Friction Coefficient} &= \mu = 0.2 \\ \text{Slide Lifting Force} &= \text{SLF} = (62.4 \times \mu \times A \times H_c) = 3,958 \text{ lbs.} \\ \text{Seal Drag (10 lb/in of slide height)} &= 780 \text{ lbs.} \end{aligned}$$

$$\text{Design Lifting Force} = (\text{Greater of Seal Drag or SLF}) + W = F = 4,473 \text{ lbs.}$$

$$\begin{aligned} \text{Electric Operator Run motor Torque} &= \text{Tr} = 293 \text{ ft.-lbs.} \\ \text{Stem Factor} &= \text{Sf} = 0.021 \\ \text{Thrust @ Run Torque} &= \text{Tr/Sf} = 13,908 \text{ lbs.} \quad \text{Acceptable, greater than F} \end{aligned}$$

$$\begin{aligned} \text{Electric Operator Stalled motor Torque} &= \text{T} = 441 \text{ ft.lb} \\ \text{Stem factor} &= \text{Sf} = 0.021 \\ \text{Design Torque} &= \text{Td} = 1.25 \times \text{T} = 551.3 \text{ ft.lb} \\ \text{Thrust @ Design Torque} &= \text{Th} = \text{Td} / \text{Sf} = 26,166 \text{ lbs.} \end{aligned}$$

$$\text{Direct stress at Design Thrust} = \sigma = \frac{\text{Th} \bullet 4}{\pi \bullet D^2} = 12,616 \text{ psi}$$

$$\text{Shear Stress at Design Thrust} = \tau = \frac{192 \bullet \text{Td}}{\pi \bullet D^3} = 7,851 \text{ psi}$$

$$\text{Combined stress} = \sigma / 2 + \sqrt{(\sigma / 2)^2 + \tau^2} = 16,380 \text{ psi} \quad \text{Acceptable, < Yield Stress (35000 psi)}$$

$$\text{Stem Buckling Load (C=2)} = \frac{2 \bullet \pi^2 \bullet E \bullet I}{l^2} = 31,094 \text{ lbs.} \quad \text{Acceptable, Greater than Th}$$

4. Yoke Design

Assumed Yoke acts as a simply supported beam under central Point Load (Roark's p.101 – 1.e)
Yoke Type = Double 7 in x 3 in Channel sections

Yoke Length = L = Aperture Width = 45 in
Yoke section Moment of Inertia = I = 45.63 in⁴
Material Modulus = E = 2.80E+07 psi (Stainless Steel Grade 316)
Yoke – neutral axis to furthest fiber = y = 3.50 in

At Design Lifting Force (F from section 3)

$$\text{Deflection} = \delta_{Lf} = \frac{F \cdot L^3}{48 \cdot E \cdot I} = 0.007 \text{ in}$$

Deflection Ratio = l / δ = **1: 6771** **Acceptable, Less than 1:360 (AWWA C.561)**

$$\text{Max bending moment at design} = M_{Lf} = \frac{F \cdot L}{4} = 50,317 \text{ in lbs.}$$

$$\text{Max Bending Stress} = \sigma_{Lf} = \frac{M \cdot y}{I} = 3,860 \text{ psi} \quad \textbf{Acceptable}$$

Less than 15000 psi

At Electric Operator Stalled motor Torque (T from section 3)

$$\text{Max bending moment} = M_{St} = \frac{T \cdot L}{Sf \cdot 4} = 235,493 \text{ in lbs.}$$

$$\text{Max Bending Stress} = \sigma_{St} = \frac{M \cdot y}{I} = 18,063 \text{ psi}$$

Reserve Factor On Yield Stress = 1.94 **Acceptable, Greater than 1.5 (AWWA C.561)**

$$\text{Deflection (at Stall)} = \frac{T \cdot L^3}{Sf \cdot 48 \cdot E \cdot I} = 0.031 \text{ in} \quad \textbf{Acceptable}$$

Less than 0.13 in

5. Loads transferred to Structure

Self contained gates do not transmit Operating loads to the mounting structure.
Loads transferred are those due to the Operating Head (Hc from section 3) acting on the gate area

$$\text{Water load} = (62.4 \times A \times Hc) = 19,788 \text{ lbs.}$$

CALCULATION SHEET

Calc Self Cont STD Electric
D. Postema February 8, 2018
Master Doc.: F030-20 Rev.1]

Sheet No. A-327-526
Project: Hilo WWTF Phase 1
Whipps #: 32752
Gate Ref: 02-GAT-1215 & 1225

Prepared By: S. Barnes
5/15/2025

Calculations

1. Allowable Leakage Rates
2. Slide design
3. Stem Design
4. Yoke Design
5. Loads transferred to Structure

1. Leakage Rates

Gate Aperture Width = 45.0 in
Gate Aperture Depth = 66.0 in
Seal Periphery = 177.0 in
Allowable Seating Leakage rate = 0.05 gal/min/ft of seal periphery
Allowable Seating Leakage = 0.74 gal/min

Allowable Unseating Seating Leakage rate = 0.05 gal/min/ft of seal periphery
Allowable Unseating Seating Leakage = 0.74 gal/min

2. Slide Design

Design Head on gate Invert = H = 22.8 ft. = 273 in
Aperture Width = L = 45 in
Density of Water = ρ = 62.4 lb/cu.ft = 0.0361 lb/cu.in
Slide section depth (Beam Width) = d = 13.5 in
Slide section Moment of Inertia = I = 11 in⁴
Slide – furthest fiber distance = y = 3.24 in
Material Modulus = E = 2.80E+07 psi (Stainless Steel Grade 316)

Assumed Slide section is a simply supported beam under uniform load (Roark's p.104 – 2.e)

$$W_a = \text{Load per unit Length} = \left(H - \frac{d}{2}\right) \cdot \rho \cdot d = 129.8 \text{ lb/in}$$

$$\text{Deflection} = \delta = \frac{5 \cdot W_a \cdot l^4}{384 \cdot E \cdot I} = 0.023 \text{ in} \quad \text{Acceptable, Less than 0.045 in}$$

$$\text{Deflection Ratio} = l / \delta = 1: 1999$$

$$\text{Bending Stress} = \sigma = \frac{W_a \cdot l^2 \cdot y}{8 \cdot I} = 9,677 \text{ psi}$$

Acceptable
Less than 15000 psi

R.F of 7.8 On Ultimate Stress

R.F of 3.6 On Yield Stress

Note: RF refers to Reserve factor (or safety factor)



$$\text{Vertical Shear through Stiffener section} = V = \frac{\rho \bullet H \bullet d \bullet l}{2}$$

$$\text{Area of slide plate section} = A' = 3.38 \text{ sq.in}$$

$$\begin{aligned} \text{Distance from slide plate neutral axis to combined section neutral axis} &= z' = 0.886 \text{ in} \\ \text{Effective width of Weld} &= b = 0.1768 \text{ in} \quad (\text{Based on a 0.25" weld}) \end{aligned}$$

$$\text{Shear Flow} = \tau = \frac{V \bullet A' \bullet z'}{I} = 814.4 \text{ lb/in}$$

$$\text{Weld Shear stress} = \tau \text{ based on skip weld, 2 on 6} = 6,911 \text{ psi} \quad \begin{array}{l} \text{Acceptable} \\ \text{Less than 10000 psi} \end{array}$$

(both sides)

3. Stem Design

$$\begin{aligned} \text{Stem Type} &= 1.75" \text{ Rolled 4tpi} \\ \text{Effective Stem Diameter} &= D = 1.625 \text{ in} \quad (\text{minor diameter, per AWWA C561}) \\ \text{Stem Radius of Gyration} &= r = 0.406 \text{ in} \\ \text{Stem Moment of Inertia} &= I = 0.342 \text{ in}^4 \\ \text{Stem Modulus Of Elasticity} &= E = 2.80\text{E}+07 \text{ psi} \\ \text{Stem Max Free Length} &= L = 73 \text{ in} \end{aligned}$$

$$L/r \text{ ratio} = 180 \quad \text{Acceptable not greater than 200}$$

$$\begin{aligned} \text{Weight of Slide \& Stem} &= W = 445 \text{ lbs.} \\ \text{Area of Gate} &= A = 20.63 \text{ sq.ft} \\ \text{Operating Head to gate c/l} &= H_c = 13.00 \text{ ft.} \\ \text{Seal Friction Coefficient} &= \mu = 0.2 \\ \text{Slide Lifting Force} &= \text{SLF} = (62.4 \times \mu \times A \times H_c) = 3,346 \text{ lbs.} \\ \text{Seal Drag (10 lb/in of slide height)} &= 660 \text{ lbs.} \end{aligned}$$

$$\text{Design Lifting Force} = (\text{Greater of Seal Drag or SLF}) + W = F = 3,791 \text{ lbs.}$$

$$\begin{aligned} \text{Electric Operator Run motor Torque} &= \text{Tr} = 293 \text{ ft.-lbs.} \\ \text{Stem Factor} &= \text{Sf} = 0.021 \\ \text{Thrust @ Run Torque} &= \text{Tr/Sf} = 13,908 \text{ lbs.} \quad \text{Acceptable, greater than F} \end{aligned}$$

$$\begin{aligned} \text{Electric Operator Stalled motor Torque} &= \text{T} = 441 \text{ ft.lb} \\ \text{Stem factor} &= \text{Sf} = 0.021 \\ \text{Design Torque} &= \text{Td} = 1.25 \times \text{T} = 551.3 \text{ ft.lb} \\ \text{Thrust @ Design Torque} &= \text{Th} = \text{Td} / \text{Sf} = 26,166 \text{ lbs.} \end{aligned}$$

$$\text{Direct stress at Design Thrust} = \sigma = \frac{\text{Th} \bullet 4}{\pi \bullet D^2} = 12,616 \text{ psi}$$

$$\text{Shear Stress at Design Thrust} = \tau = \frac{192 \bullet \text{Td}}{\pi \bullet D^3} = 7,851 \text{ psi}$$

$$\text{Combined stress} = \sigma / 2 + \sqrt{(\sigma / 2)^2 + \tau^2} = 16,380 \text{ psi} \quad \text{Acceptable, < Yield Stress (35000 psi)}$$

$$\text{Stem Buckling Load (C=2)} = \frac{2 \bullet \pi^2 \bullet E \bullet I}{l^2} = 35,500 \text{ lbs.} \quad \text{Acceptable, Greater than Th}$$

4. Yoke Design

Assumed Yoke acts as a simply supported beam under central Point Load (Roark's p.101 – 1.e)
Yoke Type = Double 6 in x 3 in Channel sections

Yoke Length = L = Aperture Width = 45 in
Yoke section Moment of Inertia = I = 31.74 in⁴
Material Modulus = E = 2.80E+07 psi (Stainless Steel Grade 316)
Yoke – neutral axis to furthest fiber = y = 3.00 in

At Design Lifting Force (F from section 3)

$$\text{Deflection} = \delta_{Lf} = \frac{F \cdot L^3}{48 \cdot E \cdot I} = 0.008 \text{ in}$$

Deflection Ratio = l / δ = **1: 5557** **Acceptable, Less than 1:360 (AWWA C.561)**

$$\text{Max bending moment at design} = M_{Lf} = \frac{F \cdot L}{4} = 42,651 \text{ in lbs.}$$

$$\text{Max Bending Stress} = \sigma_{Lf} = \frac{M \cdot y}{I} = 4,031 \text{ psi} \quad \textbf{Acceptable}$$

Less than 15000 psi

At Electric Operator Stalled motor Torque (T from section 3)

$$\text{Max bending moment} = M_{St} = \frac{T \cdot L}{Sf \cdot 4} = 235,493 \text{ in lbs.}$$

$$\text{Max Bending Stress} = \sigma_{St} = \frac{M \cdot y}{I} = 22,255 \text{ psi}$$

Reserve Factor On Yield Stress = 1.57 **Acceptable, Greater than 1.5 (AWWA C.561)**

$$\text{Deflection (at Stall)} = \frac{T \cdot L^3}{Sf \cdot 48 \cdot E \cdot I} = 0.045 \text{ in} \quad \textbf{Acceptable}$$

Less than 0.13 in

5. Loads transferred to Structure

Self contained gates do not transmit Operating loads to the mounting structure.
Loads transferred are those due to the Operating Head (Hc from section 3) acting on the gate area

$$\text{Water load} = (62.4 \times A \times Hc) = 16,731 \text{ lbs.}$$

CALCULATION SHEET

Calc Self Cont STD Electric
D. Postema February 8, 2018
Master Doc.: F030-20 Rev.1]

Sheet No. A-327-527
Project: Hilo WWTF Phase 1
Whipps #: 32752
Gate Ref: 03-FG-1110, 1120 & 1130

Prepared By: S. Barnes
5/15/2025

Calculations

1. Allowable Leakage Rates
2. Slide design
3. Stem Design
4. Yoke Design
5. Loads transferred to Structure

1. Leakage Rates

Gate Aperture Width = 24.0 in
Gate Aperture Depth = 48.0 in
Seal Periphery = 120.0 in
Allowable Seating Leakage rate = 0.05 gal/min/ft of seal periphery
Allowable Seating Leakage = 0.50 gal/min

Allowable Unseating Seating Leakage rate = 0.05 gal/min/ft of seal periphery
Allowable Unseating Seating Leakage = 0.50 gal/min

2. Slide Design

Design Head on gate Invert = **H** = 22 ft. = 264 in
Aperture Width = **L** = 24 in
Density of Water = ρ = 62.4 lb/cu.ft = 0.0361 lb/cu.in
Slide section depth (Beam Width) = **d** = 14.3 in
Slide section Moment of Inertia = **I** = 3.96 in⁴
Slide – furthest fiber distance = **y** = 2.15 in
Material Modulus = **E** = 2.80E+07 psi (Stainless Steel Grade 316)

Assumed Slide section is a simply supported beam under uniform load (Roark's p.104 – 2.e)

$$W_a = \text{Load per unit Length} = \left(H - \frac{d}{2} \right) \cdot \rho \cdot d = 132.1 \text{ lb/in}$$

$$\text{Deflection} = \delta = \frac{5 \cdot W_a \cdot l^4}{384 \cdot E \cdot I} = 0.005 \text{ in} \quad \text{Acceptable, Less than 0.024 in}$$

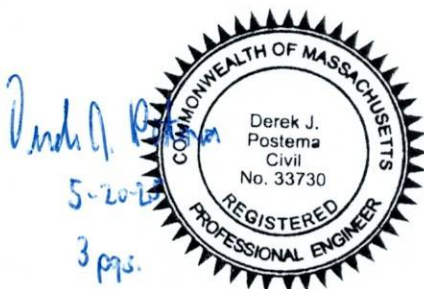
$$\text{Deflection Ratio} = l / \delta = 1: 4665.9$$

$$\text{Bending Stress} = \sigma = \frac{W_a \cdot l^2 \cdot y}{8 \cdot I} = 5,156 \text{ psi}$$

Acceptable
Less than 15000 psi

R.F of 14.5 On Ultimate Stress
R.F of 6.8 On Yield Stress

Note: RF refers to Reserve factor (or safety factor)



$$\text{Vertical Shear through Stiffener section} = V = \frac{\rho \bullet H \bullet d \bullet l}{2}$$

$$\text{Area of slide plate section} = A' = 3.56 \text{ sq.in}$$

$$\begin{aligned} \text{Distance from slide plate neutral axis to combined section neutral axis} &= z' = 0.477 \text{ in} \\ \text{Effective width of Weld} &= b = 0.1768 \text{ in} \quad (\text{Based on a 0.25" weld}) \end{aligned}$$

$$\text{Shear Flow} = \tau = \frac{V \bullet A' \bullet z'}{I} = 698.7 \text{ lb/in}$$

$$\text{Weld Shear stress} = \tau \text{ based on skip weld, 2 on 6} = 5,929 \text{ psi} \quad \begin{matrix} \text{Acceptable} \\ \text{Less than 10000 psi} \end{matrix}$$

(both sides)

3. Stem Design

$$\text{Stem Type} = 1.5" \text{ Rolled 4tpi}$$

$$\text{Effective Stem Diameter} = D = 1.375 \text{ in} \quad (\text{minor diameter, per AWWA C561})$$

$$\text{Stem Radius of Gyration} = r = 0.344 \text{ in}$$

$$\text{Stem Moment of Inertia} = I = 0.175 \text{ in}^4$$

$$\text{Stem Modulus Of Elasticity} = E = 2.80\text{E}+07 \text{ psi}$$

$$\text{Stem Max Free Length} = L = 48 \text{ in}$$

$$L/r \text{ ratio} = 140 \quad \text{Acceptable not greater than 200}$$

$$\text{Weight of Slide \& Stem} = W = 195 \text{ lbs.}$$

$$\text{Area of Gate} = A = 8.00 \text{ sq.ft}$$

$$\text{Operating Head to gate c/l} = H_c = 20.00 \text{ ft.}$$

$$\text{Seal Friction Coefficient} = \mu = 0.2$$

$$\text{Slide Lifting Force} = \text{SLF} = (62.4 \times \mu \times A \times H_c) = 1,997 \text{ lbs.}$$

$$\text{Seal Drag (10 lb/in of slide height)} = 480 \text{ lbs.}$$

$$\text{Design Lifting Force} = (\text{Greater of Seal Drag or SLF}) + W = F = 2,192 \text{ lbs.}$$

$$\text{Electric Operator Run motor Torque} = Tr = 128 \text{ ft.-lbs.}$$

$$\text{Stem Factor} = Sf = 0.019$$

$$\text{Thrust @ Run Torque} = Tr/Sf = 6,754 \text{ lbs.} \quad \text{Acceptable, greater than F}$$

$$\text{Electric Operator Stalled motor Torque} = T = 194 \text{ ft.lb}$$

$$\text{Stem factor} = Sf = 0.019$$

$$\text{Design Torque} = Td = 1.25 \times T = 242.5 \text{ ft.lb}$$

$$\text{Thrust @ Design Torque} = Th = Td / Sf = 12,796 \text{ lbs.}$$

$$\text{Direct stress at Design Thrust} = \sigma = \frac{Th \bullet 4}{\pi \bullet D^2} = 8,617 \text{ psi}$$

$$\text{Shear Stress at Design Thrust} = \tau = \frac{192 \bullet Td}{\pi \bullet D^3} = 5,701 \text{ psi}$$

$$\text{Combined stress} = \sigma / 2 + \sqrt{(\sigma / 2)^2 + \tau^2} = 11,455 \text{ psi} \quad \text{Acceptable, < Yield Stress (35000 psi)}$$

$$\text{Stem Buckling Load (C=2)} = \frac{2 \bullet \pi^2 \bullet E \bullet I}{l^2} = 42,091 \text{ lbs.} \quad \text{Acceptable, Greater than Th}$$

4. Yoke Design

Assumed Yoke acts as a simply supported beam under central Point Load (Roark's p.101 – 1.e)

Yoke Type = Double 5 in x 2.5 in Channel sections

Yoke Length = L = Aperture Width = 24 in
Yoke section Moment of Inertia = I = 17.91 in⁴
Material Modulus = E = 2.80E+07 psi (Stainless Steel Grade 316)
Yoke – neutral axis to furthest fiber = y = 2.50 in

At Design Lifting Force (F from section 3)

$$\text{Deflection} = \delta_{Lf} = \frac{F \cdot L^3}{48 \cdot E \cdot I} = 0.001 \text{ in}$$

Deflection Ratio = l / δ = **1: 19068** **Acceptable, Less than 1:360 (AWWA C.561)**

$$\text{Max bending moment at design} = M_{Lf} = \frac{F \cdot L}{4} = 13,151 \text{ in lbs.}$$

$$\text{Max Bending Stress} = \sigma_{Lf} = \frac{M \cdot y}{I} = 1,836 \text{ psi} \quad \textbf{Acceptable Less than 15000 psi}$$

At Electric Operator Stalled motor Torque (T from section 3)

$$\text{Max bending moment} = M_{St} = \frac{T \cdot L}{Sf \cdot 4} = 61,419 \text{ in lbs.}$$

$$\text{Max Bending Stress} = \sigma_{St} = \frac{M \cdot y}{I} = 8,573 \text{ psi}$$

Reserve Factor On Yield Stress = 4.0828 **Acceptable, Greater than 1.5 (AWWA C.561)**

$$\text{Deflection (at Stall)} = \frac{T \cdot L^3}{Sf \cdot 48 \cdot E \cdot I} = 0.006 \text{ in} \quad \textbf{Acceptable Less than 0.067 in}$$

5. Loads transferred to Structure

Self contained gates do not transmit Operating loads to the mounting structure.

Loads transferred are those due to the Operating Head (Hc from section 3) acting on the gate area

$$\text{Water load} = (62.4 \times A \times Hc) = 9,984 \text{ lbs.}$$

CALCULATION SHEET

Calc Self Cont STD Electric
D. Postema February 8, 2018
Master Doc.: F030-20 Rev.1]

Sheet No. A-327-528
Project: Hilo WWTF Phase 1
Whipps #: 32752
Gate Ref: 03-FG-1230

Prepared By: S. Barnes
5/15/2025

Calculations

1. Allowable Leakage Rates
2. Slide design
3. Stem Design
4. Yoke Design
5. Loads transferred to Structure

1. Leakage Rates

Gate Aperture Width = 36.0 in
Gate Aperture Depth = 36.0 in
Seal Periphery = 144.0 in
Allowable Seating Leakage rate = 0.05 gal/min/ft of seal periphery
Allowable Seating Leakage = 0.60 gal/min

Allowable Unseating Seating Leakage rate = 0.05 gal/min/ft of seal periphery
Allowable Unseating Seating Leakage = 0.60 gal/min

2. Slide Design

Design Head on gate Invert = **H** = 21.5 ft. = 258 in
Aperture Width = **L** = 36 in
Density of Water = ρ = 62.4 lb/cu.ft = 0.0361 lb/cu.in
Slide section depth (Beam Width) = **d** = 11.3 in
Slide section Moment of Inertia = **I** = 3.76 in⁴
Slide – furthest fiber distance = **y** = 2.06 in
Material Modulus = **E** = 2.80E+07 psi (Stainless Steel Grade 316)

Assumed Slide section is a simply supported beam under uniform load (Roark's p.104 – 2.e)

$$W_a = \text{Load per unit Length} = \left(H - \frac{d}{2} \right) \cdot \rho \cdot d = 102.5 \text{ lb/in}$$

$$\text{Deflection} = \delta = \frac{5 \cdot W_a \cdot l^4}{384 \cdot E \cdot I} = 0.021 \text{ in} \quad \text{Acceptable, Less than 0.036 in}$$

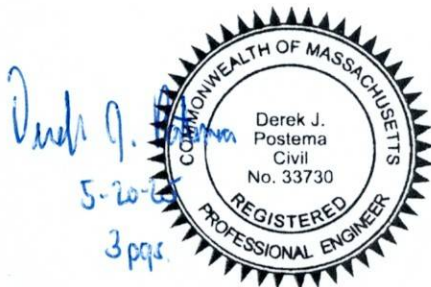
$$\text{Deflection Ratio} = l / \delta = 1: 1690$$

$$\text{Bending Stress} = \sigma = \frac{W_a \cdot l^2 \cdot y}{8 \cdot I} = 9,098 \text{ psi}$$

Acceptable
Less than 15000 psi

R.F of 8.2 On Ultimate Stress
R.F of 3.8 On Yield Stress

Note: RF refers to Reserve factor (or safety factor)



$$\text{Vertical Shear through Stiffener section} = V = \frac{\rho \bullet H \bullet d \bullet l}{2}$$

$$\text{Area of slide plate section} = A' = 2.81 \text{ sq.in}$$

$$\begin{aligned} \text{Distance from slide plate neutral axis to combined section neutral axis} &= z' = 0.566 \text{ in} \\ \text{Effective width of Weld} &= b = 0.1768 \text{ in} \quad (\text{Based on a 0.25" weld}) \end{aligned}$$

$$\text{Shear Flow} = \tau = \frac{V \bullet A' \bullet z'}{I} = 799.7 \text{ lb/in}$$

$$\text{Weld Shear stress} = \tau \text{ based on skip weld, 2 on 6} = 6,787 \text{ psi} \quad \begin{matrix} \text{Acceptable} \\ \text{Less than 10000 psi} \end{matrix}$$

(both sides)

3. Stem Design

$$\text{Stem Type} = 1.5" \text{ Rolled 4tpi}$$

$$\text{Effective Stem Diameter} = D = 1.375 \text{ in} \quad (\text{minor diameter, per AWWA C561})$$

$$\text{Stem Radius of Gyration} = r = 0.344 \text{ in}$$

$$\text{Stem Moment of Inertia} = I = 0.175 \text{ in}^4$$

$$\text{Stem Modulus Of Elasticity} = E = 2.80\text{E}+07 \text{ psi}$$

$$\text{Stem Max Free Length} = L = 55 \text{ in}$$

$$L/r \text{ ratio} = 160 \quad \text{Acceptable not greater than 200}$$

$$\text{Weight of Slide \& Stem} = W = 220 \text{ lbs.}$$

$$\text{Area of Gate} = A = 9.00 \text{ sq.ft}$$

$$\text{Operating Head to gate c/l} = H_c = 13.00 \text{ ft.}$$

$$\text{Seal Friction Coefficient} = \mu = 0.2$$

$$\text{Slide Lifting Force} = \text{SLF} = (62.4 \times \mu \times A \times H_c) = 1,460 \text{ lbs.}$$

$$\text{Seal Drag (10 lb/in of slide height + 400 corners + 20 lb/in of slide width)} = 1,480 \text{ lbs.}$$

$$\text{Design Lifting Force} = (\text{Greater of Seal Drag or SLF}) + W = F = 1,700 \text{ lbs.}$$

$$\text{Electric Operator Run motor Torque} = Tr = 128 \text{ ft.-lbs.}$$

$$\text{Stem Factor} = Sf = 0.019$$

$$\text{Thrust @ Run Torque} = Tr/Sf = 6,754 \text{ lbs.} \quad \text{Acceptable, greater than F}$$

$$\text{Electric Operator Stalled motor Torque} = T = 194 \text{ ft.lb}$$

$$\text{Stem factor} = Sf = 0.019$$

$$\text{Design Torque} = Td = 1.25 \times T = 242.5 \text{ ft.lb}$$

$$\text{Thrust @ Design Torque} = Th = Td / Sf = 12,796 \text{ lbs.}$$

$$\text{Direct stress at Design Thrust} = \sigma = \frac{Th \bullet 4}{\pi \bullet D^2} = 8,617 \text{ psi}$$

$$\text{Shear Stress at Design Thrust} = \tau = \frac{192 \bullet Td}{\pi \bullet D^3} = 5,701 \text{ psi}$$

$$\text{Combined stress} = \sigma / 2 + \sqrt{(\sigma / 2)^2 + \tau^2} = 11,455 \text{ psi} \quad \text{Acceptable, < Yield Stress (35000 psi)}$$

$$\text{Stem Buckling Load (C=2)} = \frac{2 \bullet \pi^2 \bullet E \bullet I}{l^2} = 32,058 \text{ lbs.} \quad \text{Acceptable, Greater than Th}$$

4. Yoke Design

Assumed Yoke acts as a simply supported beam under central Point Load (Roark's p.101 – 1.e)

Yoke Type = Double 5 in x 2.5 in Channel sections

Yoke Length = L = Aperture Width = 36 in
Yoke section Moment of Inertia = I = 17.91 in⁴
Material Modulus = E = 2.80E+07 psi (Stainless Steel Grade 316)
Yoke – neutral axis to furthest fiber = y = 2.50 in

At Design Lifting Force (F from section 3)

$$\text{Deflection} = \delta_{Lf} = \frac{F \cdot L^3}{48 \cdot E \cdot I} = 0.003 \text{ in}$$

Deflection Ratio = l / δ = **1: 10926** **Acceptable, Less than 1:360 (AWWA C.561)**

$$\text{Max bending moment at design} = M_{Lf} = \frac{F \cdot L}{4} = 15,300 \text{ in lbs.}$$

$$\text{Max Bending Stress} = \sigma_{Lf} = \frac{M \cdot y}{I} = 2,136 \text{ psi} \quad \textbf{Acceptable Less than 15000 psi}$$

At Electric Operator Stalled motor Torque (T from section 3)

$$\text{Max bending moment} = M_{St} = \frac{T \cdot L}{Sf \cdot 4} = 92,129 \text{ in lbs.}$$

$$\text{Max Bending Stress} = \sigma_{St} = \frac{M \cdot y}{I} = 12,859 \text{ psi}$$

Reserve Factor On Yield Stress = 2.7218 **Acceptable, Greater than 1.5 (AWWA C.561)**

$$\text{Deflection (at Stall)} = \frac{T \cdot L^3}{Sf \cdot 48 \cdot E \cdot I} = 0.020 \text{ in} \quad \textbf{Acceptable Less than 0.1 in}$$

5. Loads transferred to Structure

Self contained gates do not transmit Operating loads to the mounting structure.

Loads transferred are those due to the Operating Head (Hc from section 3) acting on the gate area

$$\text{Water load} = (62.4 \times A \times Hc) = 7,301 \text{ lbs.}$$

CALCULATION SHEET

Calc Self Cont STD Electric
D. Postema February 8, 2018
Master Doc.: F030-20 Rev.1]

Sheet No. A-327-529
Project: Hilo WWTF Phase 1
Whipps #: 32752
Gate Ref: 03-FG-1440 & 1445

Prepared By: S. Barnes
5/15/2025

Calculations

1. Allowable Leakage Rates
2. Slide design
3. Stem Design
4. Yoke Design
5. Loads transferred to Structure

1. Leakage Rates

Gate Aperture Width = 48.0 in
Gate Aperture Depth = 48.0 in
Seal Periphery = 192.0 in
Allowable Seating Leakage rate = 0.05 gal/min/ft of seal periphery
Allowable Seating Leakage = 0.80 gal/min

Allowable Unseating Seating Leakage rate = 0.05 gal/min/ft of seal periphery
Allowable Unseating Seating Leakage = 0.80 gal/min

2. Slide Design

Design Head on gate Invert = H = 22 ft. = 264 in
Aperture Width = L = 48 in
Density of Water = ρ = 62.4 lb/cu.ft = 0.0361 lb/cu.in
Slide section depth (Beam Width) = d = 15.3 in
Slide section Moment of Inertia = I = 11.3 in⁴
Slide – furthest fiber distance = y = 3.31 in
Material Modulus = E = 2.80E+07 psi (Stainless Steel Grade 316)

Assumed Slide section is a simply supported beam under uniform load (Roark's p.104 – 2.e)

$$W_a = \text{Load per unit Length} = \left(H - \frac{d}{2}\right) \cdot \rho \cdot d = 141.1 \text{ lb/in}$$

$$\text{Deflection} = \delta = \frac{5 \cdot W_a \cdot l^4}{384 \cdot E \cdot I} = 0.031 \text{ in} \quad \text{Acceptable, Less than 0.048 in}$$

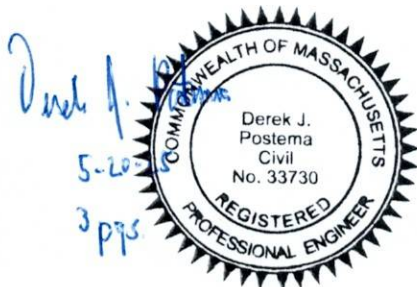
$$\text{Deflection Ratio} = l / \delta = 1: 1558$$

$$\text{Bending Stress} = \sigma = \frac{W_a \cdot l^2 \cdot y}{8 \cdot I} = 11,899 \text{ psi}$$

Acceptable
Less than 15000 psi

R.F of 6.3 On Ultimate Stress
R.F of 2.9 On Yield Stress

Note: RF refers to Reserve factor (or safety factor)



$$\text{Vertical Shear through Stiffener section} = V = \frac{\rho \bullet H \bullet d \bullet l}{2}$$

$$\text{Area of slide plate section} = A' = 3.81 \text{ sq.in}$$

$$\begin{aligned} \text{Distance from slide plate neutral axis to combined section neutral axis} &= z' = 0.814 \text{ in} \\ \text{Effective width of Weld} &= b = 0.1768 \text{ in} \quad (\text{Based on a 0.25" weld}) \end{aligned}$$

$$\text{Shear Flow} = \tau = \frac{V \bullet A' \bullet z'}{I} = 957.0 \text{ lb/in}$$

$$\text{Weld Shear stress} = \tau \text{ based on skip weld, 2 on 6} = 8,122 \text{ psi} \quad \begin{matrix} \text{Acceptable} \\ \text{Less than 10000 psi} \end{matrix}$$

(both sides)

3. Stem Design

$$\text{Stem Type} = 1.5" \text{ Rolled 4tpi}$$

$$\text{Effective Stem Diameter} = D = 1.375 \text{ in} \quad (\text{minor diameter, per AWWA C561})$$

$$\text{Stem Radius of Gyration} = r = 0.344 \text{ in}$$

$$\text{Stem Moment of Inertia} = I = 0.175 \text{ in}^4$$

$$\text{Stem Modulus Of Elasticity} = E = 2.80\text{E}+07 \text{ psi}$$

$$\text{Stem Max Free Length} = L = 49 \text{ in}$$

$$\text{L/r ratio} = 143 \quad \text{Acceptable not greater than 200}$$

$$\text{Weight of Slide \& Stem} = W = 365 \text{ lbs.}$$

$$\text{Area of Gate} = A = 16.00 \text{ sq.ft}$$

$$\text{Operating Head to gate c/l} = H_c = 9.00 \text{ ft.}$$

$$\text{Seal Friction Coefficient} = \mu = 0.2$$

$$\text{Slide Lifting Force} = \text{SLF} = (62.4 \times \mu \times A \times H_c) = 1,797 \text{ lbs.}$$

$$\text{Seal Drag (10 lb/in of slide height + 400 corners + 20 lb/in of slide width)} = 1,840 \text{ lbs.}$$

$$\text{Design Lifting Force} = (\text{Greater of Seal Drag or SLF}) + W = F = 2,205 \text{ lbs.}$$

$$\text{Electric Operator Run motor Torque} = \text{Tr} = 128 \text{ ft.-lbs.}$$

$$\text{Stem Factor} = \text{Sf} = 0.019$$

$$\text{Thrust @ Run Torque} = \text{Tr/Sf} = 6,754 \text{ lbs.} \quad \text{Acceptable, greater than F}$$

$$\text{Electric Operator Stalled motor Torque} = \text{T} = 194 \text{ ft.lb}$$

$$\text{Stem factor} = \text{Sf} = 0.019$$

$$\text{Design Torque} = \text{Td} = 1.25 \times \text{T} = 242.5 \text{ ft.lb}$$

$$\text{Thrust @ Design Torque} = \text{Th} = \text{Td} / \text{Sf} = 12,796 \text{ lbs.}$$

$$\text{Direct stress at Design Thrust} = \sigma = \frac{\text{Th} \bullet 4}{\pi \bullet D^2} = 8,617 \text{ psi}$$

$$\text{Shear Stress at Design Thrust} = \tau = \frac{192 \bullet \text{Td}}{\pi \bullet D^3} = 5,701 \text{ psi}$$

$$\text{Combined stress} = \sigma / 2 + \sqrt{(\sigma / 2)^2 + \tau^2} = 11,455 \text{ psi} \quad \text{Acceptable, < Yield Stress (35000 psi)}$$

$$\text{Stem Buckling Load (C=2)} = \frac{2 \bullet \pi^2 \bullet E \bullet I}{l^2} = 40,390 \text{ lbs.} \quad \text{Acceptable, Greater than Th}$$

4. Yoke Design

Assumed Yoke acts as a simply supported beam under central Point Load (Roark's p.101 – 1.e)
Yoke Type = Double 6 in x 3 in Channel sections

Yoke Length = L = Aperture Width = 48 in
Yoke section Moment of Inertia = I = 31.74 in⁴
Material Modulus = E = 2.80E+07 psi (Stainless Steel Grade 316)
Yoke – neutral axis to furthest fiber = y = 3.00 in

At Design Lifting Force (F from section 3)

$$\text{Deflection} = \delta_{Lf} = \frac{F \cdot L^3}{48 \cdot E \cdot I} = 0.006 \text{ in}$$

Deflection Ratio = l / δ = **1: 8398** **Acceptable, Less than 1:360 (AWWA C.561)**

$$\text{Max bending moment at design} = M_{Lf} = \frac{F \cdot L}{4} = 26,460 \text{ in lbs.}$$

$$\text{Max Bending Stress} = \sigma_{Lf} = \frac{M \cdot y}{I} = 2,501 \text{ psi} \quad \textbf{Acceptable}$$

Less than 15000 psi

At Electric Operator Stalled motor Torque (T from section 3)

$$\text{Max bending moment} = M_{St} = \frac{T \cdot L}{Sf \cdot 4} = 122,838 \text{ in lbs.}$$

$$\text{Max Bending Stress} = \sigma_{St} = \frac{M \cdot y}{I} = 11,609 \text{ psi}$$

Reserve Factor On Yield Stress = 3.01 **Acceptable, Greater than 1.5 (AWWA C.561)**

$$\text{Deflection (at Stall)} = \frac{T \cdot L^3}{Sf \cdot 48 \cdot E \cdot I} = 0.027 \text{ in} \quad \textbf{Acceptable}$$

Less than 0.13 in

5. Loads transferred to Structure

Self contained gates do not transmit Operating loads to the mounting structure.
Loads transferred are those due to the Operating Head (Hc from section 3) acting on the gate area

$$\text{Water load} = (62.4 \times A \times Hc) = 8,986 \text{ lbs.}$$

CALCULATION SHEET

Sheet No. A-327-5201
Project: Hilo WWTF Phase 1
Whipps #: 32752
Gate Ref: 03-FG-1495

Prepared By: S. Barnes
5/15/2025

Calculations

1. Allowable Leakage Rates
2. Slide design
3. Stem Design
4. Yoke Design
5. Loads transferred to Structure

1. Leakage Rates

Gate Aperture Width = 24.0 in
Gate Aperture Depth = 48.0 in
Seal Periphery = 144.0 in
Allowable Seating Leakage rate = 0.05 gal/min/ft of seal periphery
Allowable Seating Leakage = 0.60 gal/min

Allowable Unseating Seating Leakage rate = 0.05 gal/min/ft of seal periphery
Allowable Unseating Seating Leakage = 0.60 gal/min

2. Slide Design

Design Head on gate Invert = H = 22 ft. = 264 in
Aperture Width = L = 24 in
Density of Water = ρ = 62.4 lb/cu.ft = 0.0361 lb/cu.in
Slide section depth (Beam Width) = d = 15.3 in
Slide section Moment of Inertia = I = 4.02 in⁴
Slide – furthest fiber distance = y = 2.17 in
Material Modulus = E = 2.80E+07 psi (Stainless Steel Grade 316)

Assumed Slide section is a simply supported beam under uniform load (Roark's p.104 – 2.e)

$$W_a = \text{Load per unit Length} = \left(H - \frac{d}{2} \right) \cdot \rho \cdot d = 141.1 \text{ lb/in}$$

$$\text{Deflection} = \delta = \frac{5 \cdot W_a \cdot l^4}{384 \cdot E \cdot I} = 0.005 \text{ in} \quad \text{Acceptable, Less than 0.024 in}$$

$$\text{Deflection Ratio} = l / \delta = 1: 4429$$

$$\text{Bending Stress} = \sigma = \frac{W_a \cdot l^2 \cdot y}{8 \cdot I} = 5,492 \text{ psi}$$

Acceptable
Less than 15000 psi

R.F of 13.7 On Ultimate Stress

R.F of 6.4 On Yield Stress

Note: RF refers to Reserve factor (or safety factor)



$$\text{Vertical Shear through Stiffener section} = V = \frac{\rho \bullet H \bullet d \bullet l}{2}$$

$$\text{Area of slide plate section} = A' = 3.81 \text{ sq.in}$$

$$\begin{aligned} \text{Distance from slide plate neutral axis to combined section neutral axis} &= z' = 0.453 \text{ in} \\ \text{Effective width of Weld} &= b = 0.1768 \text{ in} \quad (\text{Based on a 0.25" weld}) \end{aligned}$$

$$\text{Shear Flow} = \tau = \frac{V \bullet A' \bullet z'}{I} = 749.7 \text{ lb/in}$$

$$\text{Weld Shear stress} = \tau \text{ based on skip weld, 2 on 6} = 6,362 \text{ psi} \quad \begin{array}{l} \text{Acceptable} \\ \text{Less than 10000 psi} \end{array}$$

3. Stem Design

$$\text{Stem Type} = 1.5" \text{ Rolled 4tpi}$$

$$\text{Effective Stem Diameter} = D = 1.375 \text{ in} \quad (\text{minor diameter, per AWWA C561})$$

$$\text{Stem Radius of Gyration} = r = 0.344 \text{ in}$$

$$\text{Stem Moment of Inertia} = I = 0.175 \text{ in}^4$$

$$\text{Stem Modulus Of Elasticity} = E = 2.80\text{E}+07 \text{ psi}$$

$$\text{Stem Max Free Length} = L = 49 \text{ in}$$

$$L/r \text{ ratio} = 143 \quad \text{Acceptable not greater than 200}$$

$$\text{Weight of Slide \& Stem} = W = 365 \text{ lbs.}$$

$$\text{Area of Gate} = A = 8.00 \text{ sq.ft}$$

$$\text{Operating Head to gate c/l} = H_c = 9.00 \text{ ft.}$$

$$\text{Seal Friction Coefficient} = \mu = 0.2$$

$$\text{Slide Lifting Force} = \text{SLF} = (62.4 \times \mu \times A \times H_c) = 899 \text{ lbs.}$$

$$\text{Seal Drag (10 lb/in of slide height + 400 corners + 20 lb/in of slide width)} = 1,360 \text{ lbs.}$$

$$\text{Design Lifting Force} = (\text{Greater of Seal Drag or SLF}) + W = F = 1,725 \text{ lbs.}$$

$$\text{Electric Operator Run motor Torque} = \text{Tr} = 128 \text{ ft.-lbs.}$$

$$\text{Stem Factor} = \text{Sf} = 0.019$$

$$\text{Thrust @ Run Torque} = \text{Tr/Sf} = 6,754 \text{ lbs.} \quad \text{Acceptable, greater than F}$$

$$\text{Electric Operator Stalled motor Torque} = T = 194 \text{ ft.lb}$$

$$\text{Stem factor} = \text{Sf} = 0.019$$

$$\text{Design Torque} = T_d = 1.25 \times T = 242.5 \text{ ft.lb}$$

$$\text{Thrust @ Design Torque} = T_h = T_d / \text{Sf} = 12,796 \text{ lbs.}$$

$$\text{Direct stress at Design Thrust} = \sigma = \frac{T_h \bullet 4}{\pi \bullet D^2} = 8,617 \text{ psi}$$

$$\text{Shear Stress at Design Thrust} = \tau = \frac{192 \bullet T_d}{\pi \bullet D^3} = 5,701 \text{ psi}$$

$$\text{Combined stress} = \sigma / 2 + \sqrt{(\sigma / 2)^2 + \tau^2} = 11,455 \text{ psi} \quad \text{Acceptable, < Yield Stress (35000 psi)}$$

$$\text{Stem Buckling Load (C=2)} = \frac{2 \bullet \pi^2 \bullet E \bullet I}{l^2} = 40,390 \text{ lbs.} \quad \text{Acceptable, Greater than Th}$$

4. Yoke Design

Assumed Yoke acts as a simply supported beam under central Point Load (Roark's p.101 – 1.e)

Yoke Type = Double 5 in x 2.5 in Channel sections

Yoke Length = L = Aperture Width = 24 in

Yoke section Moment of Inertia = I = 17.91 in⁴

Material Modulus = E = 2.80E+07 psi (Stainless Steel Grade 316)

Yoke – neutral axis to furthest fiber = y = 2.50 in

At Design Lifting Force (F from section 3)

$$\text{Deflection} = \delta_{Lf} = \frac{F \bullet L^3}{48 \bullet E \bullet I} = 0.001 \text{ in}$$

Deflection Ratio = l / δ = **1: 24228** **Acceptable, Less than 1:360 (AWWA C.561)**

$$\text{Max bending moment at design} = M_{Lf} = \frac{F \bullet L}{4} = 10,350 \text{ in lbs.}$$

$$\text{Max Bending Stress} = \sigma_{Lf} = \frac{M \bullet y}{I} = 1,445 \text{ psi} \quad \textbf{Acceptable}$$

Less than 15000 psi

At Electric Operator Stalled motor Torque (T from section 3)

$$\text{Max bending moment} = M_{St} = \frac{T \bullet L}{Sf \bullet 4} = 61,419 \text{ in lbs.}$$

$$\text{Max Bending Stress} = \sigma_{St} = \frac{M \bullet y}{I} = 8,573 \text{ psi}$$

Reserve Factor On Yield Stress = 4.083 **Acceptable, Greater than 1.5 (AWWA C.561)**

$$\text{Deflection (at Stall)} = \frac{T \bullet L^3}{Sf \bullet 48 \bullet E \bullet I} = 0.006 \text{ in} \quad \textbf{Acceptable}$$

Less than 0.067 in

5. Loads transferred to Structure

Self contained gates do not transmit Operating loads to the mounting structure.

Loads transferred are those due to the Operating Head (Hc from section 3) acting on the gate area

$$\text{Water load} = (62.4 \times A \times Hc) = 4,493 \text{ lbs.}$$

CALCULATION SHEET

Sheet No. A-327-5202
Project: Hilo WWTF Phase 1
Whipps #: 32752
Gate Ref: 06-FG-1410, 1420 & 1430

Prepared By: S. Barnes
5/15/2025

Calculations

1. Allowable Leakage Rates
2. Slide design
3. Stem Design
4. Yoke Design
5. Loads transferred to Structure

1. Leakage Rates

Gate Aperture Width = 30.0 in
Gate Aperture Depth = 72.0 in
Seal Periphery = 204.0 in
Allowable Seating Leakage rate = 0.05 gal/min/ft of seal periphery
Allowable Seating Leakage = 0.85 gal/min

Allowable Unseating Seating Leakage rate = 0.05 gal/min/ft of seal periphery
Allowable Unseating Seating Leakage = 0.85 gal/min

2. Slide Design

Design Head on gate Invert = **H** = 23 ft. = 276 in
Aperture Width = **L** = 30 in
Density of Water = ρ = 62.4 lb/cu.ft = 0.0361 lb/cu.in
Slide section depth (Beam Width) = **d** = 13.3 in
Slide section Moment of Inertia = **I** = 3.9 in⁴
Slide – furthest fiber distance = **y** = 2.12 in
Material Modulus = **E** = 2.80E+07 psi (Stainless Steel Grade 316)

Assumed Slide section is a simply supported beam under uniform load (Roark's p.104 – 2.e)

$$W_a = \text{Load per unit Length} = \left(H - \frac{d}{2}\right) \cdot \rho \cdot d = 128.8 \text{ lb/in}$$

$$\text{Deflection} = \delta = \frac{5 \cdot W_a \cdot l^4}{384 \cdot E \cdot I} = 0.012 \text{ in} \quad \text{Acceptable, Less than 0.03 in}$$

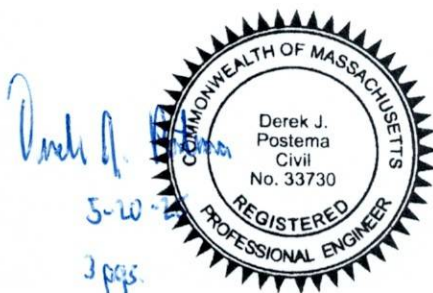
$$\text{Deflection Ratio} = l / \delta = 1: 2412$$

$$\text{Bending Stress} = \sigma = \frac{W_a \cdot l^2 \cdot y}{8 \cdot I} = 7,881 \text{ psi}$$

Acceptable
Less than 15000 psi

R.F of 9.5 On Ultimate Stress
R.F of 4.4 On Yield Stress

Note: RF refers to Reserve factor (or safety factor)



$$\text{Vertical Shear through Stiffener section} = V = \frac{\rho \bullet H \bullet d \bullet l}{2}$$

$$\text{Area of slide plate section} = A' = 3.31 \text{ sq.in}$$

$$\begin{aligned} \text{Distance from slide plate neutral axis to combined section neutral axis} &= z' = 0.503 \text{ in} \\ \text{Effective width of Weld} &= b = 0.1768 \text{ in} \quad (\text{Based on a 0.25" weld}) \end{aligned}$$

$$\text{Shear Flow} = \tau = \frac{V \bullet A' \bullet z'}{I} = 846.3 \text{ lb/in}$$

$$\text{Weld Shear stress} = \tau \text{ based on skip weld, 2 on 6} = 7,182 \text{ psi} \quad \begin{matrix} \text{Acceptable} \\ \text{Less than 10000 psi} \end{matrix}$$

(both sides)

3. Stem Design

$$\text{Stem Type} = 1.75" \text{ Rolled 4tpi}$$

$$\text{Effective Stem Diameter} = D = 1.625 \text{ in} \quad (\text{minor diameter, per AWWA C561})$$

$$\text{Stem Radius of Gyration} = r = 0.406 \text{ in}$$

$$\text{Stem Moment of Inertia} = I = 0.342 \text{ in}^4$$

$$\text{Stem Modulus Of Elasticity} = E = 2.80\text{E}+07 \text{ psi}$$

$$\text{Stem Max Free Length} = L = 72 \text{ in}$$

$$L/r \text{ ratio} = 177 \quad \text{Acceptable not greater than 200}$$

$$\text{Weight of Slide \& Stem} = W = 345 \text{ lbs.}$$

$$\text{Area of Gate} = A = 15.00 \text{ sq.ft}$$

$$\text{Operating Head to gate c/l} = H_c = 7.50 \text{ ft.}$$

$$\text{Seal Friction Coefficient} = \mu = 0.2$$

$$\text{Slide Lifting Force} = \text{SLF} = (62.4 \times \mu \times A \times H_c) = 1,404 \text{ lbs.}$$

$$\text{Seal Drag (10 lb/in of slide height + 400 corners + 20 lb/in of slide width)} = 1,720 \text{ lbs.}$$

$$\text{Design Lifting Force} = (\text{Greater of Seal Drag or SLF}) + W = F = 2,065 \text{ lbs.}$$

$$\text{Electric Operator Run motor Torque} = Tr = 293 \text{ ft.-lbs.}$$

$$\text{Stem Factor} = Sf = 0.021$$

$$\text{Thrust @ Run Torque} = Tr/Sf = 13,908 \text{ lbs.} \quad \text{Acceptable, greater than F}$$

$$\text{Electric Operator Stalled motor Torque} = T = 441 \text{ ft.lb}$$

$$\text{Stem factor} = Sf = 0.021$$

$$\text{Design Torque} = Td = 1.25 \times T = 551.3 \text{ ft.lb}$$

$$\text{Thrust @ Design Torque} = Th = Td / Sf = 26,166 \text{ lbs.}$$

$$\text{Direct stress at Design Thrust} = \sigma = \frac{Th \bullet 4}{\pi \bullet D^2} = 12,616 \text{ psi}$$

$$\text{Shear Stress at Design Thrust} = \tau = \frac{192 \bullet Td}{\pi \bullet D^3} = 7,851 \text{ psi}$$

$$\text{Combined stress} = \sigma / 2 + \sqrt{(\sigma / 2)^2 + \tau^2} = 16,380 \text{ psi} \quad \text{Acceptable, < Yield Stress (35000 psi)}$$

$$\text{Stem Buckling Load (C=2)} = \frac{2 \bullet \pi^2 \bullet E \bullet I}{l^2} = 36,493 \text{ lbs.} \quad \text{Acceptable, Greater than Th}$$

4. Yoke Design

Assumed Yoke acts as a simply supported beam under central Point Load (Roark's p.101 – 1.e)

Yoke Type = Double 5 in x 2.5 in Channel sections

Yoke Length = L = Aperture Width = 30 in

Yoke section Moment of Inertia = I = 17.91 in⁴

Material Modulus = E = 2.80E+07 psi (Stainless Steel Grade 316)

Yoke – neutral axis to furthest fiber = y = 2.50 in

At Design Lifting Force (F from section 3)

$$\text{Deflection} = \delta_{Lf} = \frac{F \bullet L^3}{48 \bullet E \bullet I} = 0.002 \text{ in}$$

Deflection Ratio = δ / L = **1: 12953** **Acceptable, Less than 1:360 (AWWA C.561)**

$$\text{Max bending moment at design} = M_{Lf} = \frac{F \bullet L}{4} = 15,488 \text{ in lbs.}$$

$$\text{Max Bending Stress} = \sigma_{Lf} = \frac{M \bullet y}{I} = 2,162 \text{ psi} \quad \textbf{Acceptable}$$

Less than 15000 psi

At Electric Operator Stalled motor Torque (T from section 3)

$$\text{Max bending moment} = M_{St} = \frac{T \bullet L}{Sf \bullet 4} = 156,995 \text{ in lbs.}$$

$$\text{Max Bending Stress} = \sigma_{St} = \frac{M \bullet y}{I} = 21,913 \text{ psi}$$

Reserve Factor On Yield Stress = 1.597 **Acceptable, Greater than 1.5 (AWWA C.561)**

$$\text{Deflection (at Stall)} = \frac{T \bullet L^3}{Sf \bullet 48 \bullet E \bullet I} = 0.023 \text{ in} \quad \textbf{Acceptable}$$

Less than 0.083 in

5. Loads transferred to Structure

Self contained gates do not transmit Operating loads to the mounting structure.

Loads transferred are those due to the Operating Head (Hc from section 3) acting on the gate area

$$\text{Water load} = (62.4 \times A \times Hc) = 7,020 \text{ lbs.}$$

CALCULATION SHEET

Sheet No. A-327-5203
Project: Hilo WWTF Phase 1
Whipps #: 32752
Gate Ref: 09-FG-1100

Prepared By: S. Barnes
5/15/2025

Calculations

1. Allowable Leakage Rates
2. Slide design
3. Stem Design
4. Yoke Design
5. Loads transferred to Structure

1. Leakage Rates

Gate Aperture Width = 16.0 in
Gate Aperture Depth = 16.0 in
Seal Periphery = 64.0 in
Allowable Seating Leakage rate = 0.05 gal/min/ft of seal periphery
Allowable Seating Leakage = 0.27 gal/min

Allowable Unseating Seating Leakage rate = 0.05 gal/min/ft of seal periphery
Allowable Unseating Seating Leakage = 0.27 gal/min

2. Slide Design

Design Head on gate Invert = H = 10.67 ft. = 128.0 in
Aperture Width = L = 16 in
Density of Water = ρ = 62.4 lb/cu.ft = 0.0361 lb/cu.in
Slide section depth (Beam Width) = d = 13 in
Slide section Moment of Inertia = I = 1.334 in⁴
Slide – furthest fiber distance = y = 2.403 in
Material Modulus = E = 2.80E+07 psi (Stainless Steel Grade 316)

Assumed Slide section is a simply supported beam under uniform load (Roark's p.104 – 2.e)

$$W_a = \text{Load per unit Length} = \left(H - \frac{d}{2} \right) \cdot \rho \cdot d = 57.0 \text{ lb/in}$$

$$\text{Deflection} = \delta = \frac{5 \cdot W_a \cdot l^4}{384 \cdot E \cdot I} = 0.001 \text{ in} \quad \text{Acceptable, Less than 0.016 in}$$

$$\text{Deflection Ratio} = l / \delta = 1: 12274$$

$$\text{Bending Stress} = \sigma = \frac{W_a \cdot l^2 \cdot y}{8 \cdot I} = 3,289 \text{ psi}$$

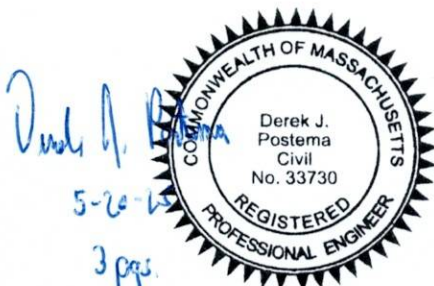
Acceptable

Less than 15000 psi

R.F of 22.8 On Ultimate Stress

R.F of 10.6 On Yield Stress

Note: RF refers to Reserve Factor (or safety factor)



$$\text{Vertical Shear through Stiffener section} = V = \frac{\rho \bullet H \bullet d \bullet l}{2} = 931 \text{ lbs.}$$

$$\text{Area of slide plate section} = A' = 3.25 \text{ sq.in}$$

$$\text{Distance from slide plate neutral axis to combined section neutral axis} = z' = 0.222 \text{ in}$$

$$\text{Effective width of Weld} = b = 0.1768 \text{ in} \quad (\text{Based on a 0.25" weld})$$

$$\text{Shear Flow} = \tau = \frac{V \bullet A' \bullet z'}{I} = 503 \text{ lbs./in}$$

$$\text{Weld Shear stress} = \tau \text{ based on skip weld, 2 on 6} = 4,272 \text{ psi} \quad \textbf{Acceptable Less than 10000 psi}$$

3. Stem Design

$$\text{Stem Type} = 1.5" \text{ Rolled 4tpi}$$

$$\text{Effective Stem Diameter} = D = 1.375 \text{ in} \quad (\text{minor diameter, per AWWA C561})$$

$$\text{Stem Radius of Gyration} = r = 0.344 \text{ in}$$

$$\text{Stem Moment of Inertia} = I = 0.175 \text{ in}^4$$

$$\text{Stem Modulus Of Elasticity} = E = 2.80\text{E}+07 \text{ psi}$$

$$\text{Stem Max Free Length} = L = 56 \text{ in}$$

$$L/r \text{ ratio} = 163 \quad \textbf{Acceptable not greater than 200}$$

$$\text{Weight of Slide \& Stem} = W = 120.0 \text{ lbs.}$$

$$\text{Area of Gate} = A = 1.8 \text{ sq.ft}$$

$$\text{Operating Head to gate c/l} = H_c = 20.0 \text{ ft.}$$

$$\text{Seal Friction Coefficient} = \mu = 0.2$$

$$\text{Slide Lifting Force} = \text{SLF} = (62.4 \times \mu \times A \times H_c) = 444 \text{ lbs.}$$

$$\text{Seal Drag (10 lb/in of slide height + 400 corners + 20 lb/in of slide width)} = 880 \text{ lbs.}$$

$$\text{Design Lifting Force} = (\text{Greater of Seal Drag or SLF}) + W = F = 1,000 \text{ lbs.}$$

Manual Operator, Ratio 2:1, 85% efficiency, 15" Handwheel

$$\text{Allowable Operating Effort} = 40 \text{ lbs.}$$

$$\text{Torque at Allowable operating effort} = T = (40 \times 2 \times 0.85 \times 0.625) = 42.5 \text{ ft.lb}$$

$$\text{Stem factor} = S_f = 0.0190$$

$$\text{Thrust @ Allowable Operating Effort} = T / S_f = 2,243 \text{ lbs.} \quad \textbf{Acceptable, greater than F}$$

$$\text{Design Torque} = T_d = 2 \times T = 85.0 \text{ ft.lb}$$

$$\text{Thrust @ Design Torque} = T_h = T_d / S_f = 4,485 \text{ lbs.}$$

$$\text{Direct stress at Design Thrust} = \sigma = \frac{T_h \bullet 4}{\pi \bullet D^2} = 3,020 \text{ psi}$$

$$\text{Shear Stress at Design Thrust} = \tau = \frac{192 \bullet T_d}{\pi \bullet D^3} = 1,998 \text{ psi}$$

$$\text{Combined stress} = \sigma / 2 + \sqrt{(\sigma / 2)^2 + \tau^2} = 4,015 \text{ psi} \quad \textbf{Acceptable, < Yield Stress (35000 psi)}$$

$$\text{Stem Buckling Load (C=2)} = \frac{2 \bullet \pi^2 \bullet E \bullet I}{l^2} = 30,924 \text{ lbs.} \quad \textbf{Acceptable, Greater than Th}$$

4. Yoke Design

Assumed Yoke acts as a simply supported beam under central Point Load (Roark's p.101 – 1.e)

Yoke Type = Double 4.5 in x 2 in Channel sections

Yoke Length = Aperture Width = L = 16.0 in
Yoke section Moment of Inertia = I = 11.7 in⁴
Material Modulus = E = 2.80E+07 psi (Stainless Steel Grade 316)
Yoke – neutral axis to furthest fiber = y = 2.3 in

At Design Lifting Force (F from section 3)

$$\text{Deflection} = \delta_{Lf} = \frac{F \bullet L^3}{48 \bullet E \bullet I} = 0.000 \text{ in}$$

Deflection Ratio = l / δ = **1: 61469** **Acceptable, Less than 1:360 (AWWA C.561)**

$$\text{Max Bending Moment at Design} = M_{Lf} = \frac{F \bullet L}{4} = 4,000 \text{ lb in}$$

$$\text{Max Bending Stress} = \sigma_{Lf} = \frac{M \bullet y}{I} = 0,769 \text{ psi} \quad \textbf{Acceptable} \quad \textbf{Less than 15000 psi}$$

At Design Thrust (T from section 3)

$$\text{Deflection} = \delta_{Dt} = \frac{T \bullet L^3}{Sf \bullet 48 \bullet E \bullet I} = 0.001 \text{ in} \quad \textbf{Acceptable} \\ \textbf{Less than 0.044 in}$$

$$\text{Max bending moment} = M_{Dt} = \frac{T \bullet L}{Sf \bullet 4} = 17,940 \text{ lb in}$$

$$\text{Max Bending Stress} = \sigma_{Dt} = \frac{M \bullet y}{I} = 3,448 \text{ psi}$$

Reserve factor on Yield Stress = 10.2 **Acceptable, Greater than 2.0 (AWWA C.561)**

Reserve factor on Ultimate Stress = 21.8 **Acceptable, Greater than 4.0 (AWWA C.561)**

5. Loads transferred to Structure

Self contained gates do not transmit Operating loads to the mounting structure.

Loads transferred are those due to the Operating Head (Hc from section 3) acting on the gate area

$$\text{Water load} = (62.4 \times A \times Hc) = 2,219 \text{ lbs.}$$

CALCULATION SHEET

Sheet No. A-327-5204
Project: Hilo WWTF Phase 1
Whipps #: 32752
Gate Ref: 02-GAT-1140, 1141 & 1142

Prepared By: S. Barnes
5/15/2025

Calculations

1. Allowable Leakage Rates
2. Slide design
3. Stem Design
4. Loads transferred to Structure

1. Leakage Rates

Gate Aperture Width = 6.0 in
Gate Aperture Depth = 6.0 in
Seal Periphery = 24.0 in
Allowable Seating Leakage rate = 0.05 gal/min/ft of seal periphery
Allowable Seating Leakage = 0.10 gal/min

Allowable Unseating Seating Leakage rate = 0.05 gal/min/ft of seal periphery
Allowable Unseating Seating Leakage = 0.10 gal/min

2. Slide Design

Design Head on gate Invert = **H** = 20.3 ft. = 243 in
Aperture Width = **L** = 6 in
Density of Water = ρ = 62.4 lb/cu.ft = 0.036 lb/cu.in
Slide section depth = **d** = 6 in

Slide plate thickness = **t** = 0.25 in
Material Modulus = **E** = 2.80E+07 psi (Stainless Steel Grade 316)

Assume gate slide section is a simply supported on all edges, uniform load (Roark's p.458 Case 1.a)
Flat Plate Design

where width to depth ratio = 1
Therefore β = 0.2874
 α = 0.0444
 q = 8.77 psi

$$\text{Max Stress} = \sigma = \frac{b \cdot q \cdot L^2}{t^2} = 1,452 \text{ psi}$$

Acceptable

R.F of 51.6 On Ultimate Stress

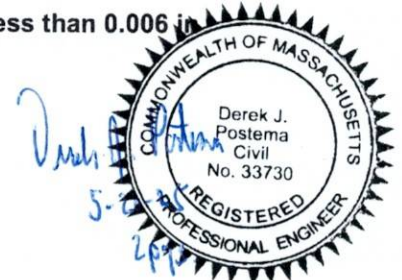
R.F of 24.1 On Yield Stress

Note: RF refers to Reserve Factor (or safety factor)

$$\text{Deflection} = \delta = \frac{\alpha \cdot q \cdot L^4}{E \cdot t^3} = 0.001 \text{ in}$$

Acceptable, Less than 0.006 in

$$\text{Deflection Ratio} = L / \delta = 1: 5200$$



3. Stem Design

Stem Type	=	1.5" Rolled 4tpi	
Effective Stem Diameter	=	D	= 1.375 in (minor diameter, per AWWA C561)
Stem Radius of Gyration	=	r	= 0.344 in
Stem Moment of Inertia	=	I	= 0.175 in ⁴
Stem Modulus Of Elasticity	=	E	= 2.80E+07 psi
Stem Max Free Length	=	L	= 48 in

L/r ratio = 140 **Acceptable not greater than 200**

Weight of Slide & Stem	=	W	= 110 lbs.	
Area of Gate	=	A	= 0.3 sq.ft	
Operating Head to gate c/l	=	Hc	= 12.75 ft.	
Seal Friction Coefficient	=	μ	= 0.2	
Slide Lifting Force	=	SLF	= (62.4 x μ x A x Hc)	= 40 lbs.
Seal Drag (10 lb/in of slide height + 400 corners + 20 lb/in of slide width)				= 580 lbs.

Design Lifting Force = (Greater of Seal Drag or SLF) + W = F = 0,690 lbs.

Manual Operator, Ratio 2:1, 85% efficiency, 15" Crank

Allowable Operating Effort	=	40 lbs.	
Torque at Allowable operating effort	=	T	= (40 x 2 x 0.85 x 1.25) = 85 ft.-lb.
Stem factor	=	Sf	= 0.019
Thrust @ Allowable Operating Effort	=	T / Sf	= 4,485 lbs. Acceptable, greater than F

Design Torque	=	Td	= 2 x T = 170.0 ft.-lb.
Thrust @ Design Torque	=	Th	= Td / Sf = 8,970 lbs.

$$\text{Direct stress at Design Thrust} = \sigma = \frac{Th \bullet 4}{\pi \bullet D^2} = 6,041 \text{ psi}$$

$$\text{Shear Stress at Design Thrust} = \tau = \frac{192 \bullet Td}{\pi \bullet D^3} = 3,997 \text{ psi}$$

$$\text{Combined stress} = \sigma / 2 + \sqrt{(\sigma / 2)^2 + \tau^2} = 8,030 \text{ psi} \quad \text{Acceptable, < Yield Stress (35000 psi)}$$

$$\text{Stem Buckling Load (C=2)} = \frac{2 \bullet \pi^2 \bullet E \bullet I}{l^2} = 42,091 \text{ lbs.} \quad \text{Acceptable, Greater than Th}$$

4. Loads transferred to Structure

Non-Self contained gates transmit Operating loads to the mounting structure, in addition to Loads due to the Operating Head (Hc from section 3) acting on the gate area.

Operating load transferred through pedestal	=	690 lbs.
Design load transferred through pedestal	=	8,970 lbs.

$$\text{Water load} = (62.4 \times A \times Hc) = 199 \text{ lbs.}$$

CALCULATION SHEET

Sheet No. A-327-5205
Project Hilo WWTF Phase 1
Whipps #: 32752
Gate Ref: 02-GAT-1240

Prepared By: S. Barnes
5/15/2025

Calculations

1. Allowable Leakage Rates
2. Slide design
3. Stem Design
4. Loads transferred to Structure

1. Leakage Rates

Gate Aperture Width = 36.0 in
Gate Aperture Depth = 36.0 in
Seal Periphery = 144.0 in
Allowable Seating Leakage rate = 0.05 gal/min/ft of seal periphery
Allowable Seating Leakage = 0.60 gal/min

Allowable Unseating Seating Leakage rate = 0.05 gal/min/ft of seal periphery
Allowable Unseating Seating Leakage = 0.60 gal/min

2. Slide Design

Design Head on gate Invert = H = 21.5 ft. = 258 in
Aperture Width = L = 36 in
Density of Water = ρ = 62.4 lb/cu.ft = 0.036 lb/cu.in
Slide section depth (Beam Width) = d = 11.3 in
Slide section Moment of Inertia = I = 4.68 in⁴
Slide – furthest fiber distance = y = 3.57 in
Material Modulus = E = 2.80E+07 psi (Stainless Steel Grade 316)

Assumed Slide section is a simply supported beam under uniform load (Roark's p.104 – 2.e)

$$W_a = \text{Load per unit Length} = \left(H - \frac{d}{2}\right) \cdot \rho \cdot d = 102.5 \text{ lb/in}$$

$$\text{Deflection} = \delta = \frac{5 \cdot W_a \cdot l^4}{384 \cdot E \cdot I} = 0.017 \text{ in} \quad \text{Acceptable, Less than 0.036 in}$$

$$\text{Deflection Ratio} = l / \delta = 1: 2104$$

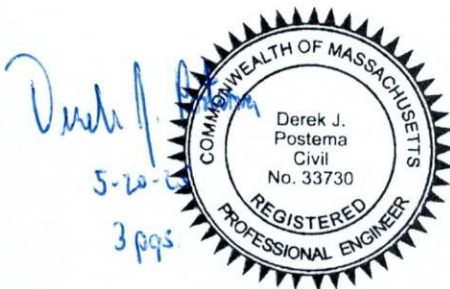
$$\text{Bending Stress} = \sigma = \frac{W_a \cdot l^2 \cdot y}{8 \cdot I} = 12,660 \text{ psi}$$

Acceptable
Less than 15000 psi

R.F of 5.9 On Ultimate Stress

R.F of 2.8 On Yield Stress

Note: RF refers to Reserve Factor (or safety factor)



$$\text{Vertical Shear through Stiffener section} = V = \frac{\rho \cdot H \cdot d \cdot l}{2} = 1886 \text{ lbs.}$$

$$\text{Area of slide plate section} = A' = 2.81 \text{ sq.in}$$

$$\text{Distance from slide plate neutral axis to combined section neutral axis} = z' = 0.557 \text{ in}$$

$$\text{Effective width of Weld} = b = 0.1768 \text{ in (Based on a 0.25" weld)}$$

$$\text{Shear Flow} = \tau = \frac{V \cdot A' \cdot z'}{I} = 632 \text{ lbs./in}$$

$$\text{Weld Shear stress} = \tau \text{ based on skip weld, 2 on 6} = 5,362 \text{ psi}$$

**Acceptable
Less than 10000 psi**

3. Stem Design

$$\text{Stem Type} = 1.5" \text{ Rolled 4tpi}$$

$$\text{Effective Stem Diameter} = D = 1.375 \text{ in (minor diameter, per AWWA C561)}$$

$$\text{Stem Radius of Gyration} = r = 0.344 \text{ in}$$

$$\text{Stem Moment of Inertia} = I = 0.175 \text{ in}^4$$

$$\text{Stem Modulus Of Elasticity} = E = 2.80\text{E}+07 \text{ psi}$$

$$\text{Stem Max Free Length} = L = 54 \text{ in}$$

$$L/r \text{ ratio} = 157 \quad \text{Acceptable not greater than 200}$$

$$\text{Weight of Slide \& Stem} = W = 250 \text{ lbs.}$$

$$\text{Area of Gate} = A = 9.00 \text{ sq.ft}$$

$$\text{Operating Head to gate c/l} = H_c = 11.50 \text{ ft}$$

$$\text{Seal Friction Coefficient} = \mu = 0.2$$

$$\text{Slide Lifting Force} = \text{SLF} = (62.4 \times \mu \times A \times H_c) = 1,292 \text{ lbs.}$$

$$\text{Seal Drag (10 lb/in of slide height + 400 corners + 20 lb/in of slide width)} = 1,480 \text{ lbs.}$$

$$\text{Design Lifting Force} = (\text{Greater of Seal Drag or SLF}) + W = F = 1,730 \text{ lbs.}$$

$$\text{Electric Operator Type} = \text{Limitorque MX--}$$

$$\text{Electric Operator Run motor Torque} = Tr = 128 \text{ ft.lb}$$

$$\text{Stem factor} = Sf = 0.019$$

$$\text{Thrust @ run Torque} = Tr/Sf = 6,754 \text{ lbs.} \quad \text{Acceptable, greater than F}$$

$$\text{Electric Operator Stalled motor Torque} = T = 194 \text{ ft.lb}$$

$$\text{Stem factor} = Sf = 0.0190$$

$$\text{Design Torque} = Td = 1.25 \times T = 242.5 \text{ ft.lbs.}$$

$$\text{Thrust @ Design Torque} = Th = Td / Sf = 12,796 \text{ lbs.}$$

$$\text{Direct stress at Design Thrust} = \sigma = \frac{Th \cdot 4}{\pi \cdot D^2} = 8,617 \text{ psi}$$

$$\text{Shear Stress at Design Thrust} = \tau = \frac{192 \cdot Td}{\pi \cdot D^3} = 5,701 \text{ psi}$$

$$\text{Combined stress} = \sigma / 2 + \sqrt{(\sigma / 2)^2 + \tau^2} = 11,455 \text{ psi} \quad \text{Acceptable, < Yield Stress (35000 psi)}$$

$$\text{Stem Buckling Load (C=2)} = \frac{2 \cdot \pi^2 \cdot E \cdot I}{l^2} = 33,257 \text{ lbs.} \quad \text{Acceptable, Greater than Th}$$

4. Loads transferred to Structure

Non-Self contained gates transmit Operating loads to the mounting structure, in addition to Loads due to the Operating Head (H_c from section 3) acting on the gate area.

Operating load transferred through pedestal	=	1,730 lbs.
Design load transferred through pedestal	=	12,796 lbs.

Water load = $(62.4 \times A \times H_c)$ = 6,458 lbs.

CALCULATION SHEET

Sheet No. A-327-5206
Project: Hilo WWTF Phase 1
Whipps #: 32752
Gate Ref: 06-FG-1515, 1525 & 1535

Prepared By: S. Barnes
5/15/2025

Calculations

1. Allowable Leakage Rates
2. Slide design
3. Stem Design
4. Loads transferred to Structure

1. Leakage Rates

Gate Aperture Width = 12.0 in
Gate Aperture Depth = 12.0 in
Seal Periphery = 48.0 in
Allowable Seating Leakage rate = 0.1 gal/min/ft of seal periphery
Allowable Seating Leakage = 0.20 gal/min

Allowable Unseating Seating Leakage rate = 0.1 gal/min/ft of seal periphery
Allowable Unseating Seating Leakage = 0.20 gal/min

2. Slide Design

Design Head on gate Invert = H = 23 ft. = 270 in
Aperture Width = L = 12 in
Density of Water = ρ = 62 lb/cu.ft = 0.036 lb/cu.in
Slide section depth (Beam Width) = d = 9.3 in
Slide section Moment of Inertia = I = 1.3 in⁴
Slide – furthest fiber distance = y = 2.3 in
Material Modulus = E = 2.80E+07 psi (Stainless Steel Grade 316)

Assumed Slide section is a simply supported beam under uniform load (Roark's p.104 – 2.e)

$$W_a = \text{Load per unit Length} = \left(H - \frac{d}{2} \right) \cdot \rho \cdot d = 88.6 \text{ lb/in}$$

$$\text{Deflection} = \delta = \frac{5 \cdot W_a \cdot l^4}{384 \cdot E \cdot I} = 0.001 \text{ in}$$

Acceptable, Less than 0.012 in

$$\text{Deflection Ratio} = l / \delta = 1: 17804$$

$$\text{Bending Stress} = \sigma = \frac{W_a \cdot l^2 \cdot y}{8 \cdot I} = 2,935 \text{ psi}$$

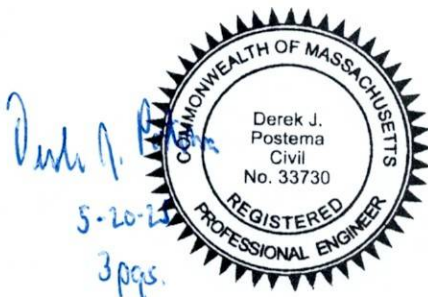
Acceptable

Less than 15000 psi

R.F of 25.6 On Ultimate Stress

R.F of 11.9 On Yield Stress

Note: RF refers to Reserve Factor (or safety factor)



$$\text{Vertical Shear through Stiffener section} = V = \frac{\rho \bullet H \bullet d \bullet l}{2} = 0,493 \text{ lbs.}$$

$$\text{Area of slide plate section} = A' = 2.31 \text{ sq.in}$$

$$\begin{aligned} \text{Distance from slide plate neutral axis to combined section neutral axis} &= z' = 0.293 \text{ in} \\ \text{Effective width of Weld} &= b = 0.1768 \text{ in} \quad (\text{Based on a 0.25" weld}) \end{aligned}$$

$$\text{Shear Flow} = \tau = \frac{V \bullet A' \bullet z'}{I} = 263 \text{ lbs./in}$$

$$\text{Weld Shear stress} = \tau \text{ based on skip weld, 2 on 6} = 2,232 \text{ psi} \quad \begin{array}{l} \text{Acceptable} \\ \text{Less than 10000 psi} \end{array}$$

(both sides)

3. Stem Design

$$\text{Stem Type} = 1.5" \text{ Rolled 4tpi}$$

$$\text{Effective Stem Diameter} = D = 1.375 \text{ in} \quad (\text{minor diameter, per AWWA C561})$$

$$\text{Stem Radius of Gyration} = r = 0.344 \text{ in}$$

$$\text{Stem Moment of Inertia} = I = 0.175 \text{ in}^4$$

$$\text{Stem Modulus Of Elasticity} = E = 2.80\text{E}+07 \text{ psi}$$

$$\text{Stem Max Free Length} = L = 66 \text{ in}$$

$$\text{L/r ratio} = 192 \quad \text{Acceptable not greater than 200}$$

$$\text{Weight of Slide \& Stem} = W = 165 \text{ lbs.}$$

$$\text{Area of Gate} = A = 1.0 \text{ sq.ft}$$

$$\text{Operating Head to gate c/l} = H_c = 20.00 \text{ ft.}$$

$$\text{Seal Friction Coefficient} = \mu = 0.2$$

$$\text{Slide Lifting Force} = \text{SLF} = (62.4 \times \mu \times A \times H_c) = 250 \text{ lbs.}$$

$$\text{Seal Drag (10 lb/in of slide height + 400 corners + 20 lb/in of slide width)} = 760 \text{ lbs.}$$

$$\text{Design Lifting Force} = (\text{Greater of Seal Drag or SLF}) + W = F = 0,925 \text{ lbs.}$$

$$\text{Manual Operator, Ratio 1:1, 76\% efficiency, 24" 'T' Key}$$

$$\text{Allowable Operating Effort} = 40 \text{ lbs.}$$

$$\text{Torque at Allowable operating effort} = T = (40 \times 1 \times 0.76 \times 1) = 30 \text{ ft.-lb.}$$

$$\text{Stem factor} = S_f = 0.019$$

$$\text{Thrust @ Allowable Operating Effort} = T / S_f = 1,604 \text{ lbs.} \quad \text{Acceptable, greater than F}$$

$$\text{Design Torque} = T_d = 2 \times T = 60.8 \text{ ft.-lb.}$$

$$\text{Thrust @ Design Torque} = T_h = T_d / S_f = 3,208 \text{ lbs.}$$

$$\text{Direct stress at Design Thrust} = \sigma = \frac{T_h \bullet 4}{\pi \bullet D^2} = 2,161 \text{ psi}$$

$$\text{Shear Stress at Design Thrust} = \tau = \frac{192 \bullet T_d}{\pi \bullet D^3} = 1,429 \text{ psi}$$

$$\text{Combined stress} = \sigma / 2 + \sqrt{(\sigma / 2)^2 + \tau^2} = 2,872 \text{ psi} \quad \text{Acceptable, < Yield Stress (35000 psi)}$$

$$\text{Stem Buckling Load (C=2)} = \frac{2 \bullet \pi^2 \bullet E \bullet I}{l^2} = 22,263 \text{ lbs.} \quad \text{Acceptable, Greater than Th}$$

4. Loads transferred to Structure

Non-Self contained gates transmit Operating loads to the mounting structure, in addition to Loads due to the Operating Head (H_c from section 3) acting on the gate area.

Operating load transferred through pedestal	=	925 lbs.
Design load transferred through pedestal	=	3,208 lbs.

Water load = $(62.4 \times A \times H_c)$ = 1,248 lbs.

CALCULATION SHEET

Sheet No. A-327-5207
Project: Hilo WWTF Phase 1
Whipps #: 32752
Gate Ref: 06-FG-1610, 1620 & 1630

Prepared By: S. Barnes
5/15/2025

Calculations

1. Allowable Leakage Rates
2. Slide design
3. Stem Design
4. Loads transferred to Structure

1. Leakage Rates

Gate Aperture Width = 12.0 in
Gate Aperture Depth = 12.0 in
Seal Periphery = 48.0 in
Allowable Seating Leakage rate = 0.1 gal/min/ft of seal periphery
Allowable Seating Leakage = 0.20 gal/min

Allowable Unseating Seating Leakage rate = 0.1 gal/min/ft of seal periphery
Allowable Unseating Seating Leakage = 0.20 gal/min

2. Slide Design

Design Head on gate Invert = **H** = 13 ft. = 150 in
Aperture Width = **L** = 12 in
Density of Water = ρ = 62 lb/cu.ft = 0.036 lb/cu.in
Slide section depth (Beam Width) = **d** = 9.3 in
Slide section Moment of Inertia = **I** = 1.3 in⁴
Slide – furthest fiber distance = **y** = 2.3 in
Material Modulus = **E** = 2.80E+07 psi (Stainless Steel Grade 316)

Assumed Slide section is a simply supported beam under uniform load (Roark's p.104 – 2.e)

$$W_a = \text{Load per unit Length} = \left(H - \frac{d}{2} \right) \cdot \rho \cdot d = 48.5 \text{ lb/in}$$

$$\text{Deflection} = \delta = \frac{5 \cdot W_a \cdot l^4}{384 \cdot E \cdot I} = 0.0004 \text{ in} \quad \text{Acceptable, Less than 0.012 in}$$

$$\text{Deflection Ratio} = l / \delta = 1: 32500$$

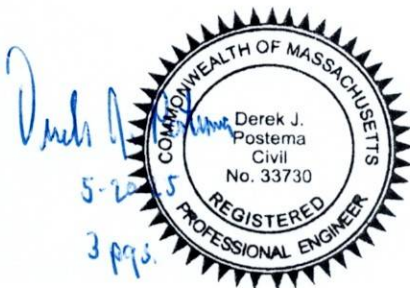
$$\text{Bending Stress} = \sigma = \frac{W_a \cdot l^2 \cdot y}{8 \cdot I} = 1,608 \text{ psi}$$

Acceptable
Less than 15000 psi

R.F of 46.7 On Ultimate Stress

R.F of 21.8 On Yield Stress

Note: RF refers to Reserve Factor (or safety factor)



$$\text{Vertical Shear through Stiffener section} = V = \frac{\rho \bullet H \bullet d \bullet l}{2} = 0,493 \text{ lbs.}$$

$$\text{Area of slide plate section} = A' = 2.31 \text{ sq.in}$$

$$\text{Distance from slide plate neutral axis to combined section neutral axis} = z' = 0.293 \text{ in}$$

$$\text{Effective width of Weld} = b = 0.1768 \text{ in (Based on a 0.25" weld)}$$

$$\text{Shear Flow} = \tau = \frac{V \bullet A' \bullet z'}{I} = 263 \text{ lbs./in}$$

$$\text{Weld Shear stress} = \tau \text{ based on skip weld, 2 on 6} = 2,232 \text{ psi}$$

**Acceptable
Less than 10000 psi**

3. Stem Design

$$\text{Stem Type} = 1.5" \text{ Rolled 4tpi}$$

$$\text{Effective Stem Diameter} = D = 1.375 \text{ in (minor diameter, per AWWA C561)}$$

$$\text{Stem Radius of Gyration} = r = 0.344 \text{ in}$$

$$\text{Stem Moment of Inertia} = I = 0.175 \text{ in}^4$$

$$\text{Stem Modulus Of Elasticity} = E = 2.80\text{E}+07 \text{ psi}$$

$$\text{Stem Max Free Length} = L = 58 \text{ in}$$

$$L/r \text{ ratio} = 169 \quad \text{Acceptable not greater than 200}$$

$$\text{Weight of Slide \& Stem} = W = 145 \text{ lbs.}$$

$$\text{Area of Gate} = A = 1.0 \text{ sq.ft}$$

$$\text{Operating Head to gate c/l} = H_c = 20.00 \text{ ft.}$$

$$\text{Seal Friction Coefficient} = \mu = 0.2$$

$$\text{Slide Lifting Force} = \text{SLF} = (62.4 \times \mu \times A \times H_c) = 250 \text{ lbs.}$$

$$\text{Seal Drag (10 lb/in of slide height + 400 corners + 20 lb/in of slide width)} = 760 \text{ lbs.}$$

$$\text{Design Lifting Force} = (\text{Greater of Seal Drag or SLF}) + W = F = 0,905 \text{ lbs.}$$

$$\text{Manual Operator, Ratio 2:1, 85\% efficiency, 15" Crank}$$

$$\text{Allowable Operating Effort} = 40 \text{ lbs.}$$

$$\text{Torque at Allowable operating effort} = T = (40 \times 2 \times 0.85 \times 1.25) = 85 \text{ ft.-lb.}$$

$$\text{Stem factor} = S_f = 0.019$$

$$\text{Thrust @ Allowable Operating Effort} = T / S_f = 4,485 \text{ lbs.} \quad \text{Acceptable, greater than F}$$

$$\text{Design Torque} = T_d = 2 \times T = 170.0 \text{ ft.-lb.}$$

$$\text{Thrust @ Design Torque} = T_h = T_d / S_f = 8,970 \text{ lbs.}$$

$$\text{Direct stress at Design Thrust} = \sigma = \frac{T_h \bullet 4}{\pi \bullet D^2} = 6,041 \text{ psi}$$

$$\text{Shear Stress at Design Thrust} = \tau = \frac{192 \bullet T_d}{\pi \bullet D^3} = 3,997 \text{ psi}$$

$$\text{Combined stress} = \sigma / 2 + \sqrt{(\sigma / 2)^2 + \tau^2} = 8,030 \text{ psi} \quad \text{Acceptable, < Yield Stress (35000 psi)}$$

$$\text{Stem Buckling Load (C=2)} = \frac{2 \bullet \pi^2 \bullet E \bullet I}{l^2} = 28,828 \text{ lbs.} \quad \text{Acceptable, Greater than Th}$$

4. Loads transferred to Structure

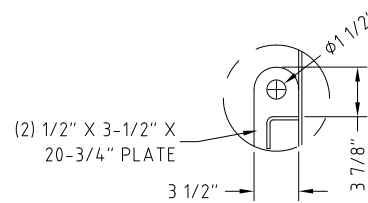
Non-Self contained gates transmit Operating loads to the mounting structure, in addition to Loads due to the Operating Head (H_c from section 3) acting on the gate area.

Operating load transferred through pedestal	=	905 lbs.
Design load transferred through pedestal	=	8,970 lbs.

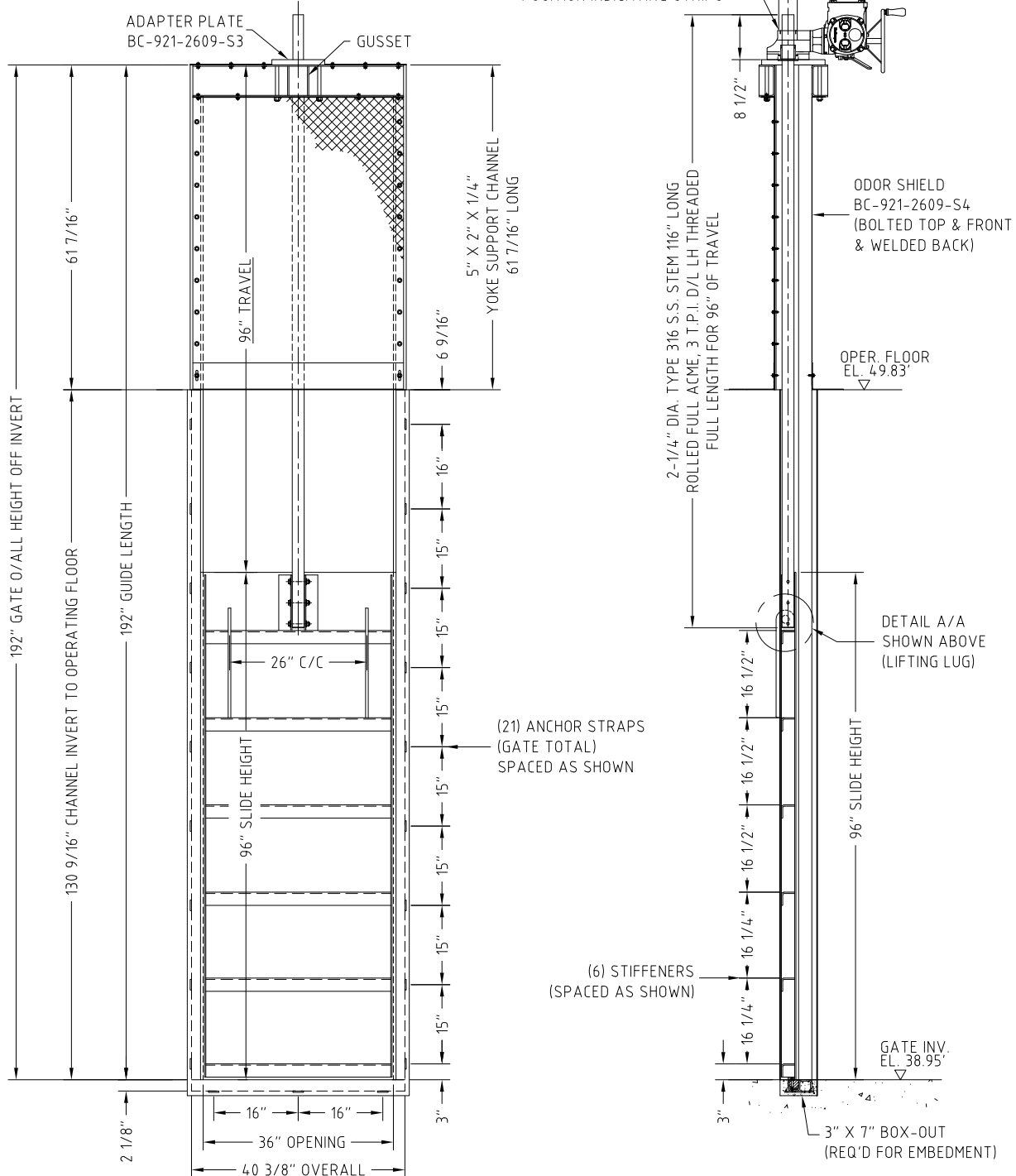
Water load = $(62.4 \times A \times H_c)$ = 1,248 lbs.

ANY OBSTRUCTIVE HANDRAILING OR
FLOORING TO BE MODIFIED BY OTHERS

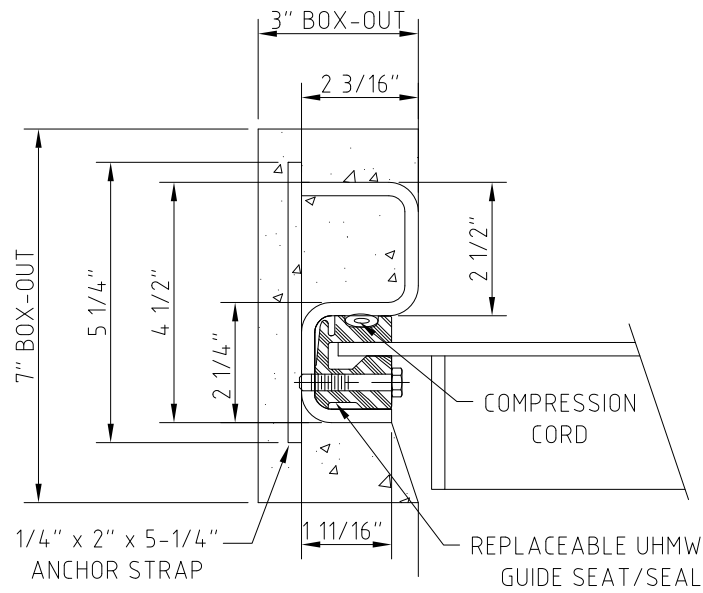
FOR GATE SECTION DETAILS: BC-921-2609-S2
YOKE: 1/4" X 3" X 6" CHANNEL
HORIZONTAL STIFFENER: 1/4" X 2-1/2" X 2-1/2" ANGLE
VERTICAL STIFFENERS: 1/4" X 2-1/2" FLAT BAR
APPROX. WEIGHT OF SLIDE: 380 LBS.
APPROX. WEIGHT OF GATE ASSEMBLY: 815 LBS.



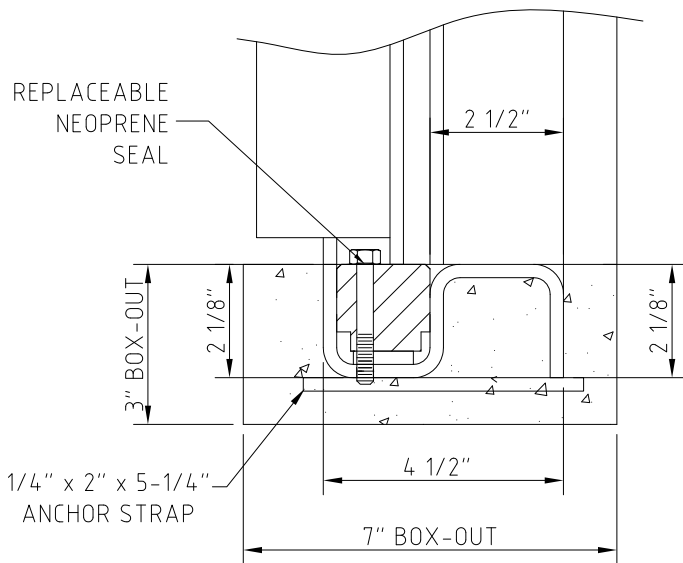
DETAIL A/A
SCALE 2:1



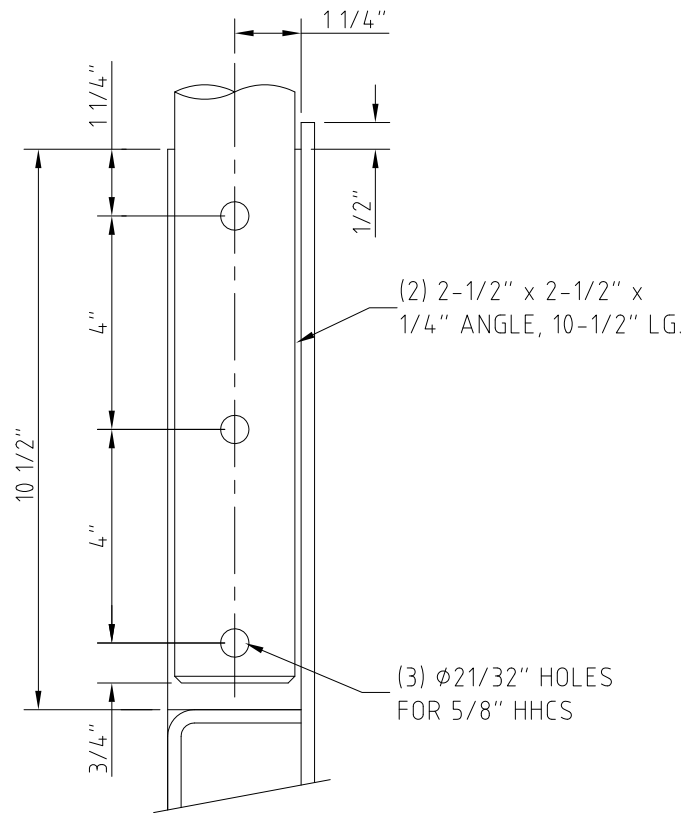
Title		SERIES 921 SLIDE GATE INSTALL. 36" x 96"		Quantity: 2	Scale: FD		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches Requested Concrete Opening Tolerances Height & Width ±1/4" Squareness ±3/8 Diagonal						
Project		HILO WWTF PHASE #1 - HILO, HAWAII		Tag No's: 02-GAT-1110 & 1120									
Item#:		A5		Seating	Unseating								
Location		SCREEN INLET #1 & #2		Design Head	20.0'	20.0'		A	Original Issue		5/6/25	KS	
Dwg. No.		BC-921-2609		S.O. No.	32752	Leakage Rate GPM/ft of seal	0.05	0.05	Rev.	Rev. Record	Date	Eng.	Chk.
										</			



EMBEDDED GUIDE SECTION

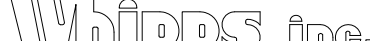


INVERT SECTION



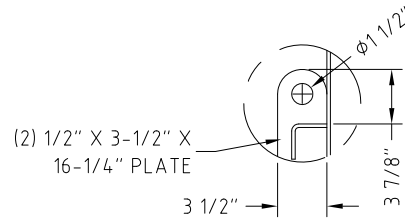
STEM CONNECTOR

Title SERIES 921 SLIDE GATE SECTION DETAILS		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches						
Material STAINLESS STEEL GRADE 316L		Scale: FD		A	Original Issue	5/6/25	KS	
				Rev.	Rev. Record	Date	Eng.	Chk.
Drg. No. BC-921-2609-S2		S.O. No. 32752		THIRD ANGLE PROJECTION		Whipps, inc.		
						370 SOUTH ATHOL RD. ATHOL, MA 01331		

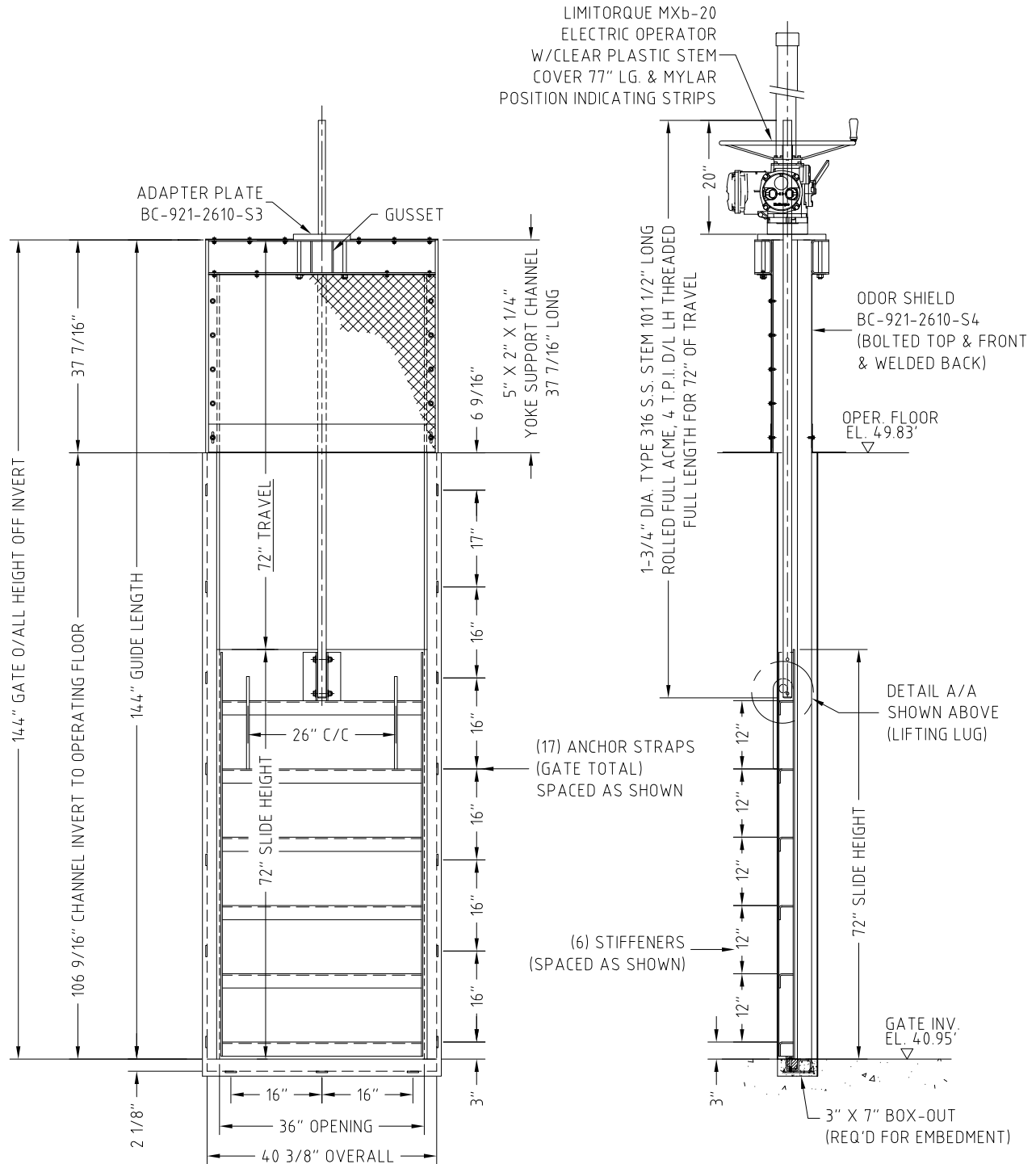
Title ODOR CONTROL COVERS		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches									
Material STAINLESS STEEL 316L		Scale: 1:20		A	Original Issue	5/6/25	KS				
				Rev.	Rev. Record	Date	Eng.	Chk.			
Drg. No.	BC-921-2609-S4	S.O. No.	32752	THIRD ANGLE PROJECTION							370 SOUTH ATHOL RD. ATHOL, MA 01331

ANY OBSTRUCTIVE HANDRAILING OR
FLOORING TO BE MODIFIED BY OTHERS

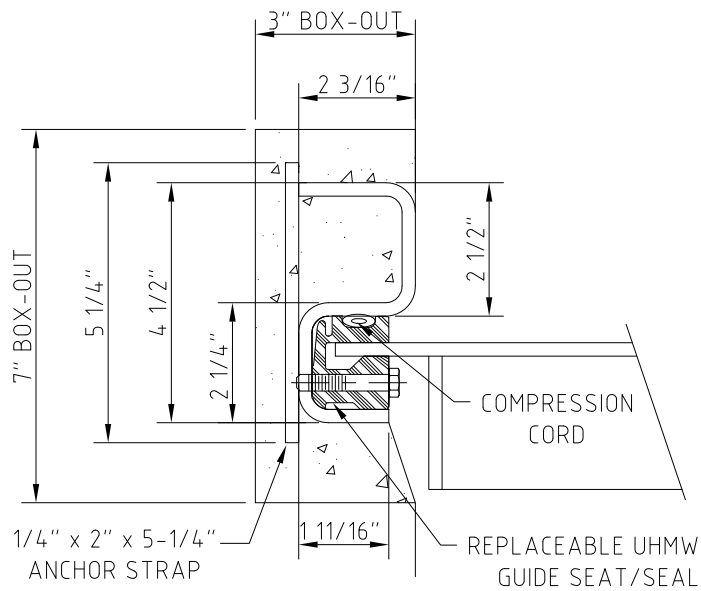
FOR GATE SECTION DETAILS: BC-921-2610-S2
YOKE: 1/4" X 3" X 6" CHANNEL
HORIZONTAL STIFFENER: 1/4" X 2-1/2" X 2-1/2" ANGLE
VERTICAL STIFFENERS: 1/4" X 2-1/2" FLAT BAR
APPROX. WEIGHT OF SLIDE: 310 LBS.
APPROX. WEIGHT OF GATE ASSEMBLY: 730 LBS.



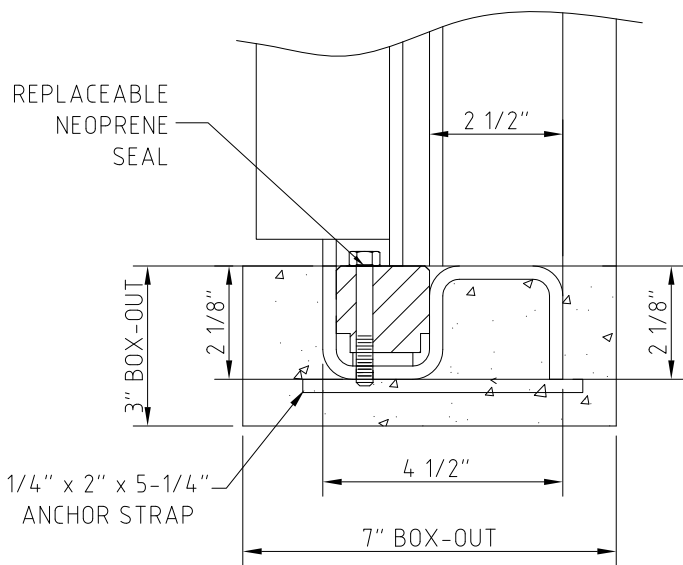
DETAIL A/A
SCALE 2:1



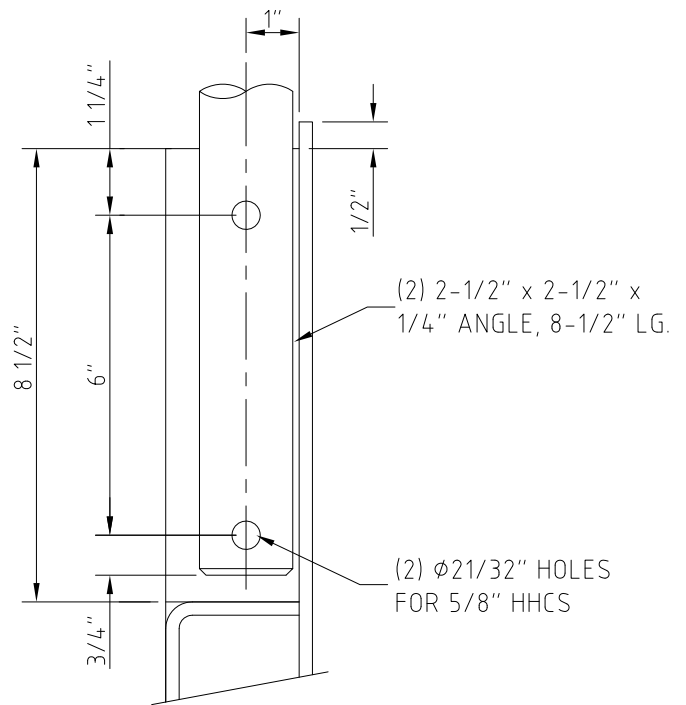
Title				SERIES 921 SLIDE GATE INSTALL. 36" x 72"		Quantity: 1		Scale: FD		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches													
						Tag No's: 02-GAT-1130						A		Original Issue		5/6/25		KS					
Project				HILO WWTF PHASE #1 - HILO, HAWAII		Item#: B5		Seating		Unseating		Requested Concrete Opening Tolerances Height & Width ±1/4" Squareness ±3/8 Diagonal		Rev.		Rev. Record		Date		Eng.		Chk.	
Location				BYPASS INLET		Design Head		20.0'		20.0'				Rev.		Rev. Record		Date		Eng.		Chk.	
Dwg. No.				BC-921-2610		S.O. No.		32752		Leakage Rate		0.05		0.05						370 SOUTH ATHOL RD. ATHOL, MA 01331			



EMBEDDED GUIDE SECTION

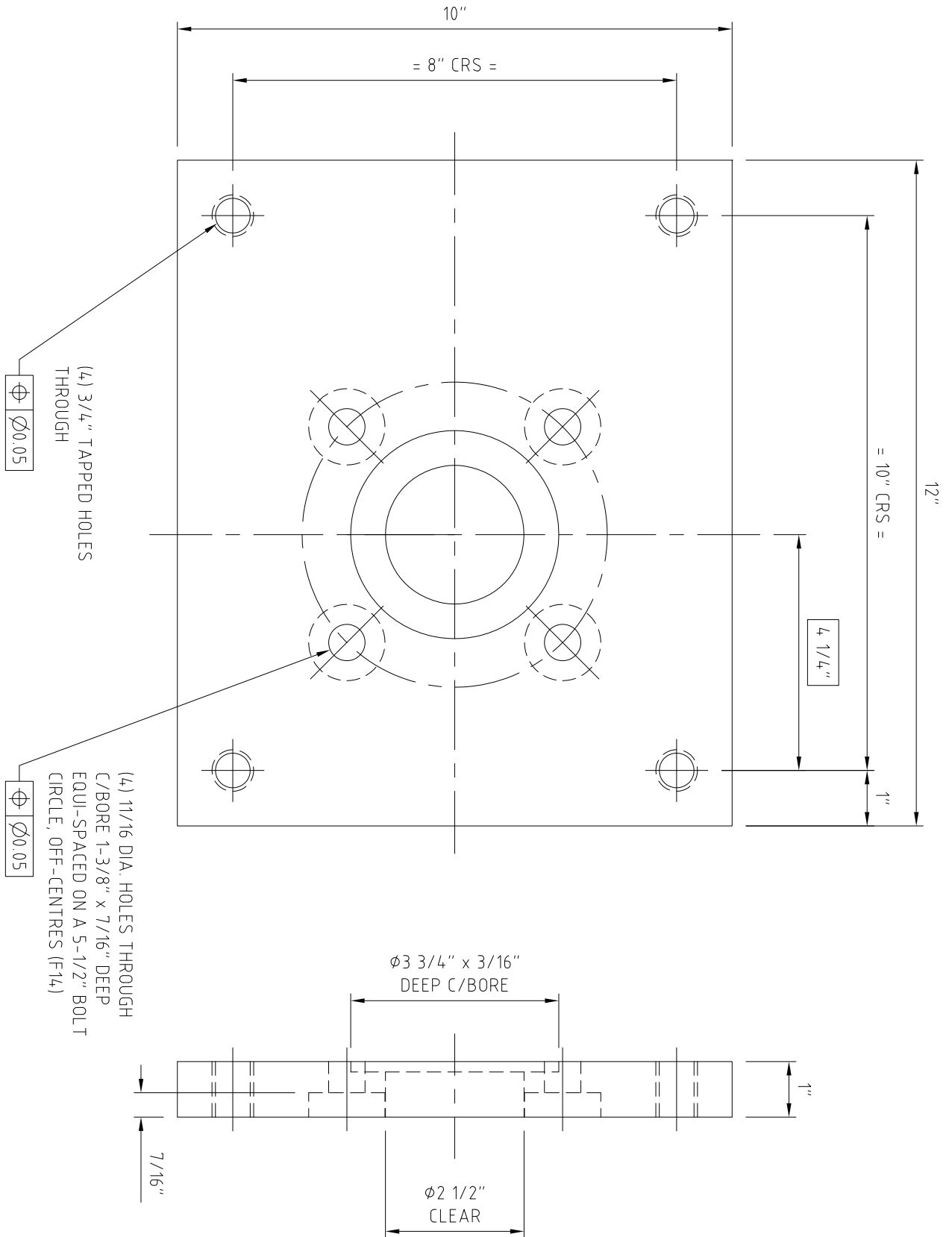


INVERT SECTION



STEM CONNECTOR

Title		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches				
Material		Scale: FD		A	Original Issue	5/6/25
				Rev.	Rev. Record	Date
Drg. No. BC-921-2610-S2		S.O. No. 32752			KS	Eng. Chk.
		THIRD ANGLE PROJECTION		Whipps, inc.		
				370 SOUTH ATHOL RD. ATHOL, MA 01331		



Title

ADAPTOR PLATE FA14

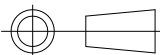
Material ALUMINUM 6061-T6

Drg. No. BC-921-2610-S3 S.O. No. 32752

DO NOT SCALE
THIS DRAWING
Work to Dimensions
Linear Dimensions are Inches

Scale: 1:2.5

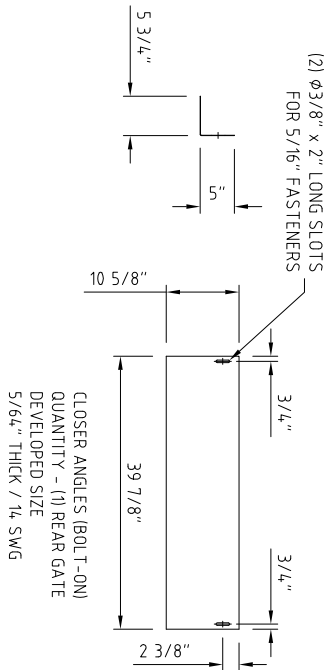
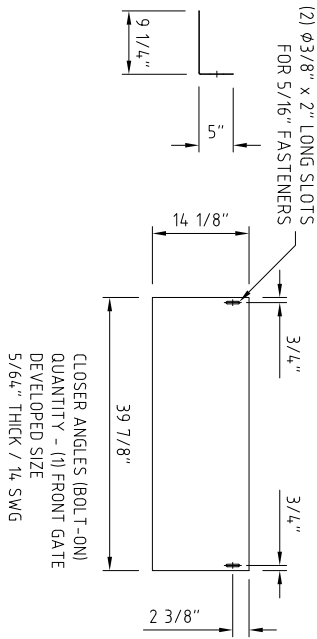
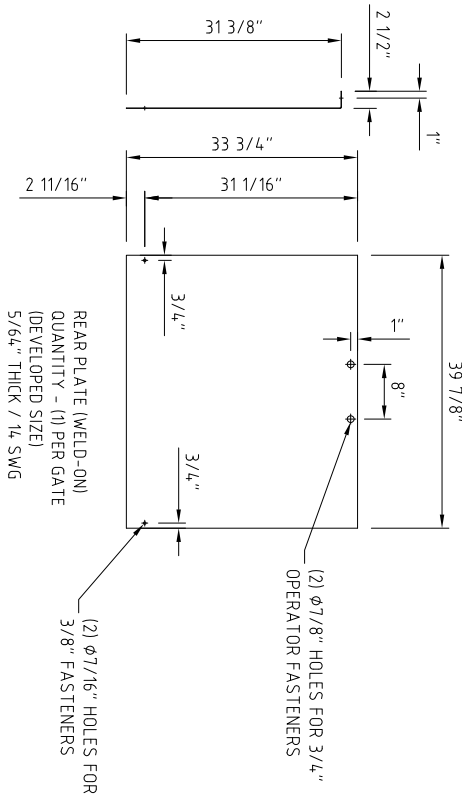
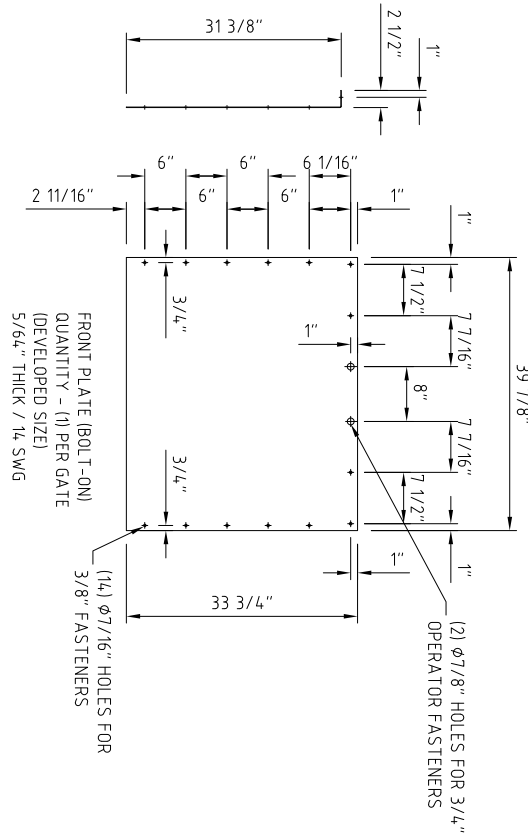
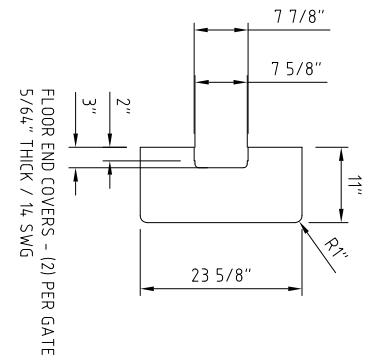
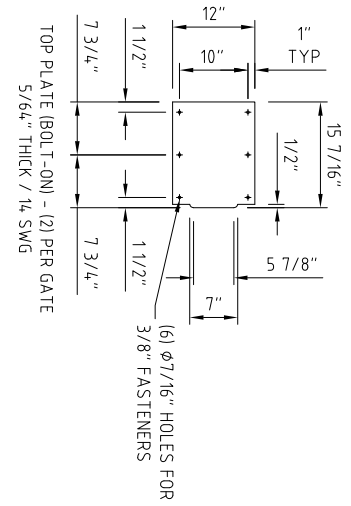
THIRD
ANGLE
PROJECTION



A	Original Issue	5/6/25	KS	
Rev.	Rev. Record	Date	Eng.	Chk.

Whipps, inc.

370 SOUTH ATHOL RD.
ATHOL, MA 01331



Title ODOR CONTROL COVERS

Material STAINLESS STEEL 316L

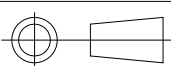
Dwg. No. BC-921-2610-S4

S.O. No. 32752

DO NOT SCALE
THIS DRAWING
Work to Dimensions
Linear Dimensions are Inches

Scale: 1:20

THIRD
ANGLE
PROJECTION

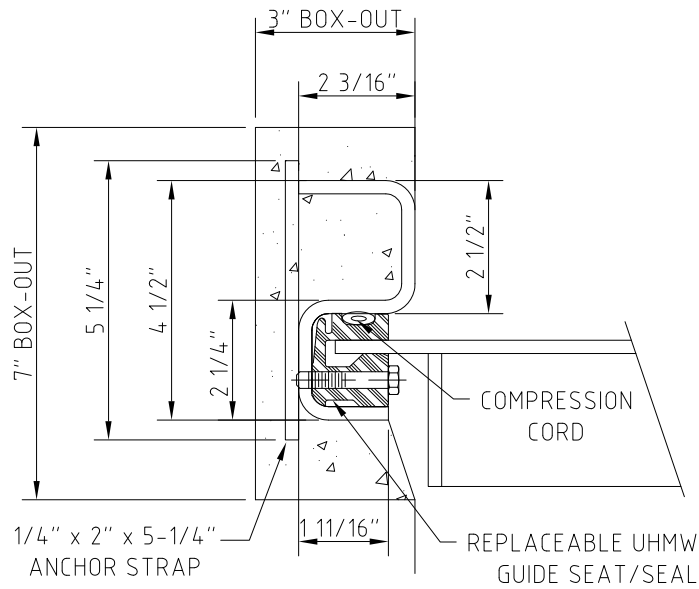


A Original Issue
Rev. Rev. Record

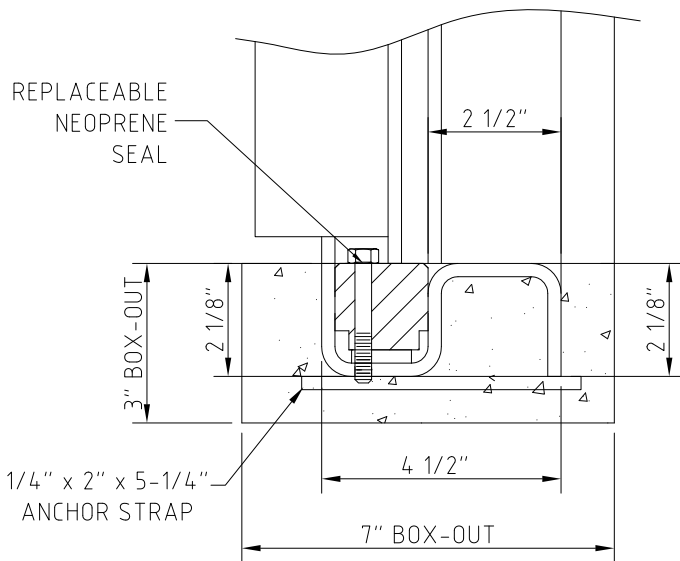
Whipps, inc.

5/6/25 KS
Date Eng. Chk.

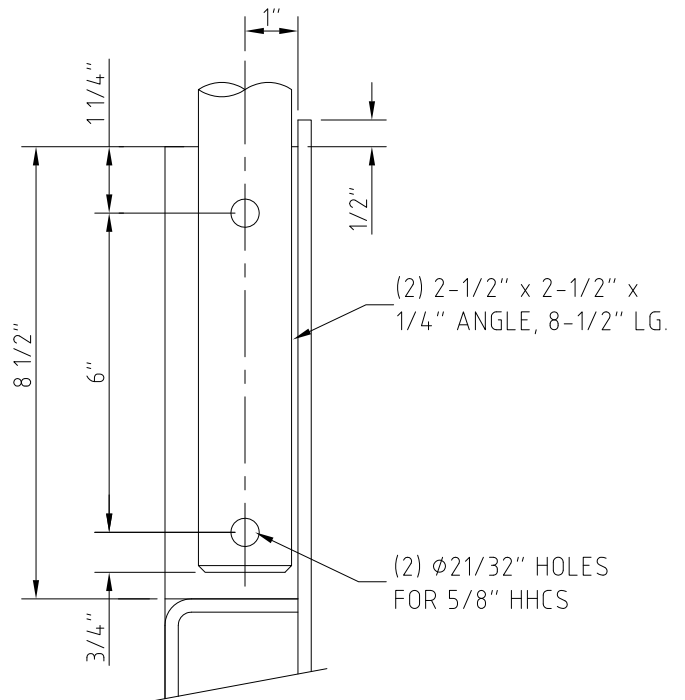
370 SOUTH ATHOL RD.
ATHOL, MA 01331



EMBEDDED GUIDE SECTION

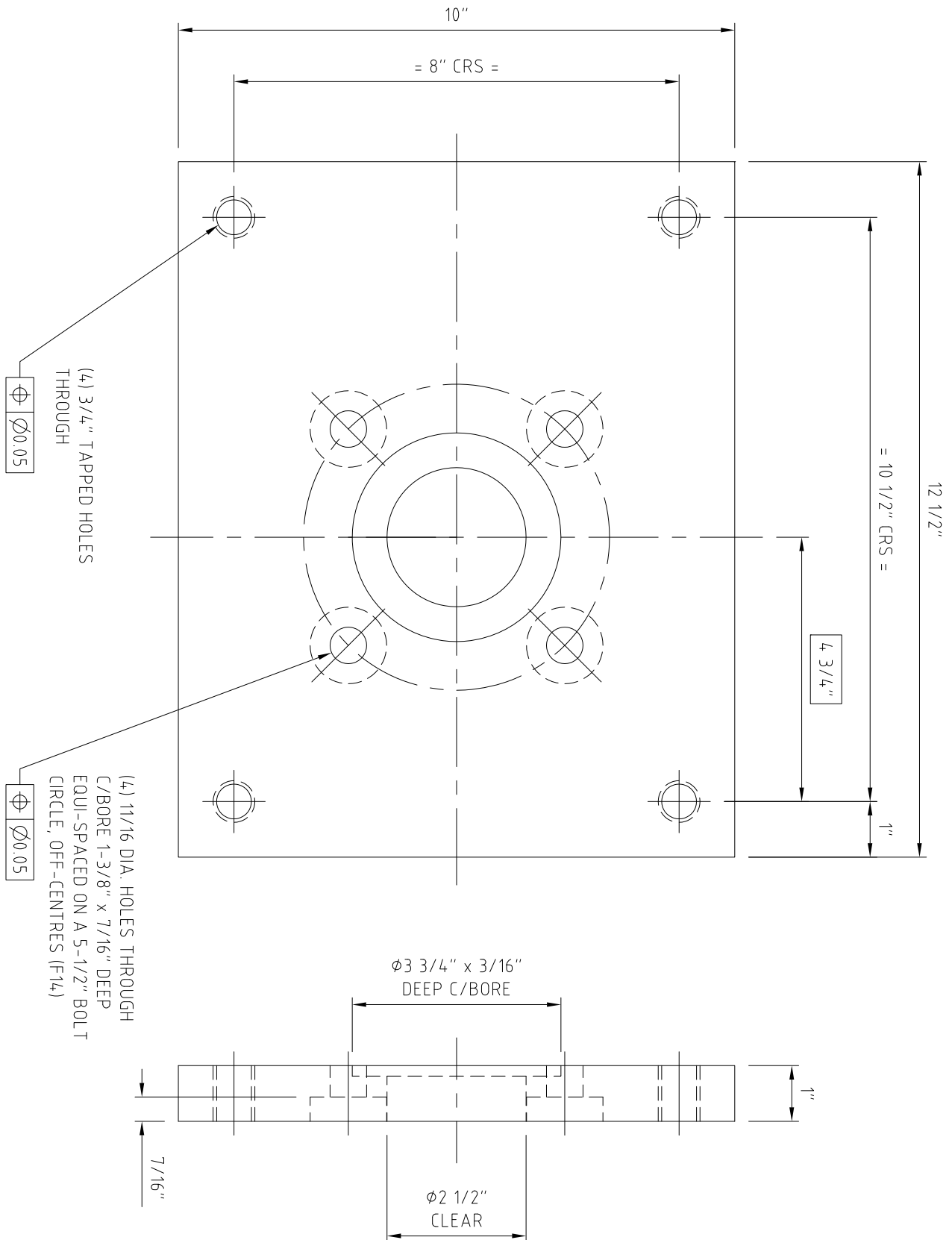


INVERT SECTION

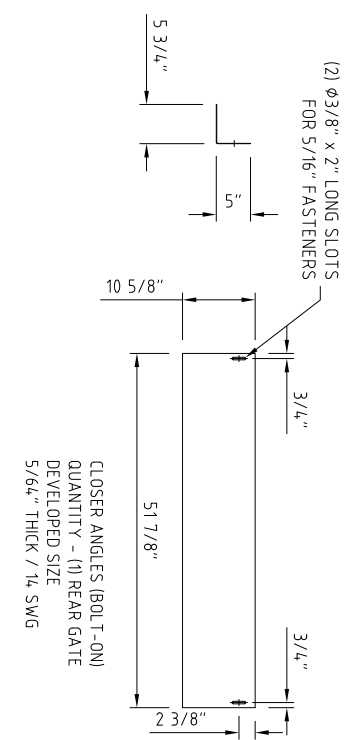
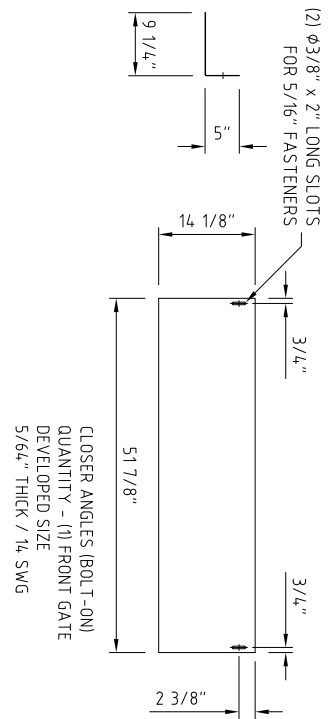
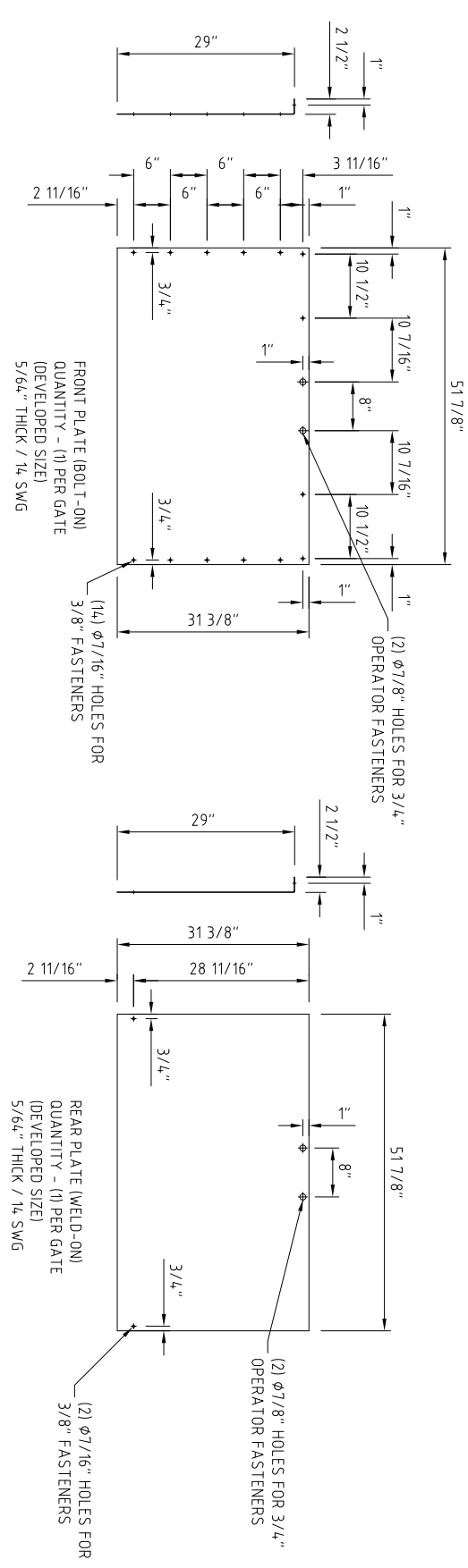
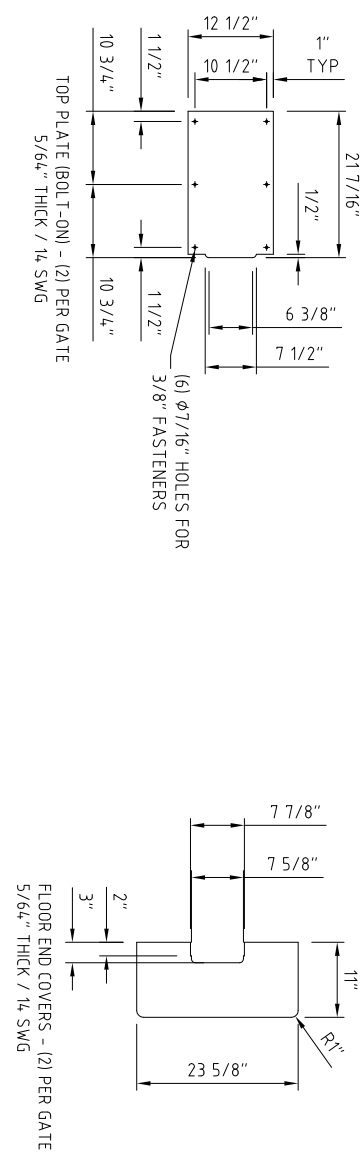


STEM CONNECTOR

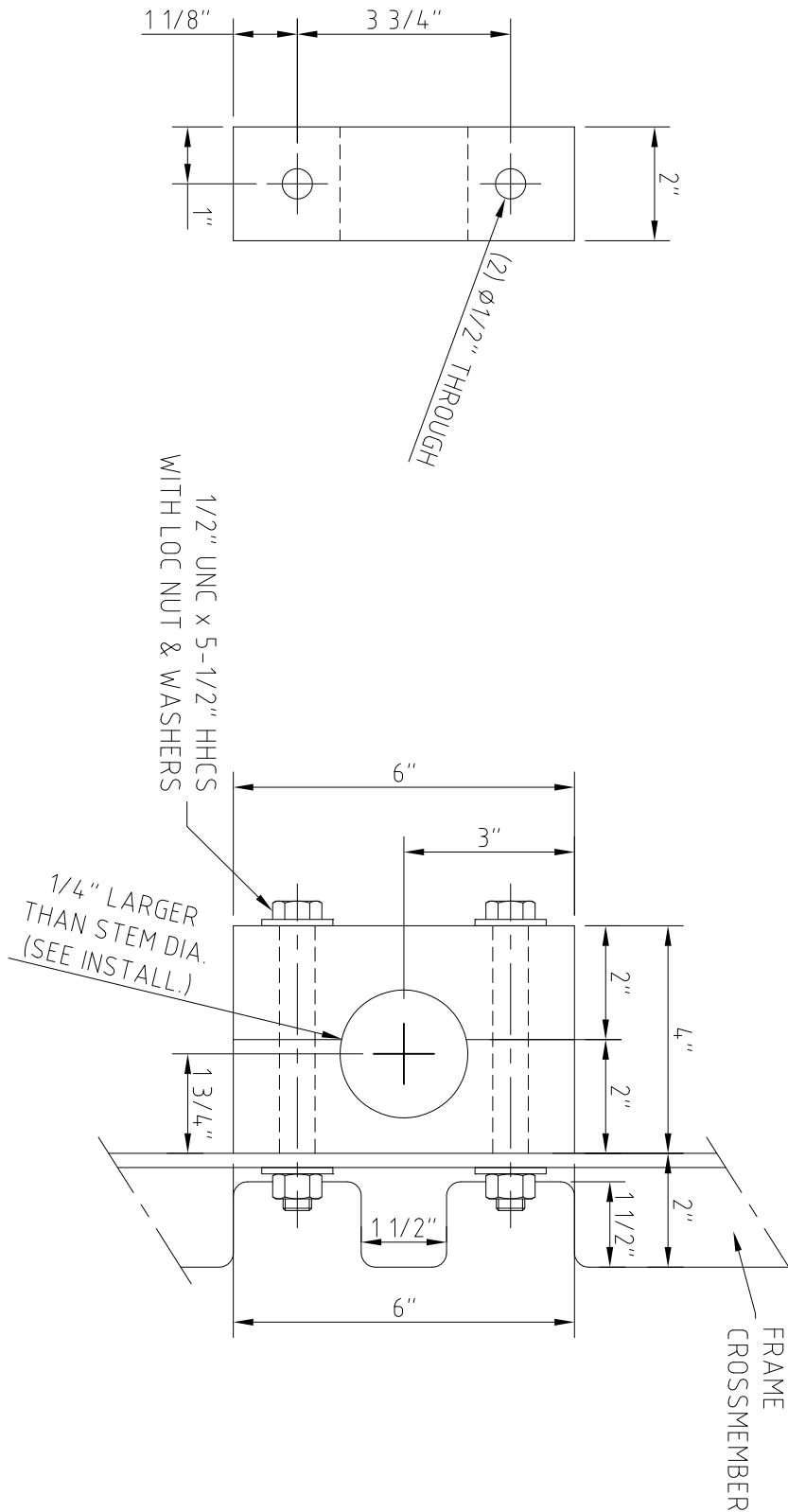
Title SERIES 921 SLIDE GATE SECTION DETAILS		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches							
Material STAINLESS STEEL GRADE 316L		Scale: FD		A	Original Issue	5/6/25	KS		
				Rev.	Rev. Record	Date	Eng.	Chk.	
Dwg. No.	BC-921-2611-S2	S.O. No.	32752	THIRD ANGLE PROJECTION		Whipps, inc.			370 SOUTH ATHOL RD. ATHOL, MA 01331



Title	DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches					
Material	Scale: 1:2.5			A	Original Issue	5/6/25
				Rev.	Rev. Record	Date
Drg. No.	S.O. No.			Whipps, inc.		
				370 SOUTH ATHOL RD. ATHOL, MA 01331		
BC-921-2611-S3			32752			
			THIRD ANGLE PROJECTION			
			KS			
			Eng.			
			Chk.			



Title ODOR CONTROL COVERS		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches						
Material STAINLESS STEEL 316L		Scale: 1:20		A	Original Issue	5/6/25	KS	
				Rev.	Rev. Record	Date	Eng.	Chk.
Drg. No. BC-921-2611-S4	S.O. No. 32752	THIRD ANGLE PROJECTION		Whipps, inc.		370 SOUTH ATHOL RD. ATHOL, MA 01331		

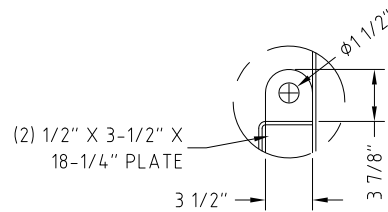


MATERIAL = UHMWPE

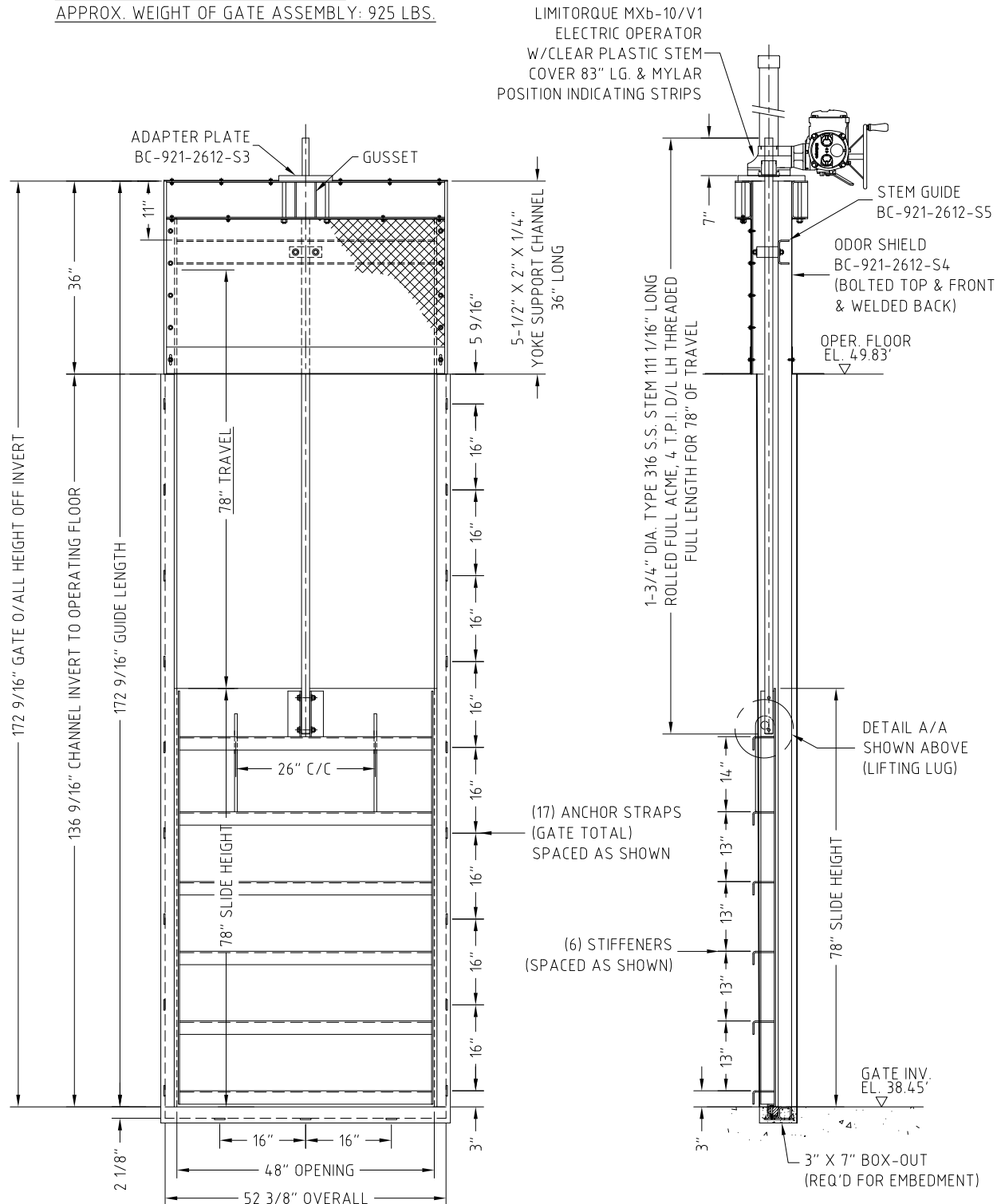
Title		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches					
		Scale: FD		A	Original Issue	5/4/12	KS
		THIRD ANGLE PROJECTION		Rev.	Rev. Record	Date	Eng. Chk.
Drg. No.	BC-921-2611-S5	S.O. No.	32752				
900 GATE STYLE STEM GUIDE DETAIL				370 SOUTH ATHOL RD. ATHOL, MA 01331			

ANY OBSTRUCTIVE HANDRAILING OR
FLOORING TO BE MODIFIED BY OTHERS

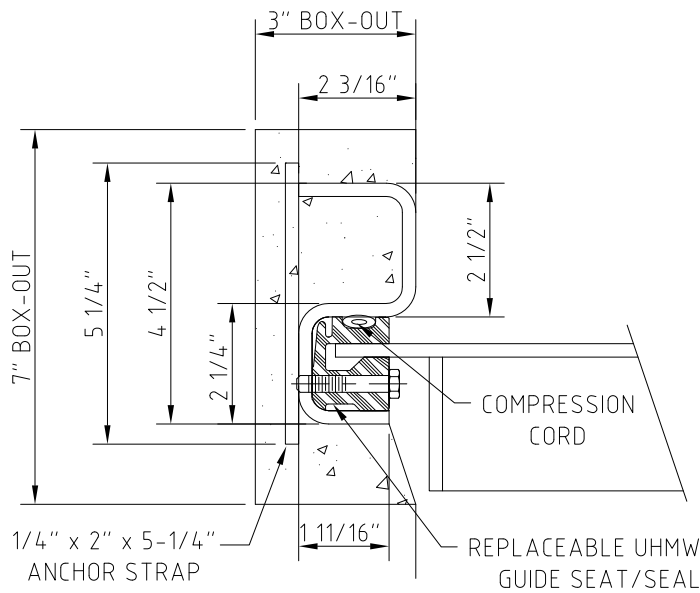
FOR GATE SECTION DETAILS: BC-921-2612-S2
YOKE: 1/4" X 3" X 7" CHANNEL
STEM GUIDE CHANNEL: 1/4" X 2" X 4-1/2" CHANNEL
HORIZONTAL STIFFENER: 1/4" X 2-1/2" X 4" ANGLE
VERTICAL STIFFENERS: 1/4" X 2-1/2" FLAT BAR
APPROX. WEIGHT OF SLIDE: 450 LBS.
APPROX. WEIGHT OF GATE ASSEMBLY: 925 LBS.



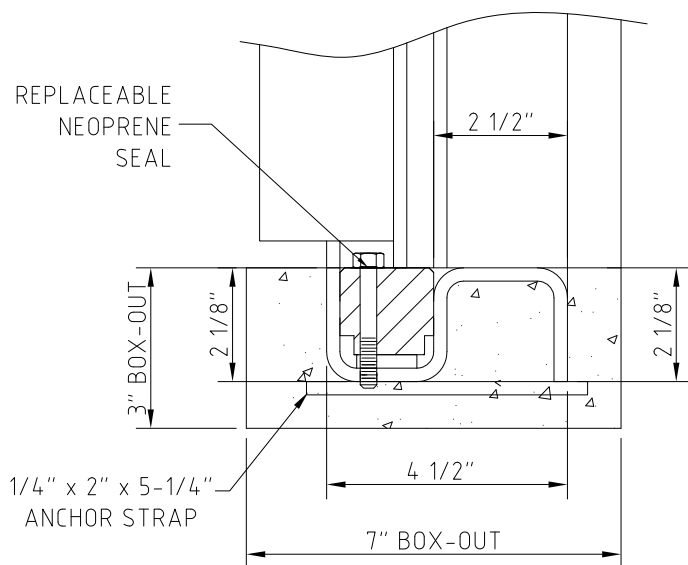
DETAIL A/A
SCALE 2:1



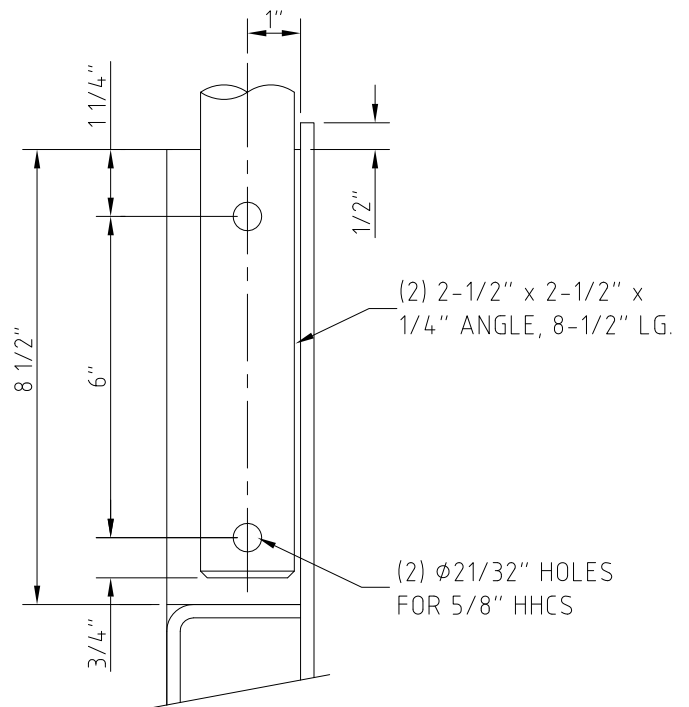
Title		SERIES 921 SLIDE GATE INSTALL. 48" x 78"		Quantity: 1	Scale: FD		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches							
Project		HILO WWTF PHASE #1 - HILO, HAWAII		Tag No's: 02-GAT-1135										
Item#:		D5		Seating	Unseating		Requested Concrete Opening Tolerances Height & Width ±1/4" Squareness ±3/8 Diagonal	A	Original Issue		5/6/25		KS	
Design Head		20.0'		20.0'				Rev.	Rev. Record		Date		Eng.	Chk.
Location		BYPASS OUTLET		Leakage Rate	0.05		0.05	THIRD ANGLE PROJECTION				Whipps, Inc.		370 SOUTH ATHOL RD. ATHOL, MA 01331
Dwg. No. BC-921-2612		S.O. No. 32752		GPM/ft of seal										01331



EMBEDDED GUIDE SECTION

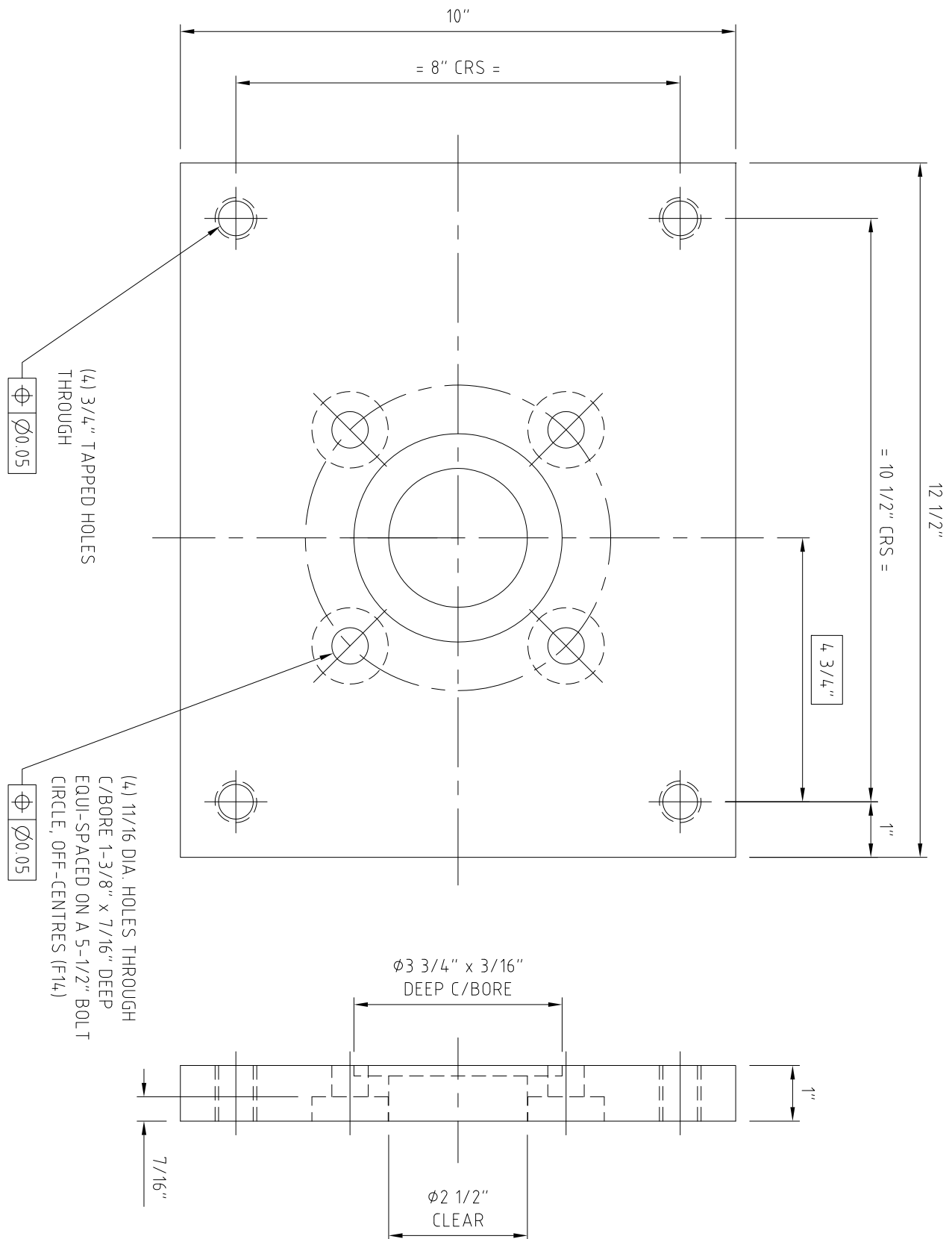


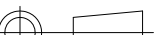
INVERT SECTION

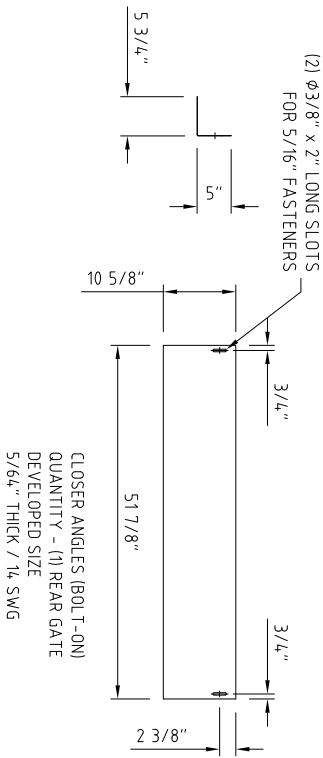
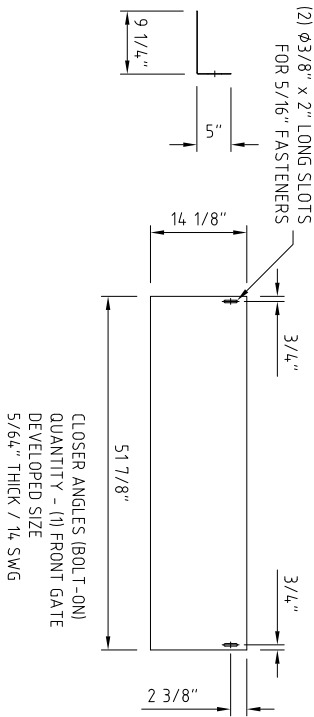
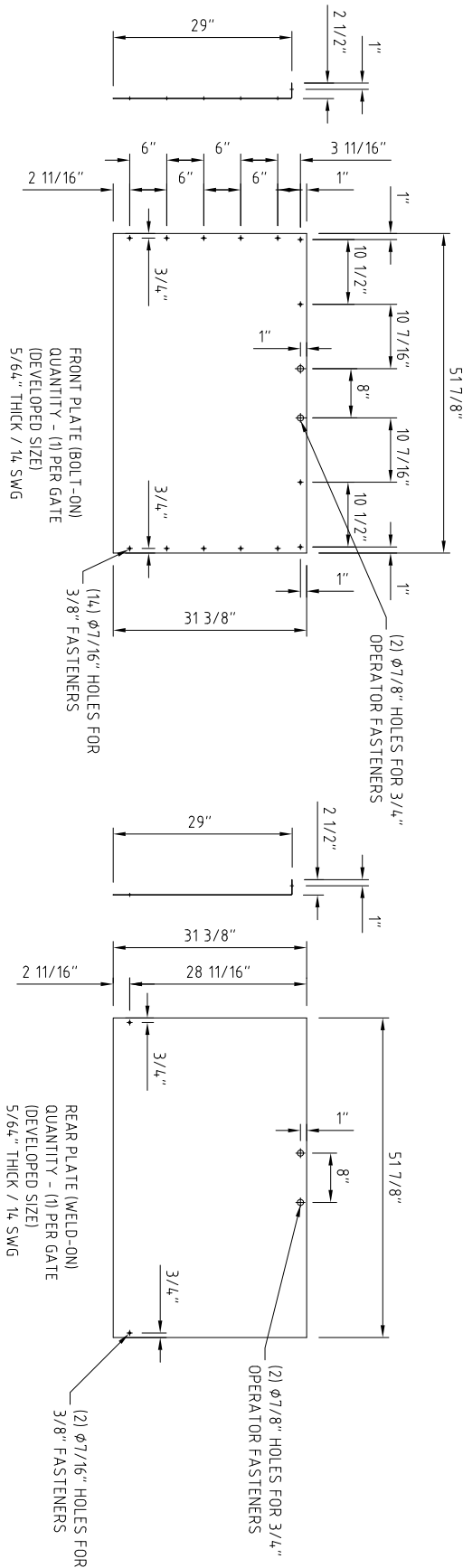
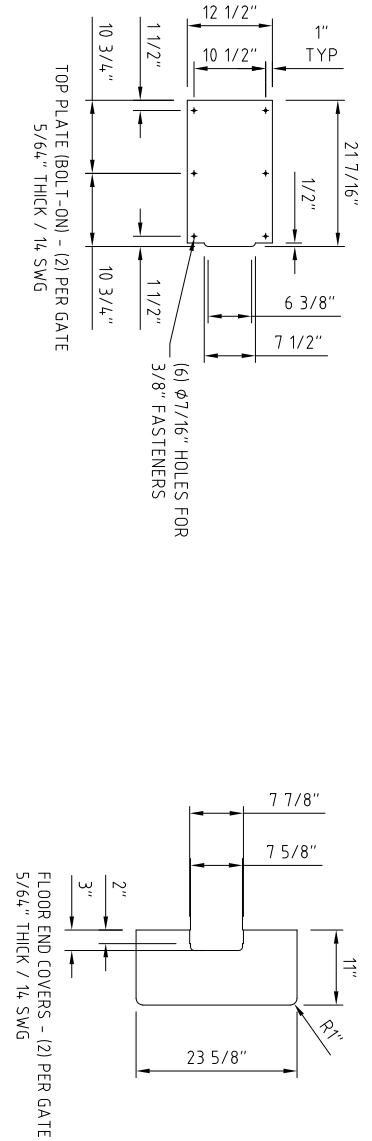


STEM CONNECTOR

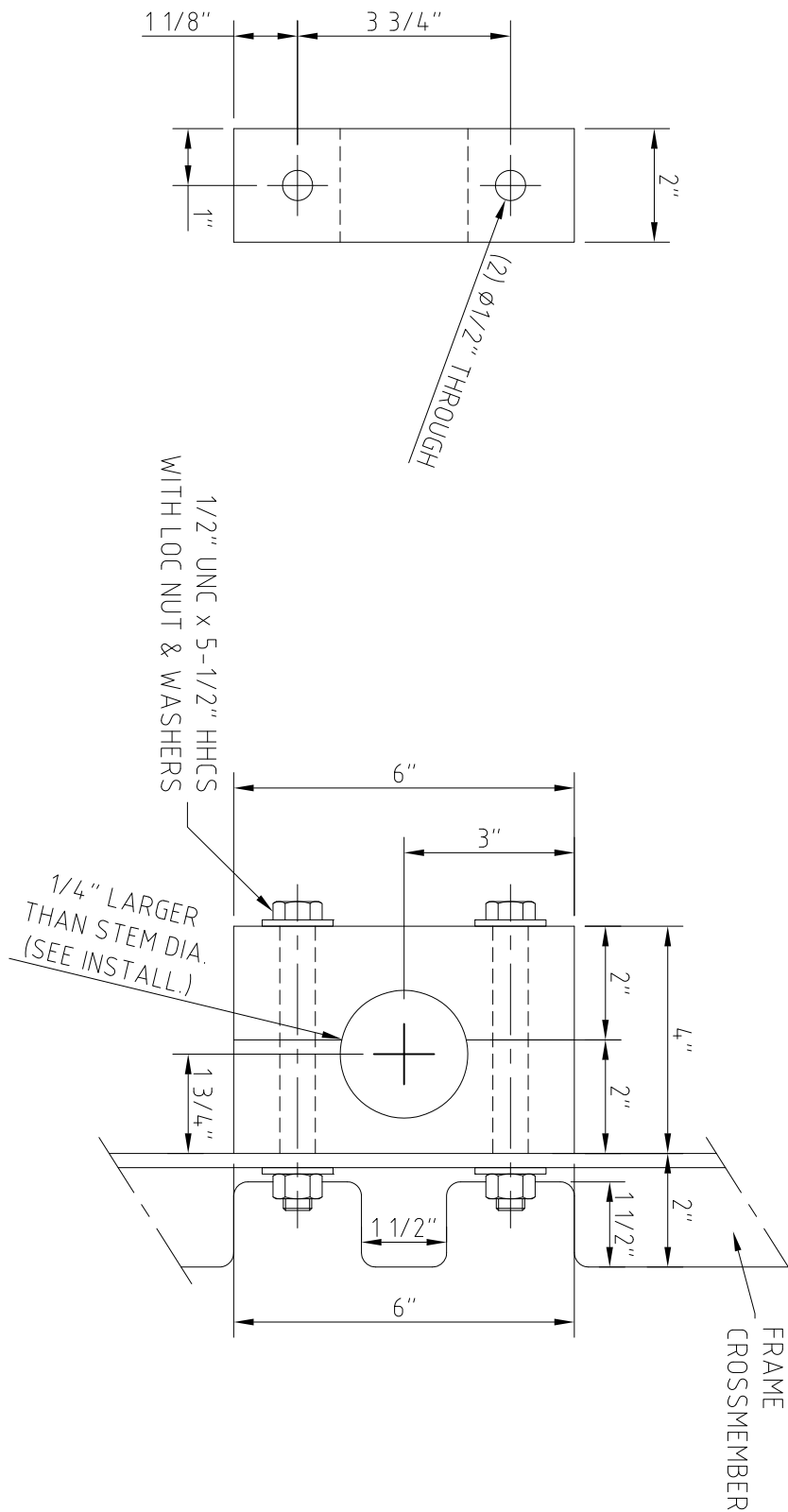
Title	DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches				
Material	Scale: FD		A	Original Issue	5/6/25
			Rev.	Rev. Record	Date
Drg. No.	S.O. No.		Date		KS
			Eng.		Chk.
BC-921-2612-S2		32752	THIRD ANGLE PROJECTION Whipps, inc. 370 SOUTH ATHOL RD. ATHOL, MA 01331		



Title ADAPTOR PLATE FA14		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches								
		Material ALUMINUM 6061-T6		Scale: 1:2.5		A	Original Issue		5/6/25	KS
Rev.	Rev. Record					Date	Eng.	Chk.		
Drg. No.	BC-921-2612-S3	S.O. No.	32752	THIRD ANGLE PROJECTION			Whipps, inc. 370 SOUTH ATHOL RD. ATHOL, MA 01331			



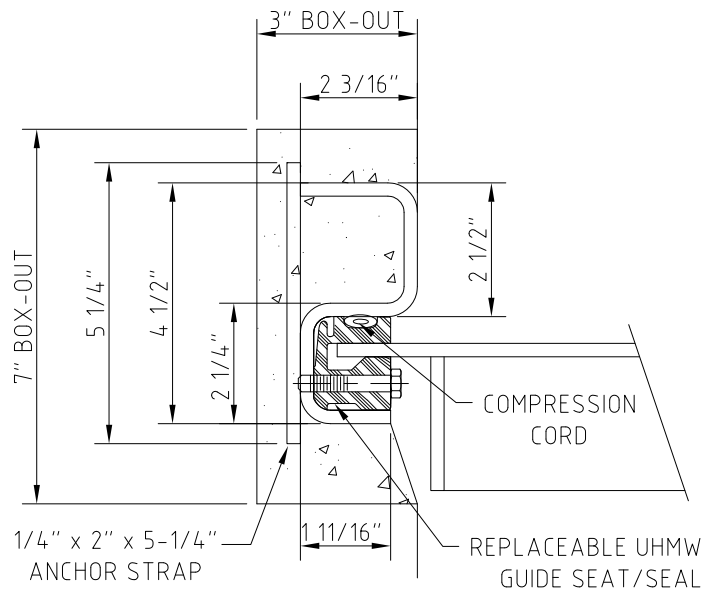
Title ODOR CONTROL COVERS		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches											
		Material STAINLESS STEEL 316L		Scale: 1:20		A	Original Issue	5/6/25	KS				
Rev.	Rev. Record					Date	Eng.	Chk.					
Drg. No.	BC-921-2612-S4	S.O. No.	32752	THIRD ANGLE PROJECTION								370 SOUTH ATHOL RD. ATHOL, MA 01331	



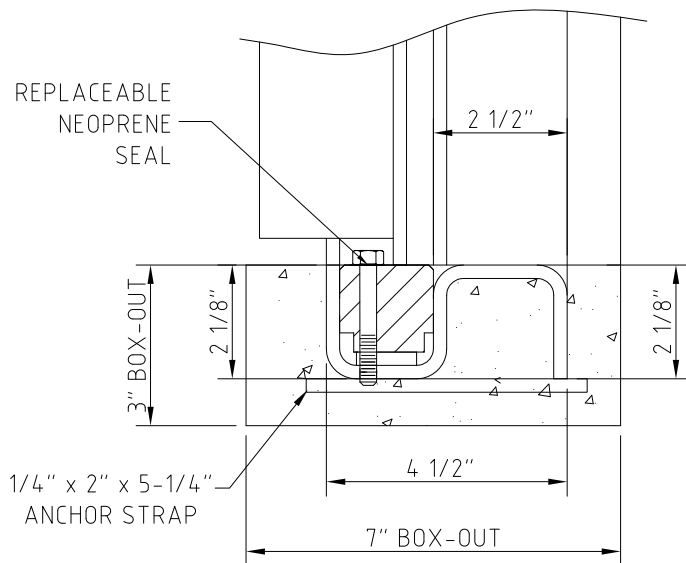
MATERIAL = UHMWPE

Title		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches				
		Scale: FD		A	Original Issue	5/4/12
		THIRD ANGLE PROJECTION		Rev.	Rev. Record	Date
						Eng.
Drg. No. BC-921-2612-S5		S.O. No. 32752		Whipps, Inc.		
				370 SOUTH ATHOL RD. ATHOL, MA 01331		

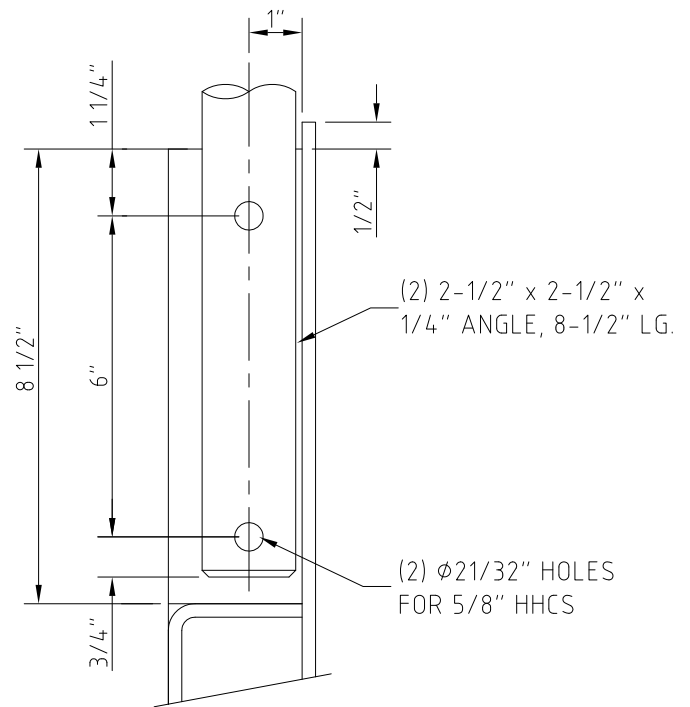
Title SERIES 921 SLIDE GATE INSTALL. 45" x 78"		Quantity: 2		Scale: FD		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches							
		Tag No's: 02-GAT-1210 & 1220											
Project HILO WWTF PHASE #1 - HILO, HAWAII		Item#: F5		Seating	Unseating			Requested Concrete Opening Tolerances Height & Width ±1/4" Squareness ±3/8 Diagonal		A	Original Issue		5/6/25
		Design Head		20.0'	20.0'	Rev.	Rev. Record			Date		Eng.	Chk.
Location GRIT INLET #1 & #2		Leakage Rate		0.05	0.05	THIRD ANGLE PROJECTION 				370 SOUTH ATHOL RD. ATHOL, MA 01331			
Dwg. No. BC-921-2613		S.O. No. 32752											



EMBEDDED GUIDE SECTION

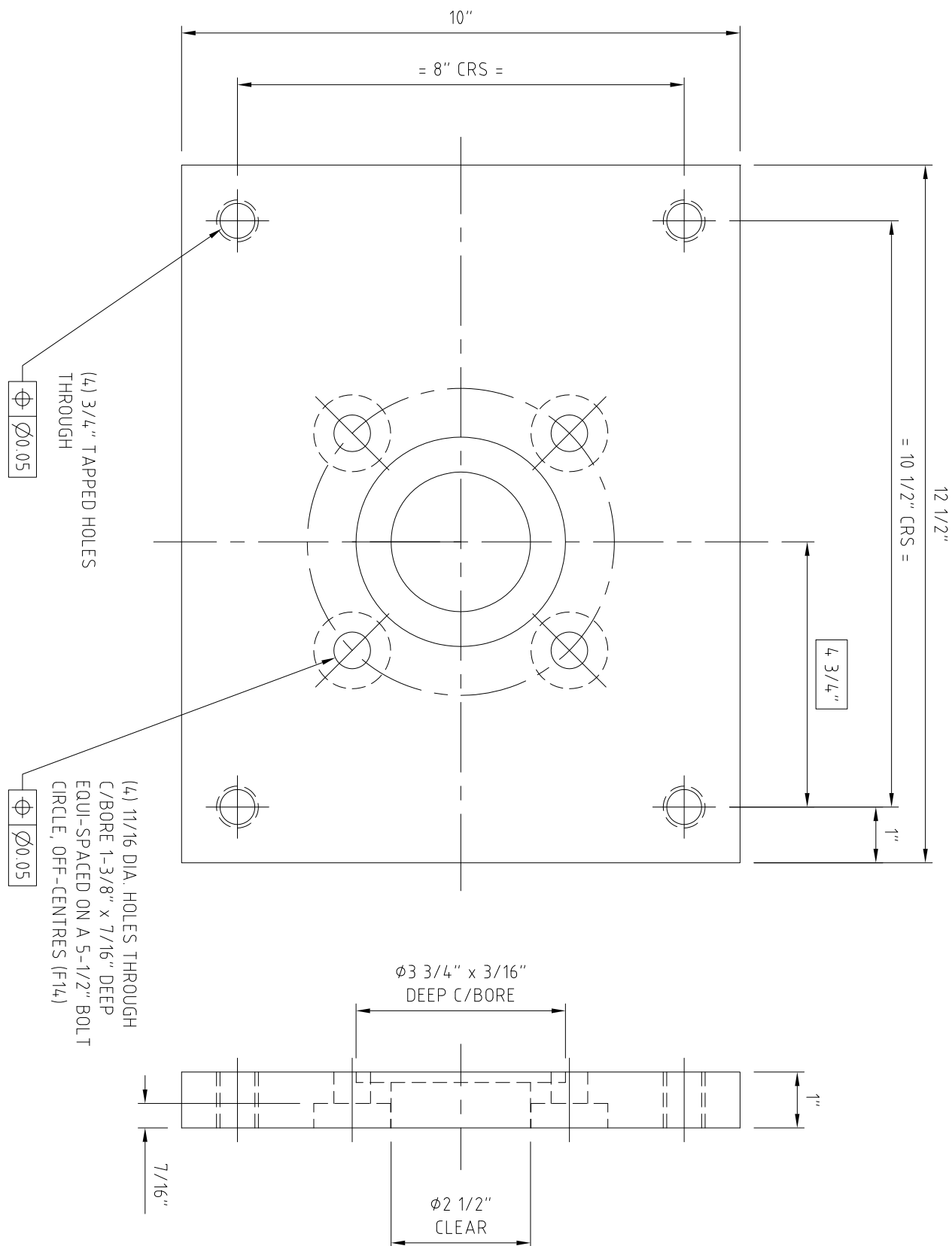


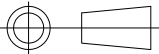
INVERT SECTION

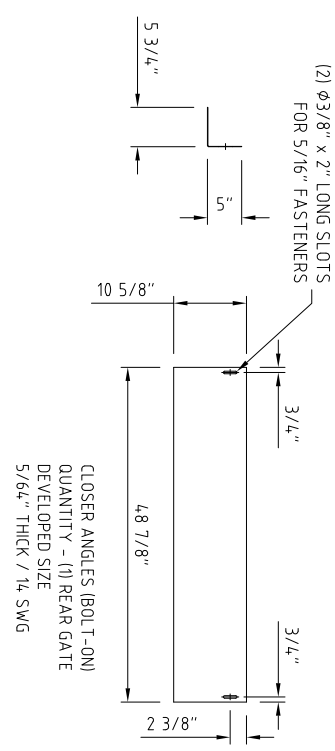
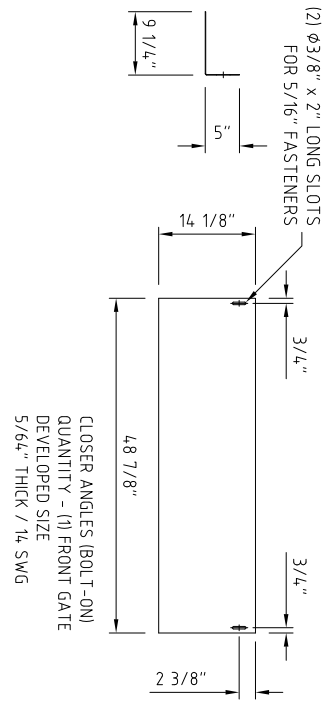
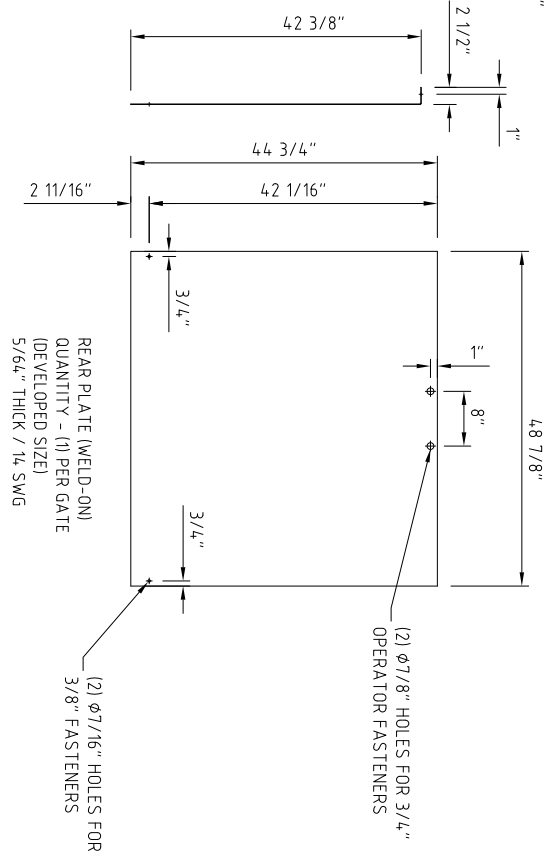
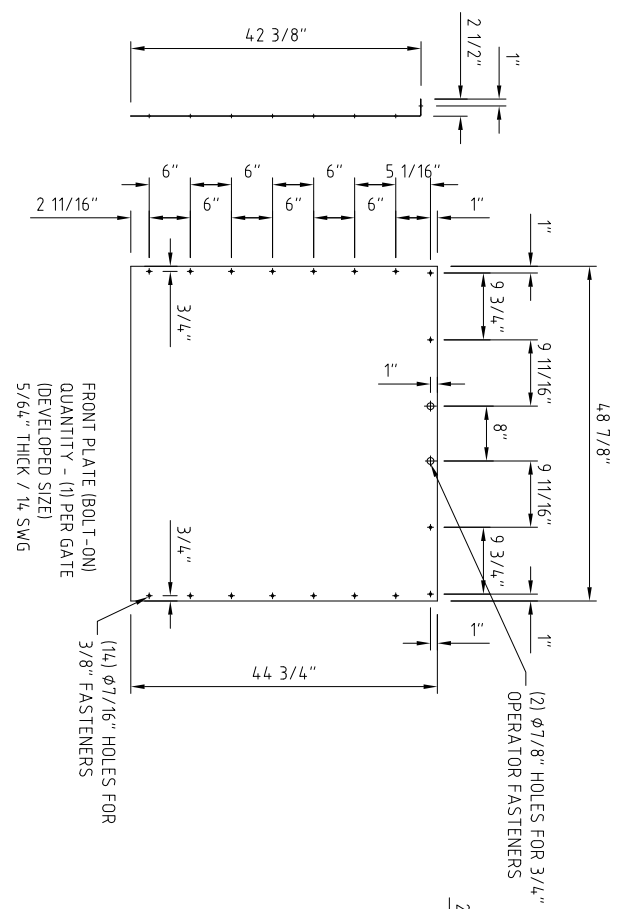
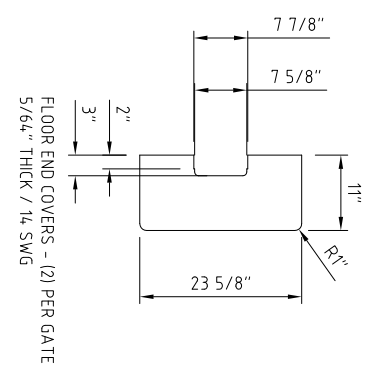
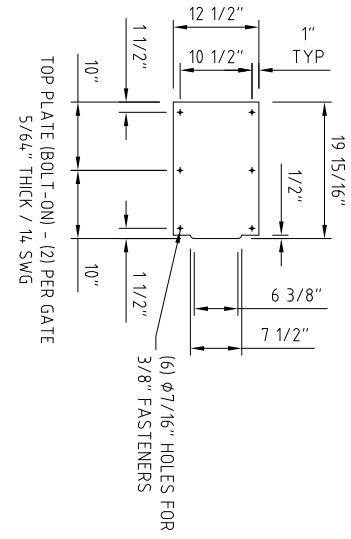


STEM CONNECTOR

Title SERIES 921 SLIDE GATE SECTION DETAILS		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches				
Material STAINLESS STEEL GRADE 316L		Scale: FD		A	Original Issue	5/6/25
				Rev.	Rev. Record	Date
Drg. No. BC-921-2613-S2		S.O. No. 32752				
		THIRD ANGLE PROJECTION		Whipps, inc.		
				370 SOUTH ATHOL RD. ATHOL, MA 01331		



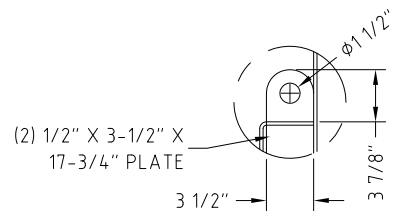
Title	DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches							
Material	Scale: 1:2.5				A	Original Issue	5/6/25	KS
					Rev.	Rev. Record	Date	Eng. Chk.
Drg. No. BC-921-2613-S3	S.O. No. 32752	THIRD ANGLE PROJECTION 		 370 SOUTH ATHOL RD. ATHOL, MA 01331				



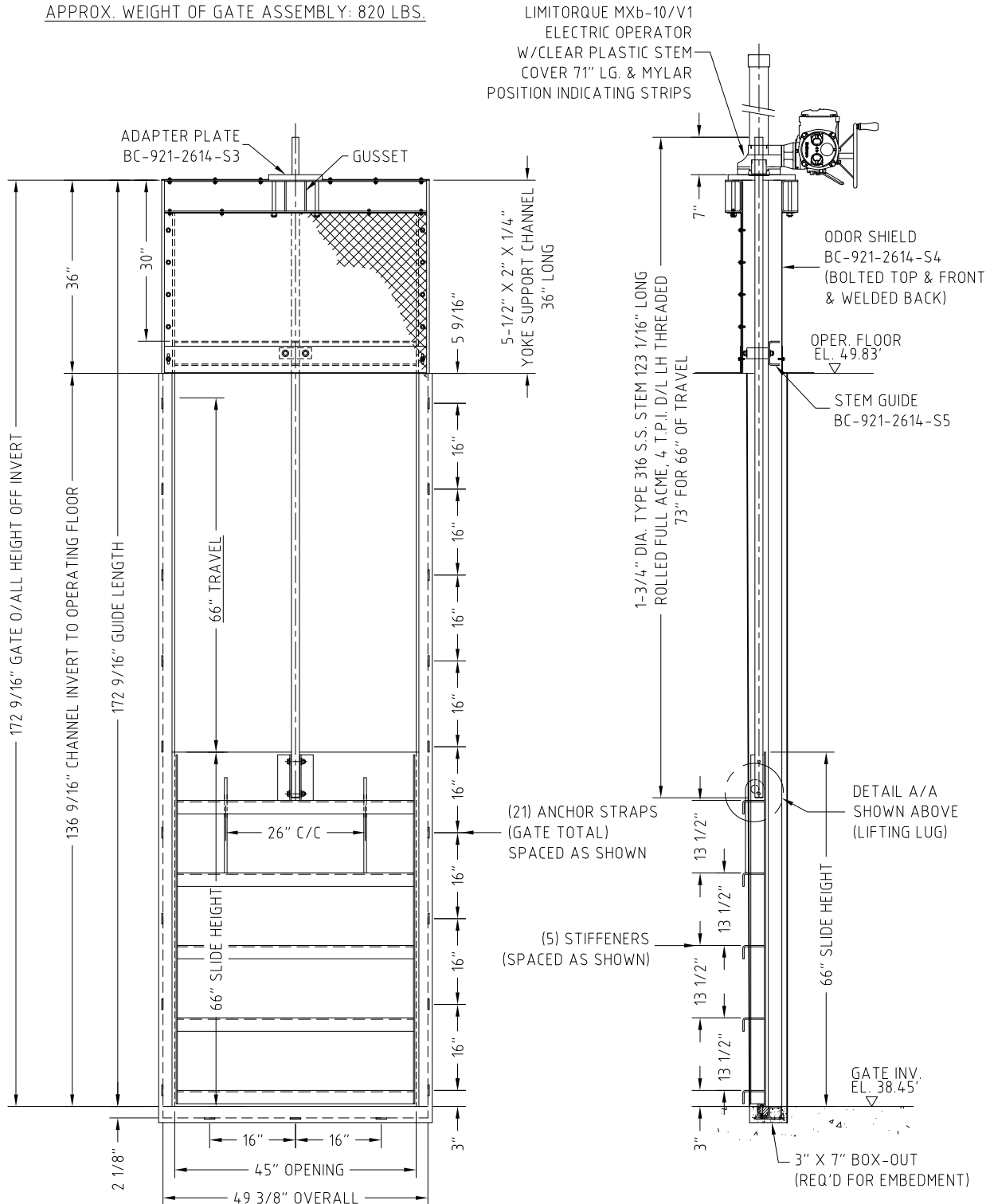
Title ODOR CONTROL COVERS		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches					
Material STAINLESS STEEL 316L		Scale: 1:20		A	Original Issue	5/6/25	KS
				Rev.	Rev. Record	Date	Eng. Chk.
Dr. No. BC-921-2613-S4		S.O. No. 32752		THIRD ANGLE PROJECTION Whipps, inc. 370 SOUTH ATHOL RD. ATHOL, MA 01331			

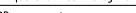
ANY OBSTRUCTIVE HANDRAILING OR
FLOORING TO BE MODIFIED BY OTHERS

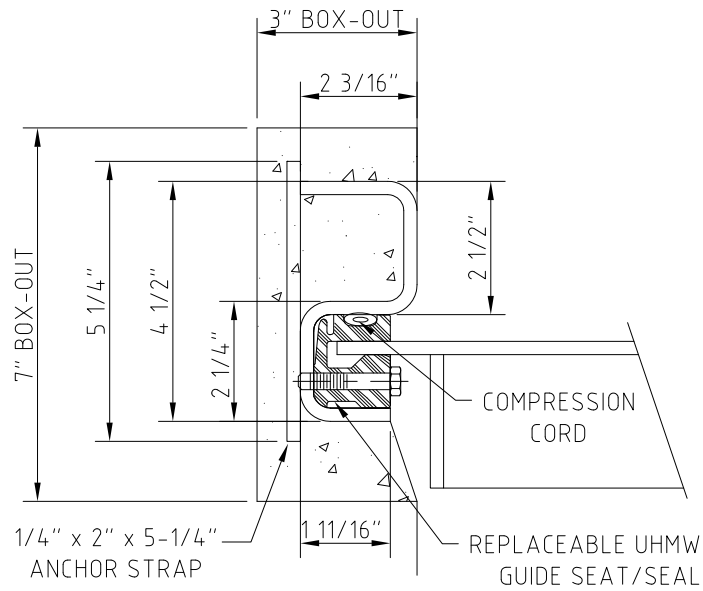
FOR GATE SECTION DETAILS: BC-921-2614-S2
YOKE: 1/4" X 3" X 6" CHANNEL
STEM GUIDE CHANNEL: 1/4" X 2" X 4-1/2" CHANNEL
HORIZONTAL STIFFENER: 1/4" X 2-1/2" X 4" ANGLE
VERTICAL STIFFENERS: 1/4" X 2-1/2" FLAT BAR
APPROX. WEIGHT OF SLIDE: 360 LBS.
APPROX. WEIGHT OF GATE ASSEMBLY: 820 LBS.



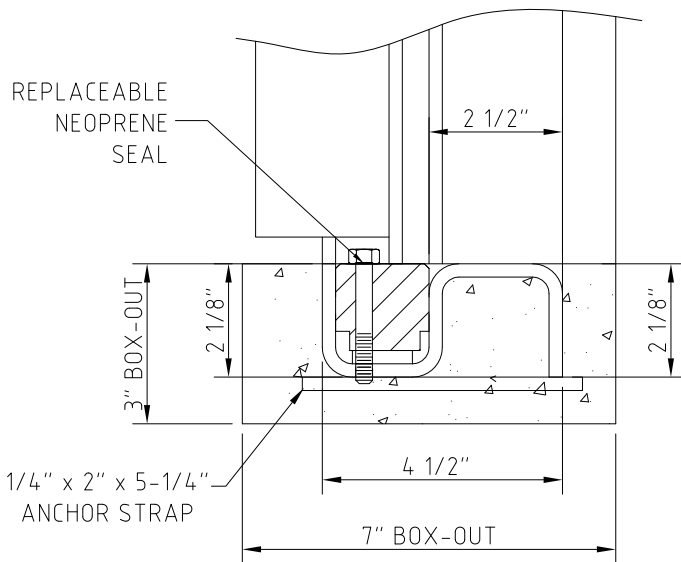
DETAIL A/A
SCALE 2:1



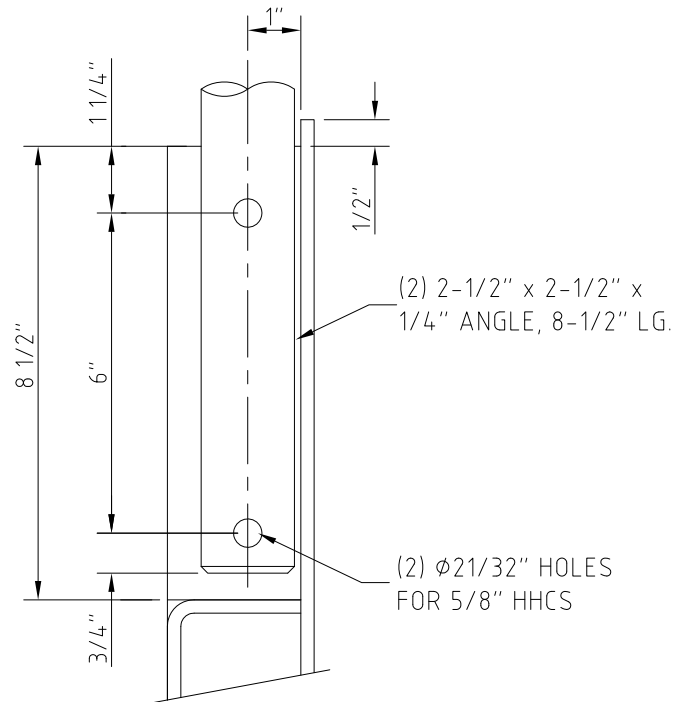
Title		SERIES 921 SLIDE GATE INSTALL. 45" x 66"		Quantity: 2	Scale: FD		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches Requested Concrete Opening Tolerances Height & Width ±1/4" Squareness ±3/8 Diagonal						
Project		HILO WWTF PHASE #1 - HILO, HAWAII		Tag No's: 02-GAT-1215 & 1225									
Item#:		G5		Seating	Unseating								
Design Head		20.0'		20.0'				A	Original Issue		5/6/25		KS
Location		GRIT OUTLET #1 & #2						Rev.	Rev. Record		Date		Eng. Chk.
Drg. No.	BC-921-2614		S.O. No.	32752		Leakage Rate	0.05	0.05				370 SOUTH ATHOL RD. ATHOL, MA 01331	
						GPM/ft of seal							



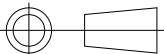
EMBEDDED GUIDE SECTION

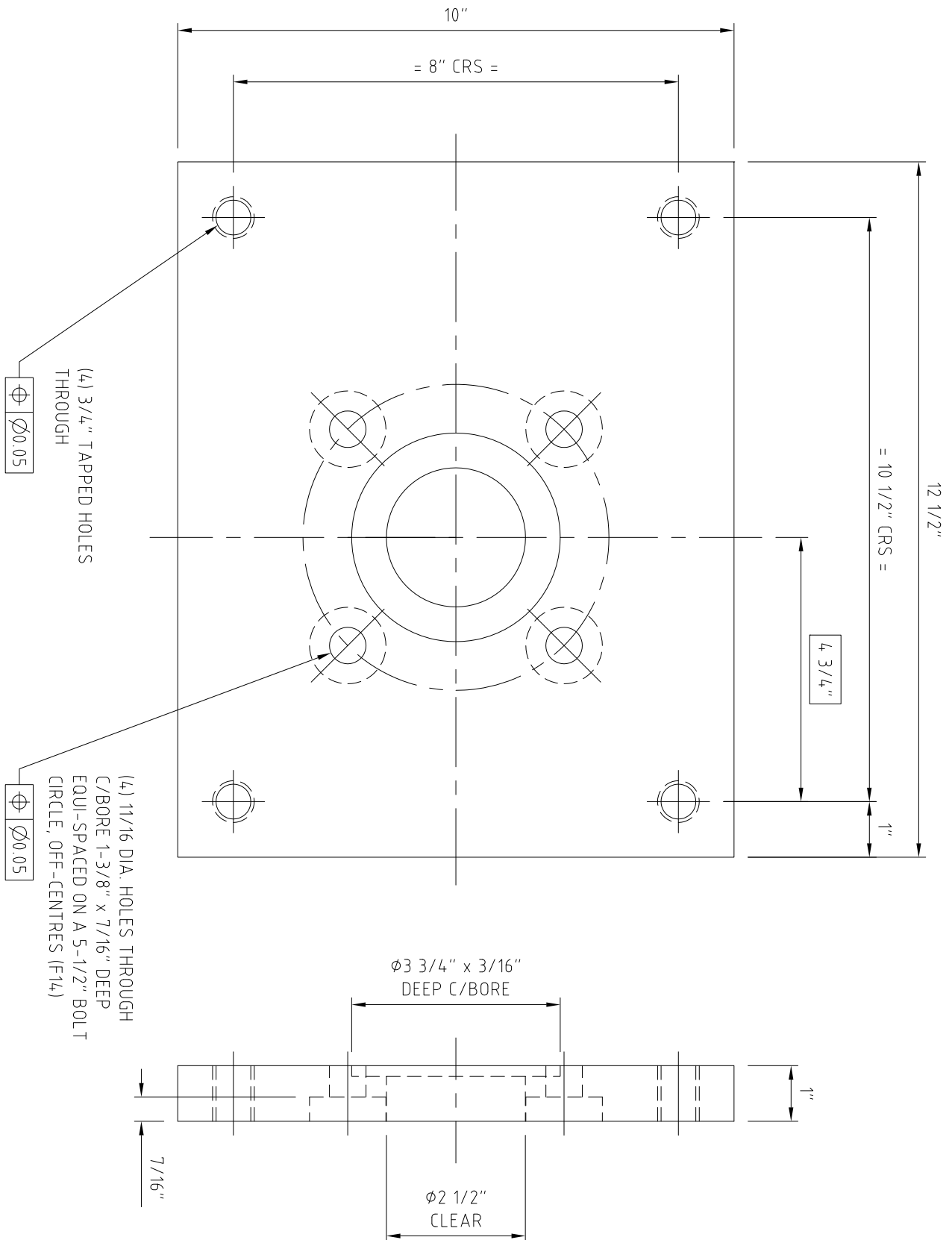


INVERT SECTION



STEM CONNECTOR

Title SERIES 921 SLIDE GATE SECTION DETAILS		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches				
Material STAINLESS STEEL GRADE 316L		Scale: FD		A	Original Issue	5/6/25
				Rev.	Rev. Record	Date
Drg. No. BC-921-2614-S2		S.O. No. 32752				
		THIRD ANGLE PROJECTION 				
				370 SOUTH ATHOL RD. ATHOL, MA 01331		



Title

ADAPTOR PLATE FA14

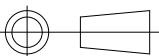
Material ALUMINUM 6061-T6

Drg. No. BC-921-2614-S3 S.O. No. 32752

DO NOT SCALE
THIS DRAWING
Work to Dimensions
Linear Dimensions are Inches

Scale: 1:2.5

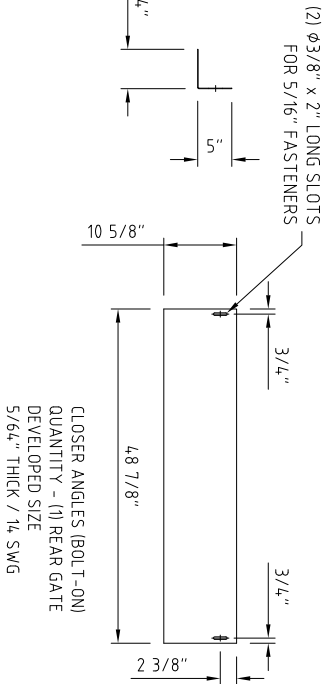
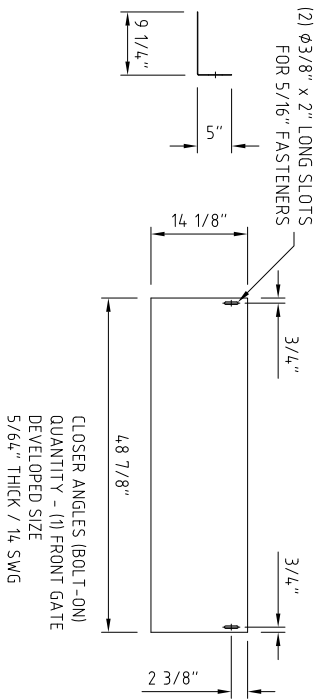
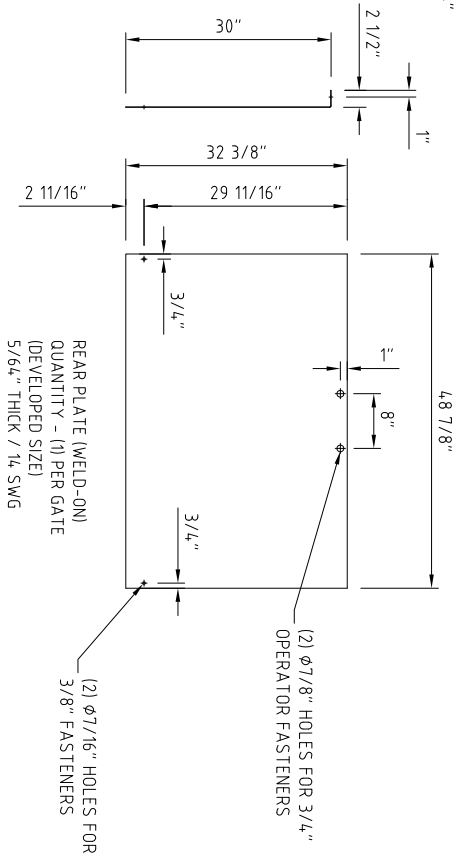
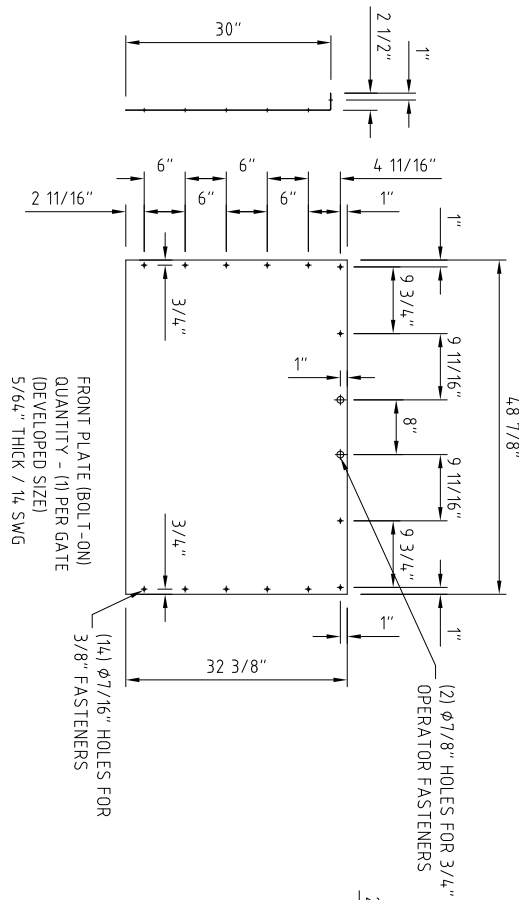
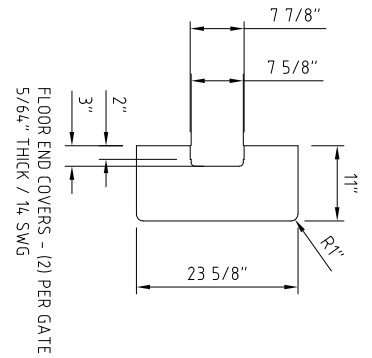
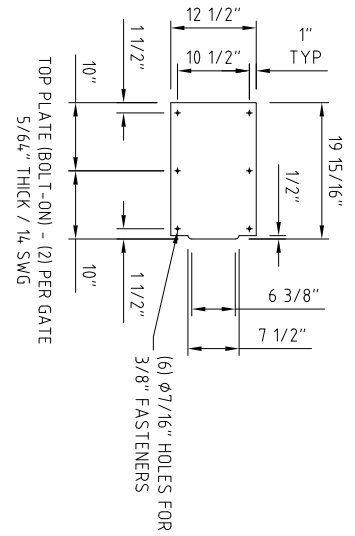
THIRD
ANGLE
PROJECTION



A	Original Issue	5/6/25	KS	
Rev.	Rev. Record	Date	Eng.	Chk.

Whipps, inc.

370 SOUTH ATHOL RD.
ATHOL, MA 01331



Title ODOR CONTROL COVERS

Material STAINLESS STEEL 316L

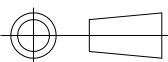
Dwg. No. BC-921-2614-S4

S.O. No. 32752

DO NOT SCALE
THIS DRAWING
Work to Dimensions
Linear Dimensions are Inches

Scale: 1:20

THIRD
ANGLE
PROJECTION



Whipps, inc.

370 SOUTH ATHOL RD.
ATHOL, MA 01331

A Original Issue

Rev. Rev. Record

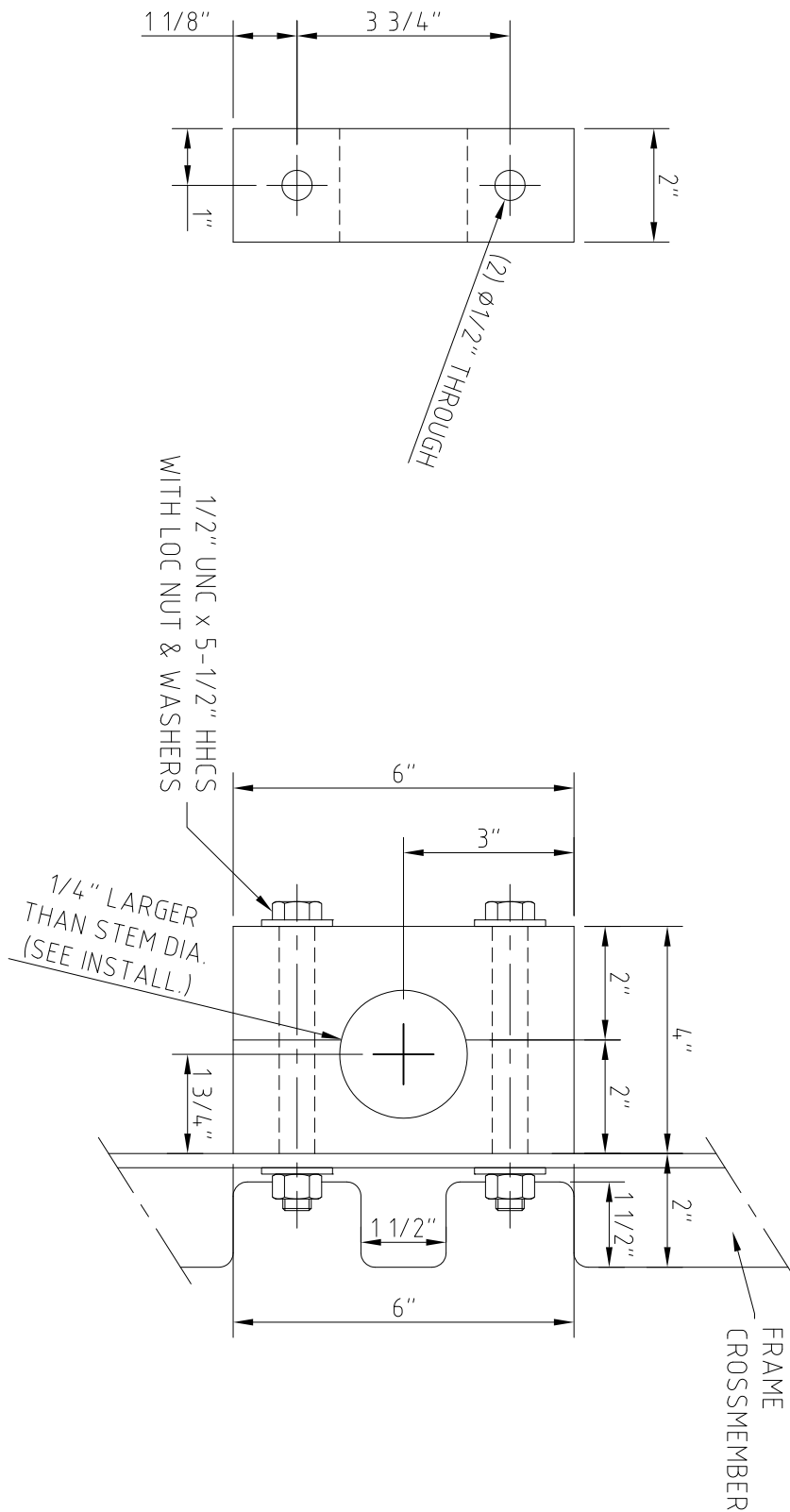
5/6/25

Date

KS

Eng.

Chk.

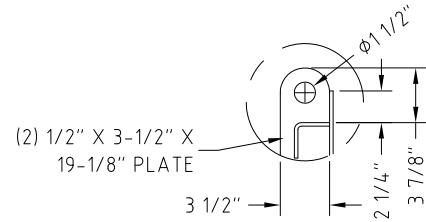


MATERIAL = UHMWPE

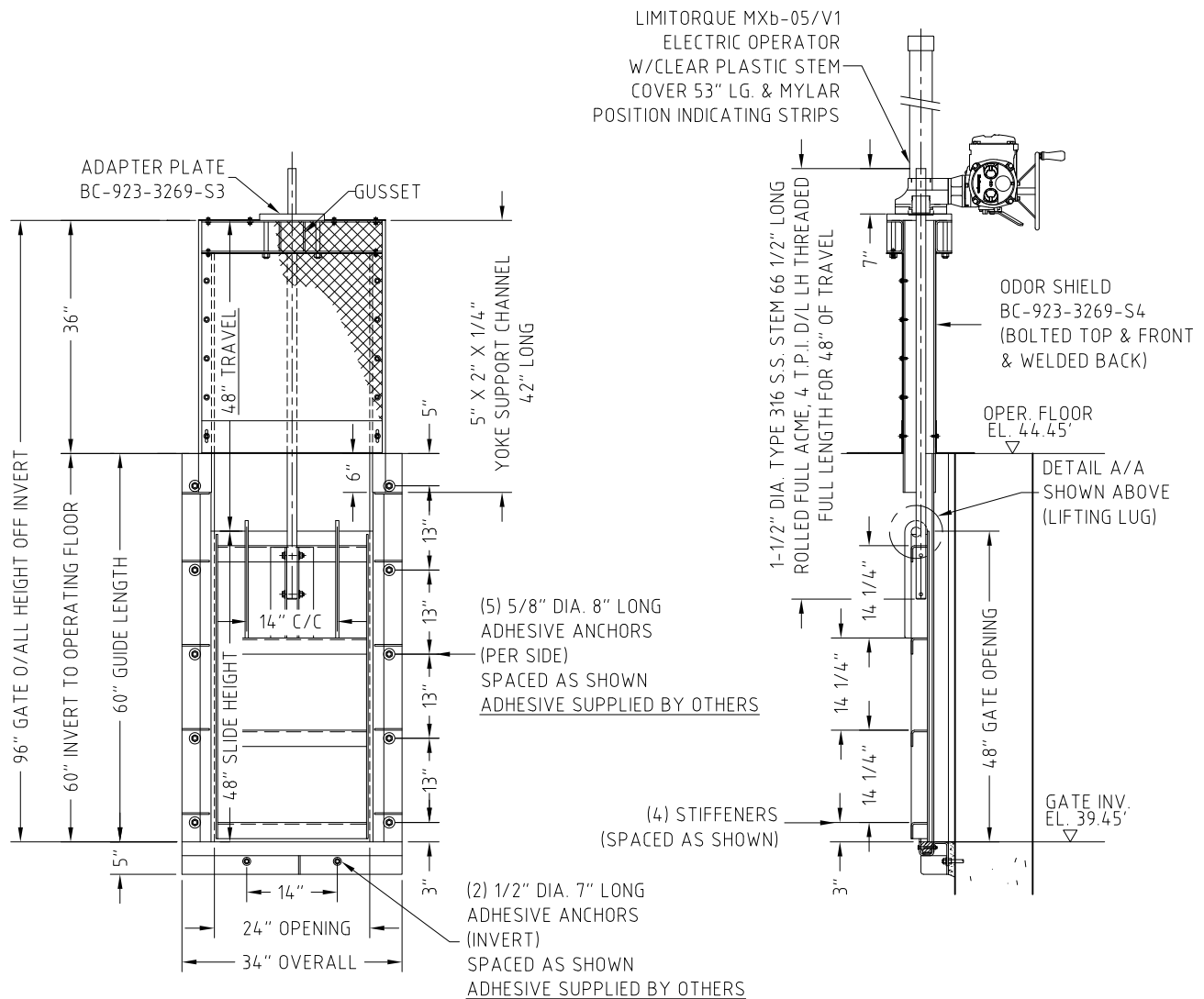
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		Scale: FD		A	Original Issue	5/4/12	KS
		THIRD ANGLE PROJECTION		Rev.	Rev. Record	Date	Eng.
							Chk.
Drg. No.	BC-921-2614-S5	S.O. No.	32752				
				370 SOUTH ATHOL RD. ATHOL, MA 01331			

ANY OBSTRUCTIVE HANDRAILING OR
FLOORING TO BE MODIFIED BY OTHERS

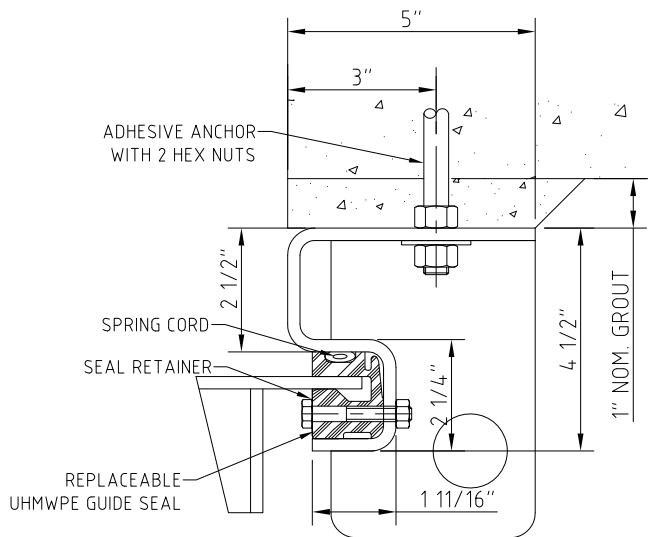
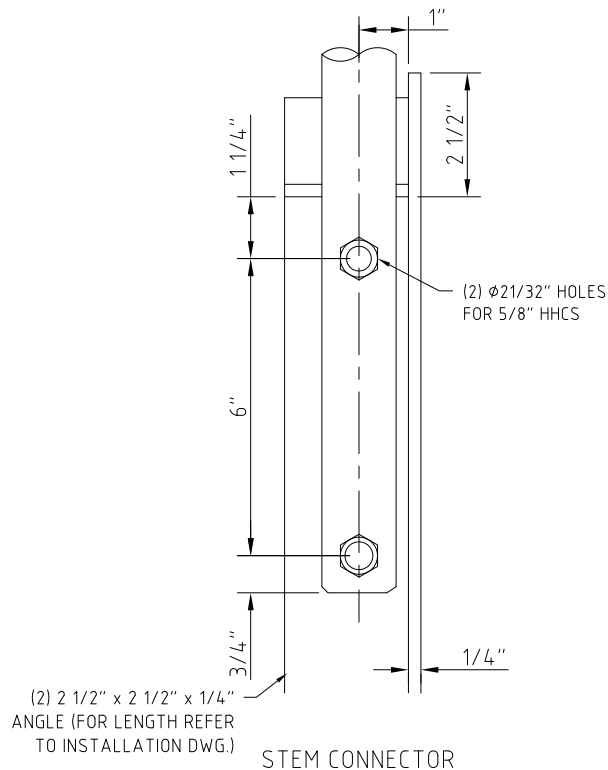
FOR GATE SECTION DETAILS: BC-923-3269-S2
YOKE: 1/4" X 2-1/2" X 5" CHANNEL
STEM GUIDE ANGLE: 1/4" X 2-1/2" X 2-1/2" ANGLE
HORIZONTAL STIFFENER: 1/4" X 2-1/2" X 2-1/2" ANGLE
VERTICAL STIFFENERS: 1/4" X 2-1/2" FLAT BAR
APPROX. WEIGHT OF SLIDE: 160 LBS.
APPROX. WEIGHT OF GATE ASSEMBLY: 520 LBS.



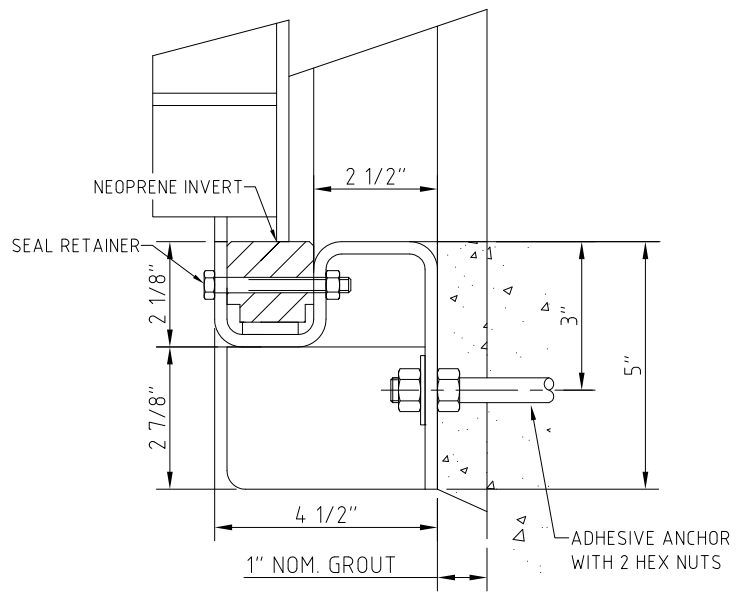
DETAIL A/A
SCALE 2:1



Title SERIES 924 SLIDE GATE INSTALL. 24" x 48"		Quantity: 3	Scale: FD		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches						
		Tag No's: 03-FG-1110, 1120 & 1130									
Project HILO WWTF PHASE #1 - HILO, HAWAII	Item#: L5		Seating	Unseating	Requested Concrete Opening Tolerances Height & Width ±1/4" Squareness ±3/8 Diagonal	A	Original Issue		5/12/25	KS	
	Design Head		20.0'	20.0'		Rev.	Rev. Record		Date	Eng.	Chk.
Location PRIMARY INLET #1, #2 & #3		Leakage Rate		0.05	0.05	THIRD ANGLE PROJECTION			370 SOUTH ATHOL RD. ATHOL, MA 01331		
		GPM/ft of seal									
Dwg. No.	BC-923-3269	S.O. No.	32752								

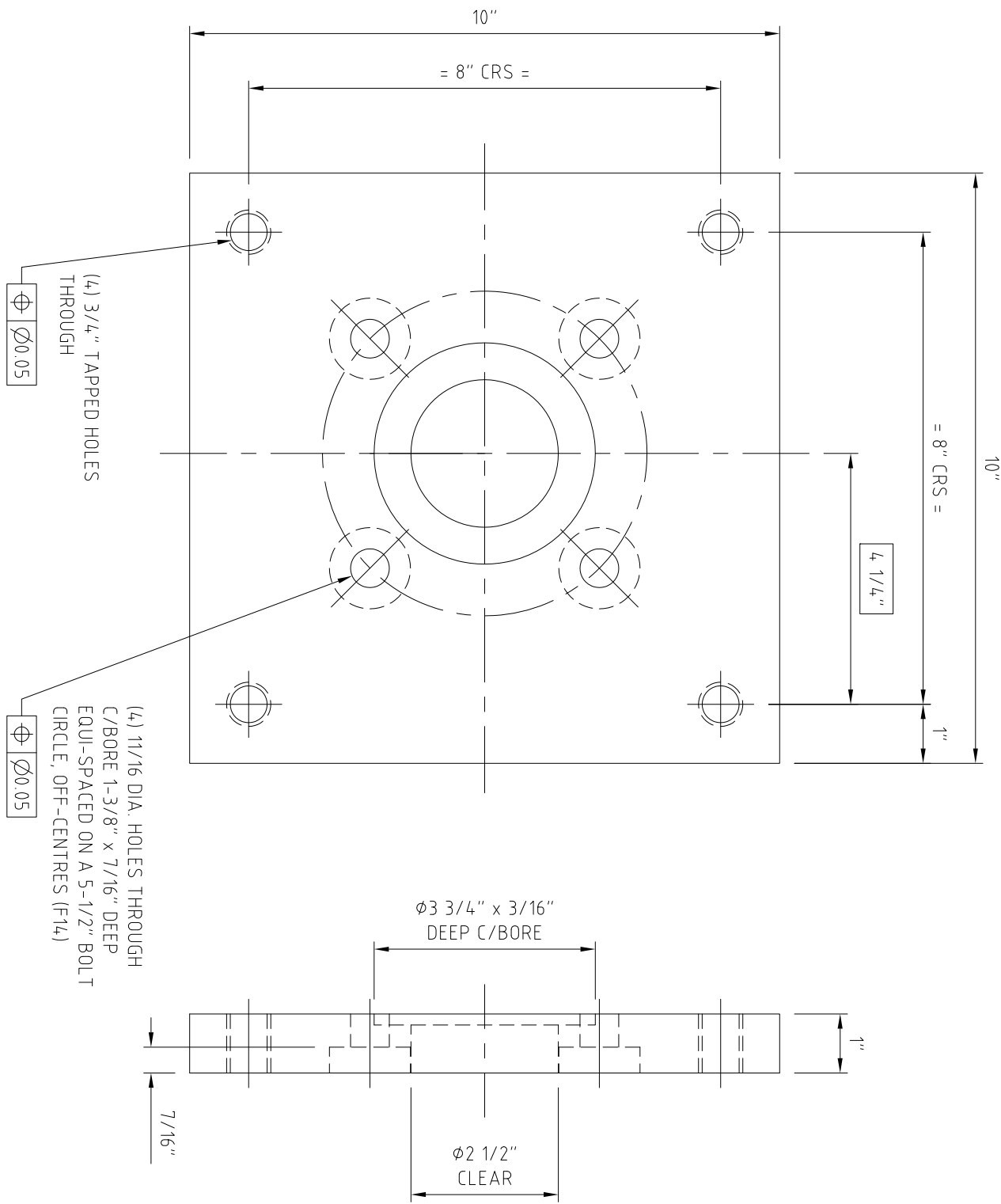


GUIDE SECTION DETAIL

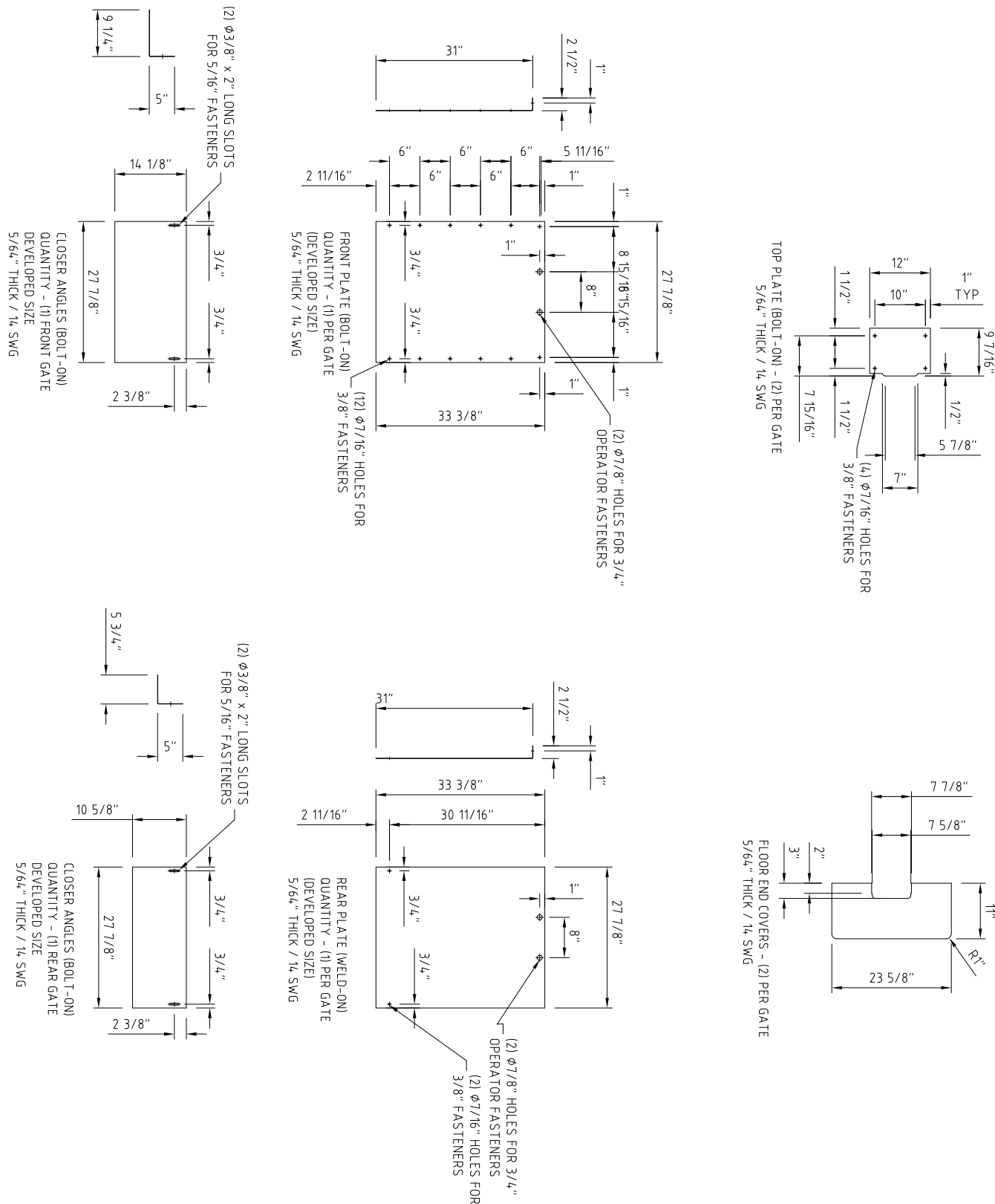


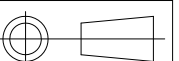
INVERT SECTION DETAIL

Title SERIES 924 SLIDE GATE SECTION DETAILS		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches							
Material STAINLESS STEEL 316L		Scale: F:D		A	Original Issue	5/12/25	KS		
				Rev.	Rev. Record	Date	Eng.	Chk.	
Dwg. No. BC-923-3269-S2		S.O. No. 32752		THIRD ANGLE PROJECTION		Whipps, Inc.			
						370 SOUTH ATHOL RD. ATHOL, MA 01331			

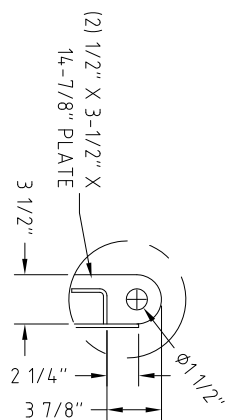


Title ADAPTOR PLATE FA14		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches						
				A	Original Issue	5/12/25	KS	
Material ALUMINUM 6061-T6		Scale: 1:2.5		Rev.	Rev. Record	Date	Eng.	Chk.
Drg. No. BC-923-3269-S3				Whipps, inc. 370 SOUTH ATHOL RD. ATHOL, MA 01331				
S.O. No. 32752		THIRD ANGLE PROJECTION						

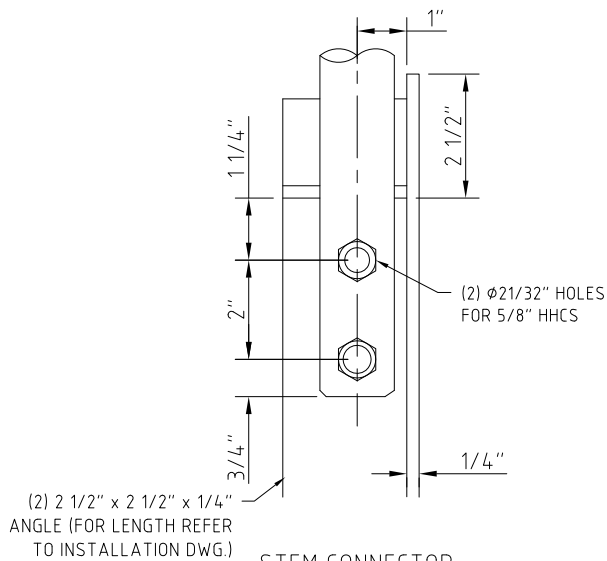


Title ODOR CONTROL COVERS		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches									
Material STAINLESS STEEL 316L		Scale: 1:20		A	Original Issue	5/12/25	KS				
				Rev.	Rev. Record	Date	Eng.	Chk.			
Drg. No.	BC-923-3269-S4	S.O. No.	32752	THIRD ANGLE PROJECTION						370 SOUTH ATHOL RD. ATHOL, MA 01331	

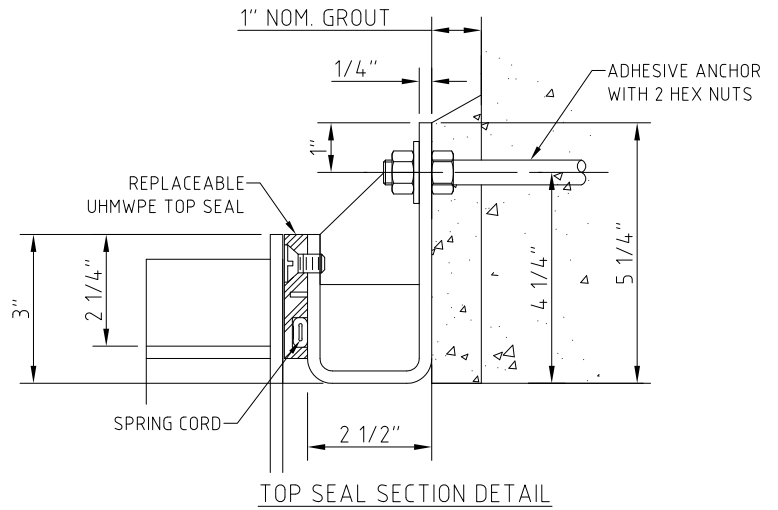
LIMITORQUE MXB-05/V1
ELECTRIC OPERATOR
W/CLEAR PLASTIC STEM
COVER 4" LG. & MYLAR
POSITION INDICATING STRIPS



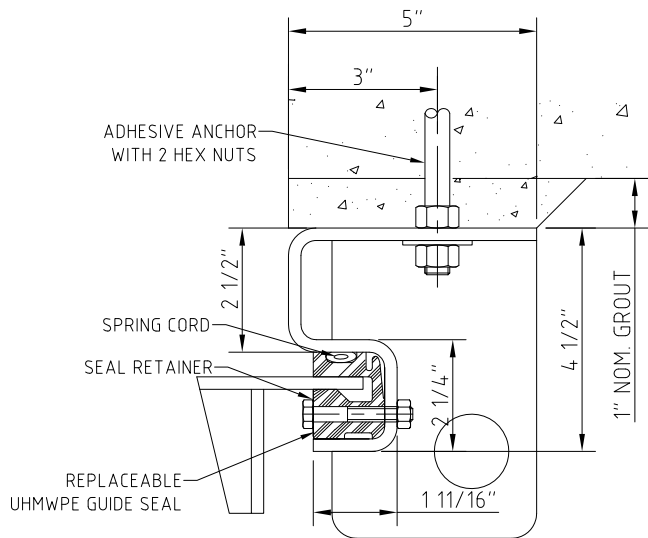
Title SERIES 924 SLIDE GATE INSTALL. 36" x 36"		Quantity: 1		Scale: FD		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches							
Project HILO WWTF PHASE #1 - HILO, HAWAII		Tag No's: 03-GAT-1230											
Location GRIT EFFLUENT BYPASS		Item#: H5		Seating	Unseating	Requested Concrete Opening Tolerances Height & Width ±1/4" Squareness ±3/8 Diagonal		A	Original Issue		5/12/25		KS
Drg. No. BC-924-3115		S.O. No. 32752		Leakage Rate GPM/ft of seal				0.05	0.05	Rev.	Rev. Record		Date
						THIRD ANGLE PROJECTION						370 SOUTH ATHOL RD. ATHOL, MA 01331	



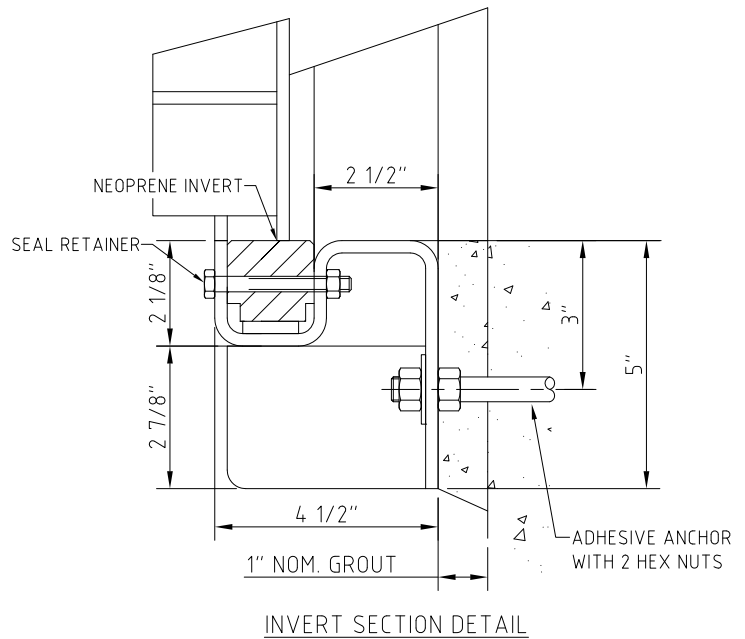
STEM CONNECTOR



TOP SEAL SECTION DETAIL

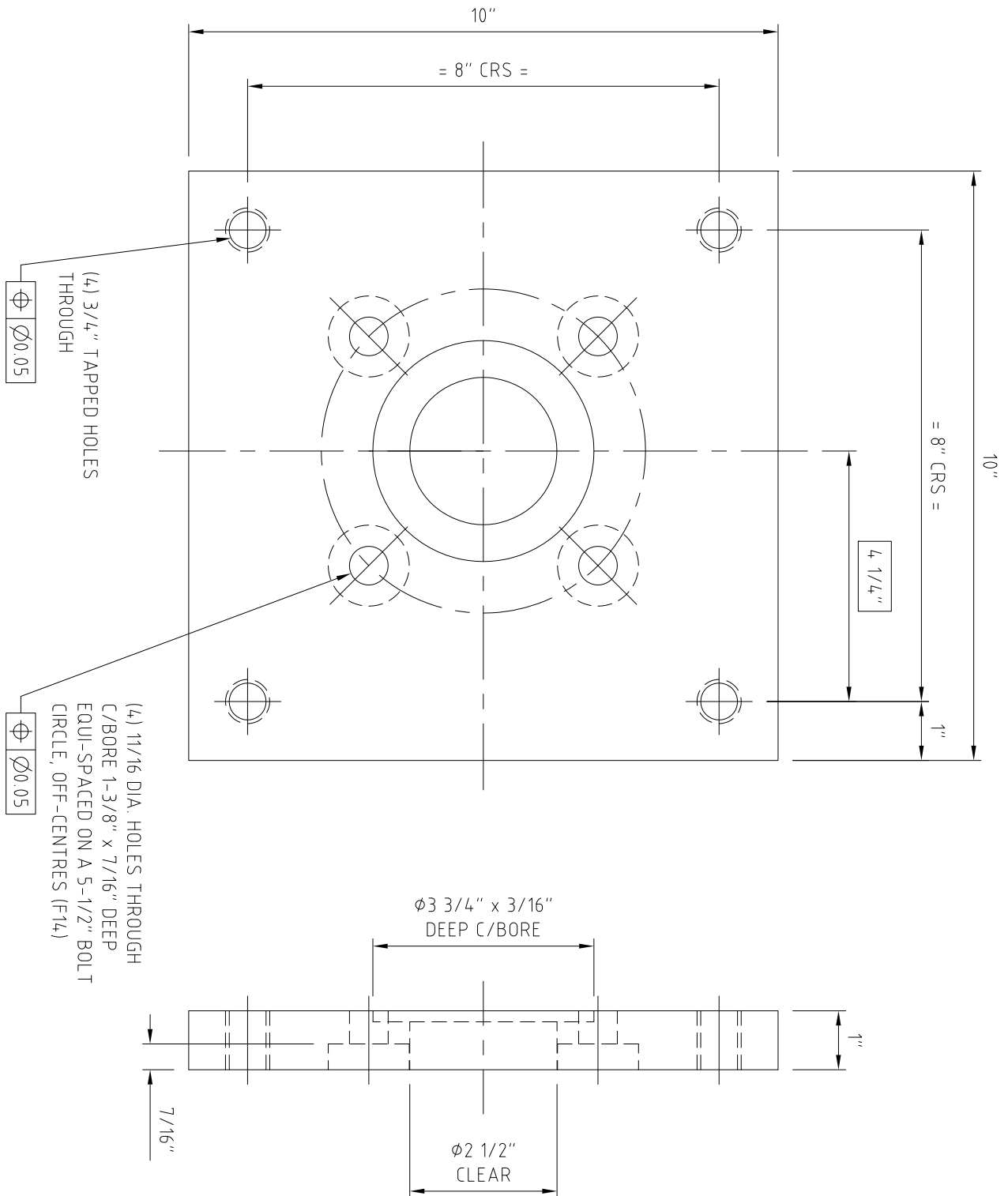


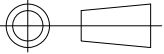
GUIDE SECTION DETAIL



INVERT SECTION DETAIL

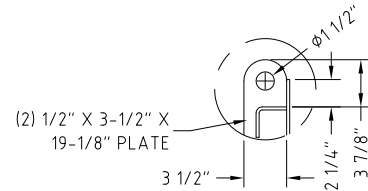
Title SERIES 924 SLIDE GATE SECTION DETAILS		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches				
Material STAINLESS STEEL 316L		Scale: F:D		A	Original Issue	5/12/25
				Rev.	Rev. Record	KS
Drg. No. BC-924-3115-S2		S.O. No. 32752		Date		Eng. Chk.
		THIRD ANGLE PROJECTION				
				370 SOUTH ATHOL RD. ATHOL, MA 01331		



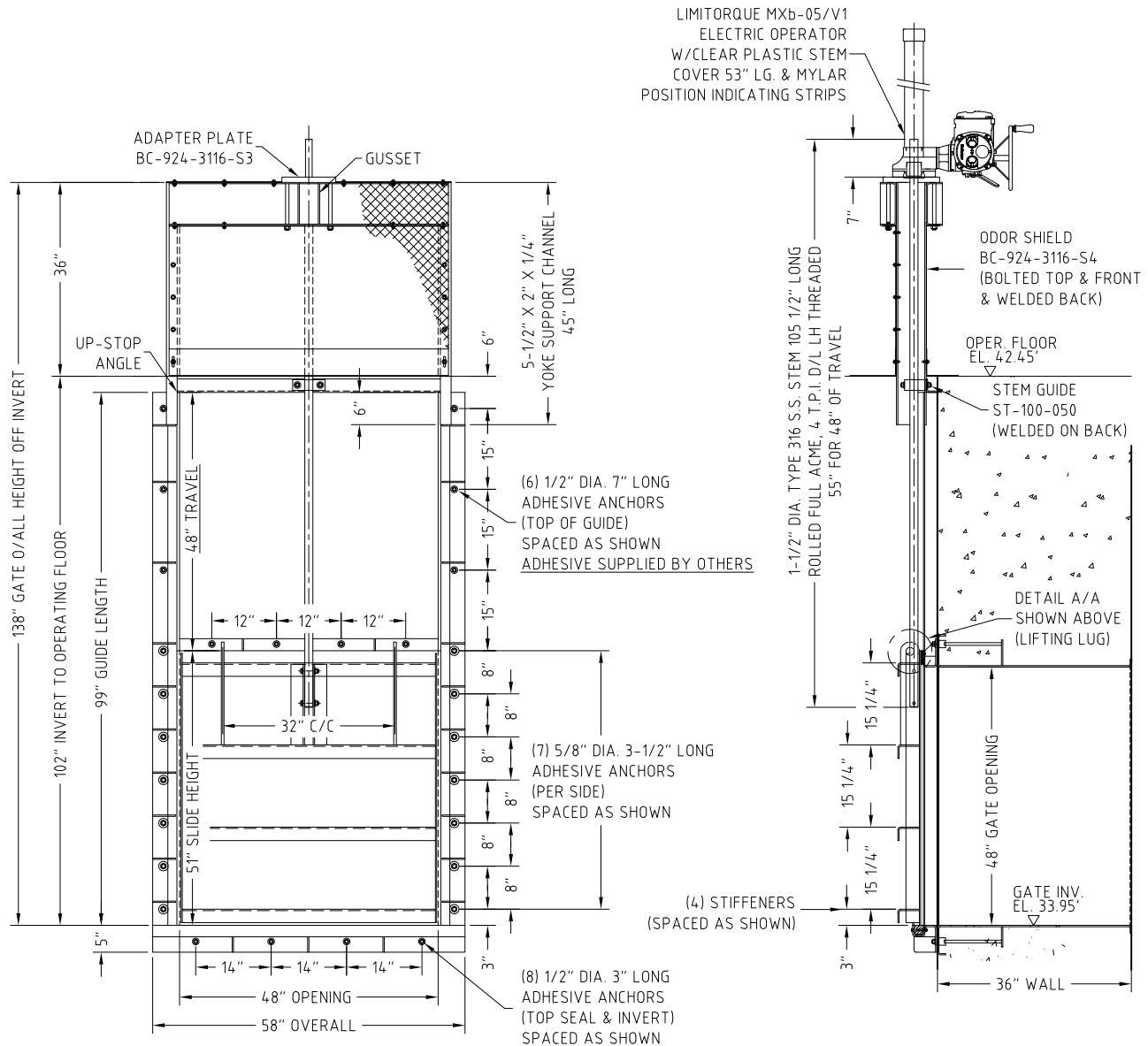
Title ADAPTOR PLATE FA14		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches						
Material ALUMINUM 6061-T6		Scale: 1:2.5		A	Original Issue	5/12/25	KS	
Drg. No. BC-924-3115-S3		S.O. No. 32752		Rev.	Rev. Record	Date	Eng.	Chk.
		THIRD ANGLE PROJECTION 		 370 SOUTH ATHOL RD. ATHOL, MA 01331				

ANY OBSTRUCTIVE HANDRAILING OR
FLOORING TO BE MODIFIED BY OTHERS

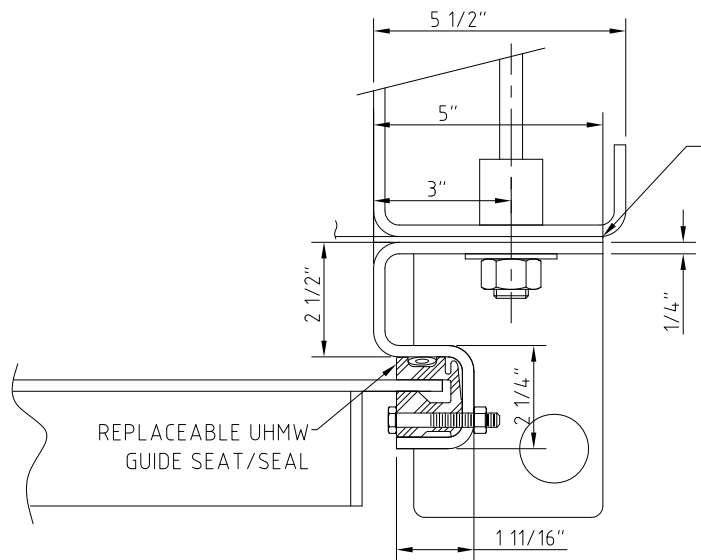
FOR GATE SECTION DETAILS: BC-924-3116-S2
FOR THIMBLE DETAILS: BC-924-3116-S5
YOKE: 1/4" X 3" X 8" CHANNEL
STEM GUIDE ANGLE: 1/4" X 2-1/2" X 2-1/2" ANGLE
HORIZONTAL STIFFENER: 1/4" X 2-1/2" X 4" ANGLE
VERTICAL STIFFENERS: 1/4" X 2-1/2" FLAT BAR
APPROX. WEIGHT OF SLIDE: 310 LBS.
APPROX. WEIGHT OF GATE ASSEMBLY: 790 LBS.



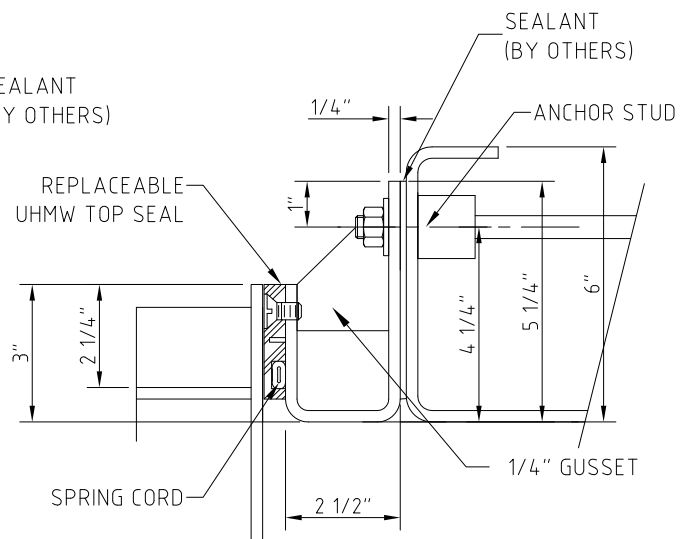
DETAIL A/A
SCALE 2:1



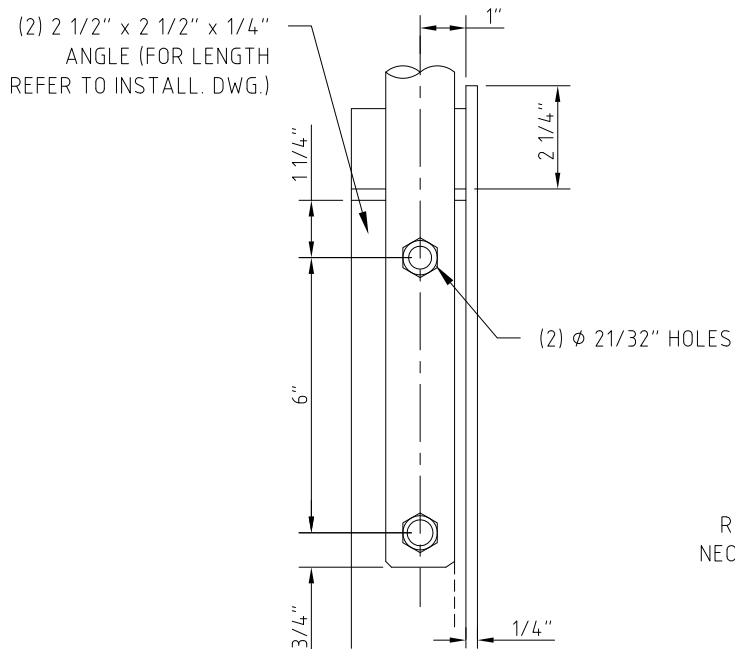
Title		SERIES 924 SLIDE GATE INSTALL. 48" x 48"		Quantity: 2	Scale: FD	DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are in inches Requested Concrete Opening Tolerances Height & Width ±1/4" Squareness ±3/8 Diagonal						
Project		HILO WWTF PHASE #1 - HILO, HAWAII		Tag No's: 03-FG-1440 & 1445								
Item#:		M5	Seating	Unseating								
Design Head		20.0'	20.0'				A	Original Issue	5/12/25	KS		
Location		PRIMARY EFFLUENT OUTLET					Rev.	Rev. Record	Date	Eng.	Chk.	
Dwg. No.	BC-924-3116	S.O. No.	32752	Leakage Rate	GPM/ft of seal	0.05	0.05					
THIRD ANGLE PROJECTION				Whipps, Inc.		370 SOUTH ATHOL RD. ATHOL, MA 01331						



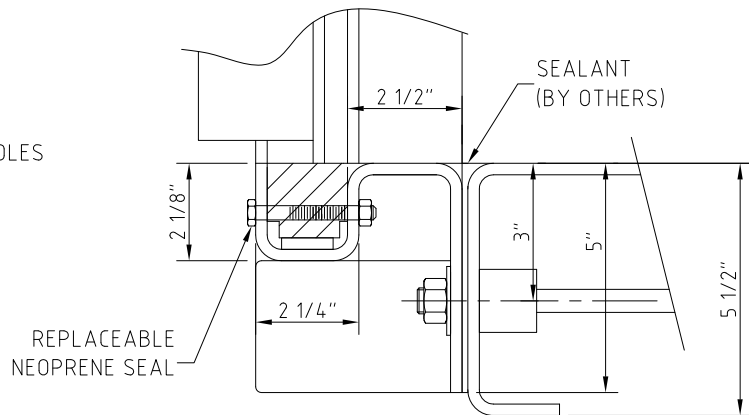
GUIDE SECTION



TOP SEAL SECTION

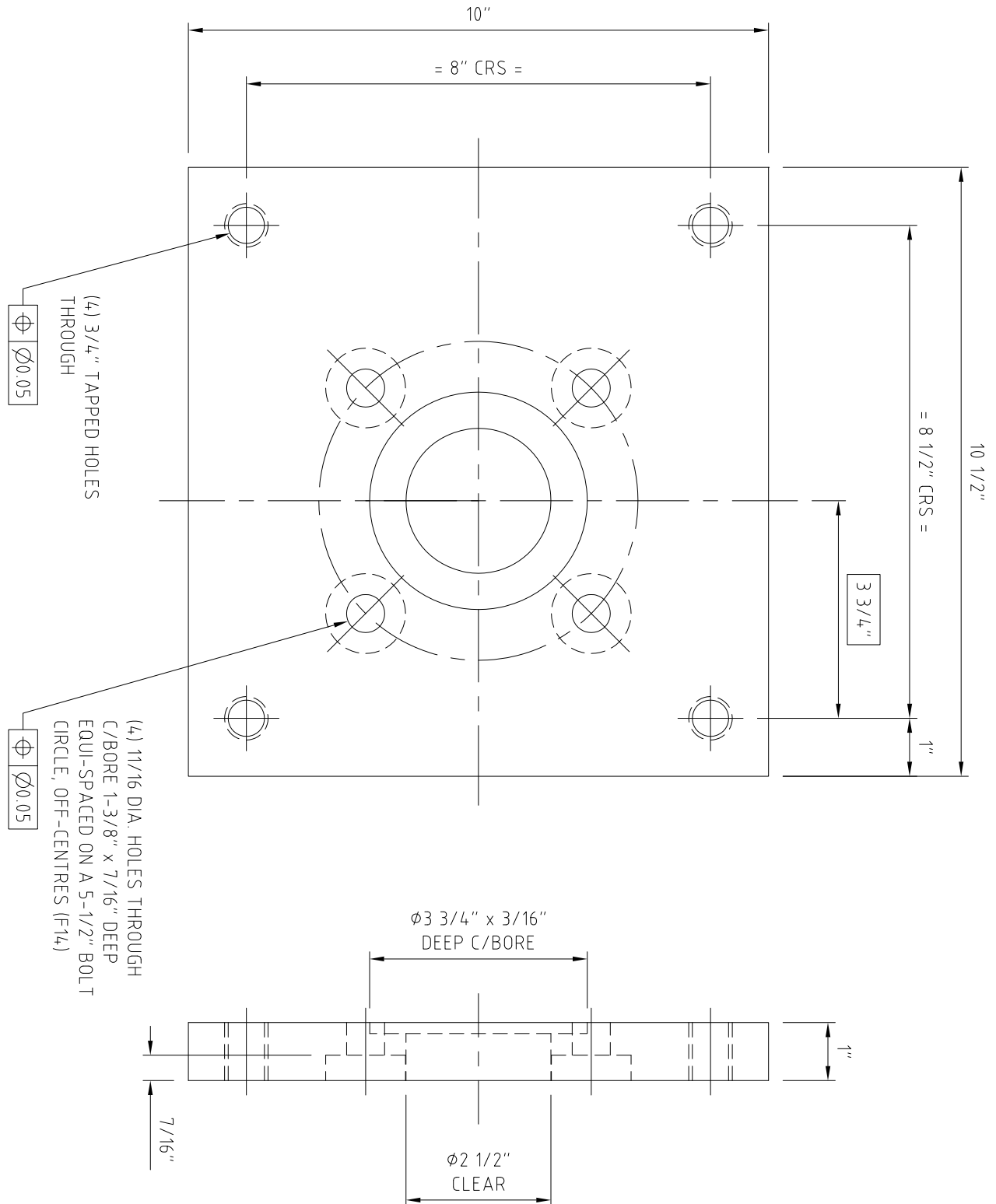


STEM CONNECTOR

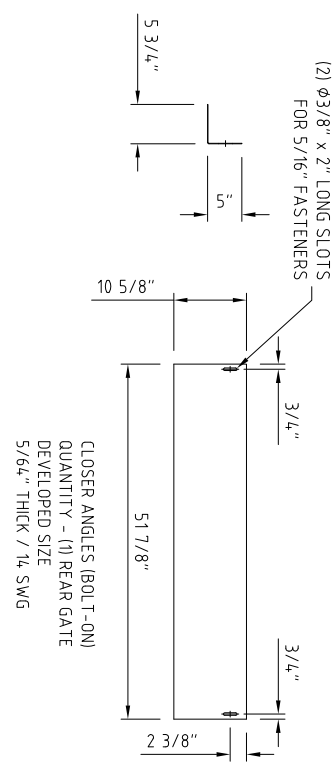
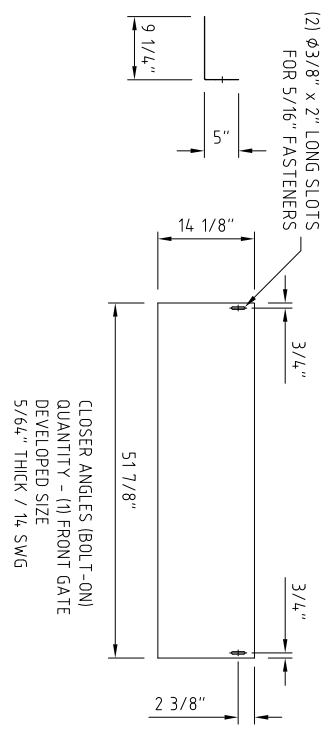
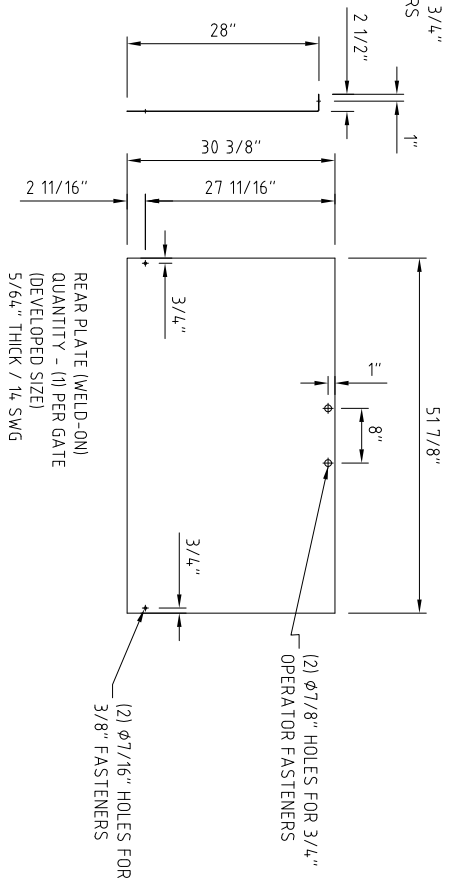
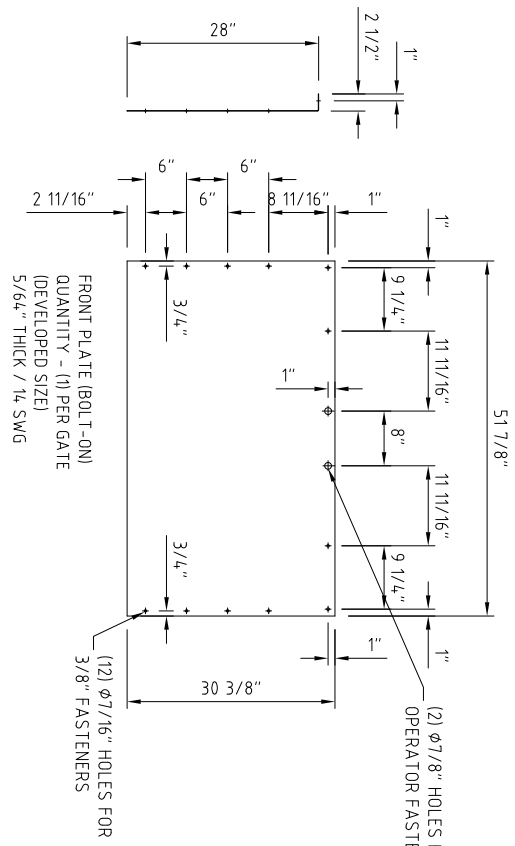
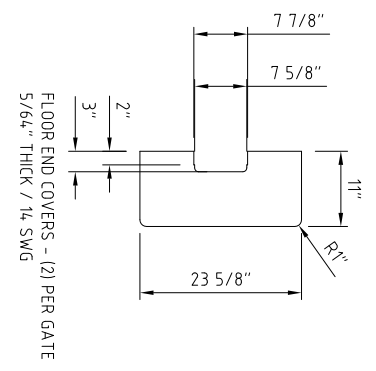
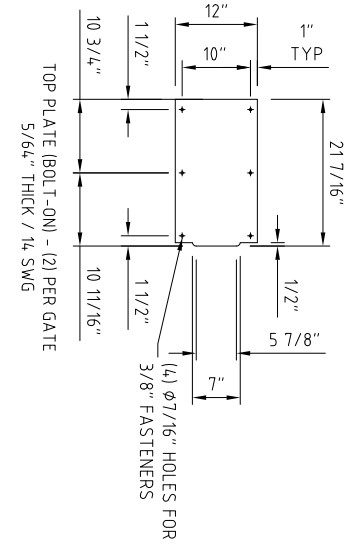


INVERT SEAL SECTION

Title SERIES 924 SLIDE GATE SECTION DETAILS		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches								
		Material STAINLESS STEEL GRADE 316L		Scale: FD		A	Original Issue		5/12/25	KS
Rev.	Rev. Record					Date	Eng.	Chk.		
Dwg. No. BC-924-3116-S2		S.O. No. 32752		THIRD ANGLE PROJECTION			Whipps, inc.			370 SOUTH ATHOL RD. ATHOL, MA 01331



Title		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches		A	Original Issue	5/12/25	KS	
Material		Scale: 1:2.5		Rev.	Rev. Record	Date	Eng.	Chk.
Drg. No. BC-924-3116-S3		S.O. No. 32752		THIRD ANGLE PROJECTION				
ADAPTOR PLATE FA14		ALUMINUM 6061-T6		Whipps, inc. 370 SOUTH ATHOL RD. ATHOL, MA 01331				



Title ODOR CONTROL COVERS

Material STAINLESS STEEL 316L

Dwg. No. BC-924-3116-S4 S.O. No. 32752

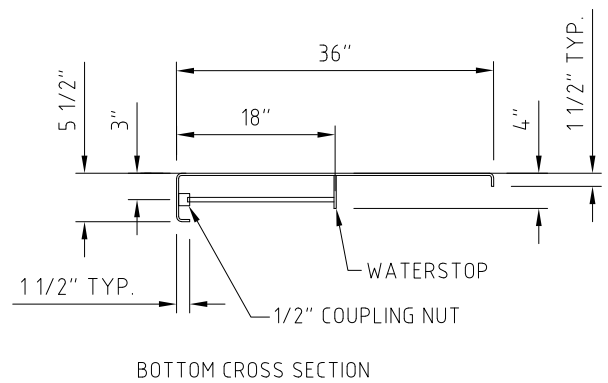
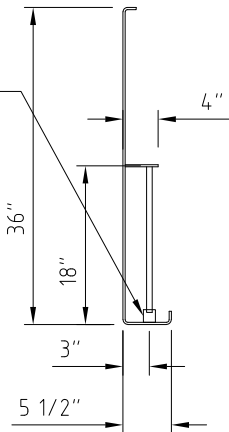
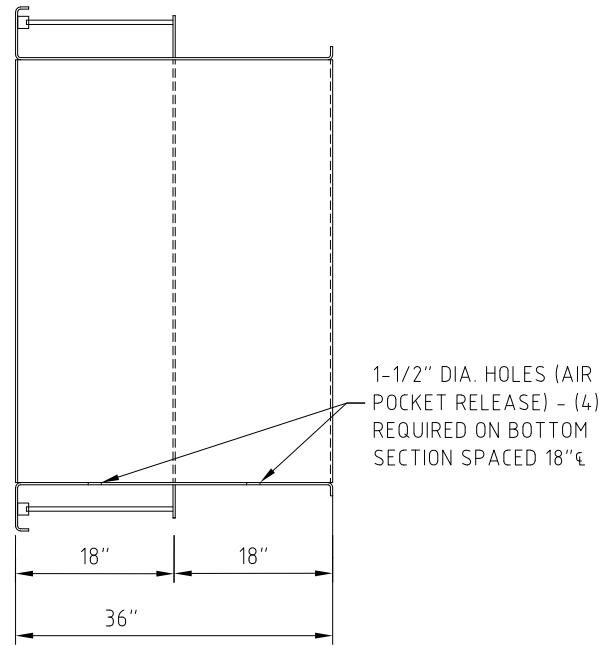
DO NOT SCALE
THIS DRAWING
Work to Dimensions
Linear Dimensions are Inches

Scale: 1:20

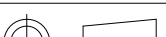
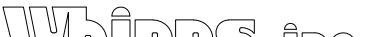
THIRD
ANGLE
PROJECTION

A	Original Issue	5/12/25	KS	
Rev.	Rev. Record	Date	Eng.	Chk.
Whipps, inc.		370 SOUTH ATHOL RD. ATHOL, MA 01331		

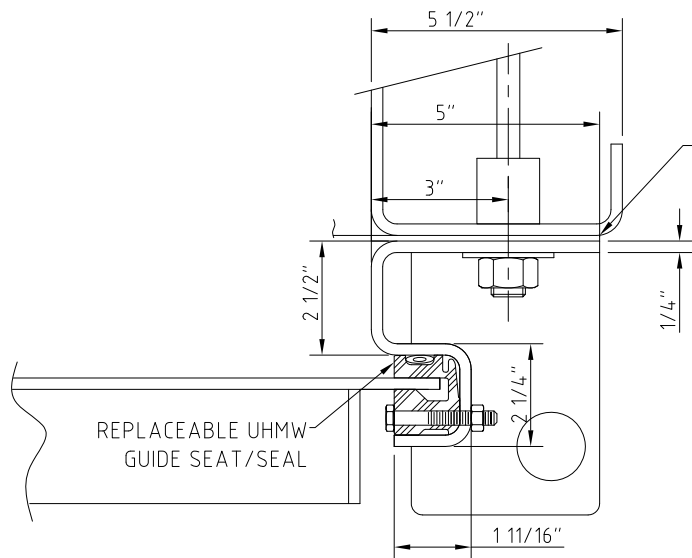
NOTE: THIMBLE TO HAVE "TOP" ,"BOTTOM"
AND CENTER LINE PERMANENTLY STAMPED
AS SHOWN



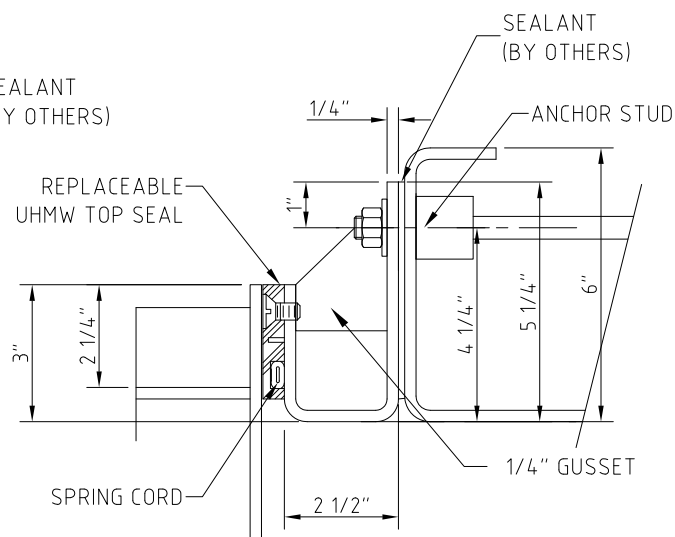
SIDE CROSS SECTION

Title		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches					
SERIES 924 48" x 48" THIMBLE DETAIL		Scale: FD					
Material	STAINLESS STEEL GRADE 316L	Item #	N5	A	Original Issue	5/12/25	KS
Rev.		Rev. Record				Date	Eng. Chk.
Drg. No. BC-924-3116-S5		S.O. No. 32752	THIRD ANGLE PROJECTION				
							
				370 SOUTH ATHOL RD. ATHOL, MA 01331			

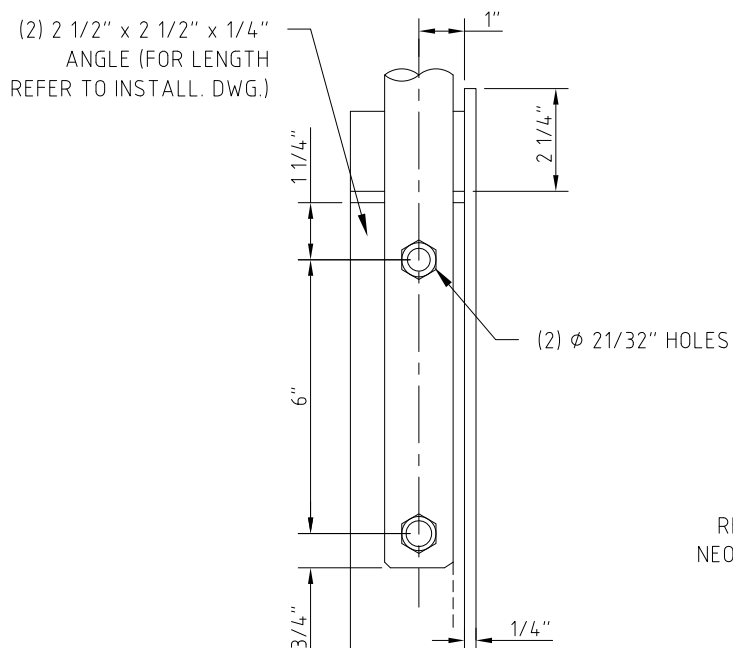
Title		Quantity: 1		Scale: FD		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches							
INSTALL. 24" x 48"		Tag No's: 03-FG-1495											
Project		Item#: P5		Seating		Unseating		Requested Concrete Opening Tolerances		A		Original Issue	
HILO WWTF PHASE #1 - HILO, HAWAII		Design Head		20.0'		20.0'		Height & Width ±1/4" Squareness ±3/8 Diagonal		Rev.		Rev. Record	
Location		Leakage Rate		0.05		0.05		THIRD ANGLE PROJECTION		Dafe		Eng. Chk	
PRIMARY EFFLUENT OVERFLOW ISOLATION		GPM/ft of seal								Whipps, Inc.		370 SOUTH ATHOL RD. ATHOL, MA 01331	
Dwg. No. BC-924-3117		S.O. 32752											



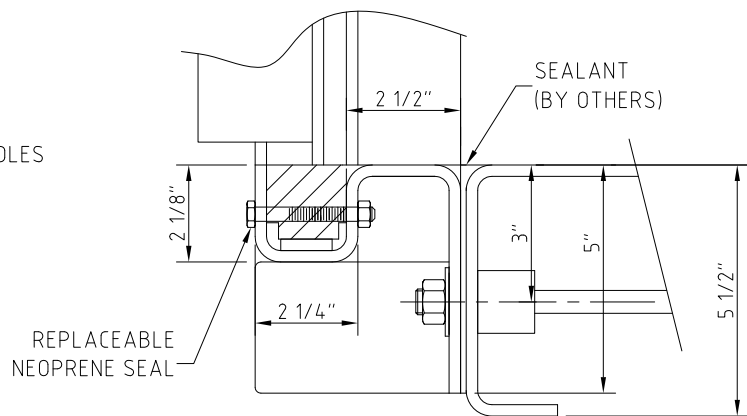
GUIDE SECTION



TOP SEAL SECTION



STEM CONNECTOR



INVERT SEAL SECTION

Title
SERIES 924 SLIDE GATE
SECTION DETAILS

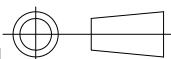
Material
STAINLESS STEEL GRADE 316L

Dwg. No. BC-924-3117-S2 S.O. No. 32752

DO NOT SCALE
THIS DRAWING
Work to Dimensions
Linear Dimensions are Inches

Scale: FD

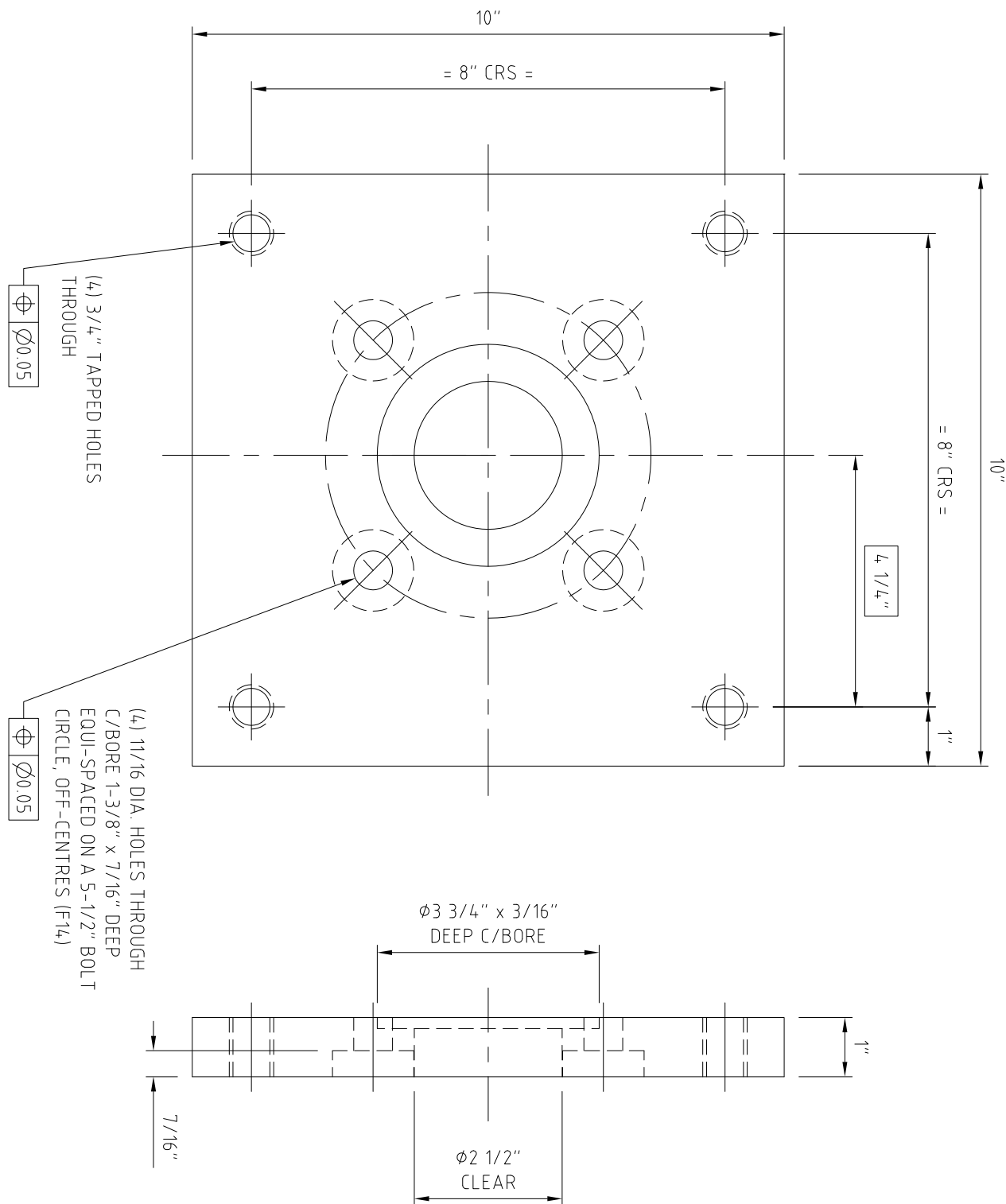
THIRD
ANGLE
PROJECTION



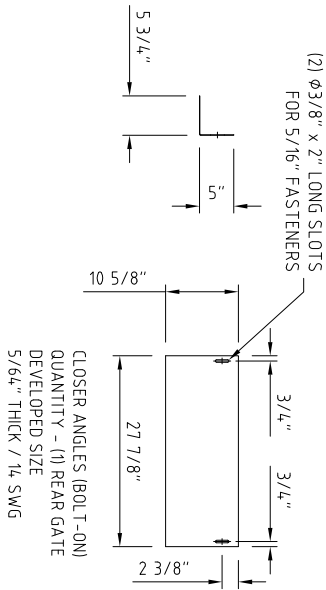
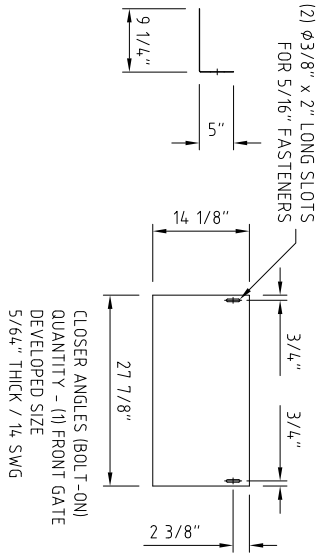
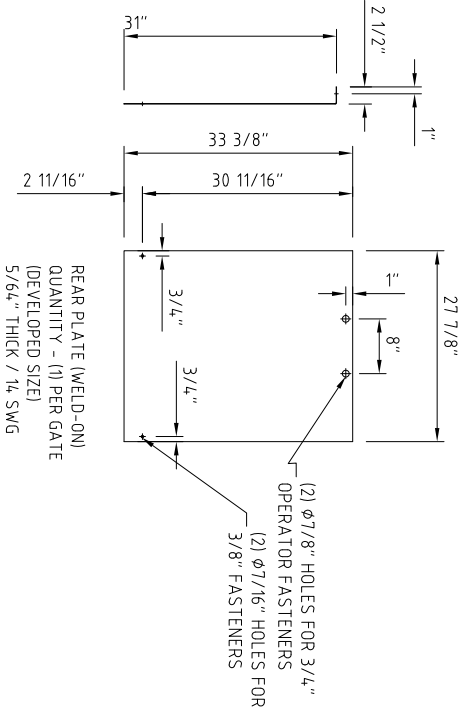
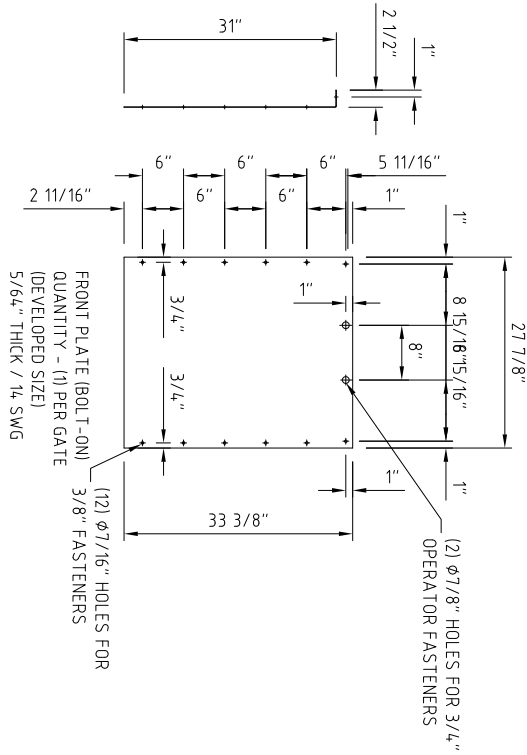
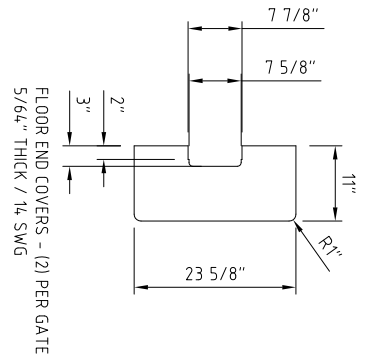
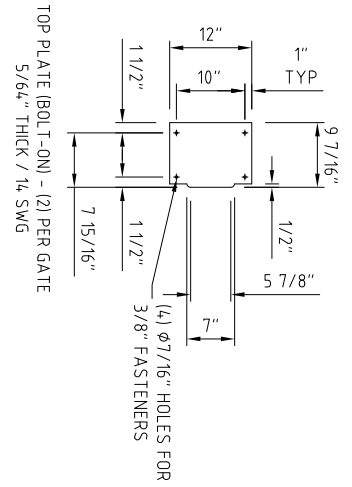
A	Original Issue	5/12/25	KS	
Rev.	Rev. Record	Date	Eng.	Chk.

Whipps, inc.

370 SOUTH ATHOL RD.
ATHOL, MA 01331



Title ADAPTOR PLATE FA14		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches						
Material ALUMINUM 6061-T6		Scale: 1:2.5		A	Original Issue	5/12/25	KS	
				Rev.	Rev. Record	Date	Eng.	Chk.
Drg. No. BC-924-3117-S3	S.O. No. 32752	THIRD ANGLE PROJECTION		Whipps, inc.		370 SOUTH ATHOL RD. ATHOL, MA 01331		



Title ODOR CONTROL COVERS

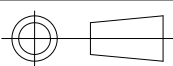
Material STAINLESS STEEL 316L

Drg. No. BC-924-3117-S4 S.O. No. 32752

DO NOT SCALE
THIS DRAWING
Work to Dimensions
Linear Dimensions are Inches

Scale: 1:20

THIRD
ANGLE
PROJECTION

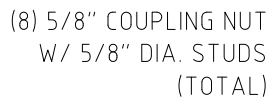
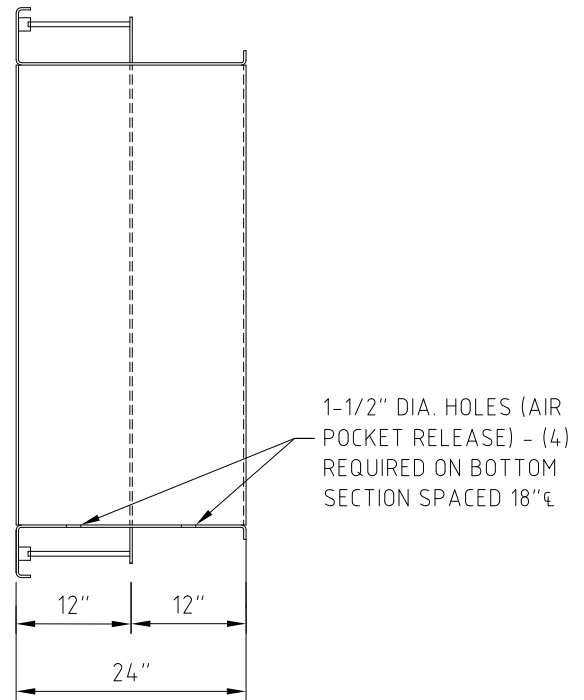


A	Original Issue	5/12/25	KS	
Rev.	Rev. Record	Date	Eng.	Chk.

Whipps, inc.

370 SOUTH ATHOL RD.
ATHOL, MA 01331

NOTE: THIMBLE TO HAVE "TOP", "BOTTOM"
AND CENTER LINE PERMANENTLY STAMPED
AS SHOWN



Technical drawing of a stepped shaft. The shaft has a total length of 5 1/2". It features a step at the left end, with a vertical height of 24" and a horizontal offset of 3". A diagonal line indicates a transition or feature, with a vertical dimension of 12" and a horizontal dimension of 4". The shaft has a central hole with a diameter of 3".

5 1/2"

3"

24"

12"

4"

1 1/2" TYP.

1 1/2" TYP.

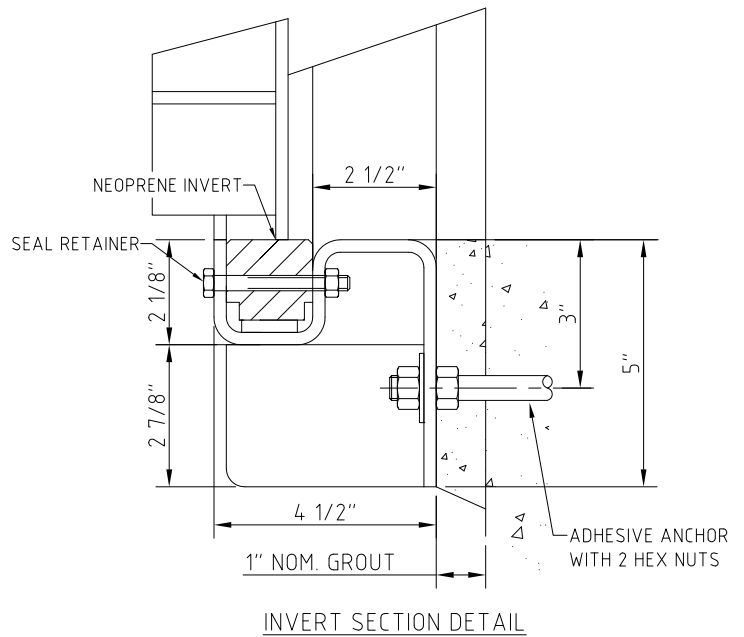
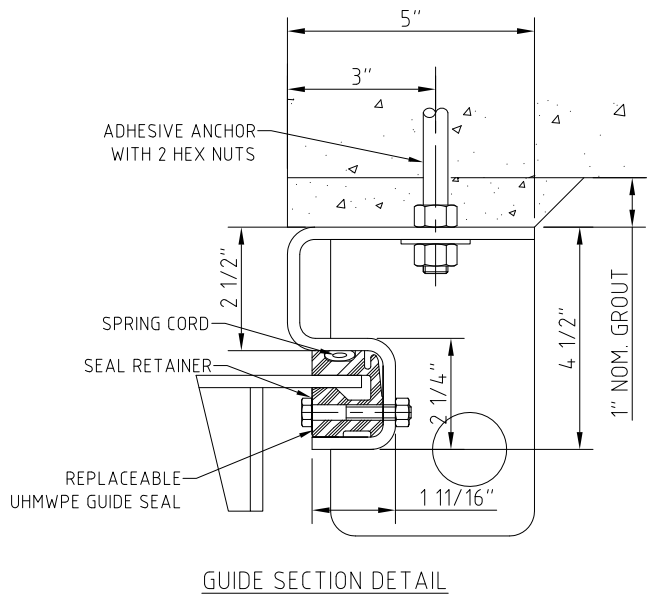
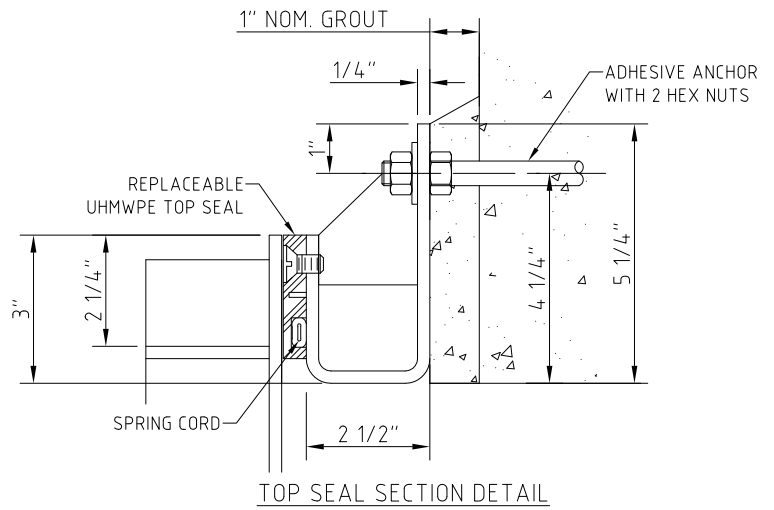
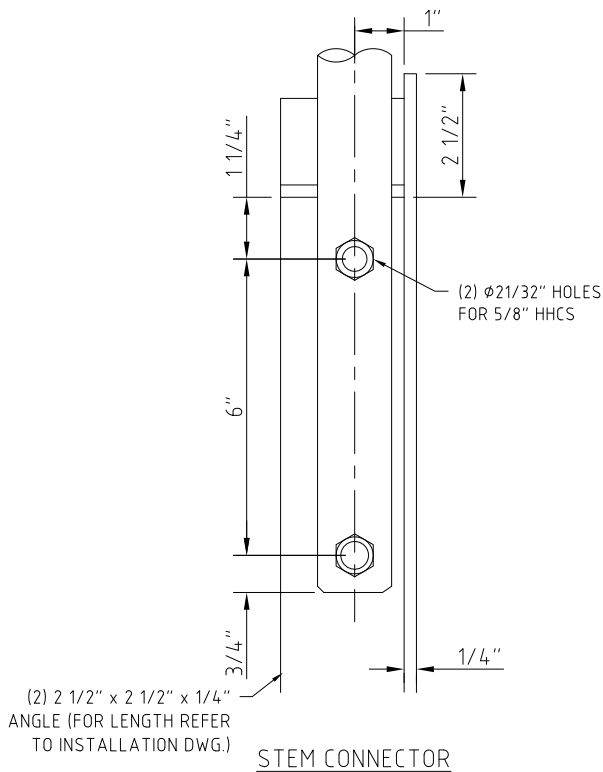
WATERSTOP

1/2" COUPLING NUT

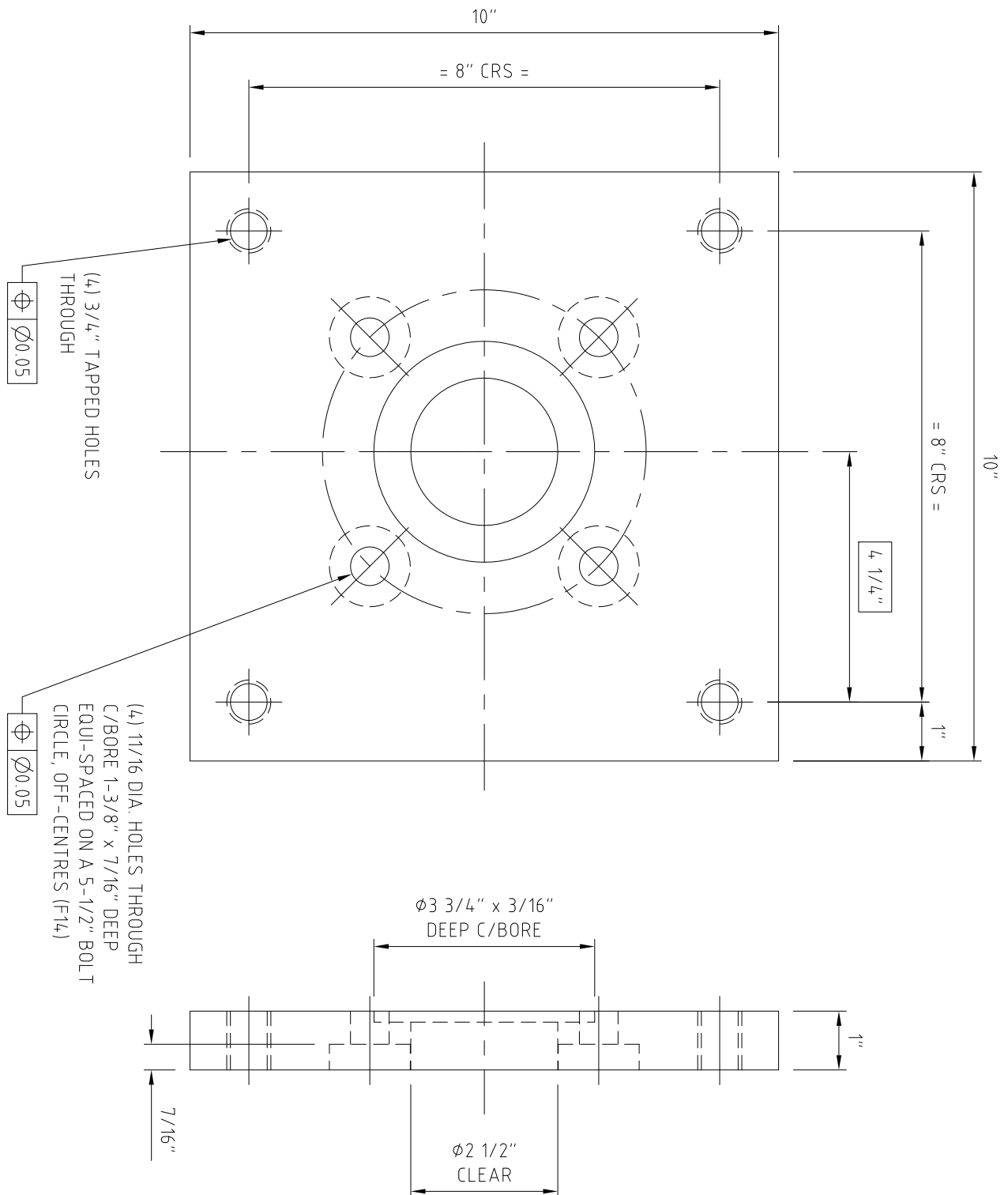
BOTTOM CROSS SECTION

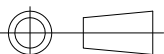
Title SERIES 924 24" x 48" THIMBLE DETAIL		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches								
Material STAINLESS STEEL GRADE 316L		Item# Q5		Scale: FD		A	Original Issue	5/12/25	KS	
						Rev.	Rev. Record	Date	Eng.	Chk.
Dwg. No. BC-924-3117-S5		S.O. No. 32752		THIRD ANGLE PROJECTION		Whipps, inc. 370 SOUTH ATHOL RD. ATHOL, MA 01331				

Title SERIES 924 SLIDE GATE INSTALL. 30" x 72"			Quantity: 3		Scale: FD		DO NOT SCALE THIS DRAWING Work To Dimensions Linear Dimensions are Inches							
Project HILO WWTF PHASE #1 - HILO, HAWAII			Tag No's: 06-FG-1410, 1420 & 1430											
Item#: R5			Seating		Unseating		Requested Concrete Opening Tolerances Height & width ±1/4" Squareness ±3/8 Diagonal		A		Original Issue		5/12/25 KS	
Location SECONDARY CLARIFIER INLET #1, #2 & #3			Design Head		20.0'		20.0'		Rev.		Rev. Record		Date Eng Chk	
Dwg. No. BC-924-3118		S.O. 32752		Leakage Rate GPM/ft of seal		0.05 0.05		THIRD ANGLE PROJECTION 				370 SOUTH ATHOL RD. ATHOL, MA 01331		



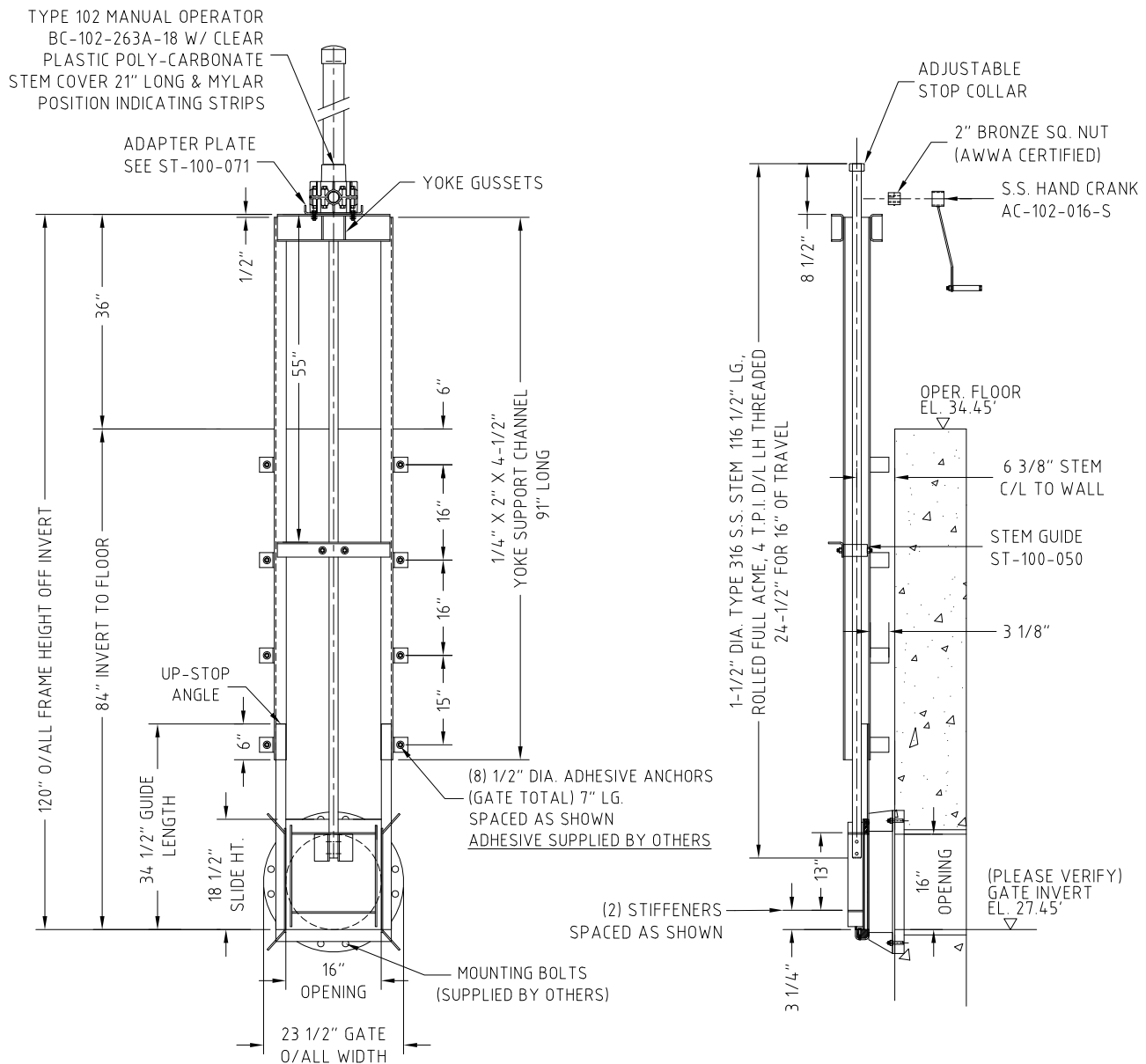
Title SERIES 924 SLIDE GATE SECTION DETAILS		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches							
Material STAINLESS STEEL 316L		Scale: F:D		A	Original Issue		5/12/25	KS	
				Rev.	Rev. Record		Date	Eng.	Chk.
Dwg. No. BC-924-3118-S2		S.O. No. 32752		THIRD ANGLE PROJECTION		Whipps, inc. 370 SOUTH ATHOL RD. ATHOL, MA 01331			
									



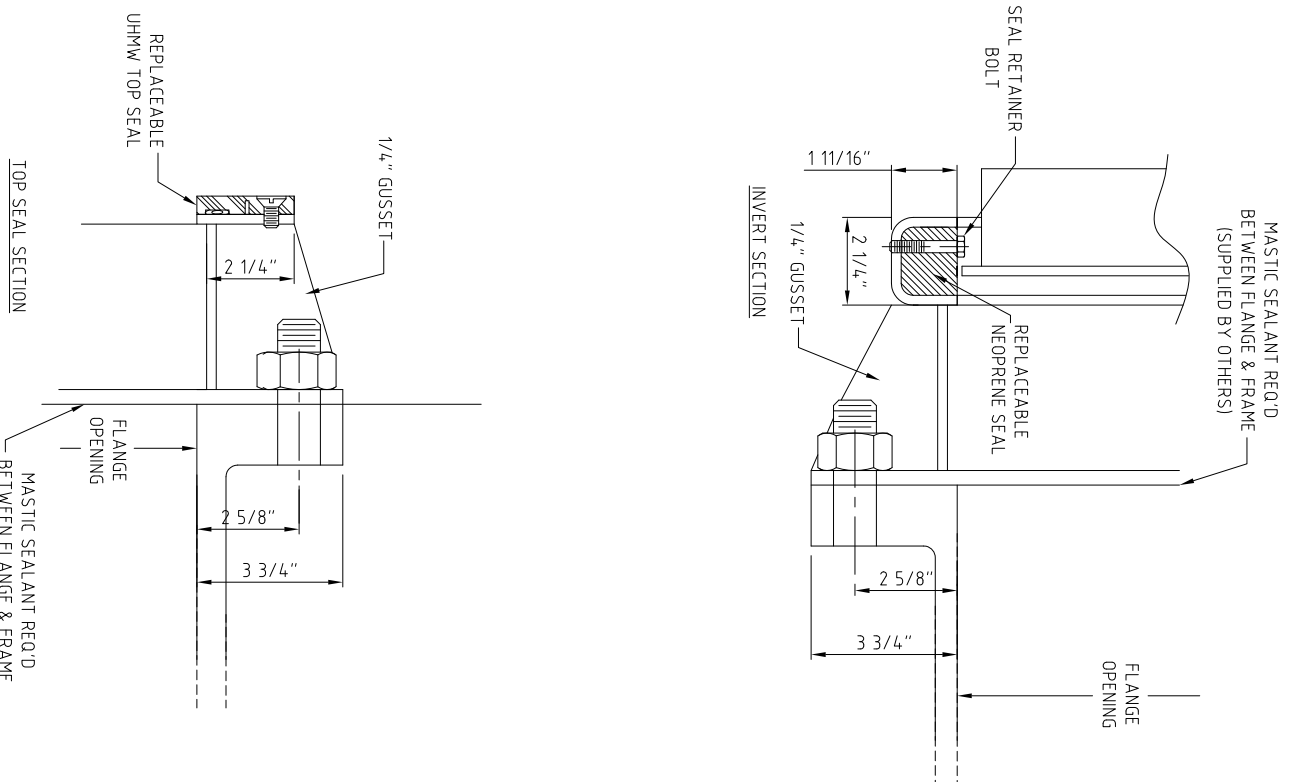
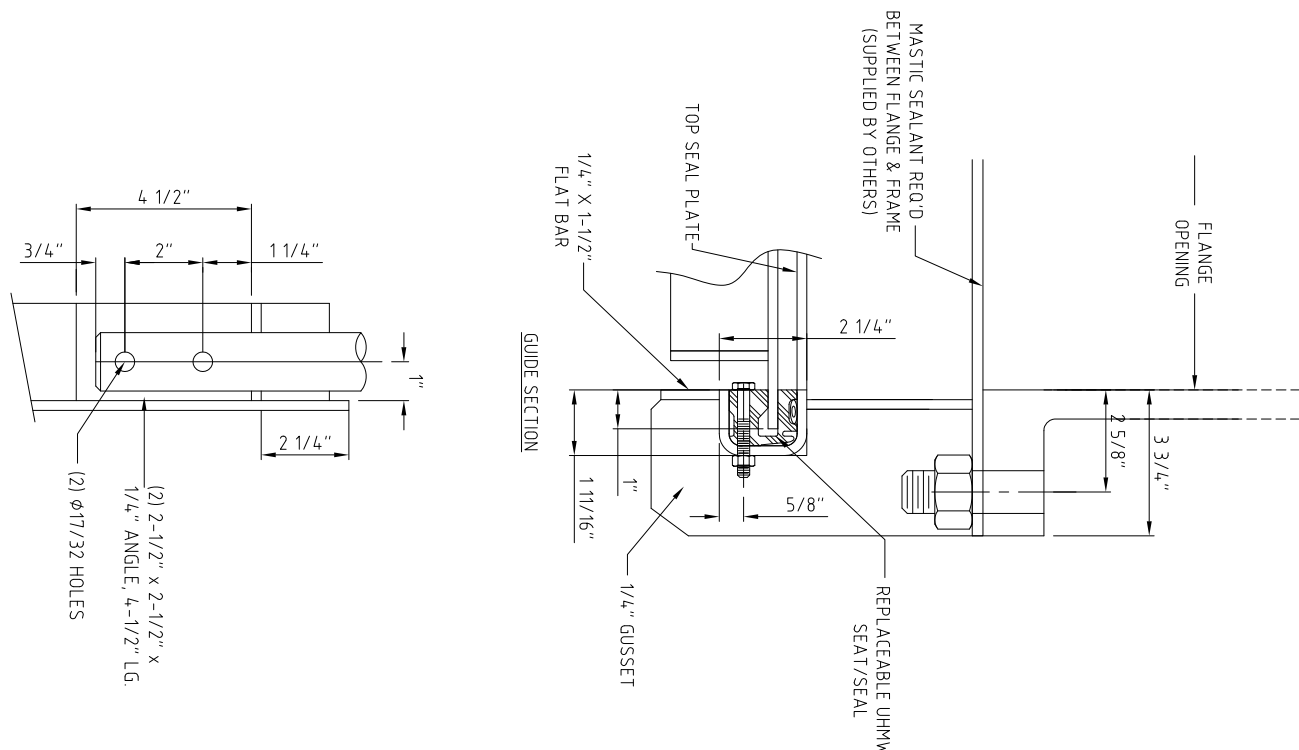
Title ADAPTOR PLATE FA14		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches						
Material ALUMINUM 6061-T6		Scale: 1:2.5		A	Original Issue	5/12/25	KS	
				Rev.	Rev. Record	Date	Eng.	Chk.
Drg. No. BC-924-3118-S3	S.O. No. 32752	THIRD ANGLE PROJECTION 		Whipps, inc. 370 SOUTH ATHOL RD. ATHOL, MA 01331				

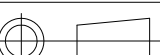
ALL OBSTRUCTIVE HAND RAILING OR FLOORING TO BE MODIFIED BY OTHERS

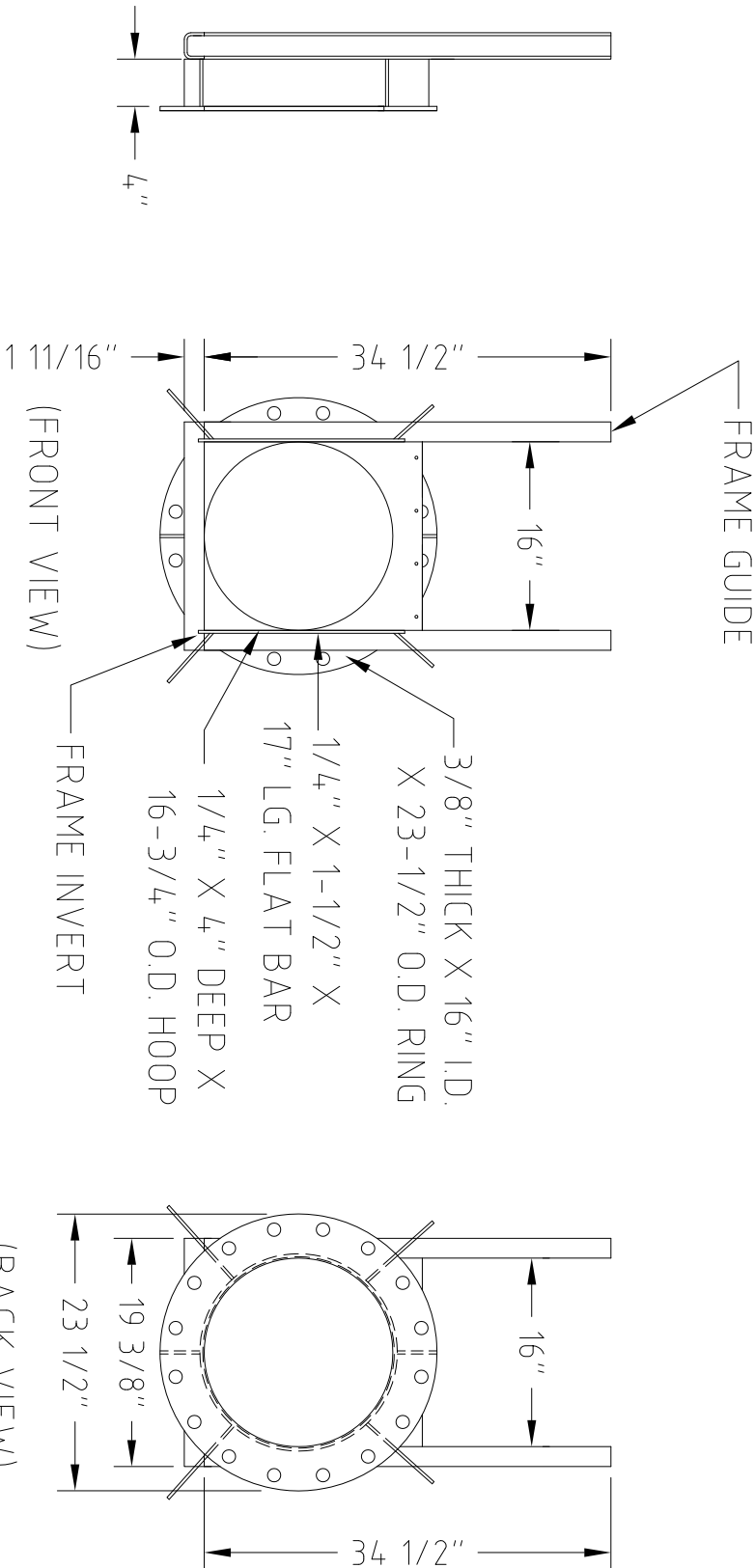
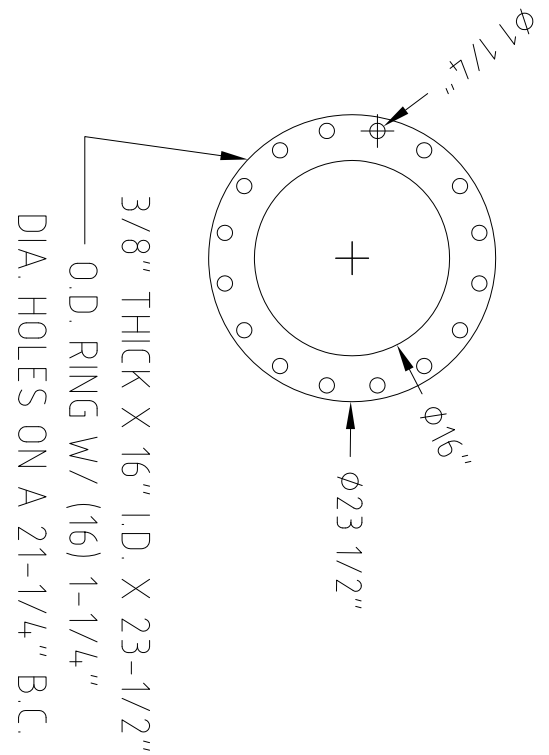
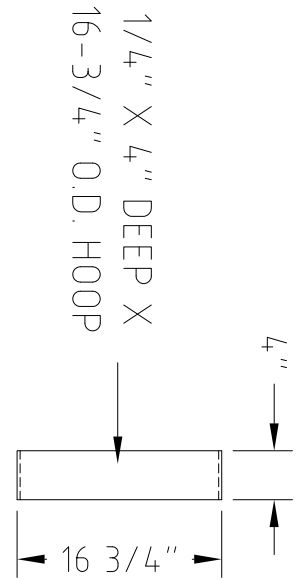
FOR GATE SECTION DETAILS: BC-924-3119-S2
 FOR FRAME DETAILS: BC-924-3119-S3
 YOKE CHANNEL: 1/4" X 2" X 4-1/2" CHANNEL
 HORIZONTAL STIFFENERS: 1/4" X 2-1/2" FLAT BAR
 VERTICAL STIFFENERS: 1/4" X 2-1/2" FLAT BAR
APPROX. WEIGHT OF SLIDE: 60 LBS.
APPROX. WEIGHT OF GATE ASSEMBLY: 420 LBS.



Title SERIES 924 SLIDE GATE INSTALL. 16" DIA.		Quantity: 1		Scale: FD		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches							
		Tag No's: 09-FG-1100											
Project HILO WWTF PHASE #1 - HILO, HAWAII		Item#: S5		Seating	Unseating	Requested Concrete Opening Tolerances Height & Width ±1/4" Squareness ±3/8 Diagonal		A	Original Issue		5/12/25		KS
		Design Head		10.0'	10.0'			Rev.	Rev. Record		Date		Eng.
Location CC TANK INLET		Leakage Rate		0.05	0.05	THIRD ANGLE PROJECTION 		 370 SOUTH ATHOL RD. ATHOL, MA 01331					
Drg. No. BC-924-3119		S.O. 32752											



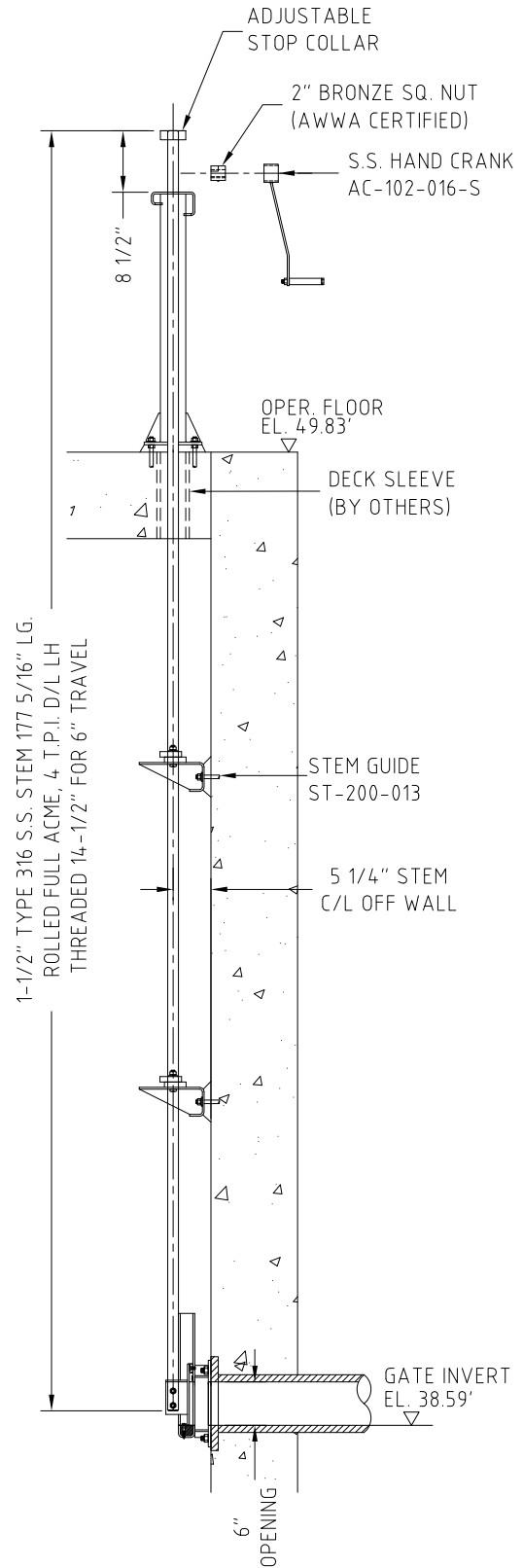
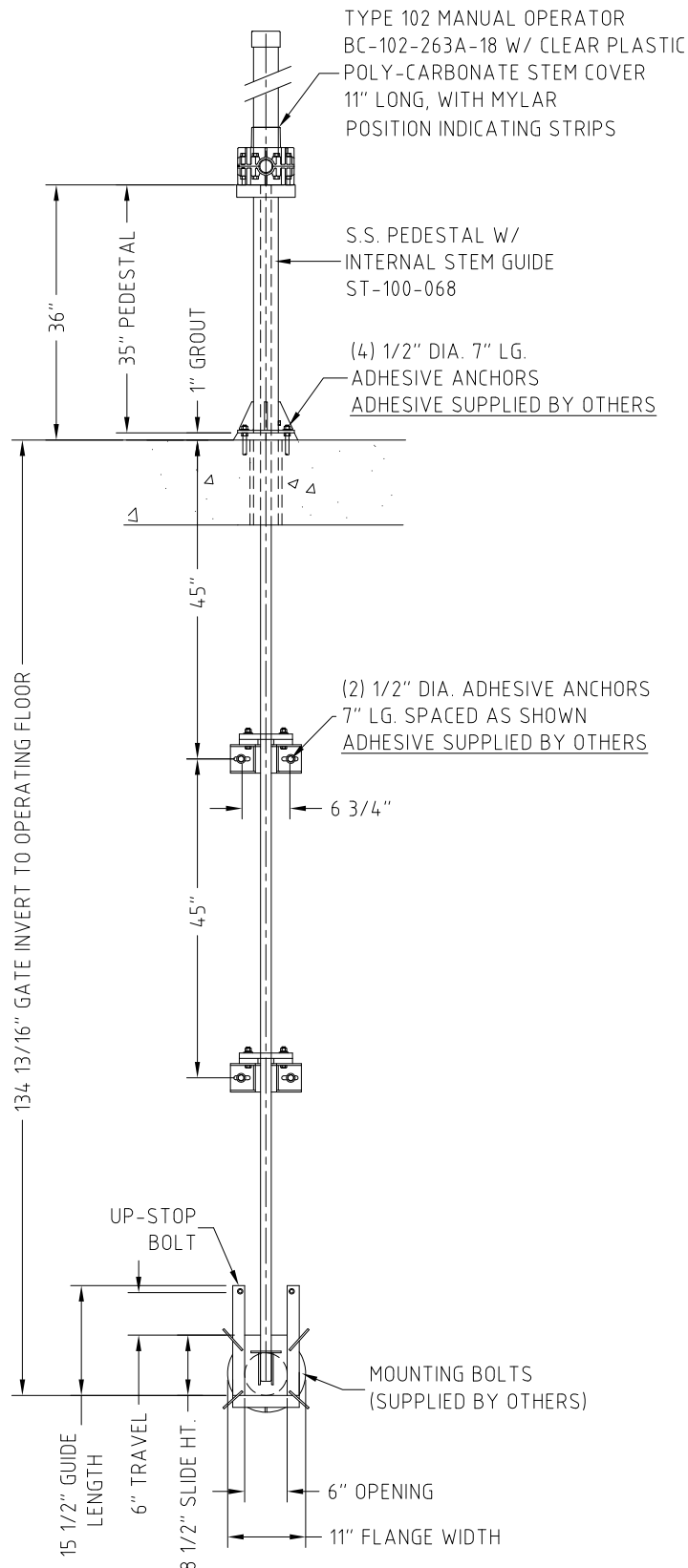
Title SERIES 924 SLIDE GATE SECTION DETAILS		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches						
		Scale: FD		A	Original Issue	5/14/25	KS	
Rev.	Rev. Record			Date	Eng.	Chk.		
Material STAINLESS STEEL GRADE 316L		THIRD ANGLE PROJECTION 		Whipps, inc. 370 SOUTH ATHOL RD. ATHOL, MA 01331				
Dwg. No. BC-924-3119-S2	S.O. No. 32752							

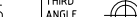
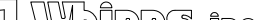


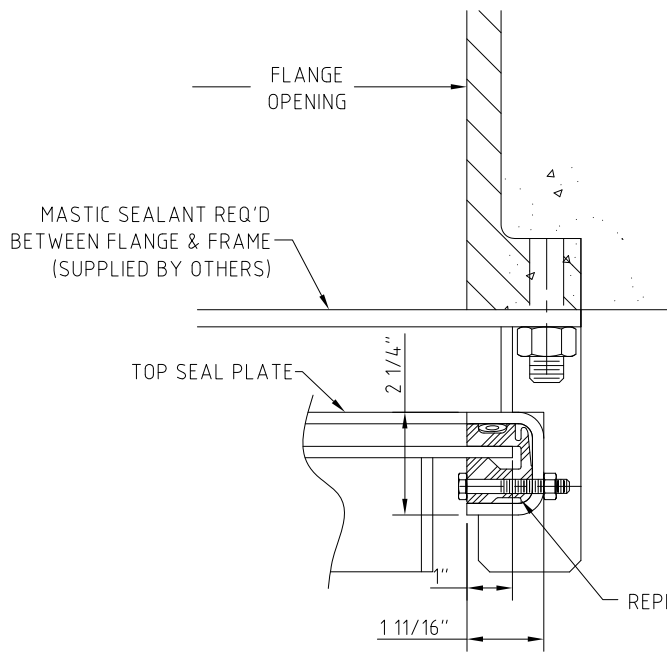
Title SERIES 924 SLIDE GATE FRAME & FLANGE DETAILS		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches					
Material STAINLESS STEEL GRADE 316L		Scale: FD		A	Original Issue	5/14/25	KS
				Rev.	Rev. Record	Date	Eng. Chk.
Drg. No. BC-924-3119-S3	S.O. No. 32752	THIRD ANGLE PROJECTION		Whipps, inc.		370 SOUTH ATHOL RD. ATHOL, MA 01331	

ALL OBSTRUCTIVE HAND RAILING OR FLOORING TO BE MODIFIED BY OTHERS

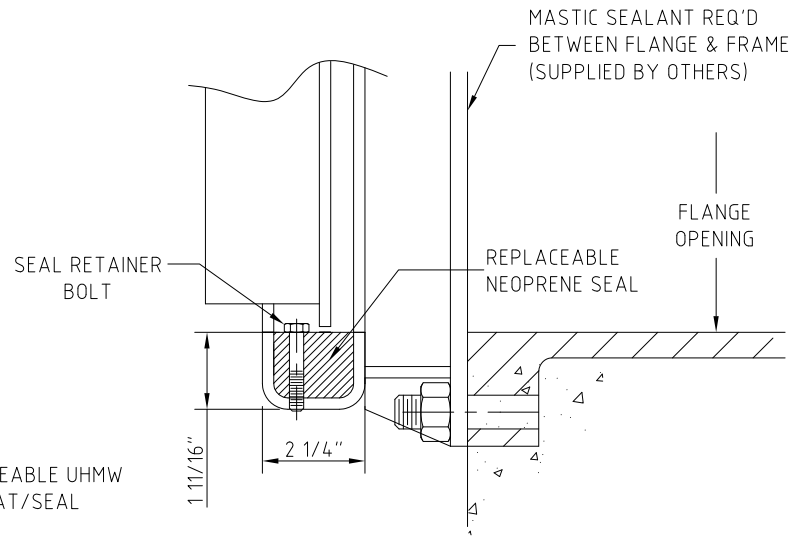
BC-924-3656-S2: GATE SECTION DETAILS
 BC-924-3656-S3: FRAME DETAILS
 HORIZONTAL STIFFENERS: 3" X 1/4" FLAT BAR
 APPROX. WEIGHT OF SLIDE: 20 LBS.
 APPROX. WEIGHT OF GATE ASSEMBLY: 70 LBS.



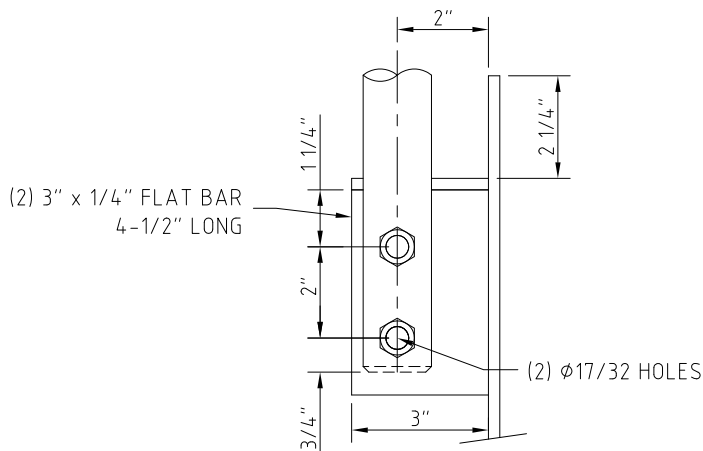
Title SERIES 925 SLIDE GATE INSTALL. 6" DIA.		Quantity: 3	Scale: FD		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches						
		Tag No's: 02-GAT-1140, 1141 & 1142									
Project HILO WWTF PHASE #1 - HILO, HAWAII		Item#: E5	Seating	Unseating	Requested Concrete Opening Tolerances Height & Width ±1/4" Squareness ±3/8 Diagonal	A	Original Issue			5/12/25	KS
		Design Head	20.0'	20.0'		Rev.	Rev. Record		Date	Eng.	Chk.
Location SCREEN AND BYPASS CHANNEL DRAIN		Leakage Rate	0.05	0.05	THIRD ANGLE PROJECTION 	 370 SOUTH ATHOL RD. ATHOL, MA 01331					
Org. No. BC-925-3656	S.O. No. 32752	GPM/ft of seal									



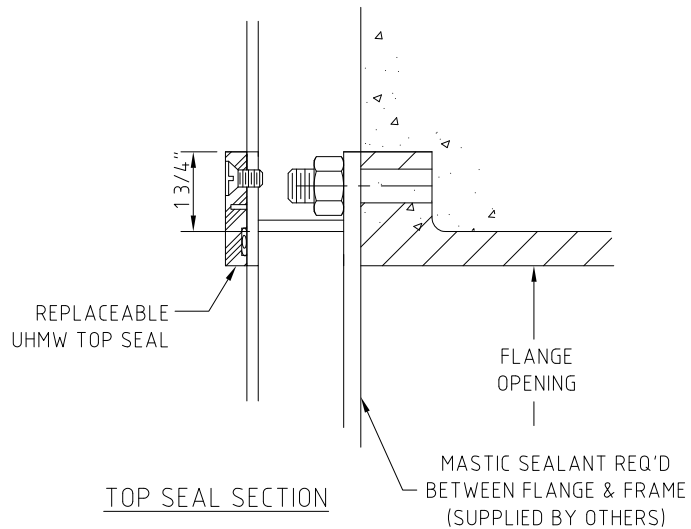
GUIDE SECTION



INVERT SECTION

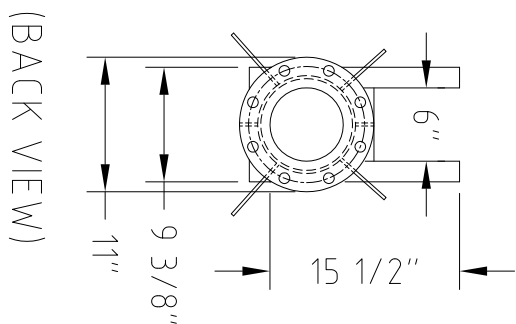
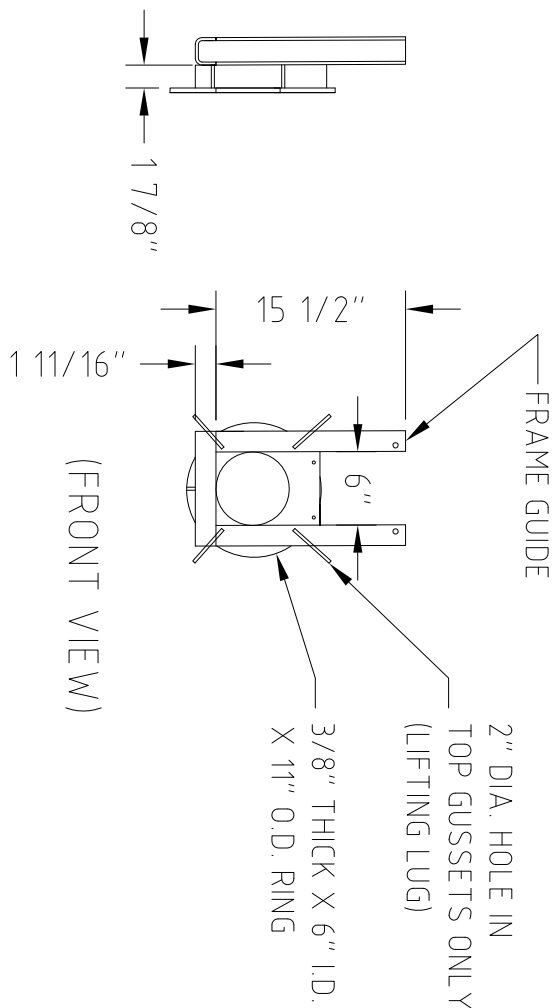
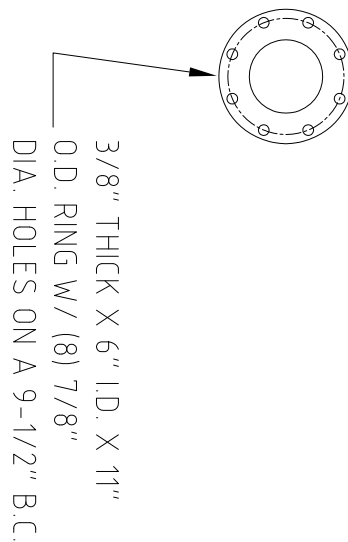
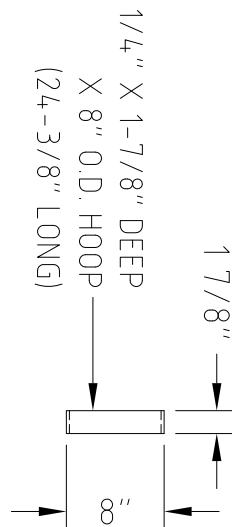


STEM CONNECTOR



TOP SEAL SECTION

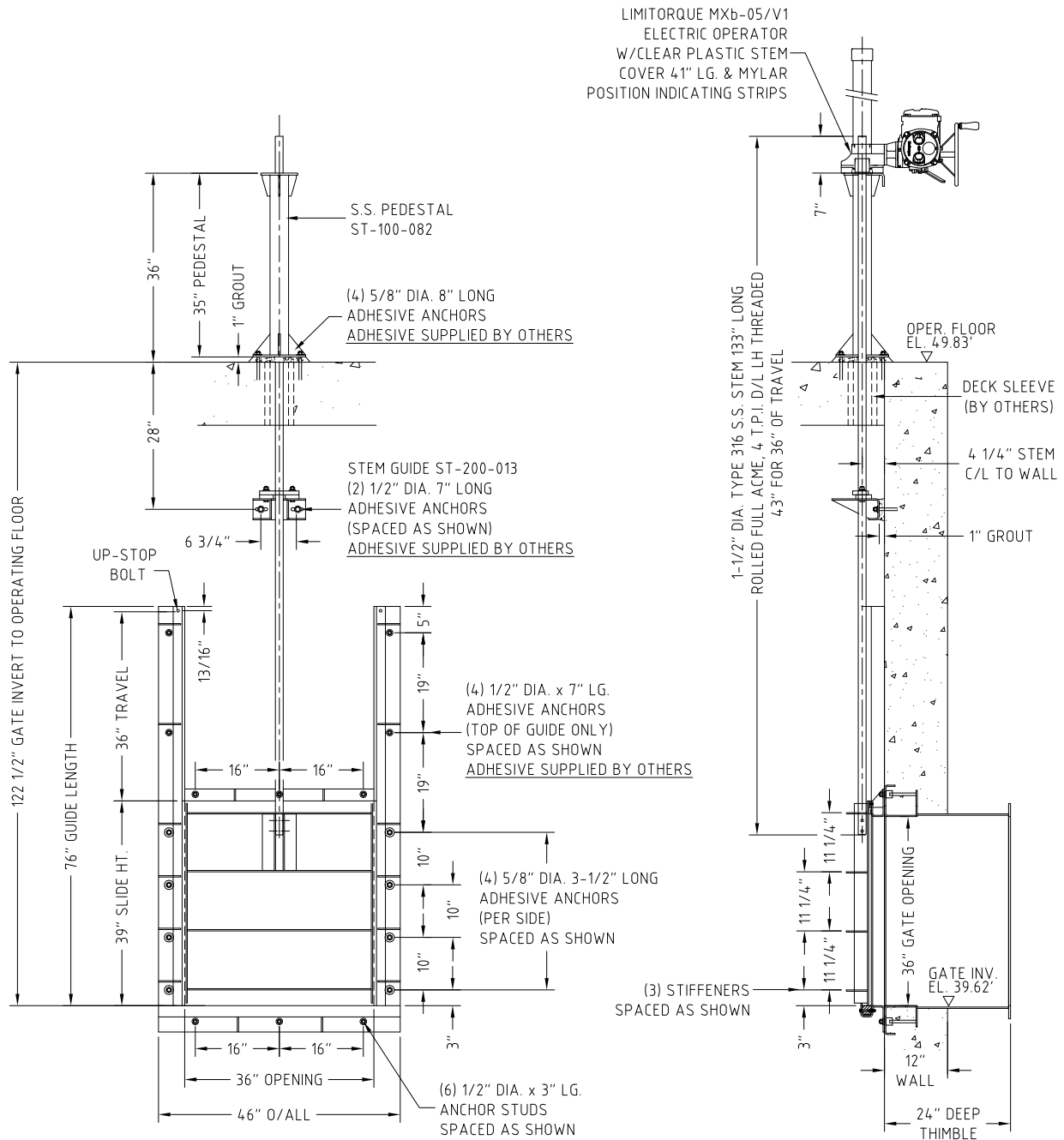
Title SERIES 925 SLIDE GATE SECTION DETAILS		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches						
		Scale: FD	A	Original Issue	5/13/25	KS		
			Rev.	Rev. Record	Date	Eng.	Chk.	
Material STAINLESS STEEL GRADE 316L		THIRD ANGLE PROJECTION	Whipps, inc. 370 SOUTH ATHOL RD. ATHOL, MA 01331					
Drg. No. BC-925-3656-S2	S.O. No. 32752							



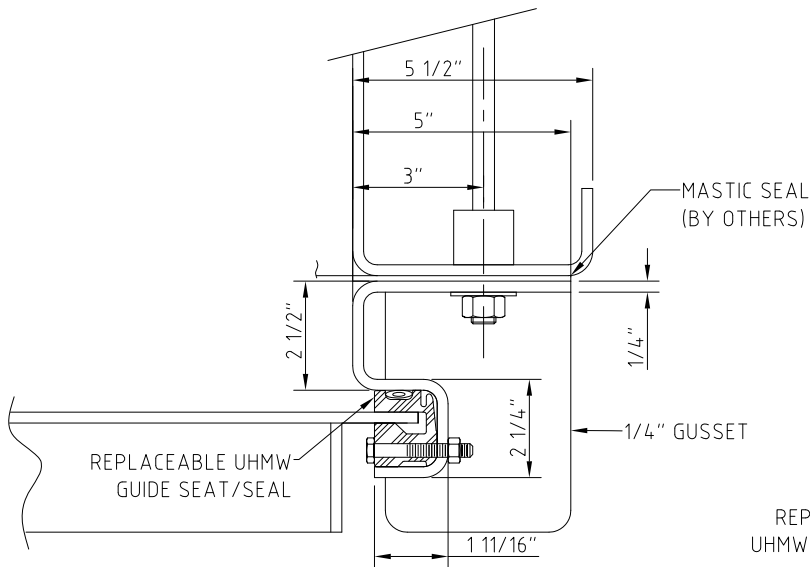
Title SERIES 925 SLIDE GATE FRAME & FLANGE DETAILS		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches				
Material STAINLESS STEEL GRADE 316L		Scale: FD		A	Original Issue	5/13/25
				Rev.	Rev. Record	Date
Drg. No. BC-925-3656-S3		S.O. No. 32752		KS Eng. Chk.		
		THIRD ANGLE PROJECTION				
				370 SOUTH ATHOL RD. ATHOL, MA 01331		

FOR GATE SECTION DETAILS: BC-925-3657-S2
 FOR THIMBLE DETAILS: BC-925-3657-S3
 HORIZONTAL STIFFENERS: 1/4" X 4" FLAT BAR
 VERTICAL STIFFENERS: 1/4" X 2-1/2" FLAT BAR
APPROX. WEIGHT OF SLIDE: 180 LBS.
APPROX. WEIGHT FOR GATE ASSEMBLY: 400 LBS.

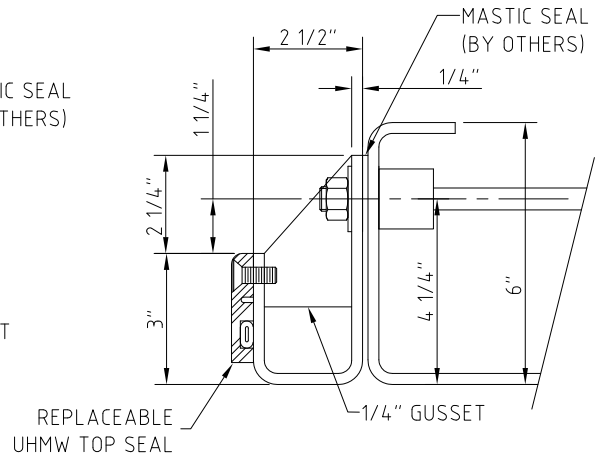
ALL OBSTRUCTIVE HAND RAILING OR
 FLOORING TO BE MODIFIED BY OTHERS



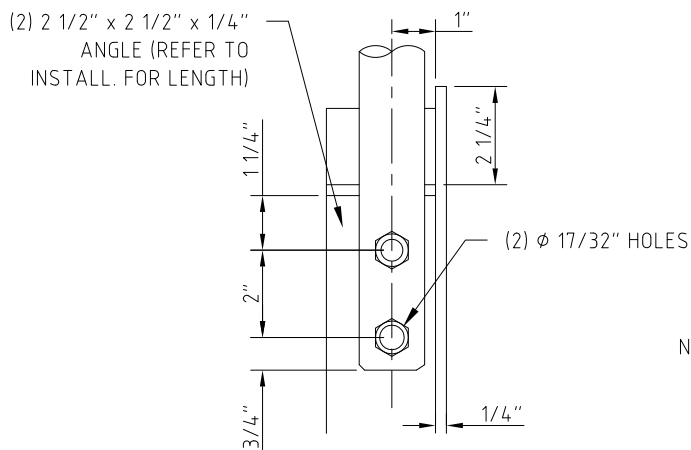
Title		SERIES 925 SLIDE GATE INSTALL. 36" x 36"		Quantity: 1		Scale: FD		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches							
Project		HILO WWTF PHASE #1 - HILO, HAWAII		Tag No's: 02-GAT-1240											
Item#:		J5		Seating		Unseating		Requested Concrete Opening Tolerances Height & Width ±1/4" Squareness ±3/8 Diagonal		A		Original Issue		5/12/25 KS	
Design Head		20.0'		20.0'		Rev.				Rev. Record		Date		Eng.	
Location		GRIT EFFLUENT BYPASS		Leakage Rate		0.05		0.05						370 SOUTH ATHOL RD. ATHOL, MA 01331	
Drg. No.		BC-925-3657		S.O. No.		32752									



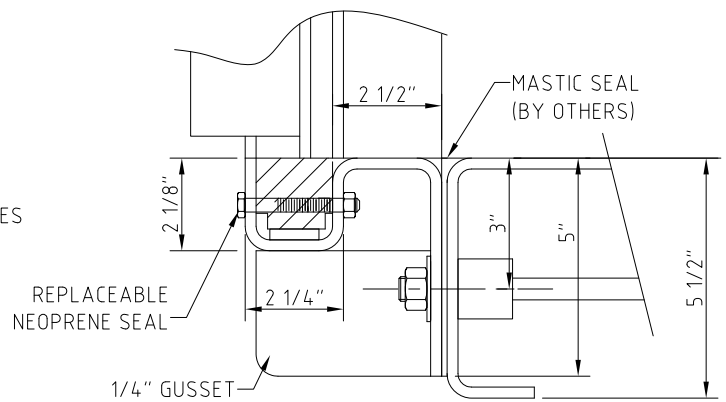
GUIDE SECTION



TOP SEAL SECTION



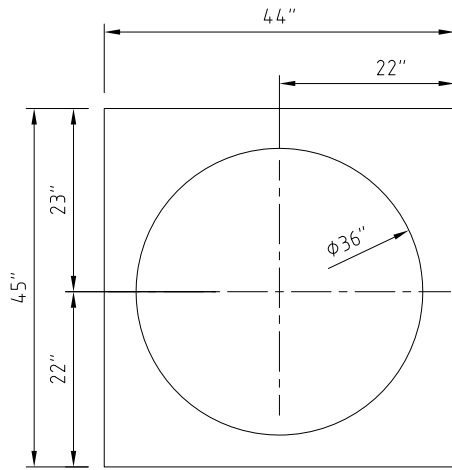
STEM CONNECTOR



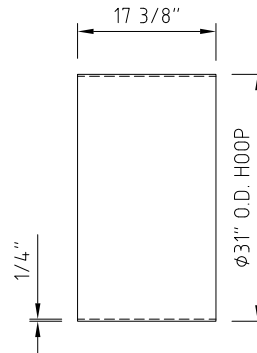
INVERT SEAL SECTION

Title	SERIES 925 SLIDE GATE SECTION DETAILS		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches						
Material	STAINLESS STEEL GRADE 316L		Scale: FD		A	Original Issue	5/13/25	KS	
					Rev.	Rev. Record	Date	Eng.	Chk.
Drg. No. BC-925-3657-S2		S.O. No. 32752	THIRD ANGLE PROJECTION		Whipps, inc.		370 SOUTH ATHOL RD. ATHOL, MA 01331		

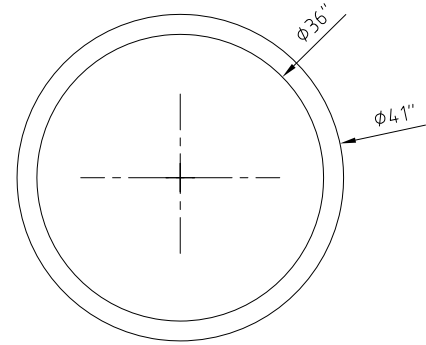
NOTE: THIMBLE TO HAVE "TOP", "BOTTOM"
AND CENTER LINE PERMANETLY STAMPED
AS SHOWN



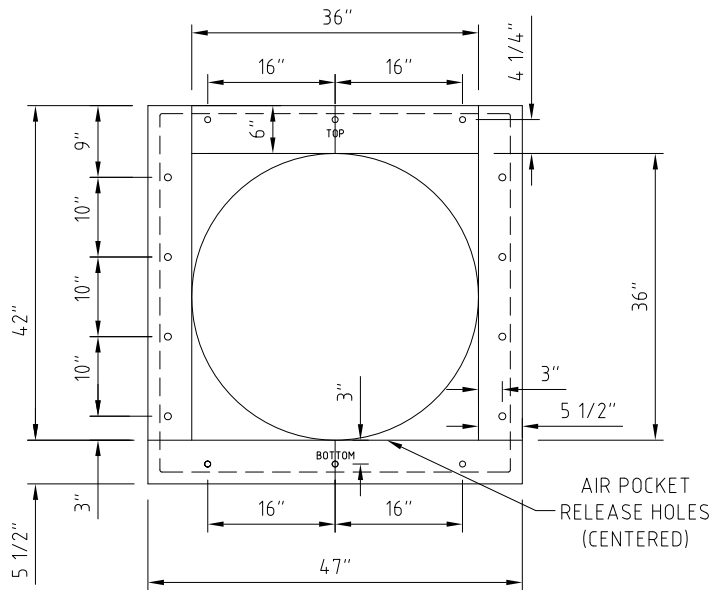
WATER STOP PLATE 1/4"



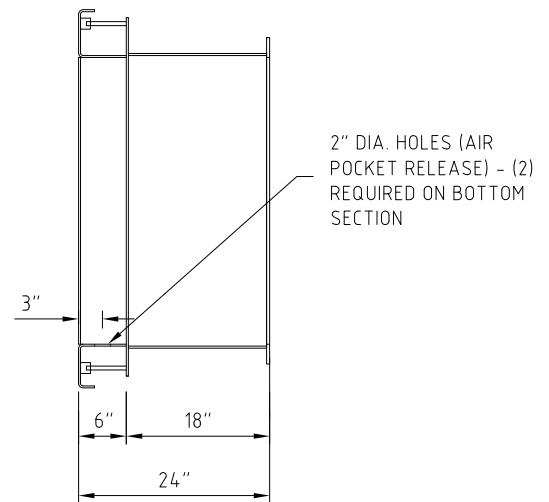
3/8" O.D. HOOP



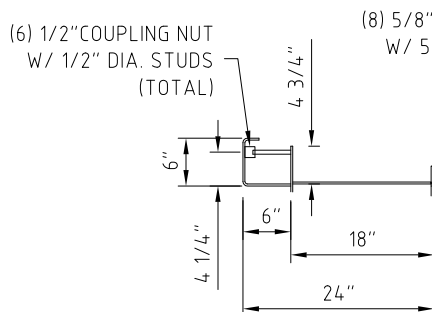
3/8" BACK RING



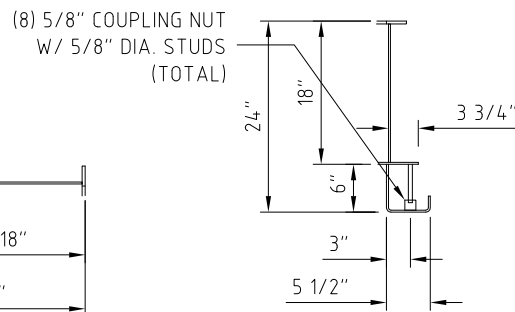
AIR POCKET
RELEASE HOLES
(CENTERED)



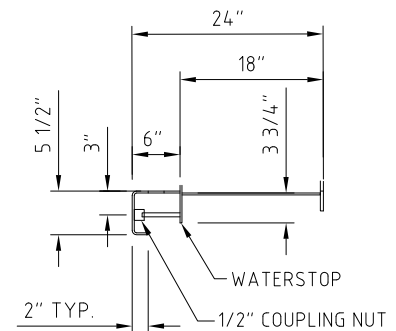
2" DIA. HOLES (AIR
POCKET RELEASE) - (2)
REQUIRED ON BOTTOM
SECTION



TOP CROSS SECTION



SIDE CROSS SECTION

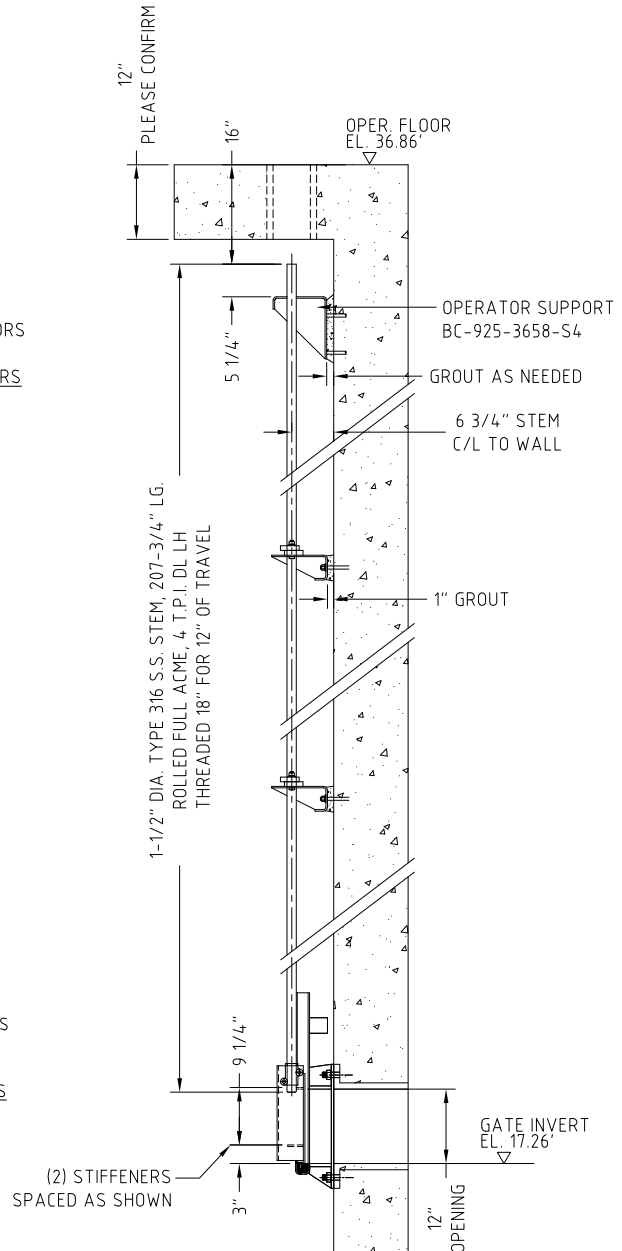
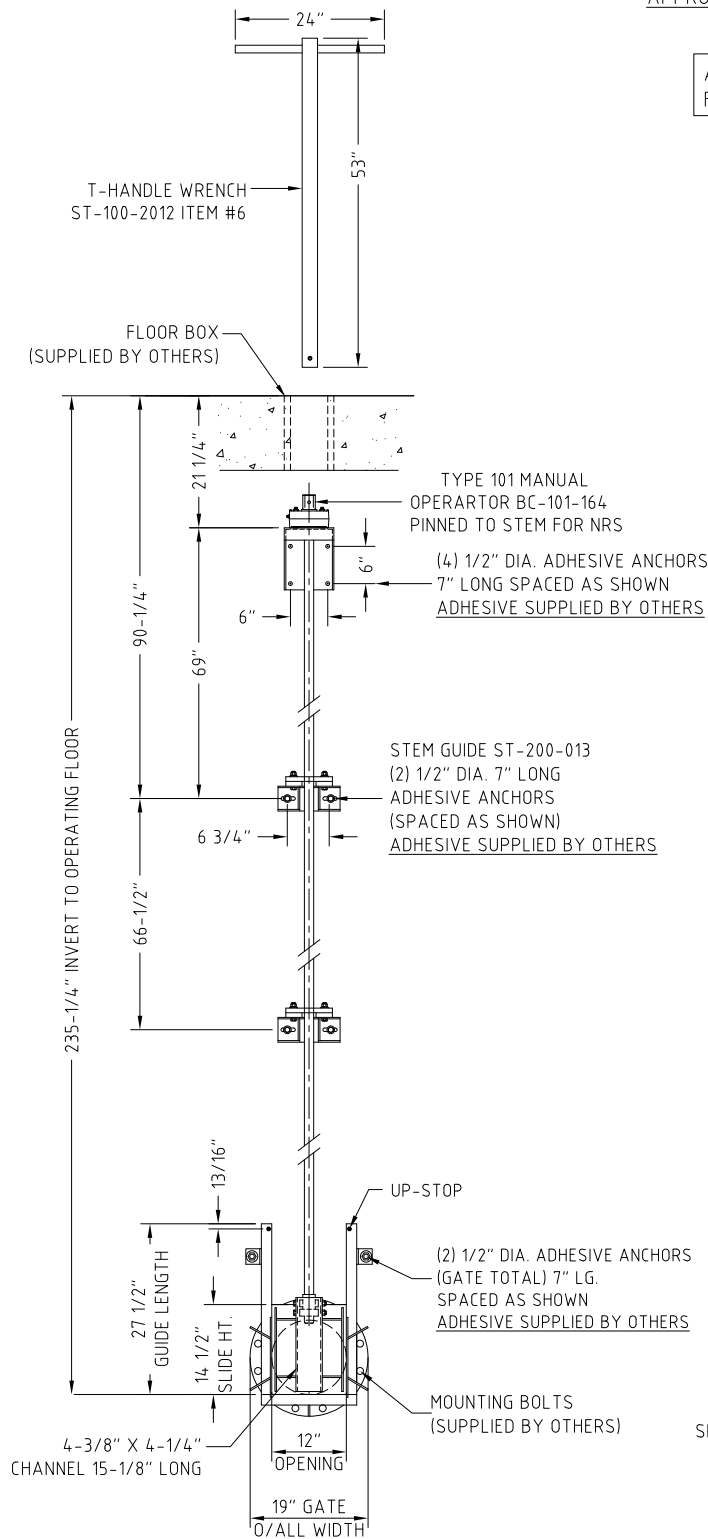


BOTTOM CROSS SECTION

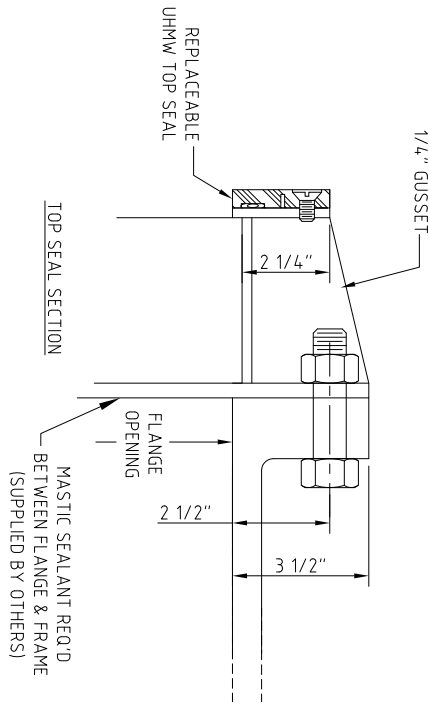
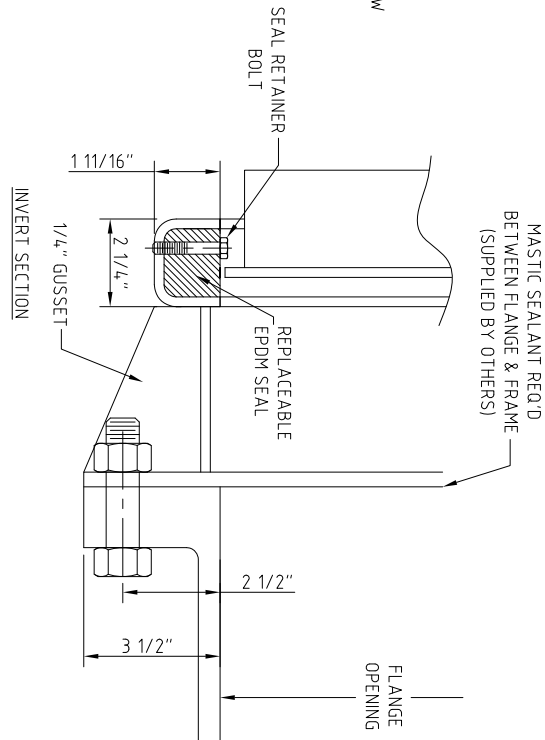
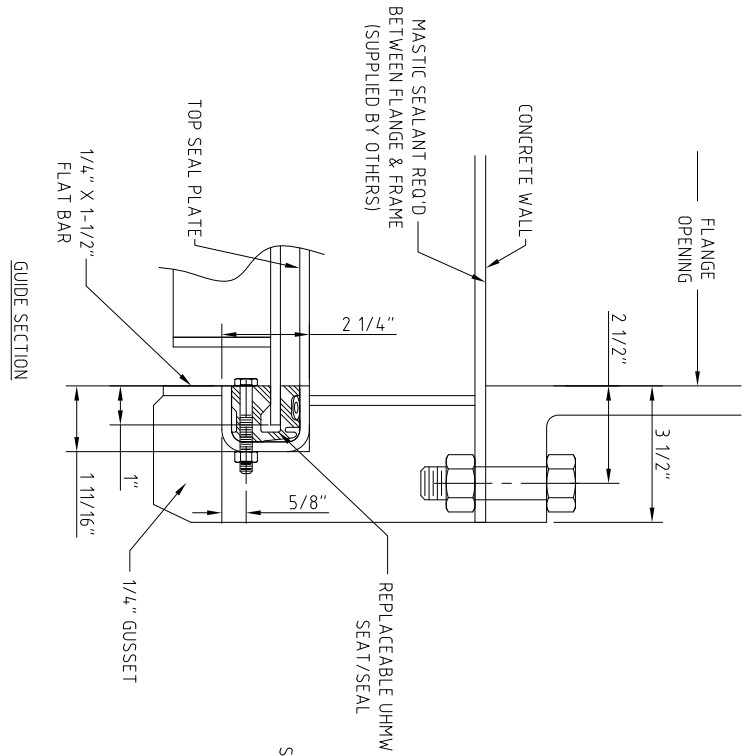
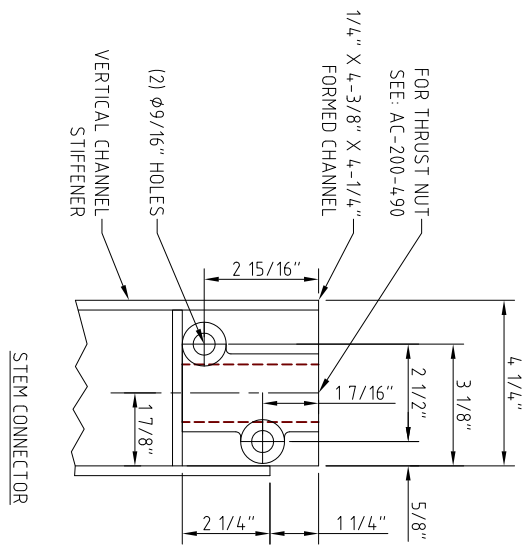
Title		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches					
SERIES 925 36" x 36"		Scale: FD		A	Original Issue	5/13/25	KS
THIMBLE DETAIL		Item# K5		Rev.	Rev. Record	Date	Eng. Chk.
Material STAINLESS STEEL GRADE 316L		THIRD ANGLE PROJECTION		Whipps, inc.			
Dwg. No. BC-925-3657-S3	S.O. No. 32752			370 SOUTH ATHOL RD. ATHOL, MA 01331			

FOR GATE SECTION DETAILS: BC-925-3658-S2
FOR FRAME DETAILS SEE: BC-925-3658-S3
HORIZONTAL STIFFENERS: 1/4" X 2-1/2" FLAT BAR
VERTICAL STIFFENERS: 1/4" X 2-1/2" FLAT BAR
APPROX. WEIGHT OF SLIDE: 35 LBS.
APPROX. WEIGHT OF GATE ASSEMBLY: 120 LBS.

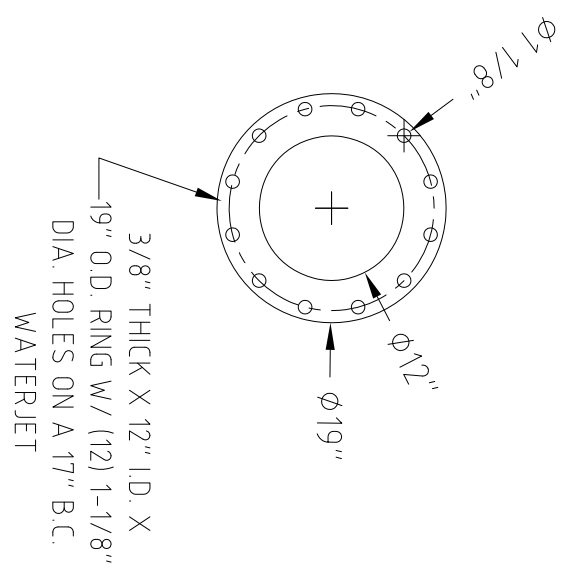
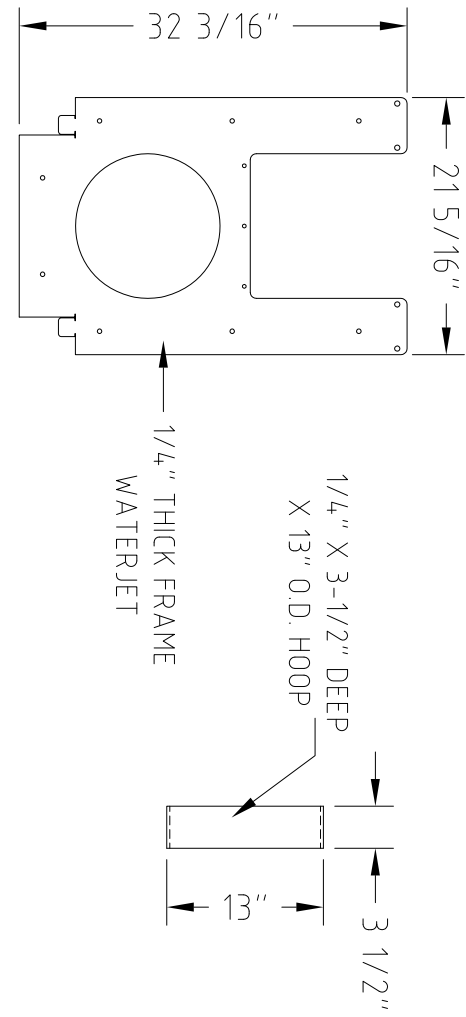
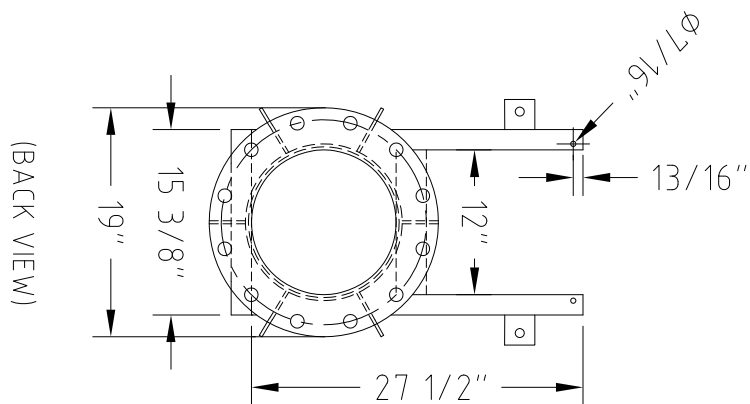
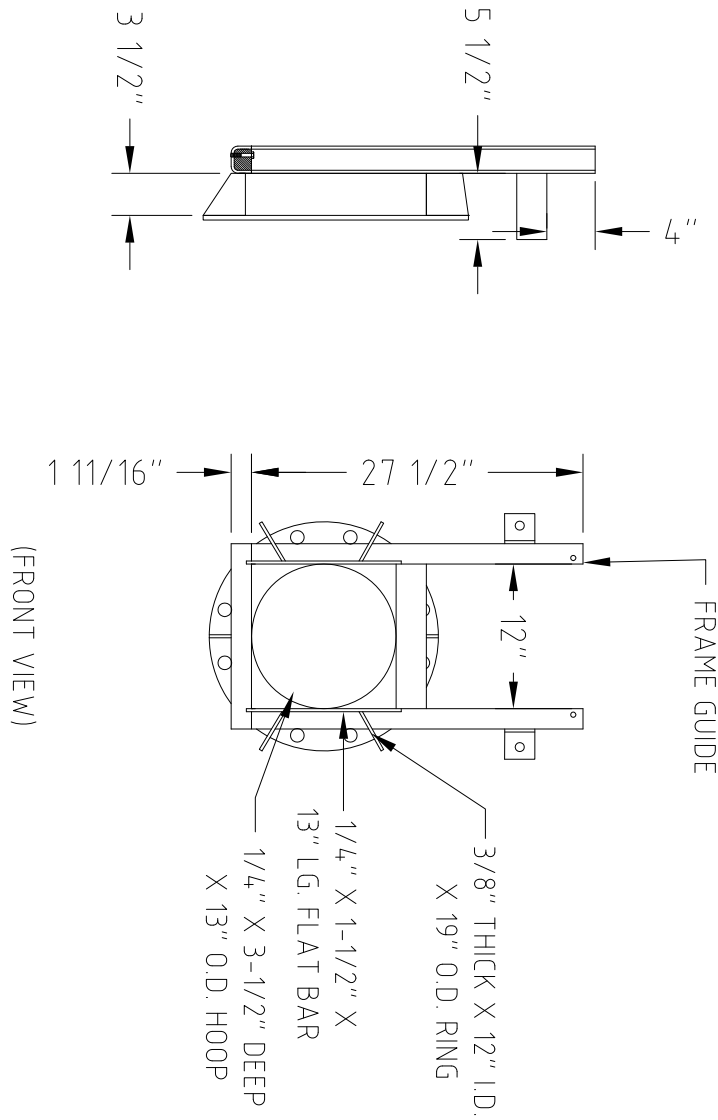
ALL OBSTRUCTIVE HAND RAILING OR
FLOORING TO BE MODIFIED BY OTHERS

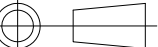


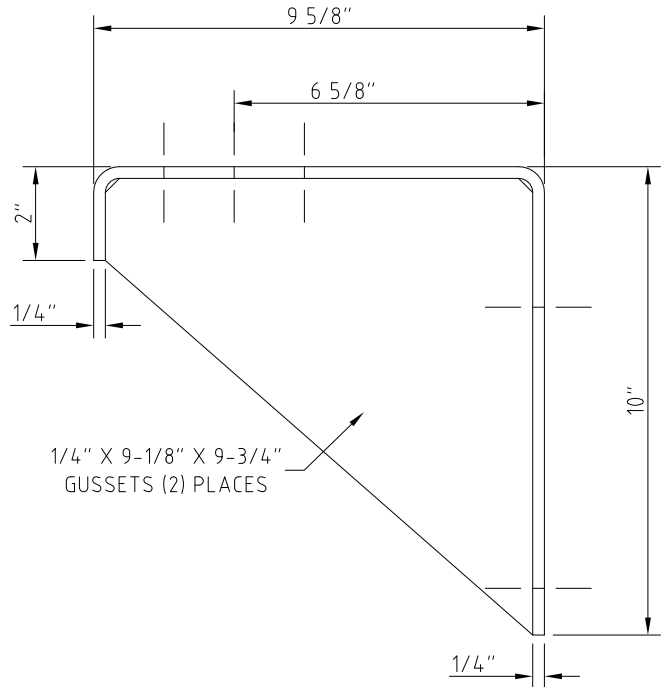
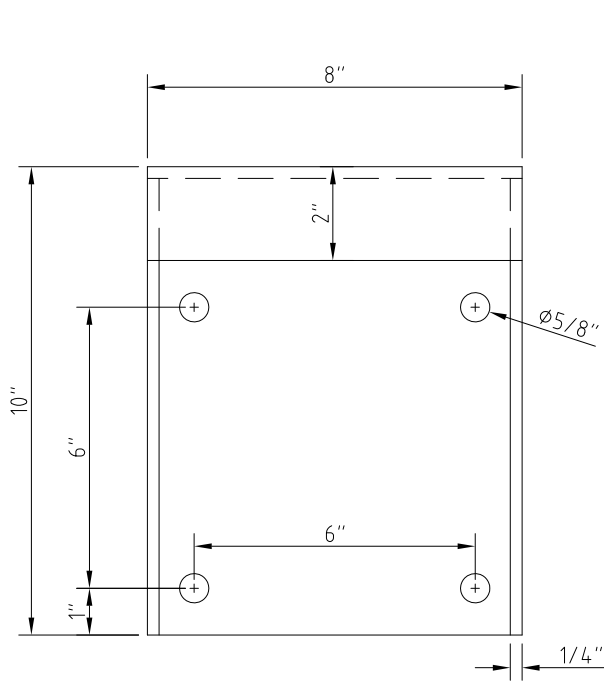
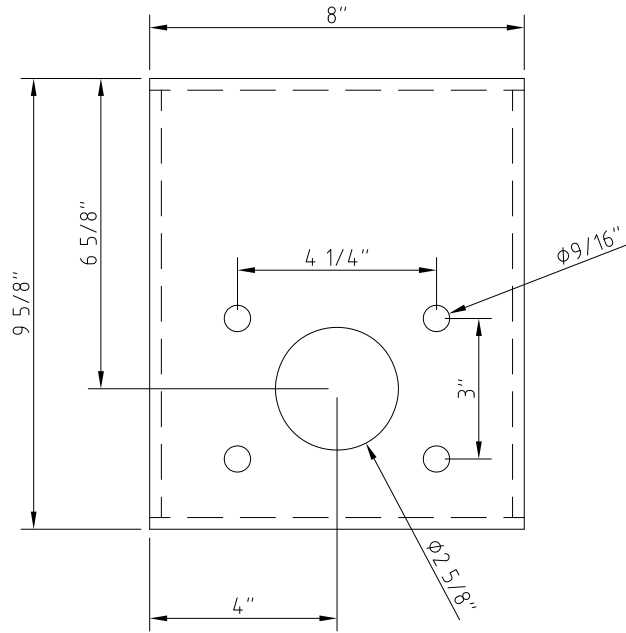
Title SERIES 925 SLIDE GATE INSTALL. 12" DIA.		Quantity: 3		Scale: FD		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches							
		Tag No's: 06-FG-1515, 1525 & 1535											
Project HILO WWTF PHASE #1 - HILO, HAWAII		Item#: U5		Seating	Unseating	Requested Concrete Opening Tolerances Height & Width ±1/4" Squareness ±3/8 Diagonal		A	Original Issue			5/12/25	KS
		Design Head		22.0'	22.0'			Rev.	Rev. Record			Date	Eng.
Location SECONDARY CLARIFIER SCUM WELL		Leakage Rate		0.05	0.05	THIRD ANGLE PROJECTION 				370 SOUTH ATHOL RD. ATHOL, MA 01331			
Dwg. No. BC-925-3658		S.O. No. 32752											



Title SERIES 925 SLIDE GATE SECTION DETAILS		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches							
Material STAINLESS STEEL GRADE 316L		Scale: FD		A	Original Issue	5/13/25	KS		
Drg. No. BC-925-3658-S2		S.O. No. 32752		Rev.	Rev. Record	Date	Eng.	Chk.	
THIRD ANGLE PROJECTION							370 SOUTH ATHOL RD. ATHOL, MA 01331		

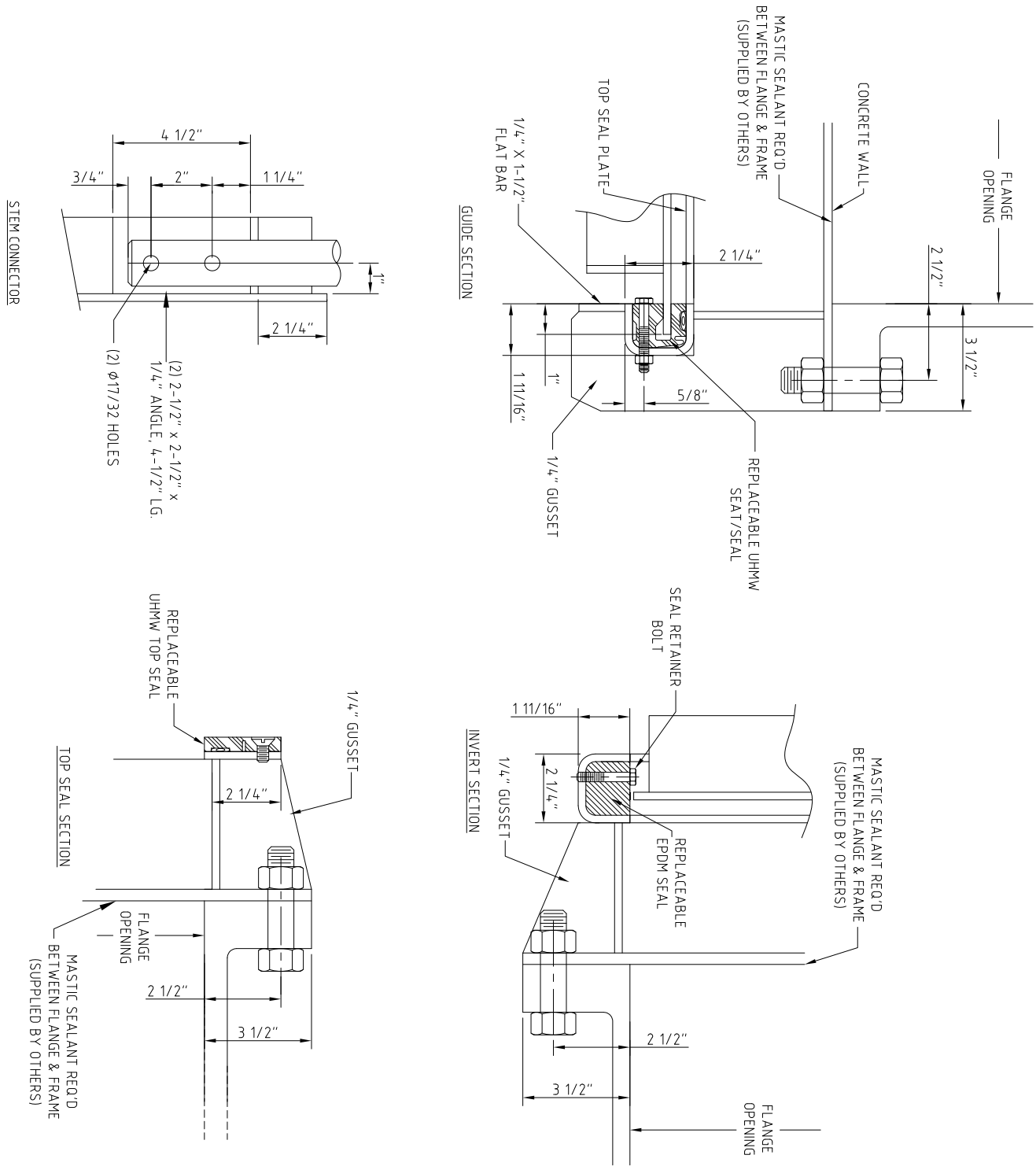


Title SERIES 925 SLIDE GATE FRAME DETAILS		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches							
Material STAINLESS STEEL GRADE 316L		Scale: FD		A	Original Issue	5/13/25	KS		
				Rev.	Rev. Record	Date	Eng.	Chk.	
Dwg. No.	BC-925-3658-S3	S.O. No.	32752	THIRD ANGLE PROJECTION			Whipps, inc.		370 SOUTH ATHOL RD. ATHOL, MA 01331

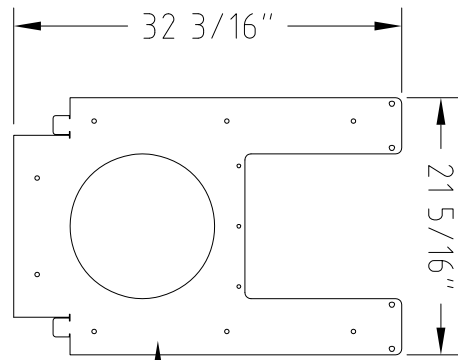


Title OPERATOR SUPPORT DETAIL		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches						
Material STAINLESS STEEL GRADE 316L		Scale: FD		A	Original Issue	5/13/25	KS	
				Rev.	Rev. Record	Date	Eng.	Chk.
Dwg. No. BC-925-3658-S4	S.O. No. 32752	THIRD ANGLE PROJECTION		Whipps, inc.				
				370 SOUTH ATHOL RD. ATHOL, MA 01331				

Title		SERIES 925 SLIDE GATE INSTALL. 12" DIA.		Quantity: 3		Scale: FD		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches											
Project		HILO WWTF PHASE #1 - HILO, HAWAII		Item#: T5		Seating Unseating				Requested Concrete Opening Tolerances Height & Width ±1/4" Squareness ±3/8 Diagonal		A	Original Issue		5/12/25 KS				
Location		SECONDARY CLARIFIER RSS WET WELL		Design Head		12.0'		12.0'		Rev.	Rev. Record		Date		Eng. Chk.				
Dwg. No.		BC-925-3659		S.O. No.		32752		Leakage Rate GPM/ft of seal		0.05		0.05						370 SOUTH ATHOL RD. ATHOL, MA 01331	

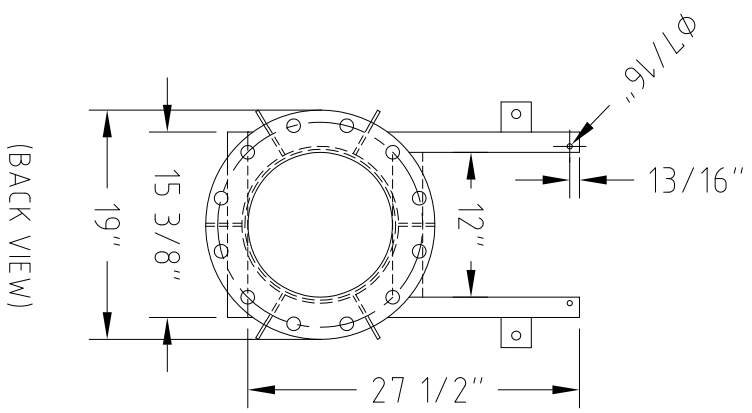
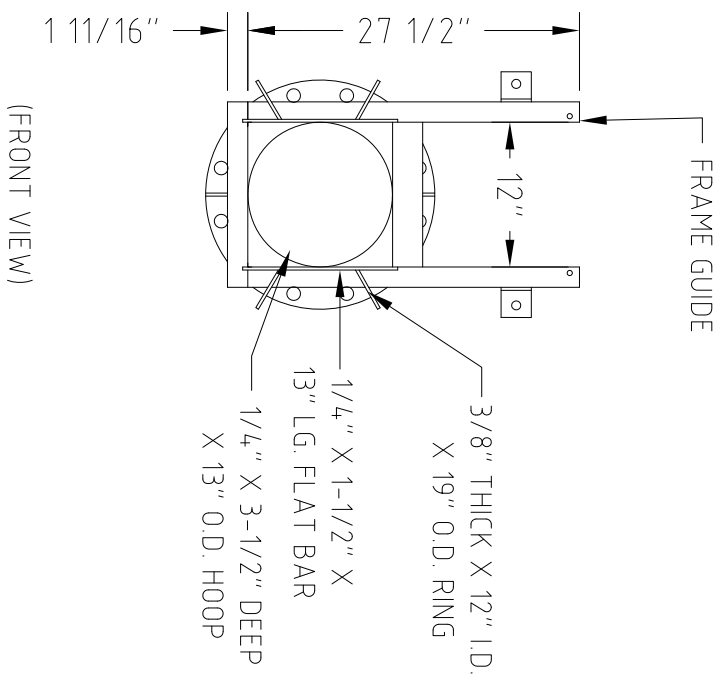
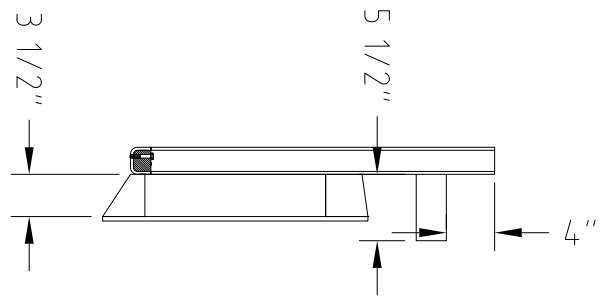
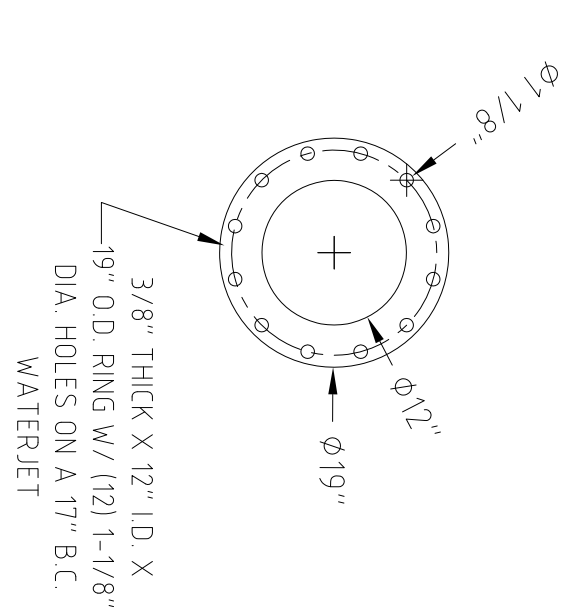
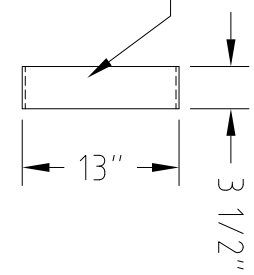


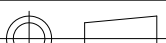
Title SERIES 925 SLIDE GATE SECTION DETAILS		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches								
Material STAINLESS STEEL GRADE 316L		Scale: FD		A	Original Issue	5/13/25	KS			
				Rev.	Rev. Record	Date	Eng.	Chk.		
Dwg. No.	BC-925-3659-S2	S.O. No.	32752						370 SOUTH ATHOL RD. ATHOL, MA 01331	
THIRD ANGLE PROJECTION										



1/4" X 3-1/2" DEEP
X 13" O.D. HOOP

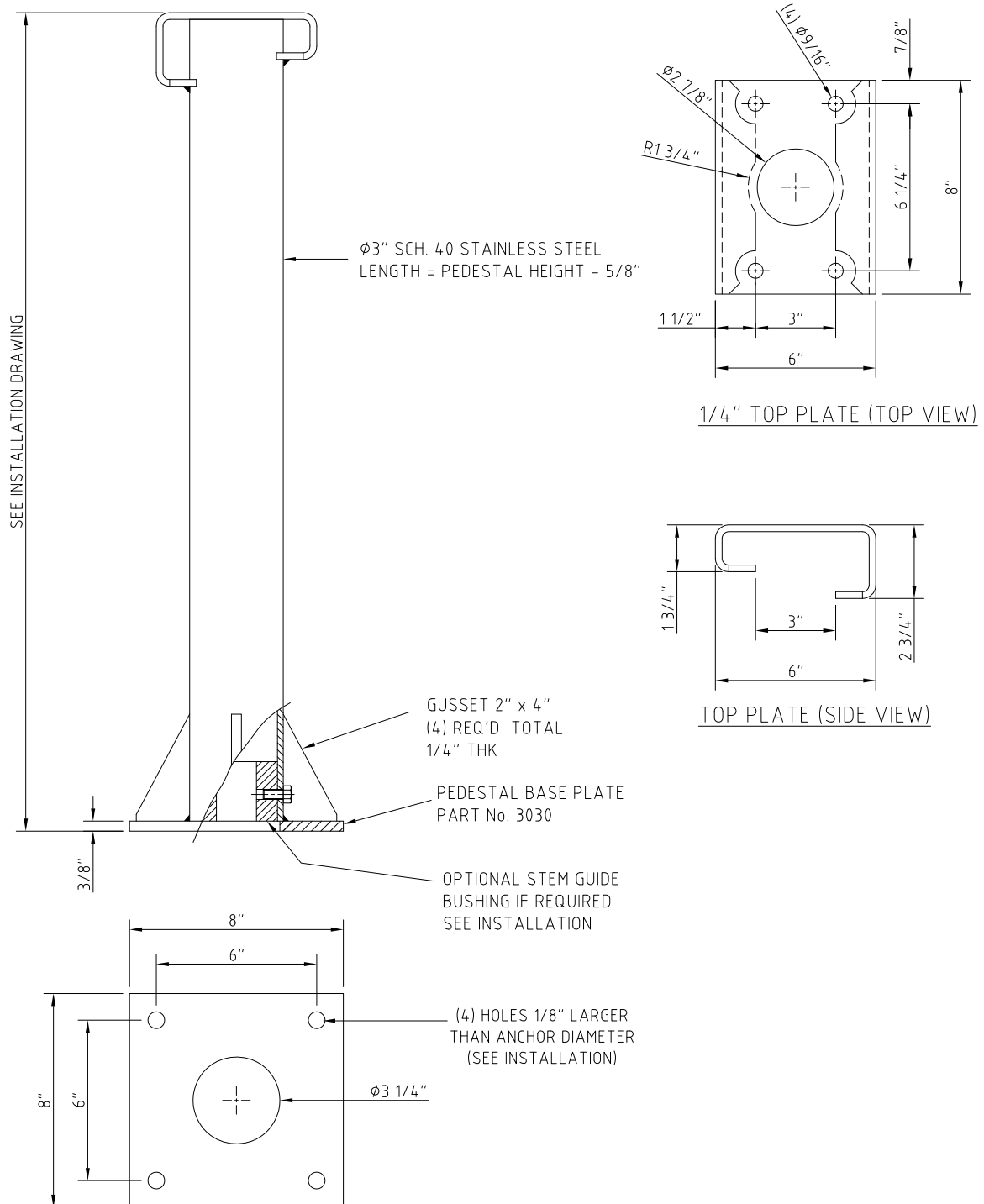
1/4" THICK FRAME
WATERJET



Title SERIES 925 SLIDE GATE FRAME DETAILS		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches							
Material STAINLESS STEEL GRADE 316L		Scale: FD		A	Original Issue	5/13/25	KS		
				Rev.	Rev. Record	Date	Eng.	Chk.	
Drg. No. BC-925-3659-S3		S.O. No. 32752		THIRD ANGLE PROJECTION 		Whipps, inc. 370 SOUTH ATHOL RD. ATHOL, MA 01331			

NOTES

1. SEE ENGINEERING DATA SHEET FOR
PEDESTAL & STEM GUIDE MATERIAL (IF REQ'D)



TITLE FLOOR MOUNT PEDESTAL TYPE 102 OPERATOR

DATE 11/13/15	SCALE 1:4	DWN BY KS	APPR BY
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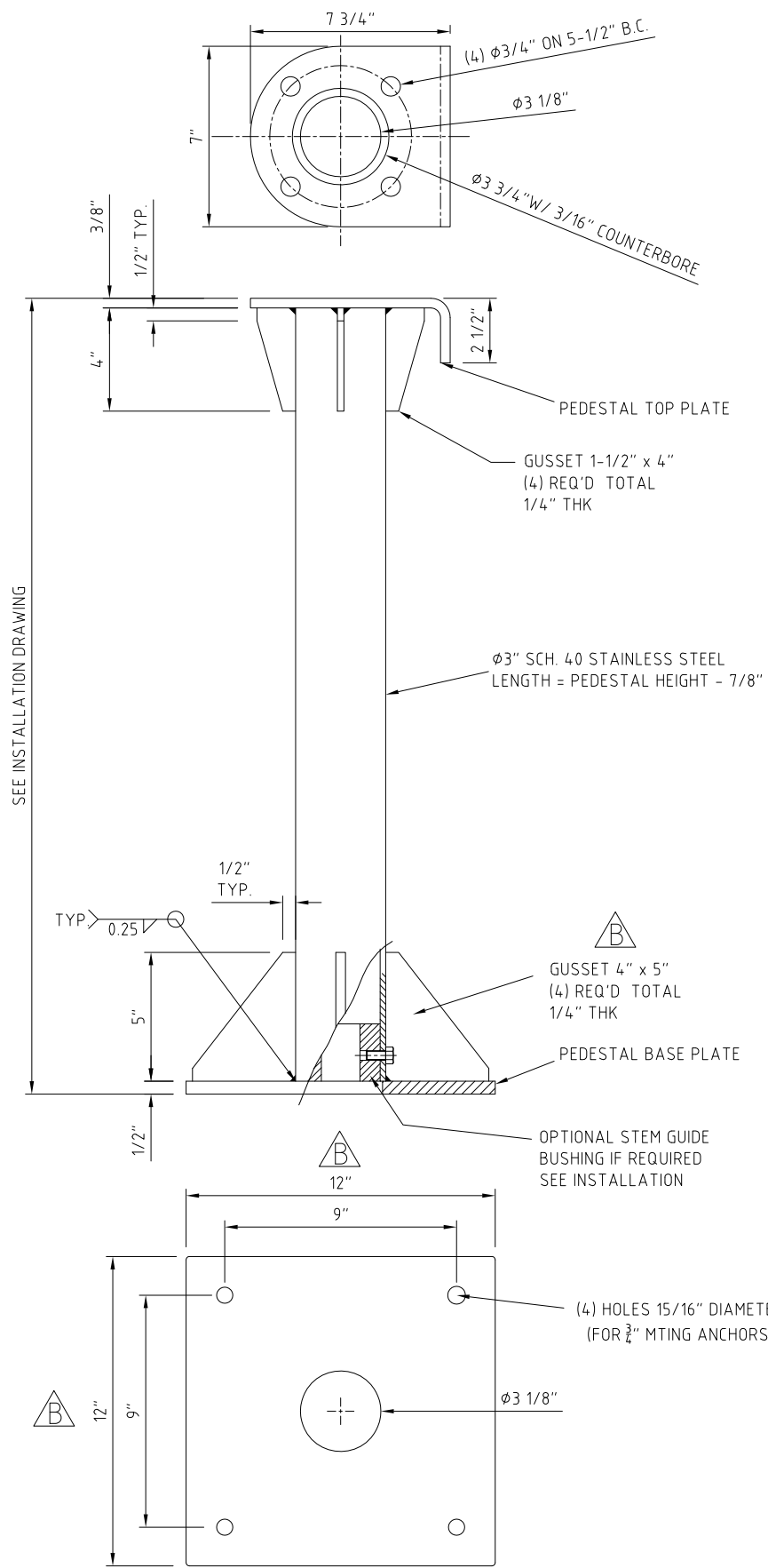
DRAWING NO. ST-100-068-A	S.O. NO.
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TOLERANCES

DATE	SYM	REVISION RECORD	AUTH	DR
2/7/25	D	Top Plate		KS
9/18/23	C	Gusset To 2" For Weld	JM	JM
6/7/19	B	TOP COPES FOR SOCKET	JM	JM

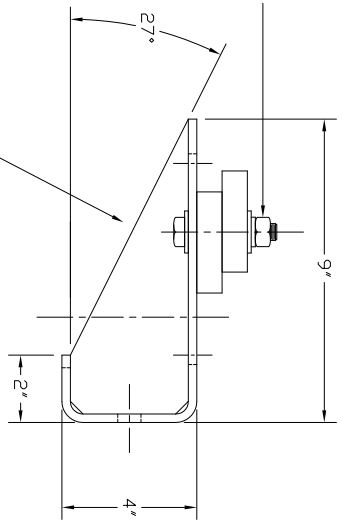
MATERIAL:

WHIPPS inc. 370 SO. ATHOL RD.
ATHOL, MA 01331



Title FLOOR MOUNT PEDESTAL FA14 (5.50" B.C.) TOP FLANGE		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches Scale: 1:7.5					
				C	Revised top plate	03/13/25	JM
Material SEE ENGINEERING SHEET		THIRD ANGLE PROJECTION		B	Revised base plate & bolt pattern	09/01/16	DJP
				A	Original Issue	08/04/16	DJP
Drg. No. ST-100-082		S.O. No. 		Rev.	Rev. Record	Date	Eng. Chk.
				Whipps, inc. 370 SOUTH ATHOL RD ATHOL, MA 01331			

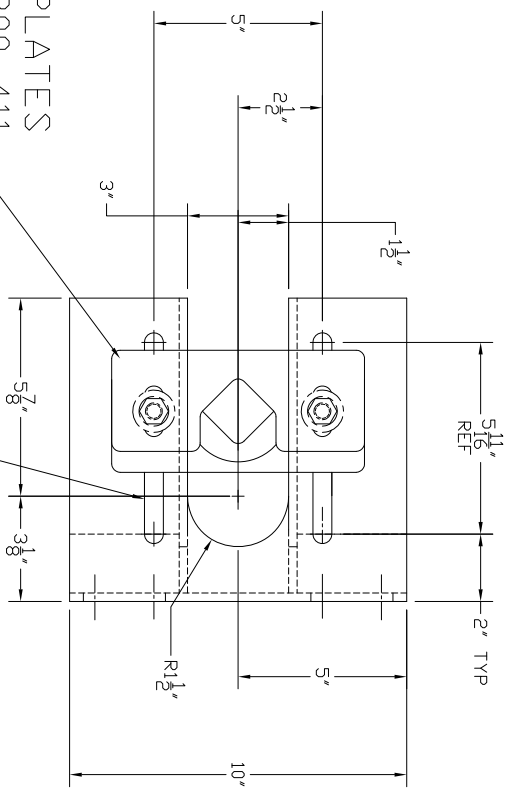
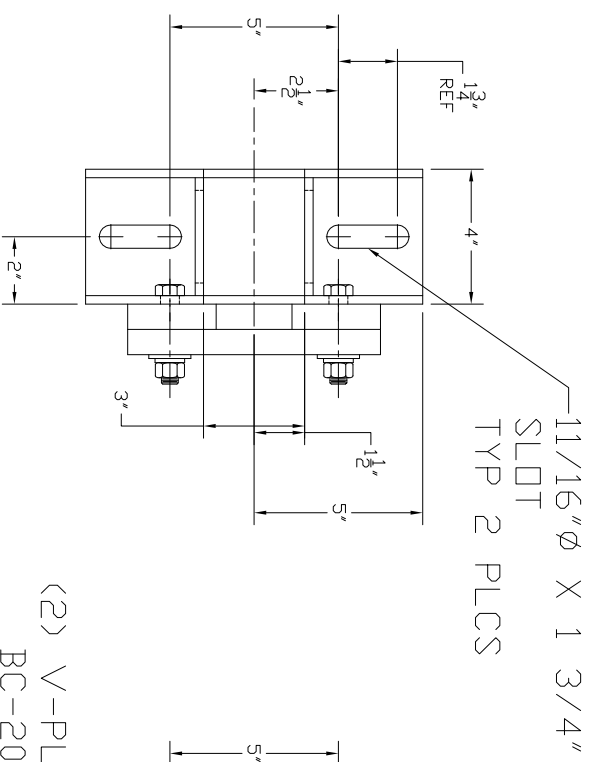
(2) 8-3/4" X 3 3/8" X 1/4" THK GUSSET



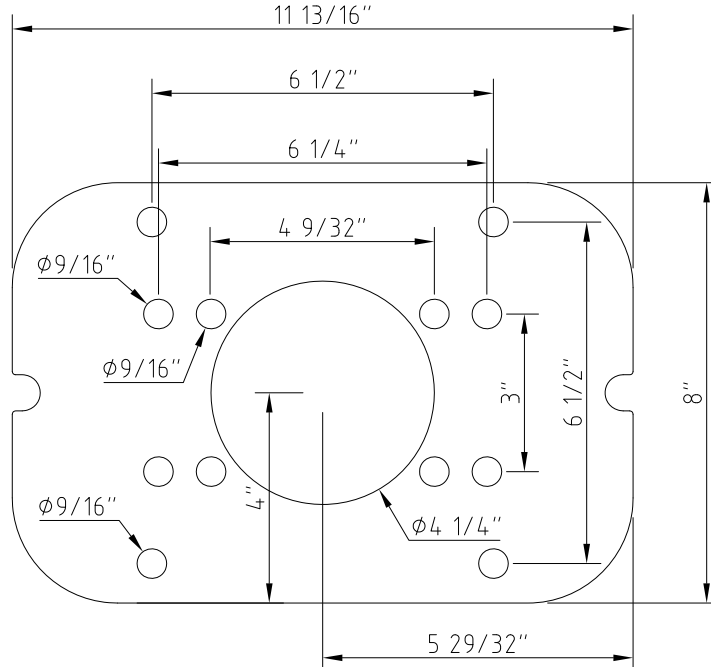
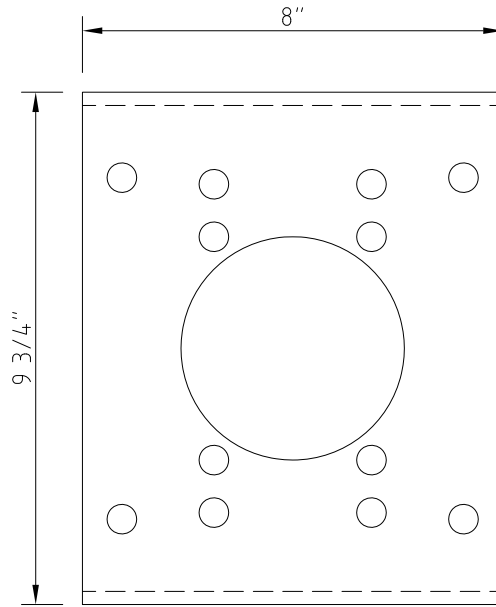
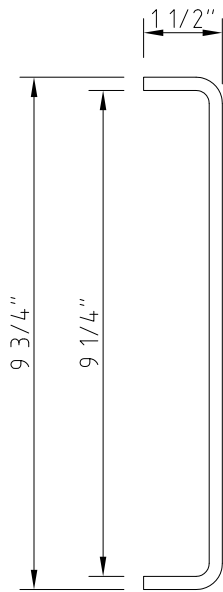
(2) 1/2"Ø X 2 3/4" S.S. SCREW

(2) V-PLATES
BC-200-411
SEE MAT'L SHT

9/16"Ø X 7 1/4" SLOT
TYP 2 PLCS

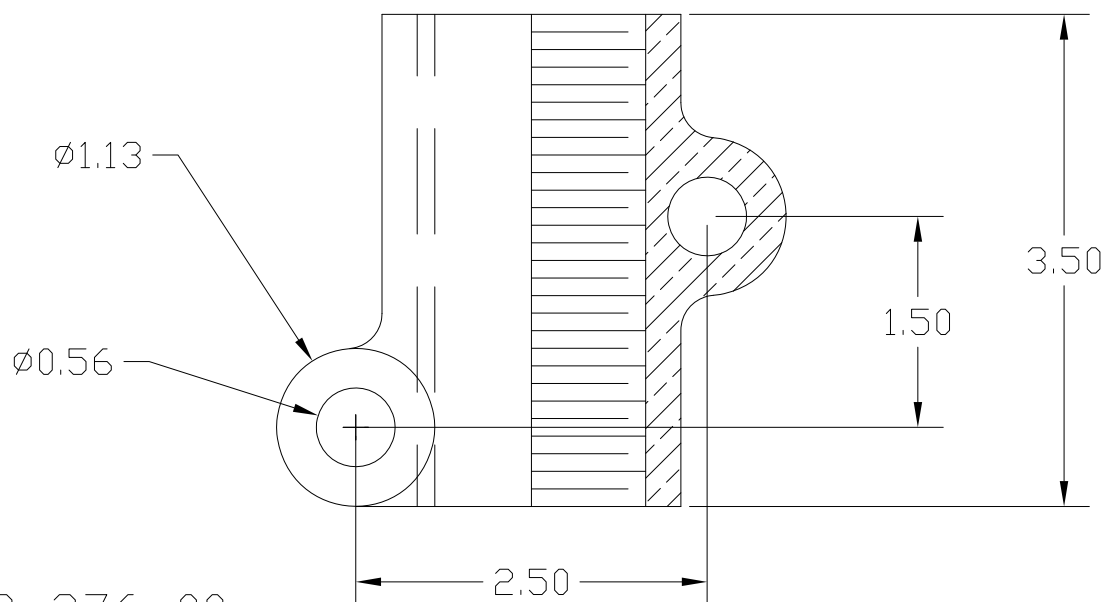
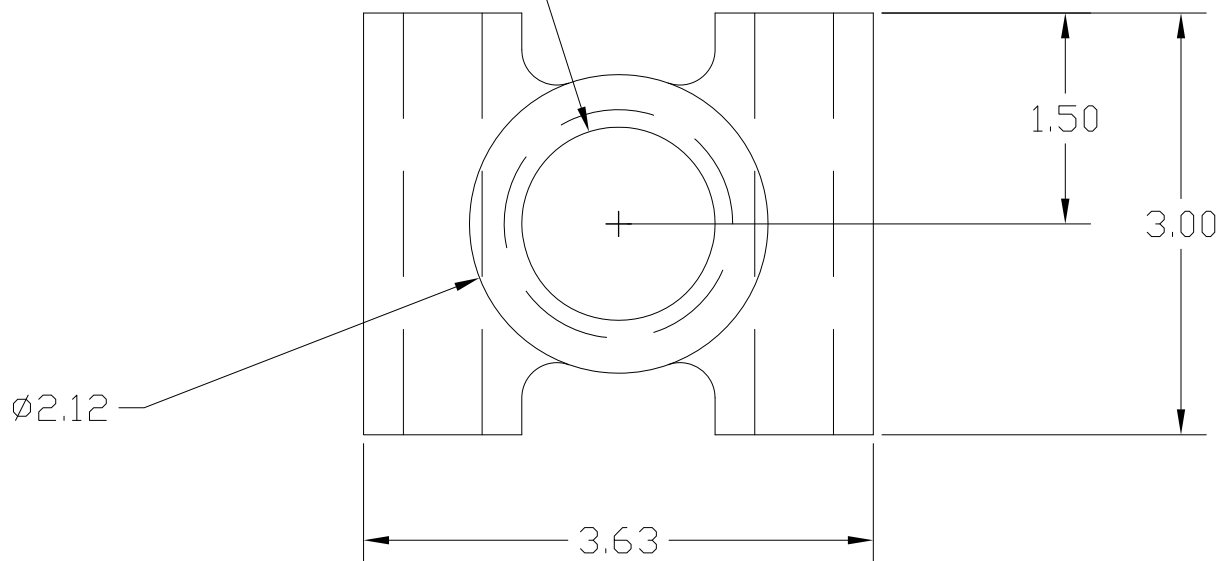


Title STEM GUIDE BRACKET DETAIL		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches		D	Rev.Anchor Slots to 11/16	6/28/24	JM	
				C	Rev. Slot For Forming	4/5/19	JM	
SEE ENG. SHEET FOR MATERIAL		Scale: FD		B	Rev. Added V-Plate	3/2/07	KS	
				A	Original Issue	2/7/07	KS	
Dr.g. ST-200-013		S.D. No.		Rev.	Rev. Record	Date	Eng.	Chk.
		THIRD ANGLE PROJECTION		Whipps, inc.				
				370 SOUTH ATHOL RD. ATHOL, MA 01331				



Title Yoke Adapter Plate Type-101/102		DO NOT SCALE THIS DRAWING Work to Dimensions Linear Dimensions are Inches				
				C	Added Grease Fitting Cope (101)	3/9/16 J.M
Material Refer To Engineering Sheet		Scale: FD		B	Enlarged Clearance Hole to 4 1/4"	1/22/16 J.M
				A	Original Issue	11/16/15 J.M
Drg. No. ST-100-071		S.O. No.		Rev.	Rev. Record	Date Eng. Chk.
		THIRD ANGLE PROJECTION		Whipps, Inc. 370 South Athol Rd. ATHOL, MA 01331		

1 1/2" ROLLED ACME MAX.
STEM CAPACITY
(1 5/8' NOM. O.D.)



PAT. NO. P-276-00

TITLE

NON-RISING STEM THRUST NUT

DATE

12/11/01

SCALE

FULL

DWN BY

G.W.

APPR BY

DATE

SYM

REVISION RECORD

AUTH

DR

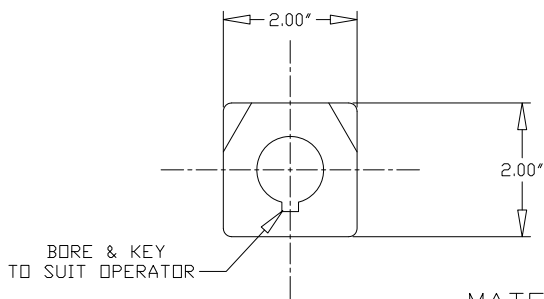
DRAWING NO.

AC-200-490

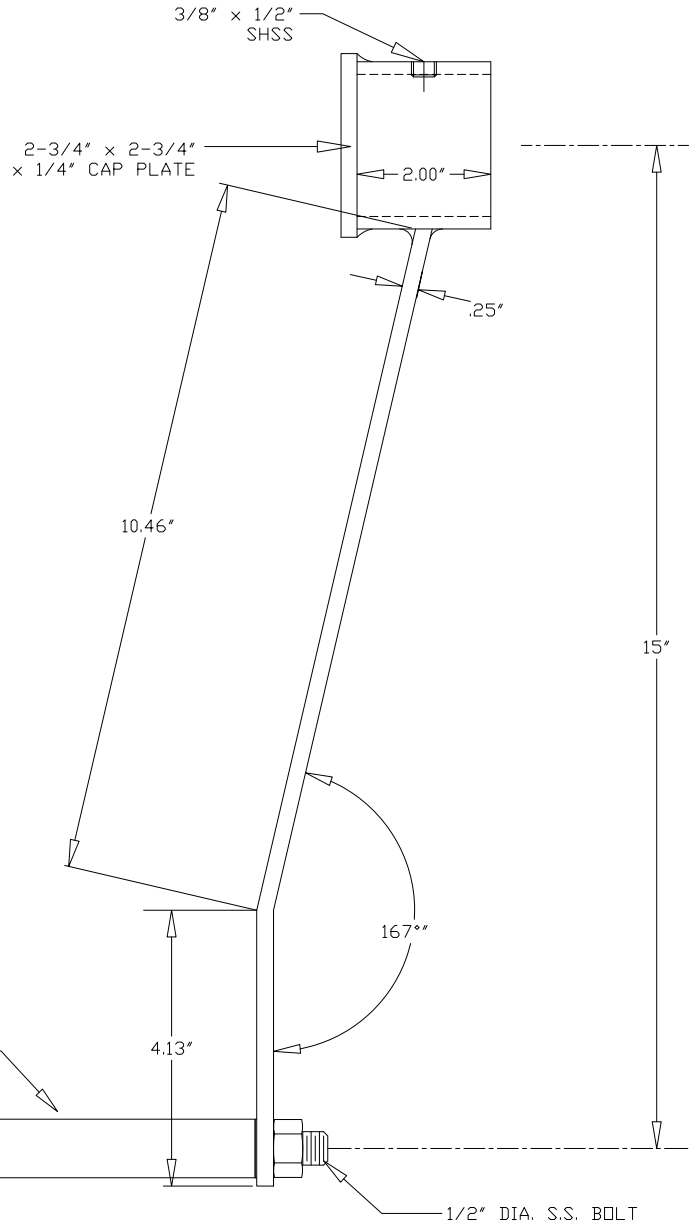
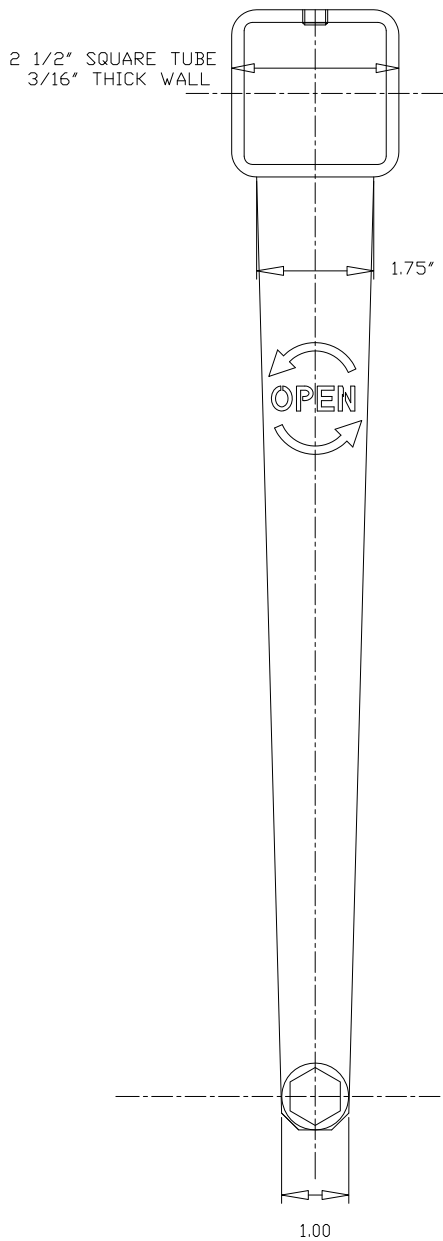
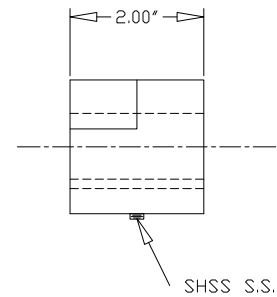
S.O. NO.

Whipps, inc.

38 SOUTH MAIN STREET
ATHOL, MA 01331



MATERIAL TYPE : BRONZE



MATERIAL TYPE : SEE ENGINEERING SHEET

TITLE STAIN. STEEL HAND CRANK ASSEMBLY
WITH 2" SQUARE BRZ. NUT

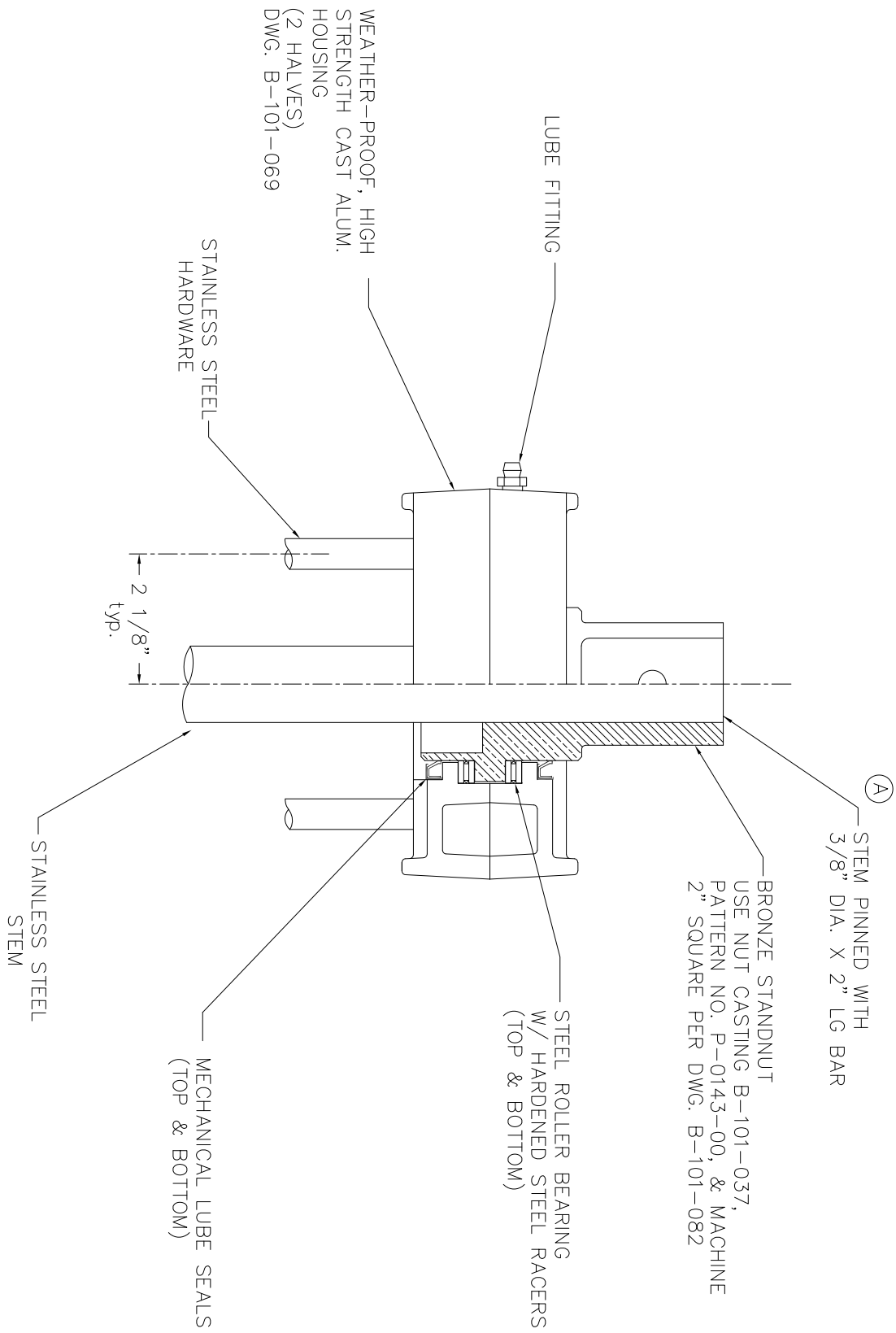
DATE 08-06-04 SCALE F / D DWN BY GW APPR BY

DRAWING NO. AC-102-016-S S.O. NO.

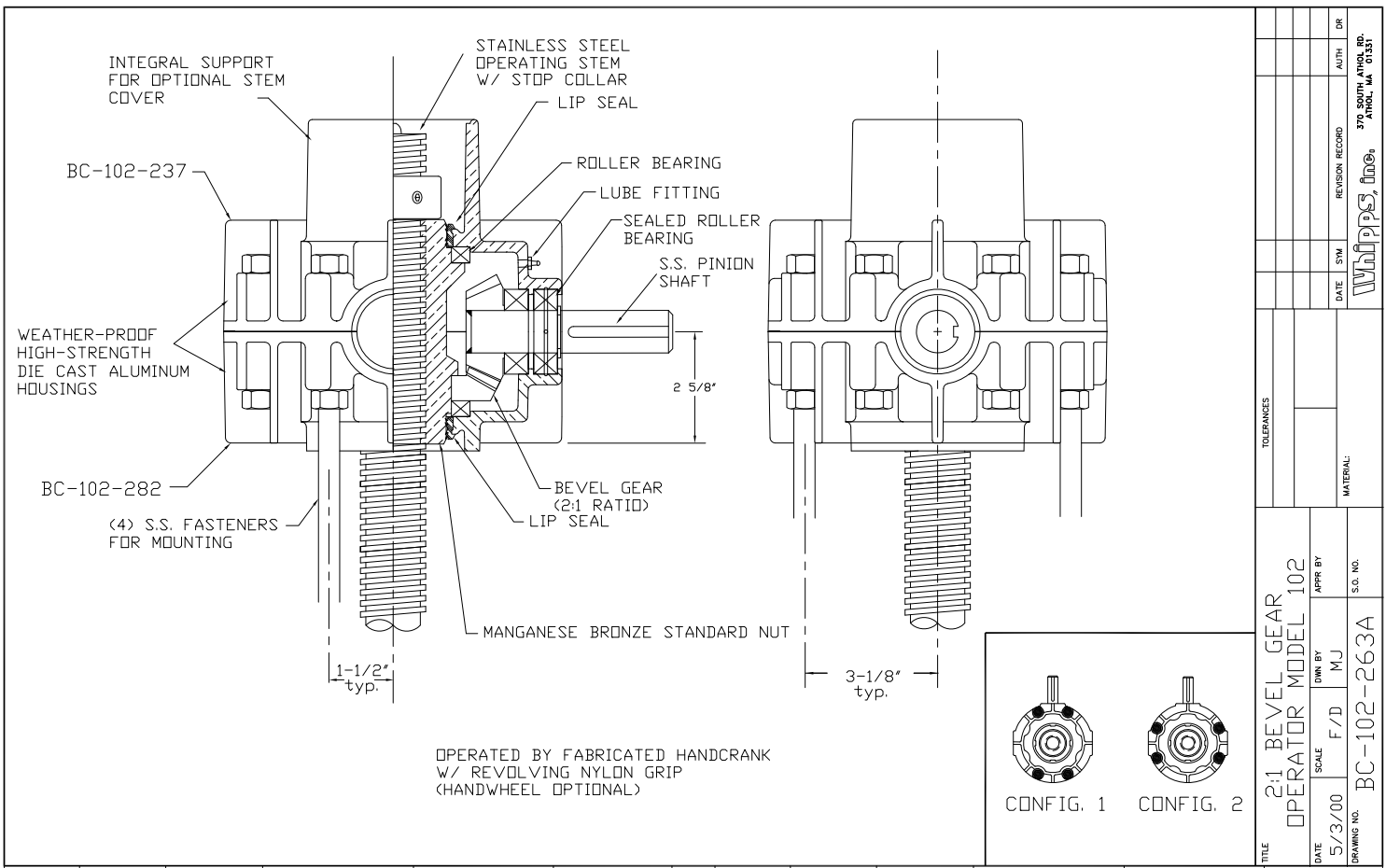
4/19/22	F	Replace Hex Head With Socket Head	JM	
6/11/18	E	Revised Cap Plate & Locking Bolt	KS	
8/01/16	D	Revised Open Arrow Direction Location	KM	
6/16/16	C	Open Arrow Direction	KS	
8/16/11	B	For Material See Engineering Sheet	JT	
1/25/11	A	Wall Thickness On Square Tube	KS	
DATE	SYM	REVISION RECORD	AUTH	DR

Whipps, inc.

38 SOUTH MAIN STREET
ATHOL, MA 01331



TITLE TYPE 101 OPERATOR FOR NON-RISING STEMS				TOLERANCES							
DATE 3/30/10	SCALE NTS	DWN BY KS	APPR BY								
DRAWING NO. BC-101-164				MATERIAL:		DATE	SYM	REVISION RECORD	AUTH	DR	
						Whipps, Inc. 370 SOUTH ATHOL RD. ATHOL, MA 01331					



ITEM	DIA	CONFIG	NUT ORIENTATION	MOUNTING STUD	PINION	ITEM	DIA	CONFIG	NUT ORIENTATION	MOUNTING STUD	PINION
1	1.38"	1	STD	STD	STD	33	1.75"	1	STD	STD	STD
2	1.38"	2	STD	STD	STD	34	1.75"	2	STD	STD	STD
3	1.38"	1	STD	STD	SEE BOM	35	1.75"	1	STD	STD	SEE BOM
4	1.38"	2	STD	STD	SEE BOM	36	1.75"	2	STD	STD	SEE BOM
5	1.38"	1	STD	SEE BOM	STD	37	1.75"	1	STD	SEE BOM	STD
6	1.38"	2	STD	SEE BOM	STD	38	1.75"	2	STD	SEE BOM	STD
7	1.38"	1	STD	SEE BOM	SEE BOM	39	1.75"	1	STD	SEE BOM	SEE BOM
8	1.38"	2	STD	SEE BOM	SEE BOM	40	1.75"	2	STD	SEE BOM	SEE BOM
9	1.38"	1	REVERSED	STD	STD	41	1.75"	1	REVERSED	STD	STD
10	1.38"	2	REVERSED	STD	STD	42	1.75"	2	REVERSED	STD	STD
11	1.38"	1	REVERSED	STD	SEE BOM	43	1.75"	1	REVERSED	STD	SEE BOM
12	1.38"	2	REVERSED	STD	SEE BOM	44	1.75"	2	REVERSED	STD	SEE BOM
13	1.38"	1	REVERSED	SEE BOM	STD	45	1.75"	1	REVERSED	SEE BOM	STD
14	1.38"	2	REVERSED	SEE BOM	STD	46	1.75"	2	REVERSED	SEE BOM	STD
15	1.38"	1	REVERSED	SEE BOM	SEE BOM	47	1.75"	1	REVERSED	SEE BOM	SEE BOM
16	1.38"	2	REVERSED	SEE BOM	SEE BOM	48	1.75"	2	REVERSED	SEE BOM	SEE BOM
17	1.50"	1	STD	STD	STD	49	2.00"	1	STD	STD	STD
18	1.50"	2	STD	STD	STD	50	2.00"	2	STD	STD	STD
19	1.50"	1	STD	STD	SEE BOM	51	2.00"	1	STD	STD	SEE BOM
20	1.50"	2	STD	STD	SEE BOM	52	2.00"	2	STD	STD	SEE BOM
21	1.50"	1	STD	SEE BOM	STD	53	2.00"	1	STD	SEE BOM	STD
22	1.50"	2	STD	SEE BOM	STD	54	2.00"	2	STD	SEE BOM	STD
23	1.50"	1	STD	SEE BOM	SEE BOM	55	2.00"	1	STD	SEE BOM	SEE BOM
24	1.50"	2	STD	SEE BOM	SEE BOM	56	2.00"	2	STD	SEE BOM	SEE BOM
25	1.50"	1	REVERSED	STD	STD	57	2.00"	1	REVERSED	STD	STD
26	1.50"	2	REVERSED	STD	STD	58	2.00"	2	REVERSED	STD	STD
27	1.50"	1	REVERSED	STD	SEE BOM	59	2.00"	1	REVERSED	STD	SEE BOM
28	1.50"	2	REVERSED	STD	SEE BOM	60	2.00"	2	REVERSED	STD	SEE BOM
29	1.50"	1	REVERSED	SEE BOM	STD	61	2.00"	1	REVERSED	SEE BOM	STD
30	1.50"	2	REVERSED	SEE BOM	STD	62	2.00"	2	REVERSED	SEE BOM	STD
31	1.50"	1	REVERSED	SEE BOM	SEE BOM	63	2.00"	1	REVERSED	SEE BOM	SEE BOM
32	1.50"	2	REVERSED	SEE BOM	SEE BOM	64	2.00"	2	REVERSED	SEE BOM	SEE BOM

ADHESIVE ANCHOR RECOMMENDATIONS

ADHESIVE SUPPLIED BY INSTALLER TO BE:

- **HILTI** HIT RE500 V3 EPOXY ADHESIVE
- **ITW RAMSET / REDHEAD** EPCON CERAMIC 6 EPOXY ADHESIVE
- OR APPROVED EQUAL

Whipps Inc. Supplies the Threaded Rod, Nuts & Washers required for installation.

Unless otherwise noted, the embedment depth required for each size anchor will be

Anchor Diameter	Embedment Depth
1/2"	4-1/2"
5/8"	6"
3/4"	7"
7/8"	8"
1"	9"

Whipps, inc.

Installation Instructions

Fabricated Sluice Gate, Slide Gate and Weir Gate

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This manual should be read carefully before installation, operation, and maintenance of Whipps, Inc. equipment

Introduction

This manual describes the recommended methods of installation, initial operation and maintenance for Whipps, Inc. fabricated sluice gates, slide gates, weir gates, operating mechanisms and related components. This manual should be used in conjunction with the approved installation drawings provided by Whipps, Inc. No special tools are required.

Whipps, Inc. gates are custom built to meet the requirements of each specific application. The gates provided have low leakage characteristics. However, care must be taken in the handling, storage and installation of the equipment to ensure that it will function as intended and restrict leakage within the specified parameters.

The information in this manual is intended only as a recommendation for the proper and satisfactory installation of our equipment. Whipps, Inc. assumes no liability, expressed or implied, for the interpretation of the recommendations or faulty installation of the gates. Whipps, Inc.'s responsibility is limited to defects in manufacturing.

Handling and Storage

To prevent personal injury or equipment damage, follow standard safety procedures when handling equipment and be sure rigging equipment is properly set and in safe working condition.

When unloading the equipment from the box trailer or flat bed truck, use care during removal and storage. If the equipment has been shipped mounted to a wooden skid, lift the skidded material from the bottom.

If damage has occurred in transit, file the necessary report with the freight carrier. Please take photos of the damage if possible and contact Whipps, Inc. immediately.

Thoroughly review the packing list and compare the items on the list to the equipment received.

Although Whipps Inc. gates are durable and heavily constructed, care is necessary during storage, handling and installation. Stem threads and hoists have precision surfaces that should be protected from damage.

Equipment should be stored on planks or timbers on a flat surface to keep them off the ground and to prevent distortion. Equipment should be covered with tarps to protect the equipment from foreign matter while stored. Where there are several medium or small gates and where storage space is limited, it may be necessary to stack the gates with heavy timber blocking placed between the gates to prevent damage. When stacking equipment, take care to avoid damaging operator pinion shafts or other components that may extend upward or outward.

If electric actuators or hydraulic cylinders are provided, extra care is required to protect this precision equipment. This equipment should be stored indoors in accordance with the original manufacturer's instructions. For electric actuators, this may include the energizing of heaters upon receipt of units to prevent corrosion of controls. For hydraulic cylinders, this includes storing cylinders vertically to prevent damage to seals.

To prevent bending when handling and storing, stems should be supported over their full length. They should be stored on a flat surface and the threaded portion should be covered and protected from damage. Couplings and thrust nuts (when applicable) may be shipped on the stems and may require removal prior to installation. Stop collars and anchor bolt hardware is normally shipped in a bag or box. Operating mechanisms should be handled and treated as precision machinery and protected accordingly. See actuator O&M manual for storage instructions.

Installation

Installation - General

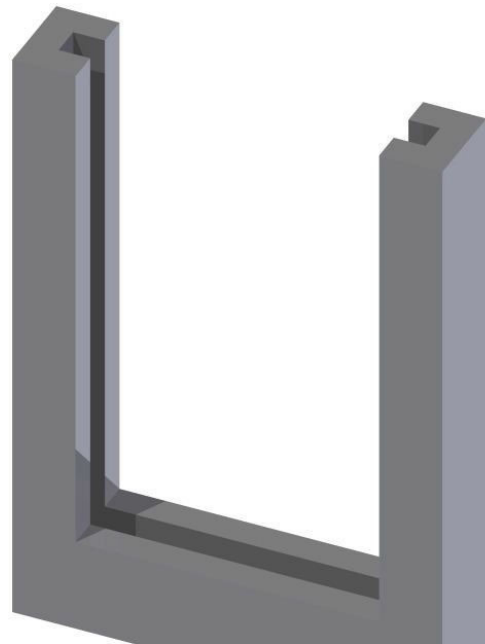
The most important aspects of a gate installation are listed in this section. If these recommendations are followed, a proper gate installation is assured.

Carefully review the installation drawing for each gate prior to installation to confirm proper setting and component location. If the installation drawings are not available, please contact Whipps Inc. at 978-249-7924 or www.whipps.com.

If up stop bolts (upward opening gates) or down stop bolts (downward opening gates) have been removed from the side frames to facilitate installation, they must be re-installed.

Installation - Embedded Frames

1. On gates with embedded side frames and/or an embedded invert member, box-outs or recesses are required in the channel walls and/or the channel floor during the concrete pour. The box-outs shall be of sufficient size to accommodate the gate. See installation drawings for dimensional information.
2. The frame must be well supported prior to the addition of grout to prevent distortion. Distortion of the frame will cause excessive operating effort and / or excessive leakage.
3. Care should be taken to keep the seals and slide free from grout.
4. Follow grout or concrete manufacturer requirements while filling and curing box-outs. Protect curing product from freezing as directed by manufacturer, and ensure products are mixed thoroughly and in correct proportion/ condition.



Installation – Gate and/or Components Mounted with Anchor Bolts

When anchor bolts are furnished for mounting the gate or components such as pedestals, stem guides and/or wall brackets, the location and projection of the anchor bolts will be shown on the installation drawing. In most cases, epoxy or wedge type anchor bolts will be utilized.

When hook type anchor bolts are utilized, the anchor bolts should be placed in the holes drilled in the forms at locations indicated on the drawings. The hook ends of the anchor bolts should then be wired to the opposite form or to reinforcing rods to hold the bolts firmly in place.

Where gates are mounted with anchor bolts it is necessary that a uniform grout pad (non-shrink grout) or a resilient gasket be placed between the flange of the gate and the concrete wall. This grout or gasket is necessary to serve as a seal between the gate and wall and the type will be indicated on the installation drawings. The projection of the anchor bolts, shown on the installation drawings, includes provisions for the grout or gasket. Grout pads might also be required for pedestals, stem guides or wall brackets.

When a gasket is utilized to seal between the gate and the wall, the wall will need to be straight and plumb. If the wall is not straight and plumb, leakage can occur between the gate and the wall. Removal of the gate, modifications to the wall and re-installation of the gate may be required to rectify this situation.

Gates should not be mounted directly to a wall without grout or a gasket as this will result in leakage between the gate and wall.

1. All anchor bolts should be checked prior to installation to ensure that the threads are undamaged. Anchor bolts should be installed as recommended by the anchor bolt manufacturer.

2. **Do not install the gates without mounting the jacking nuts on the anchor bolts as shown on the installation drawing.** If the jacking nuts are not installed and the outside nuts are over tightened, frame distortion can occur and this can lead to excessive leakage. Frame distortion can pull the seal away from the slide thus creating a path for leakage.



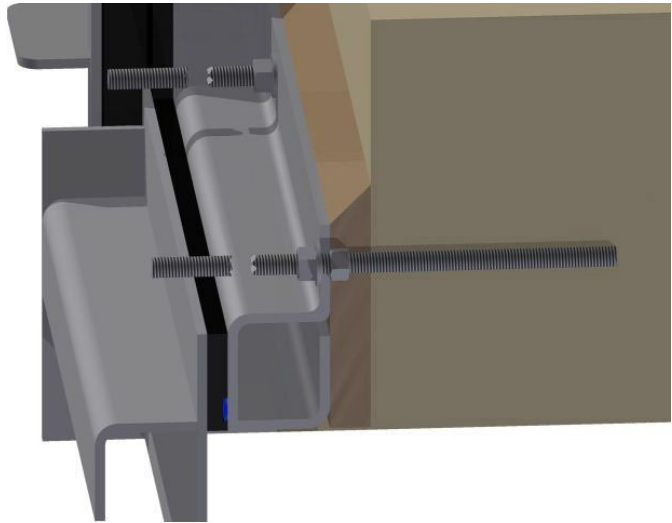
3. Two nuts will be provided for each anchor bolt that requires a jacking nut. Refer to the installation drawings for details. The jacking nut should be installed on the anchor bolt prior to mounting the gate, leaving approximately 1 inch for the insertion of grout. The jacking nut needs to be positioned to ensure that the gate will be mounted true and plumb, even if the concrete wall is not straight.
4. After anchor bolt and jacking nut installation in the wall, the anchor should be lifted and carefully set in place in such a way as to not damage the threads on the bolts. After the gate is mounted on the anchor bolts, attach the other nuts to the anchor bolts. The use of the double nut arrangement helps to ensure that the gate will be mounted straight and plumb and can be firmly tightened into position without distortion.

5. Horizontal Invert and Top Seals – Special Care:

Wall mounted gates that have a horizontal invert seal, or horizontal top seal: extra care will be required to insure the correct seal compress against the slide plate to insure minimum amount of leakage. This may require the gate frame with horizontal seals to be jacked away from the concrete wall to apply more force on the seal that is in contact with the slide plate. This is accomplished by using the jacking nut that was installed with the anchor studs behind the horizontal seal. When properly adjusted, one should not be able to slide a 0.004" feeler gauge between the seal and the gate disc. **This is one of the most important procedures on installing gates with a horizontal seal member.**

6. With the gate flange located approximately 1 inch from the wall, forms should be mounted around the flange and a non-shrink grout should be placed between the flange and the concrete wall. The grout needs to be completely applied around the perimeter of the gate as shown on the installation drawings. All voids should be filled with grout to ensure leakage cannot occur between the gate and wall. Dry packing grout will not produce a proper water seal and should not be considered.
7. Care should be taken to avoid getting grout on the seals or the slide. Extra care should be taken on gates with top seals. All grout that adheres to the seals or the slide should be removed before operating the gate. This could cause extensive seal damage.

8. **Closely review the installation drawings, as it might be necessary to grind or cut off a portion of the anchor studs to provide clearance for unimpeded vertical travel of the slide.** Check the projection of the anchor bolt across the top of the opening on upward opening gates with top seals and check the projection across the bottom of the opening on downward opening gates. Where shown, the anchor bolt should be cut down to the nut.

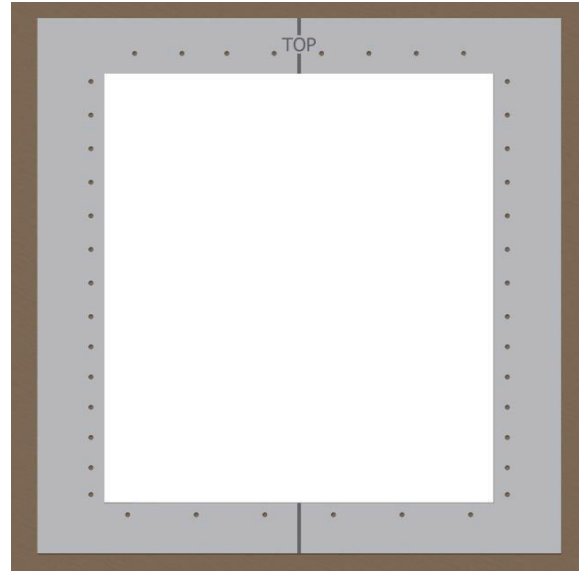


9. If any upstop bolts (upward opening gates) or downstop bolts (downward opening gates) were removed from the side frames to facilitate installation, re-installed them now.

Installation - Wall Thimbles

1. The front face of the rectangular, square or circular wall thimbles are marked with vertical centerlines and with "TOP" stamped on the top of the wall thimble. **Wall thimbles should be set in place with the "TOP" mark up and top and bottom centerline marks plumb.**

2. After being set at the proper elevation, the wall thimble must be internally braced to carry the weight of the concrete. Care should be used in placement of the braces so as not to distort the wall thimble. Gate attachment hardware will be misaligned if the wall thimble is distorted.
3. The wall thimble should be firmly supported on the form. Forms should be supported and stiffened against movement. If forms move, they will distort the wall thimble mounting flange and the gate may leak.
4. The tapped holes in the face of the wall thimble must be plugged or capped to prevent concrete from entering the holes.
5. After the concrete has hardened and the forms removed, the front surface of the wall thimble should be thoroughly cleaned. Make sure to remove all concrete that has flowed onto the surface from the edges. All tapped holes should be inspected and cleaned of concrete if necessary.



Installation - Gate Mounted to New Wall Thimble

1. The face of the wall thimble should be thoroughly cleaned and all wall thimble studs in place. Care should be taken to prevent damage to the studs during installation.
2. A gasket material is required between the surface of the wall thimble and the mounting flange of the gate. Mastic is normally used for this purpose and should be applied in accordance with the label directions.
3. If a gasket material other than mastic is used, it should be installed over the studs to provide a smooth mounting surface for the gate. If the gasket is other than one piece, the gasket joints should be aligned in accordance with the match markings and cemented with a liquid-type gasket material. When applying gasket materials, care should be taken to ensure that excessive amounts of lumpy, dried materials are not present when the gate is drawn tightly and evenly to the wall thimble.
4. The mounting flange of the gate should be thoroughly cleaned.
5. The gate can then be lifted and set over the studs and the nuts put in place and tightened. Care should be taken during this process to help ensure that the threads on the studs are not damaged. The sequence of tightening should be done in multiple passes by applying

progressively larger force each pass. Equal torque should be applied to all nuts so that the gate is firmly and evenly tightened to the mounting flange without distortion.

Installation - Gate Mounted to an Existing Wall Thimble

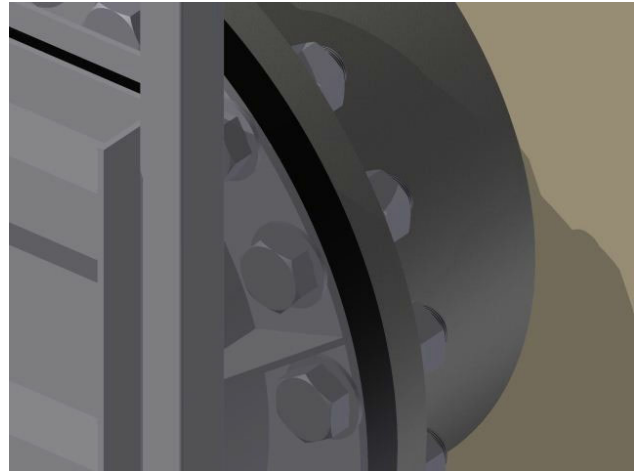
See instructions for “Installation - Gate Mounted to a New Wall Thimble” after a close inspection of the existing wall thimble once the front flange is accessible. If the mounting flange of the existing wall thimble is damaged, contact the factory prior to installation.

Installation – Gate Mounted to a Pipe Flange

Where gates are mounted on pipe flanges, the procedure is the same as when the gate is mounted on a wall thimble. The type of attachment hardware shall be as shown on the installation drawings.

Consult the factory for assistance if the flange on which the gate is to be installed is damaged or unusable for any reason.

Nut Tightening Torque



Proper tightening of the nuts on anchor bolts holding the gate to the wall or studs holding the gate to the wall thimble may prevent serious problems in operation or performance of the gate. Tabulated below are recommended torque values for common fastener sizes. Coat all anchor bolts with approved anti-gall compound (fastorq faslube AG or approved equivalent.)

DIAMETER (in.)	TORQUE (ft.-lb.)
1/2	35
5/8	75
3/4	100
7/8	150
1	200

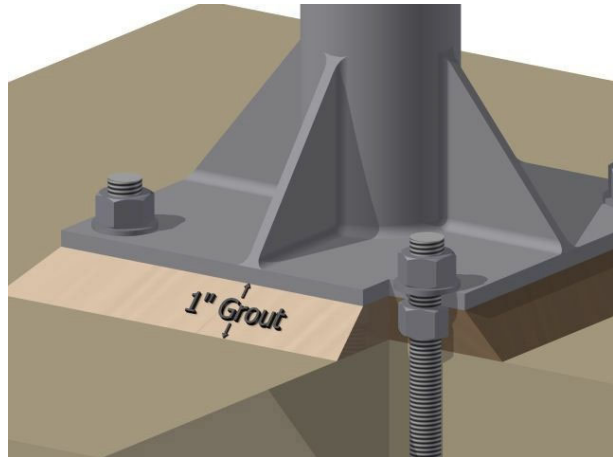
Installation – Assembly

On non-self contained gates, some field assembly is required. Refer to the installation drawings for the location and position of all components.

When assembling gates that have dual stems. Be sure that the stems are installed straight and plumb. When the operators are installed, it is important that both stems be in proper time and the top of the slide be level.

All pedestals are identified by the installation drawing and/or drawing number and should be used with the proper gate and stem.

1. After the stem has been completely assembled and positioned in place, the operator can be lowered onto the stem and turned into position by operation of the handwheel or crank.
2. Jacking nuts should be placed on the anchor bolts between the operating floor and the base of the pedestal so that it is plumb and the base is approximately 1" above the operating floor.
3. Approximately 1" of grout should then be placed between the pedestal base and the operating floor.
4. After the grout has hardened, the outside anchor nuts should be tightened firmly in place.
5. For manual operators, after the operator has been installed, tension should be applied to the stem by the handle or crank in a direction that would normally open the gate. However, the gate should not be opened. The intent is merely to apply tension that will result in a straight stem.
6. For electric actuators, the gate should be opened with the manual handwheel at least 3 inches before using the electric controls. In this manner, the proper phasing and direction of rotation of the motor can be determined without damaging the gate assembly. Once the unit has been installed, the manufacturer's directions should be followed closely in setting the closing and opening limit switches. The torque switches have been properly set at the factory and should not need adjustment. Follow the manufacturer's instructions if it appears that adjustment is necessary.
7. The stem guide, when applicable, should be anchor bolted to the wall in accordance with the installation drawings with uniform clearance possible between the stem and the stem guide bushing.
8. The stem should be thoroughly cleaned and lubricated with high pressure type industrial grease, such as Mobil Centaur Moly or equal. See lubrication chart.
9. The gates should be placed in the fully closed position. On upward opening gates, the slide should be lowered so that there is minimum compression of the slide onto the invert seal. On downward opening gates, the slide should be positioned as shown on the installation drawing.



10. Stop collars are provided on manually operated gates. The stop collar should be threaded onto the top of the stem only after the operator has been installed and the gate is in the fully closed position. Set the stop collar so there is approximately 1/16" of clearance between the bottom of the stop collar and the top of the operator nut. Set screws should then be tightened to hold the stop collar in place.
11. Stem covers should be attached immediately after gates are installed. See electric actuator O&M for factory recommendations on mounting and sealing threaded stem cover adaptors. Seal connection between stem cover and stem cover adapter using silicone following stem cover installation. Preventing water from intruding the actuator and debris build up on the threaded stem. Mylar indicating strips, positioned to indicate the location of the gate in both the fully open and closed positions.
12. The crank or handwheel should turn easily. If there is any binding or noise during operation, check to be sure that the stem guides, pedestal and stem are properly aligned and the stem threads are lubricated. No more than 40-foot pounds of force should be applied while turning.

Hydraulic Cylinder Operators

Hydraulic cylinders should be stored in the vertical position and filled with hydraulic fluid. If it is necessary to store them horizontally for a short period, they should be rotated every two weeks to help prevent damage to the seals.

1. Hydraulic cylinders should be mounted on the anchor bolts in such a way that the piston rod and stem are in proper alignment and there is sufficient clearance for piping, fitting, etc.
2. The coupling between the piston rod and the stem should be screwed into place and locked.
3. With the gate in the closed position, the piston should be lowered so that it is in contact with the bottom head of the cylinder.
4. With the piston in this position, the thrust nut should be adjusted on the stem so that it is in contact with the bottom of the thrust nut pocket. Set screws should be tightened to lock it in place. In most cases, the top area of the piston is larger than the underside. Therefore, if pressure applied to both surfaces is the same, more force will be applied in the closing direction than in the opening direction. For that reason, pressure-reducing valves should be provided in the line to the top of the cylinder to lower the pressure to that required to properly close the gate. In this way, full operating pressure can be applied to the bottom of the piston resulting in more opening than closing force. All piping should be thoroughly flushed and cleaned prior to making connection to the hydraulic cylinder.
5. Operate the hydraulic motor by moving lever valve either up or down for desired direction. Lever should be in the CLOCKWISE position to raise the gate and COUNTER CLOCKWISE to lower the gate.

Prior to Operating

1. Clean both sides of the slide, the guides, seals and stem of all grout, sand, paint and other debris.
2. Check to make sure that stem guides are positioned correctly and are securely fastened.
3. Clean and lubricate the stem threads. See lubrication chart.
4. Adjust stop collars and/or limit switches.
5. Install stem covers. Seal connection between stem cover and stem cover adapter using silicone following stem cover installation.

Operating Instructions

Whipps, Inc. fabricated gates are constructed to operate satisfactorily under the specified operating conditions. These gates should be operated with care so as not to exceed the specified conditions. If, in the operation of the gate, an obstruction is met, either in the opening or closing direction, the obstruction should be removed before continuing in the operation. When the gate is fully opened or closed, excessive force should not be placed on the handwheel, crank or gate stem by personnel in an effort to move the gate further. There should never be a need for a pipe extension or other additional leverage applied to the handwheel or crank. If excessive force is required, a thorough visual inspection of the gate assembly and stem is strongly recommended.

If a problem arises in the operation of the gate, such as an unusual head condition or evidence of excessive corrosion, the factory should be consulted before the gate is used or operated.

Installation Inspection Check List

Manually Operated Gates

1. Check hoist, stem guide, and gate attaching bolts for proper tightness.
2. Apply tension to stem and check stem guides for proper alignment. There must be a uniform clearance between the operating stem and stem guides.
3. Visually inspect gate seals, including the invert seal, and both sides of the slide. Thoroughly clean off foreign matter. Whipps recommends exercising the gates every month during routine inspection and maintenance.
4. Visually inspect the threaded portion of the stem. It must be clean and free of foreign matter, including dirt or sand, and lubricated with suitable industrial grease. If a wire brush is used to clean the stem, use only a stainless steel type. Do not use a carbon steel brush!
5. Adjust stem stop collar to within 1/16" of the top of the hoist operating nut and lock in place.

6. Stem covers should be attached immediately after gates are installed. See electric actuator O&M for factory recommendations on mounting and sealing threaded stem cover adaptors. Preventing water from intruding the actuator and debris build up on the threaded stem. Seal connection between stem cover and stem cover adapter using silicone following stem cover installation. Mylar indicating strips, positioned to indicate the location of the gate in both the fully open and closed positions.

Maintenance Instructions

Gates

Gates should be visually inspected at regular intervals (at least every three months or as needed) for signs of misalignment, damage or corrosive attack. Drain the channel to inspect the equipment for obstructions, debris, and material that has the potential to impact the equipment operation. If any of these things are observed during the inspection, they should be removed. Corrosion, when it occurs, is most prominent at the water line.

Please note gates with non-rising stems typically require additional maintenance. If the water level rises to the threaded portion of the stem, the threads may become coated with grit or debris. If the threads become grit laden, the following procedure is recommended to prolong the useful service life of the operating nut (mounted on the slide):

1. The threaded portion of stem should be cleaned and re-greased. The stem must remain free of grit and be sufficiently lubricated to prevent accelerated wear to the operating nut (mounted on the slide).
2. Whipps does require gates have occasional operation during the course of a year. If allowable the gates should be exercised at least once every 3 months or as needed, keeping the guides clean and free of debris.

Manual Operators

At least every 3 months or as needed, grease fittings (if applicable) should be lubricated with a small amount of heavy duty grease which will not harden in cold weather nor become liquid in warm weather.

See Lubrication Chart. Some manual operators may be permanently sealed, and these units will not have lubrication fittings. Whipps recommends exercising the gates every 3 months or as needed during routine inspection and maintenance.

Electric and Hydraulic Operators

Periodic maintenance schedules should be set-up in accordance with the original manufacturer's operation and maintenance manual. Do not operate with portable an electric or hydraulic operator until all stop collars have been placed in the proper position and set into place.

1. Stem covers should be attached immediately after gates are installed. See electric actuator O&M for factory recommendations on mounting and sealing threaded stem cover adaptors. Seal connection between stem cover and stem cover adapter using silicone following stem cover installation. Preventing water from intruding the actuator and debris build up on the threaded stem. Mylar indicating strips, positioned to indicate the location of the gate in both the fully open and closed positions.

Modulating Electric Operators

These operators can cause accelerated wear in the operating nut since the stem and gates are operated more frequently and at times continuously.

1. The threaded portion of the stem must be clean and greased always.
2. The operating nuts should be removed and inspected for wear after the first six months of operation and every year thereafter.
3. Replace bronze operating nut as soon as noticeable wear is evident.

Operating Stems

It is important that operating stems be periodically cleaned and greased. Even though some environmental conditions are more severe than others and the use of pipe covers will protect stems, they still need to be cleaned and greased at least once every six months, more often if the grease becomes dirty. This is especially important on large gates and/or frequently operated gates such as gates with modulating electric actuators. See Lubrication Chart.

Installation Drawings

The drawings submitted by Whipps, Inc. for approval and/or field use, are planned so that the installation drawing is the master reference.

The drawings depict as much as possible of the structure surrounding the supplied equipment. The location of embedded material such as anchor bolts and wall thimbles are shown. The identification of fasteners and components (studs, anchor bolts, gate assemblies, hoists, stems, stem guides, stem couplings, adaptor plates, wall thimbles, thrust nuts, stop collars and other equipment) is done by calling out physical sizes and/or assembly or detail drawing numbers. More information is available on the detail drawings, which have been included with the installation drawing.

Spare Parts

Whipps, Inc. does not typically recommend stocking spare parts by customers or owners since the equipment is designed for a very long service life when recommended maintenance procedures are followed.

If a repair part is required, contact the PARTS DEPARTMENT at Whipps, Inc. at 978-249-7924 or www.whipps.com with as much of the following information as possible:

1. Plant name and location.

2. Original (four or five digit) shop order number which is indicated on correspondence and installation drawings.
3. The installation drawing number, and a description of the part, with any other available drawing numbers or the size (width x height) and location of the gate in the plant.
4. Description of damage and cause. (Digital photos of damage are useful.)
5. Approximate delivery requirements.

Gate Shipment Policy

The equipment furnished on this order has been inspected prior to leaving the factory and has been accepted by the freight carrier. Please check the packing list accompanying the shipment for shortages and examine the equipment for damages prior to accepting the shipment. Before handling, storing or installing this equipment, read the installation manual that accompanies the shipment.

Missing Equipment

If there is missing equipment, report it to the Whipps, Inc. Service Department upon receiving the equipment. Claims for shortages of equipment shown on the packing list will not be accepted unless a claim is filed within (30) thirty days of the shipment of the equipment.

Damage in Transit

If the equipment has been damaged in transit, the purchaser is responsible for filing the claim with the transport company. Please contact Whipps, Inc. for assistance in filing the claim.

Installation, Inspection and Adjustment

Installation supervision, inspection of installed equipment, setting of limit switches and certification of satisfactory initial operation are not included unless specifically indicated on the customer's purchase order and accepted by the company. Otherwise, Whipps, Inc. will provide this service at the standard published charges.

Field Issues

If trouble develops either in the installation, operation or performance of the equipment, the installation manual and drawings should be checked to determine if the equipment has been installed properly. If proper performance or operation cannot be obtained and assistance from the factory is desired, please contact Whipps Inc or the local representative. Arrangements will

be made to send a service technician to the job site if required. The service technician will make a thorough examination of the problem and if the equipment is faulty in workmanship or material, the necessary repairs will be made by the factory at no cost to the purchaser if within the warranty period.

If, however, the problem is due to faulty installation or adjustment, the cost of the field service will be charged to the purchaser.

If repairs are made in the field by the purchaser or authorized by the purchaser, back charges for these repairs will not be accepted by Whipps, Inc. unless we have been notified prior to the incurring of these costs and has accepted the responsibility for these repairs.

Whipps, Inc. will not be liable for contingent costs or costs of delays due to the faculty equipment and the repairs thereof.

Field Service Policy

Field service charges begin the time of departure from the Whipps facility and accrue until the return of the service person; it includes a daily rate plus travel and subsistence expenses. Premium day and hour rates will be charge on Saturdays, Sundays, and Holidays and for time spent before 6 a.m. or after 5 p.m., or over eight hours per day. A schedule of Field Service charges is available from the Whipps, Inc. Field Service Department.

In an EMERGENCY SITUATION, check the following:

Check the operator where the operator and stem interface. Check the operating nut to ensure that the nut is intact. If the internal threads on the nut have been stripped, proceed with replacement of the nut. If a stripped nut has caused the slide to drop and the slide needs to be raised, a mechanical means of lifting the slide will be required. Once lifted, the slide should be secured in place.

If the operating nut is intact, check the connection between the lower portion of the stem and the slide. On gates with rising stems, this connection is typically a bolted connection. On gates with non-rising stems, there may be a bronze thrust nut. If a thrust nut is used, check the internal threading on the thrust nut to ensure it has not been stripped. If a stripped nut has caused the slide to drop and the slide needs to be raised, a mechanical means of lifting the slide will be required. The stem may need to be removed prior to raising the slide. Once lifted, the slide should be secured in place.

If the gate is outfitted with electric actuator and the actuator is not functioning, please consult the troubleshooting portion of the O&M manual.

If there are no apparent problems with the operating nut, thrust nut or actuator, visually inspect the gate to ensure there are no obstructions preventing operation.

Further Questions?

Please contact Whipps, Inc. if unforeseen problems occur that are not covered in this manual.



WATER CONTROL GATES

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Tel (978) 249-7924 Fax (978) 249-3072
www.whipps.com

Test Procedure	
Document Title:	Series 900 & Series 800 Gate Shop Leak Test Procedure
Procedure No:	043-01
Document Owner:	Quality Manager
Date Created:	10/21/2016

1. Purpose/Objectives:

This procedure describes the process steps and procedure to be followed for performing shop leakage tests in accordance with the most current revision of AWWA C561 Fabricated Stainless-Steel Slide Gates & AWWA C562 Fabricated Aluminum Slide Gates.

2. Scope:

This procedure applies to all Whipps, Inc. Series 900 Stainless Steel Gates & Series 800 Aluminum Gates compliant with AWWA C561 & C562. AWWA C561 & C562 do not require that all compliant gates be shop leak tested, but only when a shop leakage test is specified in the purchaser's requirements and when such tests are feasible.

3. Related Documents:

ANSI/AWWA C561 & C562
F043-01 Slide/Sluice Gate Shop Leakage Test (Form)

4. Procedures:

4.1 Open Channel Gates w/ No Top Seal – Model Numbers: 921, 923, 923-C, 923-D, 923-D-I, 951, 953, 953-C, 953-D-I, 821, 822, 823, 823-C, 823-C-I, 823-D, 823-D-I

- 4.1.1 All testing will be performed with the gate only, excluding all components of the gate actuator system.
- 4.1.2 All gates will be tested for unseating head only.
- 4.1.3 All gates will be tested in the vertical position.

- 4.1.4 Individually tested gates will be mounted to a suitable backing plate or test fixture.
- 4.1.5 When testing multiple gates of the same size gates may be bolted back to back and tested simultaneously.
- 4.1.6 The test gate will be filled with water to a level equal with the top of the gate waterway opening.
- 4.1.7 Once the gate is filled, and once any leakage has stabilized, leakage will be collected for a time interval of at least one (1) minute.
- 4.1.8 Total amount of collected leakage will be measured utilizing graduated containers and will be compared to the specified maximum allowable leakage (gpm/ft of perimeter seal).
- 4.1.9 If the measured leakage of the gate exceeds the maximum allowable leakage the water will be drained, if necessary, from the test assembly. Adjustments or repairs will then be made to the appropriate gate component and the gate will be re-tested.
- 4.1.10 If the measured leakage is equal to or less than the maximum allowable leakage the gate will be considered as having passed the leakage test.

4.2 Full Aperture Sealing Gates – Model Numbers: 923-D, 923-D-I, 924, 925, 935, 953-D-I, 954, 955, 955-I, 975, 823-D, 823-D-I, 824, 825

- 4.2.1 All testing will be performed with the gate only, excluding all components of the gate actuator system.
- 4.2.2 All gates will be tested for unseating head only, with the test pressure applied in the unseating direction only.
- 4.2.3 All gates will be tested in the horizontal position.
- 4.2.4 Individually tested gates will be mounted to a suitable backing plate or test fixture.
- 4.2.5 When testing multiple gates of the same size gates may be bolted back to back and tested simultaneously.
- 4.2.6 The test gate will be completely filled with water utilizing an inlet line and valve attached to the backing plate or test fixture. When necessary, the inlet line and valve will be attached temporarily to the actual test gate.

- 4.2.7 While filling the test gate care will be taken to bleed all trapped air from the test gate and test apparatus. Trapped air will be eradicated either by means of a bleeder vent attached to the test apparatus, or by leaving the gate partially opened until it can be determined with certainty that the gate is full of water, and then closing the gate.
- 4.2.8 Once the gate and test apparatus are full of water the specified design unseating head is applied and measured during the test by attaching a calibrated pressure gauge to the inlet line. Test pressure will be equal to $0.433 \times$ feet of design seating head (ex. 0.433×6 feet of design head = 2.6 psi test pressure).
- 4.2.9 Once the gate is filled and at the designated test pressure, and once any leakage has stabilized, leakage will be collected for a time interval of at least one (1) minute.
- 4.2.10 Total amount of collected leakage will be measured utilizing graduated containers and will be compared to the specified maximum allowable leakage (gpm/ft of perimeter seal).
- 4.2.11 If the measured leakage of the gate exceeds the maximum allowable leakage the water will be drained, if necessary, from the test assembly. Adjustments or repairs will then be made to the appropriate gate component and the gate will be re-tested.
- 4.2.12 If the measured leakage is equal to or less than the maximum allowable leakage the gate will be considered as having passed the leakage test.

5. Records:

Results of all tests will be recorded on either form F043-04 "Water Testing Record" or form F043-01 "Slide/Sluice Gate Shop Leakage Test" and will be included as part of the permanent order documentation. At the request of the customer, or to satisfy contract requirements, the leakage test record will be submitted.

Revision Control			
Rev	Change description	Date	By
	Original Release	10/21/2016	RHT II
1	Changed wording of 4.2.2 & 4.2.8	07/23/2019	RHT II
2	Added 935 to paragraph 4.2	05/03/2022	RHT II
3	Revised procedure to include all Whipps 800 Series gates	06/29/2022	RHT II

Document Approval		
Who	Signatures	Date
Manufacturing Director:	Timothy M. Clark	07/23/2019
Engineering Director:	Mark Johnson	07/23/2019
Quality Manager:	Richard H. Truesdell II	07/23/2019



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Test Procedure	
Document Title:	Slide/Sluice Gate Shop Performance/Operational Test Procedure
Procedure No:	043-03
Document Owner:	Quality Manager
Date Created:	08/23/2018

1. Purpose/Objectives:

This procedure describes the process steps and procedure to be followed for performing shop performance/operational tests in accordance with the most current revision of AWWA C561 Fabricated Stainless-Steel Slide Gates.

2. Scope:

This procedure applies, where feasible, to all Whipps, Inc. aluminum and stainless steel Slide/Sluice Gates, compliant with AWWA C561. AWWA C561 requires that all compliant gates be shop performance tested to ensure that the gate is functional.

3. Related Documents:

ANSI/AWWA C561

F043-03 Operational/Functional Shop Gate Test Form

4. Procedure:

- 4.1 All testing will be performed with the gate only, excluding all components of the gate actuator system. The only exception being gates that have inter-connecting shafts. Gates with inter-connecting shafts will be tested with all the components of the gate actuator system, excluding the electric motor actuator.
- 4.2 All gates will be tested in the horizontal position.
- 4.3 All gates will be tested without water load on the gate.
- 4.4 Individually tested gates will be securely mounted to a hydraulically operated test fixture, or some other suitable test fixture.
- 4.5 The test will begin with the gate leaf/disc fully inserted into the gate frame and with the gate in the fully closed position.

- 4.6 From the closed position, operate the gate leaf/disc to the fully opened position and then back to the fully closed position, through one (1) complete open/close cycle.
- 4.7 The gate leaf/disc must move freely in the frame/guide assembly with no evidence of binding or interference.
- 4.8 If the gate leaf/disc does not move freely in the frame/guide assembly with no evidence of binding or interference, adjustments or repairs will then be made to the appropriate gate component and the gate will be re-tested.

5. Records:

Results of all tests will be recorded on form F043-05 "Operational Testing Record" or form F043-03 "Operational / Functional Shop Gate Test" and will be included as part of the permanent order documentation. At the request of the customer, or to satisfy contract requirements, the operational test record will be submitted.

Revision Control			
Rev	Change description	Date	By
	Original Release	08/23/2018	RHT II
01	Added reference to Performance/Operational Test Certificate of Compliance	10/31/2018	RHT II
02	Added "where feasible" – 2. Scope	01/23/2018	RHT II
03	Revised Related Documents - Added reference to F043-03 Operational/Functional Shop Gate Test Form	02/15/2022	RHT II
04	Added paragraph 4.8	05/03/2022	RHT II
05	Added inter-connecting shaft wording to paragraph 4.1	08/25/2022	RHT II

Document Approval		
Who	Signatures	Date
Quality Manager:	Richard H. Truesdell II	08/23/2018
Manufacturing Director:	Timothy M. Clark	08/23/2018
Engineering Director:	Mark Johnson	08/23/2018

Field Leak Test Procedure

- (1) The slide gate to be tested should be inspected to ensure it has been installed in accordance with Whipps installation instructions. The slide should be fully raised and guides inspected to ensure grout has not ingressed into the compression cord groove & all debris has been removed from the sealing surfaces.
- (2) The threaded portion of the stem should be inspected to ensure it has been properly lubricated in line with Whipps O&M instructions. The slide should then be lowered & forced into the invert seal with the equivalent thrust to applying 40 lbs force to the 'T'-bar / handwheel / handcrank etc.
- (3) Once the required test head level has been reached, incoming flow may be stopped and a stopwatch started.
- (4) The slide gate should now be visually inspected for evidence of excessive leakage. No slide gate is drip-tight, each has an allowable leakage rate calculated by multiplying the wetted seal perimeter length in feet by the applicable leakage rate allowance in the gate specification.

Eg. ANSI/AWWA C561 specifies seating / unseating leakage should not exceed 0.10 gpm/ft. Therefore, for example a 4' x 4' gate has a leakage allowance of; $4' + 4' + 4' + 4' = 16' \times 0.10 = 1.6 \text{ gpm}$

- (5) The leakage rate is determined by collecting the leakage passing between the slide and gate seals. This can be done by directly collecting the leakage or creating a reservoir to collect the leakage and collecting after the predetermined test time has expended. If not all leakage can be contained, the remainder should be estimated.
- (6) Test data should be recorded on a report form so that direct comparison between allowable & recorded leakage values can be compared.

- (7) If recorded leakage values appear excessive, visual inspection of the gate should be made to ensure a comparable rate of leakage is observed, otherwise alternative sources of leakage should be investigated, eg. porous concrete etc.
- (8) Test results should be verified & signed off by a witness.



Limitorque

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Customer Data Sheet

SALES ORDER NO. 209929	ORDER LINE 001	DATE ORDERED 05/06/25
CUSTOMER PURCHASE ORDER 32752EL		CUSTOMER NO. 93481
OPERATOR AHIPKINS		CONFIGURATION 229877.0
SOLD TO	WHIPPS INC PO BOX 1058 370 SOUTH ATHOL ROAD ATHOL, MA 01331-0000 UNITED STATES	

END USER **HILO, HI**
END PROJECT **HILO WWTF**

ACTUATOR INFORMATION

Electric Actuator Type MX-10 Series B
Enclosure Type XP-FM-C&D NEMA 3,4,4X,6,IP68
Electric Output RPM 100
Valve Torque 99 Ft Lbs 134 Nm
MX/Gear Box Max Rated Torque 427 Ft Lbs 579 Nm
MX/Gear Box Stall Torque 643 Ft Lbs 872 Nm
Operating Time for 90 degrees 1 Sec
Handwheel TOP Mounted 18 Inch
Open Direction CCW
Lube SHC-632
Gear Box Type V3
Gear Box Ratio 5:1
Gear Box SGA Ratio NA
Gear Box Position NA
Gear Box Base MSS-FA16
Gear Box Output RPM 20.00
Outline Drawing 03-654-0096
Sun Shade Drawing NA
Wiring Diagram WD-MXb-041100-1101
Disconnect Diagram NA
Conduit Entries 1-1/4,1-1/2,1-1/4,NONE
2-Speed Timer N Quik Board N
Control Options Network:DDC-TCP Digital:R4 Analog:NONE
Main Board BLUETOOTH/UPS Wire Controls 3 WIRE MAINT
Disconnect Switch NONE
Warranty STANDARD

MOTOR INFORMATION

Motor Size 9 FT-LBS
Motor Voltage 480
Mtr Phase/Frequency Hz 3/60
Motor Horsepower 0.49
Motor RPM 1800
Locked Rotor Amps (LRA) 7.0
Full Load Amps (FLA) 1.7

Instruction Manual VAIOM000071 on the internet www.limitorque.com
Qualifications: The MX complies with all the test parameters identified in Limitorque MX qualification testing bulletin AIBR000087. Installation of the MX in environments outside the parameters identified in bulletin AIBR000087 voids the warranty.



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TAGGING AND INFORMATION: 02-GAT-1110 & 02-GAT-1120
TAG: A5



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SALES ORDER NO. 209929	ORDER LINE 002	DATE ORDERED 05/06/25
CUSTOMER PURCHASE ORDER 32752EL		CUSTOMER NO. 93481
OPERATOR AHIPKINS		CONFIGURATION 229878.0
SOLD TO	WHIPPS INC PO BOX 1058 370 SOUTH ATHOL ROAD ATHOL, MA 01331-0000 UNITED STATES	

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ACTUATOR INFORMATION

Electric Actuator Type MX-20 SERIES B
Enclosure Type XP-FM-C&D NEMA 3,4,4X,6,IP68
Electric Output RPM 26
Valve Torque 52 Ft Lbs 71 Nm
MX Set Torque 90 Ft Lbs 122 Nm
MX Requested Max Torque 157 Ft Lbs 213 Nm
MX Max Rated Torque 157 Ft Lbs 213 Nm
MX Stall Torque 338 Ft Lbs 458 Nm
Handwheel TOP Mounted 24 Inch
Open Direction CCW
Lube SHC-632
MX Base Type MSS-FA14
MX Drive Nut Type A1
Outline Drawing 03-653-0006
Mounting Base Dwg 03-615-0008-1
Sun Shade Drawing NA
Wiring Diagram WD-MXb-141110-1101
Disconnect Diagram NA
Conduit Entries 1-1/4,1-1/2,1-1/4,NONE
2-Speed Timer N Quik Board N
Control Options: Network:DDC-TCP Digital:R4 Analog:ANALOG BOARD
Main Board :BLUETOOTH/UPS Wire Controls 3 WIRE MAINT
Disconnect Switch NONE
Warranty STANDARD
MOTOR INFORMATION
Motor Size 10 FT-LBS
Motor Voltage 480
Mtr Phase/Frequency Hz 3/60
Motor Horsepower 0.54
Motor RPM 1800
Locked Rotor Amps (LRA) 8.5
Full Load Amps (FLA) 1.4

Instruction Manual VAIOM000071 on internet @ www.limitorque.com
Qualifications: The MX complies with all the test parameters identified in Limitorque bulletin AIBR000087.
Installation of the MX in environments outside the parameters identified in bulletin AIBR000087 voids the warranty.

TAGGING AND INFORMATION:



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SALES ORDER NO. 209929	ORDER LINE 002	DATE ORDERED 05/06/25
CUSTOMER PURCHASE ORDER 32752EL		CUSTOMER NO. 93481
OPERATOR AHIPKINS		CONFIGURATION 229878.0
SOLD TO		WHIPPS INC PO BOX 1058 370 SOUTH ATHOL ROAD ATHOL, MA 01331-0000 UNITED STATES

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TAGGING AND INFORMATION: 02-GAT-1130
TAG: B5



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SALES ORDER NO. 209929	ORDER LINE 003	DATE ORDERED 05/06/25
CUSTOMER PURCHASE ORDER 32752EL		CUSTOMER NO. 93481
OPERATOR AHIPKINS		CONFIGURATION 229879.0
SOLD TO	WHIPPS INC PO BOX 1058 370 SOUTH ATHOL ROAD ATHOL, MA 01331-0000 UNITED STATES	

END USER **HILO, HI**
END PROJECT **HILO WWTF**

ACTUATOR INFORMATION

Electric Actuator Type MX-10 Series B
Enclosure Type XP-FM-C&D NEMA 3,4,4X,6,IP68
Electric Output RPM 100
Valve Torque 78 Ft Lbs 106 Nm
MX/Gear Box Max Rated Torque 293 Ft Lbs 397 Nm
MX/Gear Box Stall Torque 441 Ft Lbs 598 Nm
Operating Time for 90 degrees 1 Sec
Handwheel TOP Mounted 18 Inch
Open Direction CCW
Lube SHC-632
Gear Box Type V1
Gear Box Ratio 3.5:1
Gear Box SGA Ratio NA
Gear Box Position NA
Gear Box Base MSS-FA14
Gear Box Output RPM 28.57
Outline Drawing 03-654-0094
Sun Shade Drawing NA
Wiring Diagram WD-MXb-041100-1101
Disconnect Diagram NA
Conduit Entries 1-1/4,1-1/2,1-1/4,NONE
2-Speed Timer N Quik Board N
Control Options Network:DDC-TCP Digital:R4 Analog:NONE
Main Board BLUETOOTH/UPS Wire Controls 3 WIRE MAINT
Disconnect Switch NONE
Warranty STANDARD

MOTOR INFORMATION

Motor Size 9 FT-LBS
Motor Voltage 480
Mtr Phase/Frequency Hz 3/60
Motor Horsepower 0.49
Motor RPM 1800
Locked Rotor Amps (LRA) 7.0
Full Load Amps (FLA) 1.7

Instruction Manual VAIOM000071 on the internet www.limitorque.com
Qualifications: The MX complies with all the test parameters identified in Limitorque MX qualification testing bulletin AIBR000087. Installation of the MX in environments outside the parameters identified in bulletin AIBR000087 voids the warranty.



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Customer Data Sheet

SALES ORDER NO. 209929	ORDER LINE 003	DATE ORDERED 05/06/25
CUSTOMER PURCHASE ORDER 32752EL		CUSTOMER NO. 93481
OPERATOR AHIPKINS		CONFIGURATION 229879.0
SOLD TO		WHIPPS INC PO BOX 1058 370 SOUTH ATHOL ROAD ATHOL, MA 01331-0000 UNITED STATES

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TAGGING AND INFORMATION: 02-GAT-1115 & 02-GAT-1125
TAG: C5



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Customer Data Sheet

SALES ORDER NO. 209929	ORDER LINE 004	DATE ORDERED 05/06/25
CUSTOMER PURCHASE ORDER 32752EL		CUSTOMER NO. 93481
OPERATOR AHIPKINS		CONFIGURATION 229880.0
SOLD TO	WHIPPS INC PO BOX 1058 370 SOUTH ATHOL ROAD ATHOL, MA 01331-0000 UNITED STATES	

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END PROJECT **HILO WWTF**

ACTUATOR INFORMATION

Electric Actuator Type MX-10 Series B
Enclosure Type XP-FM-C&D NEMA 3,4,4X,6,IP68
Electric Output RPM 100
Valve Torque 86 Ft Lbs 117 Nm
MX/Gear Box Max Rated Torque 293 Ft Lbs 397 Nm
MX/Gear Box Stall Torque 441 Ft Lbs 598 Nm
Operating Time for 90 degrees 1 Sec
Handwheel TOP Mounted 18 Inch
Open Direction CCW
Lube SHC-632
Gear Box Type V1
Gear Box Ratio 3.5:1
Gear Box SGA Ratio NA
Gear Box Position NA
Gear Box Base MSS-FA14
Gear Box Output RPM 28.57
Outline Drawing 03-654-0094
Sun Shade Drawing NA
Wiring Diagram WD-MXb-041100-1101
Disconnect Diagram NA
Conduit Entries 1-1/4,1-1/2,1-1/4,NONE
2-Speed Timer N Quik Board N
Control Options Network:DDC-TCP Digital:R4 Analog:NONE
Main Board BLUETOOTH/UPS Wire Controls 3 WIRE MAINT
Disconnect Switch NONE
Warranty STANDARD

MOTOR INFORMATION

Motor Size 9 FT-LBS
Motor Voltage 480
Mtr Phase/Frequency Hz 3/60
Motor Horsepower 0.49
Motor RPM 1800
Locked Rotor Amps (LRA) 7.0
Full Load Amps (FLA) 1.7

Instruction Manual VAIOM000071 on the internet www.limitorque.com
Qualifications: The MX complies with all the test parameters identified in Limitorque MX qualification testing bulletin AIBR000087. Installation of the MX in environments outside the parameters identified in bulletin AIBR000087 voids the warranty.



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Customer Data Sheet

SALES ORDER NO. 209929	ORDER LINE 004	DATE ORDERED 05/06/25
CUSTOMER PURCHASE ORDER 32752EL		CUSTOMER NO. 93481
OPERATOR AHIPKINS		CONFIGURATION 229880.0
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TAGGING AND INFORMATION: 02-GAT-1135
TAG: D5



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SALES ORDER NO. 209929	ORDER LINE 005	DATE ORDERED 05/06/25
CUSTOMER PURCHASE ORDER 32752EL		CUSTOMER NO. 93481
OPERATOR AHIPKINS		CONFIGURATION 229881.0
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ACTUATOR INFORMATION

Electric Actuator Type MX-10 Series B
Enclosure Type XP-FM-C&D NEMA 3,4,4X,6,IP68
Electric Output RPM 100
Valve Torque 81 Ft Lbs 110 Nm
MX/Gear Box Max Rated Torque 293 Ft Lbs 397 Nm
MX/Gear Box Stall Torque 441 Ft Lbs 598 Nm
Operating Time for 90 degrees 1 Sec
Handwheel TOP Mounted 18 Inch
Open Direction CCW
Lube SHC-632
Gear Box Type V1
Gear Box Ratio 3.5:1
Gear Box SGA Ratio NA
Gear Box Position NA
Gear Box Base MSS-FA14
Gear Box Output RPM 28.57
Outline Drawing 03-654-0094
Sun Shade Drawing NA
Wiring Diagram WD-MXb-041100-1101
Disconnect Diagram NA
Conduit Entries 1-1/4,1-1/2,1-1/4,NONE
2-Speed Timer N Quik Board N
Control Options Network:DDC-TCP Digital:R4 Analog:NONE
Main Board BLUETOOTH/UPS Wire Controls 3 WIRE MAINT
Disconnect Switch NONE
Warranty STANDARD
MOTOR INFORMATION
Motor Size 9 FT-LBS
Motor Voltage 480
Mtr Phase/Frequency Hz 3/60
Motor Horsepower 0.49
Motor RPM 1800
Locked Rotor Amps (LRA) 7.0
Full Load Amps (FLA) 1.7

Instruction Manual VAIOM000071 on the internet www.limitorque.com
Qualifications: The MX complies with all the test parameters identified in Limitorque MX qualification testing bulletin AIBR000087. Installation of the MX in environments outside the parameters identified in bulletin AIBR000087 voids the warranty.

Customer Data Sheet

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SALES ORDER NO. 209929	ORDER LINE 005	DATE ORDERED 05/06/25
CUSTOMER PURCHASE ORDER 32752EL		CUSTOMER NO. 93481
OPERATOR AHIPKINS		CONFIGURATION 229881.0
SOLD TO	WHIPPS INC PO BOX 1058 370 SOUTH ATHOL ROAD ATHOL, MA 01331-0000 UNITED STATES	

END USER **HILO, HI**
END PROJECT **HILO WWTF**

TAGGING AND INFORMATION: 02-GAT-1210 & 02-GAT-1220
TAG: F5



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Customer Data Sheet

SALES ORDER NO. 209929	ORDER LINE 006	DATE ORDERED 05/06/25
CUSTOMER PURCHASE ORDER 32752EL		CUSTOMER NO. 93481
OPERATOR AHIPKINS		CONFIGURATION 229882.0
SOLD TO	WHIPPS INC PO BOX 1058 370 SOUTH ATHOL ROAD ATHOL, MA 01331-0000 UNITED STATES	

END USER **HILO, HI**
END PROJECT **HILO WWTF**

ACTUATOR INFORMATION

Electric Actuator Type MX-10 Series B
Enclosure Type XP-FM-C&D NEMA 3,4,4X,6,IP68
Electric Output RPM 100
Valve Torque 72 Ft Lbs 98 Nm
MX/Gear Box Max Rated Torque 293 Ft Lbs 397 Nm
MX/Gear Box Stall Torque 441 Ft Lbs 598 Nm
Operating Time for 90 degrees 1 Sec
Handwheel TOP Mounted 18 Inch
Open Direction CCW
Lube SHC-632
Gear Box Type V1
Gear Box Ratio 3.5:1
Gear Box SGA Ratio NA
Gear Box Position NA
Gear Box Base MSS-FA14
Gear Box Output RPM 28.57
Outline Drawing 03-654-0094
Sun Shade Drawing NA
Wiring Diagram WD-MXb-041100-1101
Disconnect Diagram NA
Conduit Entries 1-1/4,1-1/2,1-1/4,NONE
2-Speed Timer N Quik Board N
Control Options Network:DDC-TCP Digital:R4 Analog:NONE
Main Board BLUETOOTH/UPS Wire Controls 3 WIRE MAINT
Disconnect Switch NONE
Warranty STANDARD

MOTOR INFORMATION

Motor Size 9 FT-LBS
Motor Voltage 480
Mtr Phase/Frequency Hz 3/60
Motor Horsepower 0.49
Motor RPM 1800
Locked Rotor Amps (LRA) 7.0
Full Load Amps (FLA) 1.7

Instruction Manual VAIOM000071 on the internet www.limitorque.com
Qualifications: The MX complies with all the test parameters identified in Limitorque MX qualification testing bulletin AIBR000087. Installation of the MX in environments outside the parameters identified in bulletin AIBR000087 voids the warranty.



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Customer Data Sheet

SALES ORDER NO. 209929	ORDER LINE 006	DATE ORDERED 05/06/25
CUSTOMER PURCHASE ORDER 32752EL		CUSTOMER NO. 93481
OPERATOR AHIPKINS		CONFIGURATION 229882.0
SOLD TO		WHIPPS INC PO BOX 1058 370 SOUTH ATHOL ROAD ATHOL, MA 01331-0000 UNITED STATES

END USER **HILO, HI**
END PROJECT **HILO WWTF**

TAGGING AND INFORMATION: 02-GAT-1215 & 02-GAT-1225
TAG:G5



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Customer Data Sheet

SALES ORDER NO. 209929	ORDER LINE 007	DATE ORDERED 05/06/25
CUSTOMER PURCHASE ORDER 32752EL		CUSTOMER NO. 93481
OPERATOR AHIPKINS		CONFIGURATION 229883.0
SOLD TO	WHIPPS INC PO BOX 1058 370 SOUTH ATHOL ROAD ATHOL, MA 01331-0000 UNITED STATES	

END USER **HILO, HI**
END PROJECT **HILO WWTF**

ACTUATOR INFORMATION

Electric Actuator Type MX-05 Series B
Enclosure Type XP-FM-C&D NEMA 3,4,4X,6,IP68
Electric Output RPM 100
Valve Torque 37 Ft Lbs 50 Nm
MX/Gear Box Max Rated Torque 128 Ft Lbs 174 Nm
MX/Gear Box Stall Torque 194 Ft Lbs 263 Nm
Operating Time for 90 degrees 1 Sec
Handwheel TOP Mounted 12 Inch
Open Direction CCW
Lube SHC-632
Gear Box Type V1
Gear Box Ratio 3.5:1
Gear Box SGA Ratio NA
Gear Box Position NA
Gear Box Base MSS-FA14
Gear Box Output RPM 28.57
Outline Drawing 03-654-0094
Sun Shade Drawing NA
Wiring Diagram WD-MXb-041100-1101
Disconnect Diagram NA
Conduit Entries 1-1/4,1-1/2,1-1/4,NONE
2-Speed Timer N Quik Board N
Control Options Network:DDC-TCP Digital:R4 Analog:NONE
Main Board BLUETOOTH/UPS Wire Controls 3 WIRE MAINT
Disconnect Switch NONE
Warranty STANDARD

MOTOR INFORMATION

Motor Size 4 FT-LBS
Motor Voltage 480
Mtr Phase/Frequency Hz 3/60
Motor Horsepower 0.22
Motor RPM 1800
Locked Rotor Amps (LRA) 3.2
Full Load Amps (FLA) 0.9

Instruction Manual VAION000071 on the internet www.limitorque.com
Qualifications: The MX complies with all the test parameters identified in Limitorque MX qualification testing bulletin AIBR000087. Installation of the MX in environments outside the parameters identified in bulletin AIBR000087 voids the warranty.

Customer Data Sheet

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SALES ORDER NO. 209929	ORDER LINE 007	DATE ORDERED 05/06/25
CUSTOMER PURCHASE ORDER 32752EL		CUSTOMER NO. 93481
OPERATOR AHIPKINS		CONFIGURATION 229883.0
SOLD TO		WHIPPS INC PO BOX 1058 370 SOUTH ATHOL ROAD ATHOL, MA 01331-0000 UNITED STATES

END USER **HILO, HI**
END PROJECT **HILO WWTF**

TAGGING AND INFORMATION: 03-GAT-1230
TAG: H5



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Customer Data Sheet

SALES ORDER NO. 209929	ORDER LINE 008	DATE ORDERED 05/06/25
CUSTOMER PURCHASE ORDER 32752EL		CUSTOMER NO. 93481
OPERATOR AHIPKINS		CONFIGURATION 229884.0
SOLD TO	WHIPPS INC PO BOX 1058 370 SOUTH ATHOL ROAD ATHOL, MA 01331-0000 UNITED STATES	

END USER **HILO, HI**
END PROJECT **HILO WWTF**

ACTUATOR INFORMATION

Electric Actuator Type MX-05 Series B
Enclosure Type XP-FM-C&D NEMA 3,4,4X,6,IP68
Electric Output RPM 100
Valve Torque 36 Ft Lbs 49 Nm
MX/Gear Box Max Rated Torque 128 Ft Lbs 174 Nm
MX/Gear Box Stall Torque 194 Ft Lbs 263 Nm
Operating Time for 90 degrees 1 Sec
Handwheel TOP Mounted 12 Inch
Open Direction CCW
Lube SHC-632
Gear Box Type V1
Gear Box Ratio 3.5:1
Gear Box SGA Ratio NA
Gear Box Position NA
Gear Box Base MSS-FA14
Gear Box Output RPM 28.57
Outline Drawing 03-654-0094
Sun Shade Drawing NA
Wiring Diagram WD-MXb-041100-1101
Disconnect Diagram NA
Conduit Entries 1-1/4,1-1/2,1-1/4,NONE
2-Speed Timer N Quik Board N
Control Options Network:DDC-TCP Digital:R4 Analog:NONE
Main Board BLUETOOTH/UPS Wire Controls 3 WIRE MAINT
Disconnect Switch NONE
Warranty STANDARD
MOTOR INFORMATION
Motor Size 4 FT-LBS
Motor Voltage 480
Mtr Phase/Frequency Hz 3/60
Motor Horsepower 0.22
Motor RPM 1800
Locked Rotor Amps (LRA) 3.2
Full Load Amps (FLA) 0.9

Instruction Manual VAIOM000071 on the internet www.limitorque.com
Qualifications: The MX complies with all the test parameters identified in Limitorque MX qualification testing bulletin AIBR000087. Installation of the MX in environments outside the parameters identified in bulletin AIBR000087 voids the warranty.



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Customer Data Sheet

SALES ORDER NO. 209929	ORDER LINE 008	DATE ORDERED 05/06/25
CUSTOMER PURCHASE ORDER 32752EL		CUSTOMER NO. 93481
OPERATOR AHIPKINS		CONFIGURATION 229884.0
SOLD TO		WHIPPS INC PO BOX 1058 370 SOUTH ATHOL ROAD ATHOL, MA 01331-0000 UNITED STATES

END USER **HILO, HI**
END PROJECT **HILO WWTF**

TAGGING AND INFORMATION: 02-GAT-1240
TAG: J5



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Customer Data Sheet

SALES ORDER NO. 209929	ORDER LINE 009	DATE ORDERED 05/06/25
CUSTOMER PURCHASE ORDER 32752EL		CUSTOMER NO. 93481
OPERATOR AHIPKINS		CONFIGURATION 229885.0
SOLD TO	WHIPPS INC PO BOX 1058 370 SOUTH ATHOL ROAD ATHOL, MA 01331-0000 UNITED STATES	

END USER **HILO, HI**
END PROJECT **HILO WWTF**

ACTUATOR INFORMATION

Electric Actuator Type MX-05 Series B
Enclosure Type XP-FM-C&D NEMA 3,4,4X,6,IP68
Electric Output RPM 100
Valve Torque 13 Ft Lbs 18 Nm
MX/Gear Box Max Rated Torque 128 Ft Lbs 174 Nm
MX/Gear Box Stall Torque 194 Ft Lbs 263 Nm
Operating Time for 90 degrees 1 Sec
Handwheel TOP Mounted 12 Inch
Open Direction CCW
Lube SHC-632
Gear Box Type V1
Gear Box Ratio 3.5:1
Gear Box SGA Ratio NA
Gear Box Position NA
Gear Box Base MSS-FA14
Gear Box Output RPM 28.57
Outline Drawing 03-654-0094
Sun Shade Drawing NA
Wiring Diagram WD-MXb-041100-1101
Disconnect Diagram NA
Conduit Entries 1-1/4,1-1/2,1-1/4,NONE
2-Speed Timer N Quik Board N
Control Options Network:DDC-TCP Digital:R4 Analog:NONE
Main Board BLUETOOTH/UPS Wire Controls 3 WIRE MAINT
Disconnect Switch NONE
Warranty STANDARD

MOTOR INFORMATION

Motor Size 4 FT-LBS
Motor Voltage 480
Mtr Phase/Frequency Hz 3/60
Motor Horsepower 0.22
Motor RPM 1800
Locked Rotor Amps (LRA) 3.2
Full Load Amps (FLA) 0.9

Instruction Manual VAIOM000071 on the internet www.limitorque.com
Qualifications: The MX complies with all the test parameters identified in Limitorque MX qualification testing bulletin AIBR000087. Installation of the MX in environments outside the parameters identified in bulletin AIBR000087 voids the warranty.



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SALES ORDER NO. 209929	ORDER LINE 009	DATE ORDERED 05/06/25
CUSTOMER PURCHASE ORDER 32752EL		CUSTOMER NO. 93481
OPERATOR AHIPKINS		CONFIGURATION 229885.0
SOLD TO		WHIPPS INC PO BOX 1058 370 SOUTH ATHOL ROAD ATHOL, MA 01331-0000 UNITED STATES

END USER **HILO, HI**
END PROJECT **HILO WWTF**

TAGGING AND INFORMATION: 03-FG-1110, 03-FG-1120 & 03-FG-1130
TAG: L5



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SALES ORDER NO. 209929	ORDER LINE 010	DATE ORDERED 05/06/25
CUSTOMER PURCHASE ORDER 32752EL		CUSTOMER NO. 93481
OPERATOR AHIPKINS		CONFIGURATION 229886.0
SOLD TO	WHIPPS INC PO BOX 1058 370 SOUTH ATHOL ROAD ATHOL, MA 01331-0000 UNITED STATES	

END USER **HILO, HI**
END PROJECT **HILO WWTF**

ACTUATOR INFORMATION

Electric Actuator Type MX-05 Series B
Enclosure Type XP-FM-C&D NEMA 3,4,4X,6,IP68
Electric Output RPM 100
Valve Torque 47 Ft Lbs 64 Nm
MX/Gear Box Max Rated Torque 128 Ft Lbs 174 Nm
MX/Gear Box Stall Torque 194 Ft Lbs 263 Nm
Operating Time for 90 degrees 1 Sec
Handwheel TOP Mounted 12 Inch
Open Direction CCW
Lube SHC-632
Gear Box Type V1
Gear Box Ratio 3.5:1
Gear Box SGA Ratio NA
Gear Box Position NA
Gear Box Base MSS-FA14
Gear Box Output RPM 28.57
Outline Drawing 03-654-0094
Sun Shade Drawing NA
Wiring Diagram WD-MXb-041100-1101
Disconnect Diagram NA
Conduit Entries 1-1/4,1-1/2,1-1/4,NONE
2-Speed Timer N Quik Board N
Control Options Network:DDC-TCP Digital:R4 Analog:NONE
Main Board BLUETOOTH/UPS Wire Controls 3 WIRE MAINT
Disconnect Switch NONE
Warranty STANDARD

MOTOR INFORMATION

Motor Size 4 FT-LBS
Motor Voltage 480
Mtr Phase/Frequency Hz 3/60
Motor Horsepower 0.22
Motor RPM 1800
Locked Rotor Amps (LRA) 3.2
Full Load Amps (FLA) 0.9

Instruction Manual VAIOM000071 on the internet www.limitorque.com
Qualifications: The MX complies with all the test parameters identified in Limitorque MX qualification testing bulletin AIBR000087. Installation of the MX in environments outside the parameters identified in bulletin AIBR000087 voids the warranty.



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Customer Data Sheet

SALES ORDER NO. 209929	ORDER LINE 010	DATE ORDERED 05/06/25
CUSTOMER PURCHASE ORDER 32752EL		CUSTOMER NO. 93481
OPERATOR AHIPKINS		CONFIGURATION 229886.0
SOLD TO		WHIPPS INC PO BOX 1058 370 SOUTH ATHOL ROAD ATHOL, MA 01331-0000 UNITED STATES

END USER **HILO, HI**
END PROJECT **HILO WWTF**

TAGGING AND INFORMATION: 03-FG-1440 & 03-FG-1445
TAG: M5



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Customer Data Sheet

SALES ORDER NO. 209929	ORDER LINE 011	DATE ORDERED 05/06/25
CUSTOMER PURCHASE ORDER 32752EL		CUSTOMER NO. 93481
OPERATOR AHIPKINS		CONFIGURATION 229887.0
SOLD TO	WHIPPS INC PO BOX 1058 370 SOUTH ATHOL ROAD ATHOL, MA 01331-0000 UNITED STATES	

END USER **HILO, HI**
END PROJECT **HILO WWTF**

ACTUATOR INFORMATION

Electric Actuator Type MX-05 Series B
Enclosure Type XP-FM-C&D NEMA 3,4,4X,6,IP68
Electric Output RPM 100
Valve Torque 33 Ft Lbs 45 Nm
MX/Gear Box Max Rated Torque 128 Ft Lbs 174 Nm
MX/Gear Box Stall Torque 194 Ft Lbs 263 Nm
Operating Time for 90 degrees 1 Sec
Handwheel TOP Mounted 12 Inch
Open Direction CCW
Lube SHC-632
Gear Box Type V1
Gear Box Ratio 3.5:1
Gear Box SGA Ratio NA
Gear Box Position NA
Gear Box Base MSS-FA14
Gear Box Output RPM 28.57
Outline Drawing 03-654-0094
Sun Shade Drawing NA
Wiring Diagram WD-MXb-041100-1101
Disconnect Diagram NA
Conduit Entries 1-1/4,1-1/2,1-1/4,NONE
2-Speed Timer N Quik Board N
Control Options Network:DDC-TCP Digital:R4 Analog:NONE
Main Board BLUETOOTH/UPS Wire Controls 3 WIRE MAINT
Disconnect Switch NONE
Warranty STANDARD
MOTOR INFORMATION
Motor Size 4 FT-LBS
Motor Voltage 480
Mtr Phase/Frequency Hz 3/60
Motor Horsepower 0.22
Motor RPM 1800
Locked Rotor Amps (LRA) 3.2
Full Load Amps (FLA) 0.9

Instruction Manual VAIOM000071 on the internet www.limitorque.com
Qualifications: The MX complies with all the test parameters identified in Limitorque MX qualification testing bulletin AIBR000087. Installation of the MX in environments outside the parameters identified in bulletin AIBR000087 voids the warranty.



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Customer Data Sheet

SALES ORDER NO. 209929	ORDER LINE 011	DATE ORDERED 05/06/25
CUSTOMER PURCHASE ORDER 32752EL		CUSTOMER NO. 93481
OPERATOR AHIPKINS		CONFIGURATION 229887.0
SOLD TO		WHIPPS INC PO BOX 1058 370 SOUTH ATHOL ROAD ATHOL, MA 01331-0000 UNITED STATES

END USER **HILO, HI**
END PROJECT **HILO WWTF**

TAGGING AND INFORMATION: 03-FG-1495
TAG: P5



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Customer Data Sheet

SALES ORDER NO. 209929	ORDER LINE 012	DATE ORDERED 05/06/25
CUSTOMER PURCHASE ORDER 32752EL		CUSTOMER NO. 93481
OPERATOR AHIPKINS		CONFIGURATION 229888.0
SOLD TO	WHIPPS INC PO BOX 1058 370 SOUTH ATHOL ROAD ATHOL, MA 01331-0000 UNITED STATES	

END USER **HILO, HI**
END PROJECT **HILO WWTF**

ACTUATOR INFORMATION

Electric Actuator Type MX-10 Series B
Enclosure Type XP-FM-C&D NEMA 3,4,4X,6,IP68
Electric Output RPM 100
Valve Torque 49 Ft Lbs 66 Nm
MX/Gear Box Max Rated Torque 293 Ft Lbs 397 Nm
MX/Gear Box Stall Torque 441 Ft Lbs 598 Nm
Operating Time for 90 degrees 1 Sec
Handwheel TOP Mounted 18 Inch
Open Direction CCW
Lube SHC-632
Gear Box Type V1
Gear Box Ratio 3.5:1
Gear Box SGA Ratio NA
Gear Box Position NA
Gear Box Base MSS-FA14
Gear Box Output RPM 28.57
Outline Drawing 03-654-0094
Sun Shade Drawing NA
Wiring Diagram WD-MXb-041100-1101
Disconnect Diagram NA
Conduit Entries 1-1/4,1-1/2,1-1/4,NONE
2-Speed Timer N Quik Board N
Control Options Network:DDC-TCP Digital:R4 Analog:NONE
Main Board BLUETOOTH/UPS Wire Controls 3 WIRE MAINT
Disconnect Switch NONE
Warranty STANDARD
MOTOR INFORMATION
Motor Size 9 FT-LBS
Motor Voltage 480
Mtr Phase/Frequency Hz 3/60
Motor Horsepower 0.49
Motor RPM 1800
Locked Rotor Amps (LRA) 7.0
Full Load Amps (FLA) 1.7

Instruction Manual VAIOM000071 on the internet www.limitorque.com
Qualifications: The MX complies with all the test parameters identified in Limitorque MX qualification testing bulletin AIBR000087. Installation of the MX in environments outside the parameters identified in bulletin AIBR000087 voids the warranty.



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Customer Data Sheet

SALES ORDER NO. 209929	ORDER LINE 012	DATE ORDERED 05/06/25
CUSTOMER PURCHASE ORDER 32752EL		CUSTOMER NO. 93481
OPERATOR AHIPKINS		CONFIGURATION 229888.0
SOLD TO		WHIPPS INC PO BOX 1058 370 SOUTH ATHOL ROAD ATHOL, MA 01331-0000 UNITED STATES

END USER **HILO, HI**
END PROJECT **HILO WWTF**

TAGGING AND INFORMATION: 03-FG-1410, 03-GAT-1420 & 03-FG-1430
TAG: R5



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Customer Data Sheet

SALES ORDER NO. 209929	ORDER LINE 013	DATE ORDERED 05/06/25
CUSTOMER PURCHASE ORDER 32752EL		CUSTOMER NO. 93481
OPERATOR AHIPKINS		CONFIGURATION 229889.0
SOLD TO	WHIPPS INC PO BOX 1058 370 SOUTH ATHOL ROAD ATHOL, MA 01331-0000 UNITED STATES	

END USER **HILO, HI**
END PROJECT **HILO WWTF**

ACTUATOR INFORMATION

Electric Actuator Type MX-20 SERIES B
Enclosure Type XP-FM-C&D NEMA 3,4,4X,6,IP68
Electric Output RPM 26
Valve Torque 10 Ft Lbs 14 Nm
MX Set Torque 90 Ft Lbs 122 Nm
MX Requested Max Torque 225 Ft Lbs 305 Nm
MX Max Rated Torque 225 Ft Lbs 305 Nm
MX Stall Torque 338 Ft Lbs 458 Nm
Handwheel TOP Mounted 24 Inch
Open Direction CCW
Lube SHC-632
MX Base Type MSS-FA14
MX Drive Nut Type B4
Outline Drawing 03-653-0006
Mounting Base Dwg 03-615-0008-1
Sun Shade Drawing NA
Wiring Diagram WD-MXb-041110-1101
Disconnect Diagram NA
Conduit Entries 1-1/4,1-1/2,1-1/4,NONE
2-Speed Timer N Quik Board N
Control Options: Network:DDC-TCP Digital:R4 Analog:ANALOG BOARD
Main Board :BLUETOOTH/UPS Wire Controls 3 WIRE MAINT
Disconnect Switch NONE
Warranty STANDARD
MOTOR INFORMATION
Motor Size 10 FT-LBS
Motor Voltage 480
Mtr Phase/Frequency Hz 3/60
Motor Horsepower 0.54
Motor RPM 1800
Locked Rotor Amps (LRA) 8.5
Full Load Amps (FLA) 1.4

Instruction Manual VAIOM000071 on internet @ www.limitorque.com
Qualifications: The MX complies with all the test parameters identified in Limitorque bulletin AIBR000087.
Installation of the MX in environments outside the parameters identified in bulletin AIBR000087 voids the warranty.

TAGGING AND INFORMATION:

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SALES ORDER NO. 209929	ORDER LINE 013	DATE ORDERED 05/06/25
CUSTOMER PURCHASE ORDER 32752EL		CUSTOMER NO. 93481
OPERATOR AHIPKINS		CONFIGURATION 229889.0
SOLD TO	WHIPPS INC PO BOX 1058 370 SOUTH ATHOL ROAD ATHOL, MA 01331-0000 UNITED STATES	

END USER **HILO, HI**
END PROJECT **HILO WWTF**



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Customer Data Sheet

SALES ORDER NO. 209929	ORDER LINE 014	DATE ORDERED 05/06/25
CUSTOMER PURCHASE ORDER 32752EL		CUSTOMER NO. 93481
OPERATOR AHIPKINS		CONFIGURATION 229890.0
SOLD TO	WHIPPS INC PO BOX 1058 370 SOUTH ATHOL ROAD ATHOL, MA 01331-0000 UNITED STATES	

END USER **HILO, HI**
END PROJECT **HILO WWTF**

ACTUATOR INFORMATION

Electric Actuator Type MX-10 SERIES B
Enclosure Type XP-FM-C&D NEMA 3,4,4X,6,IP68
Electric Output RPM 100
Valve Torque 10 Ft Lbs 14 Nm
MX Set Torque 36 Ft Lbs 49 Nm
MX Requested Max Torque 89 Ft Lbs 121 Nm
MX Max Rated Torque 89 Ft Lbs 121 Nm
MX Stall Torque 134 Ft Lbs 182 Nm
Handwheel TOP Mounted 18 Inch
Open Direction CCW
Lube SHC-632
MX Base Type MSS-FA10
MX Drive Nut Type B4
Outline Drawing 03-653-0003
Mounting Base Dwg 03-615-0005-1
Sun Shade Drawing NA
Wiring Diagram WD-MXb-041100-1101
Disconnect Diagram NA
Conduit Entries 1-1/4,1-1/2,1-1/4,NONE
2-Speed Timer N Quik Board N
Control Options: Network:DDC-TCP Digital:R4 Analog:NONE
Main Board :BLUETOOTH/UPS Wire Controls 3 WIRE MAINT
Disconnect Switch NONE
Warranty STANDARD
MOTOR INFORMATION
Motor Size 9 FT-LBS
Motor Voltage 480
Mtr Phase/Frequency Hz 3/60
Motor Horsepower 0.49
Motor RPM 1800
Locked Rotor Amps (LRA) 7.0
Full Load Amps (FLA) 1.7

Instruction Manual VAIOM000071 on internet @ www.limitorque.com
Qualifications: The MX complies with all the test parameters identified in Limitorque bulletin AIBR000087.
Installation of the MX in environments outside the parameters identified in bulletin AIBR000087 voids the warranty.

TAGGING AND INFORMATION:

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SALES ORDER NO. 209929	ORDER LINE 014	DATE ORDERED 05/06/25
CUSTOMER PURCHASE ORDER 32752EL		CUSTOMER NO. 93481
OPERATOR AHIPKINS		CONFIGURATION 229890.0
SOLD TO		WHIPPS INC PO BOX 1058 370 SOUTH ATHOL ROAD ATHOL, MA 01331-0000 UNITED STATES

END USER **HILO, HI**
END PROJECT **HILO WWTF**



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Customer Data Sheet

SALES ORDER NO. 209929	ORDER LINE 015	DATE ORDERED 05/06/25
CUSTOMER PURCHASE ORDER 32752EL		CUSTOMER NO. 93481
OPERATOR AHIPKINS		CONFIGURATION 229891.0
SOLD TO	WHIPPS INC PO BOX 1058 370 SOUTH ATHOL ROAD ATHOL, MA 01331-0000 UNITED STATES	

END USER **HILO, HI**
END PROJECT **HILO WWTF**

ACTUATOR INFORMATION

Electric Actuator Type MX-05 SERIES B
Enclosure Type XP-FM-C&D NEMA 3,4,4X,6,IP68
Electric Output RPM 100
Valve Torque 5 Ft Lbs 7 Nm
MX Set Torque 16 Ft Lbs 22 Nm
MX Requested Max Torque 39 Ft Lbs 53 Nm
MX Max Rated Torque 39 Ft Lbs 53 Nm
MX Stall Torque 59 Ft Lbs 80 Nm
Handwheel TOP Mounted 12 Inch
Open Direction CCW
Lube SHC-632
MX Base Type MSS-FA10
MX Drive Nut Type B4
Outline Drawing 03-653-0001
Mounting Base Dwg 03-615-0002-1
Sun Shade Drawing NA
Wiring Diagram WD-MXb-041100-1101
Disconnect Diagram NA
Conduit Entries 1-1/4,1-1/2,1-1/4,NONE
2-Speed Timer N Quik Board N
Control Options: Network:DDC-TCP Digital:R4 Analog:NONE
Main Board :BLUETOOTH/UPS Wire Controls 3 WIRE MAINT
Disconnect Switch NONE
Warranty STANDARD
MOTOR INFORMATION
Motor Size 4 FT-LBS
Motor Voltage 480
Mtr Phase/Frequency Hz 3/60
Motor Horsepower 0.22
Motor RPM 1800
Locked Rotor Amps (LRA) 3.2
Full Load Amps (FLA) 0.9

Instruction Manual VAIOM000071 on internet @ www.limitorque.com
Qualifications: The MX complies with all the test parameters identified in Limitorque bulletin AIBR000087.
Installation of the MX in environments outside the parameters identified in bulletin AIBR000087 voids the warranty.

TAGGING AND INFORMATION:

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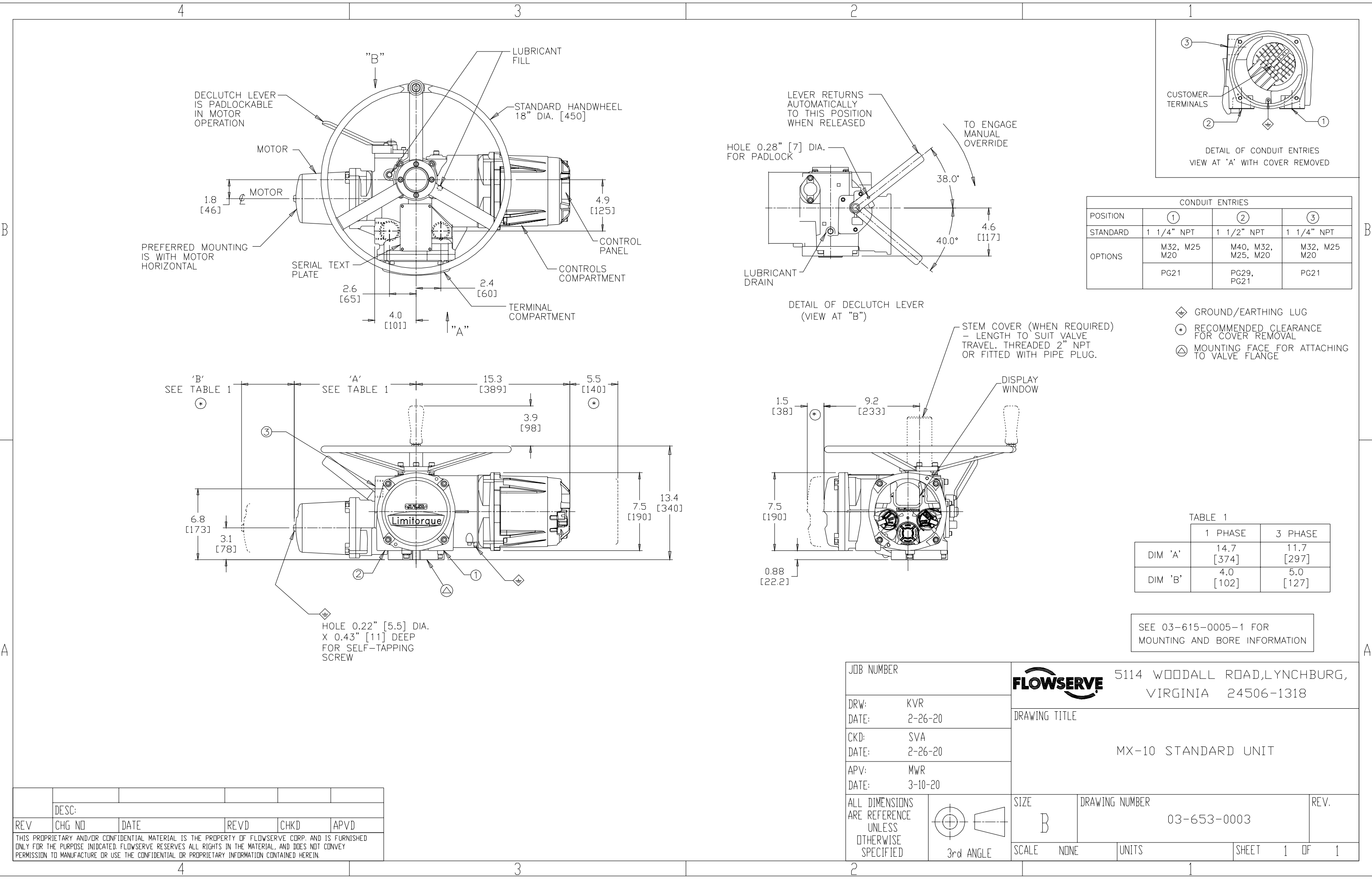
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Lynchburg, VA 24502-2248
434-845-9736 www.flowserve.com

Phone: 434-528-4400
Fax:

SALES ORDER NO. 209929	ORDER LINE 015	DATE ORDERED 05/06/25
CUSTOMER PURCHASE ORDER 32752EL		CUSTOMER NO. 93481
OPERATOR AHIPKINS		CONFIGURATION 229891.0
SOLD TO		WHIPPS INC PO BOX 1058 370 SOUTH ATHOL ROAD ATHOL, MA 01331-0000 UNITED STATES

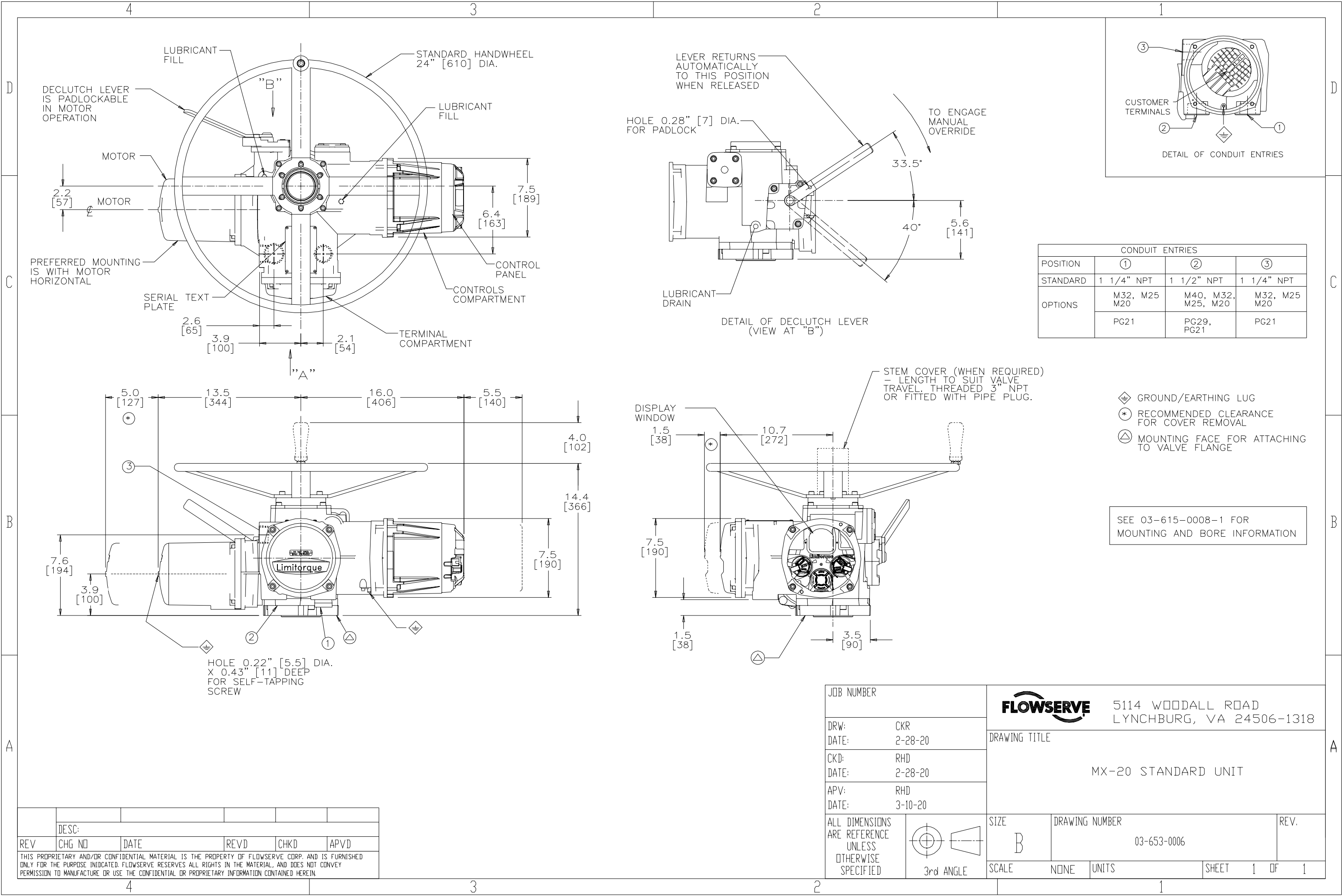
END USER **HILO, HI**
END PROJECT **HILO WWTF**

	DESC:				
REV	CHG NO	DATE	REVD	CHKD	APVD
THIS PROPRIETARY AND/OR CONFIDENTIAL MATERIAL IS THE PROPERTY OF FLOWSERVICE CORP. AND IS FURNISHED ONLY FOR THE PURPOSE INDICATED. FLOWSERVICE RESERVES ALL RIGHTS IN THE MATERIAL, AND DOES NOT CONVEY PERMISSION TO MANUFACTURE OR USE THE CONFIDENTIAL OR PROPRIETARY INFORMATION CONTAINED HEREIN.					



	DESC:				
REV	CHG NO	DATE	REVD	CHKD	APVD
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FLOWSERVE 5114 WOODALL ROAD,LYNCHBURG, VIRGINIA 24506-1318



DESC:					
REV	CHG NO	DATE	REVD	CHKD	APVD
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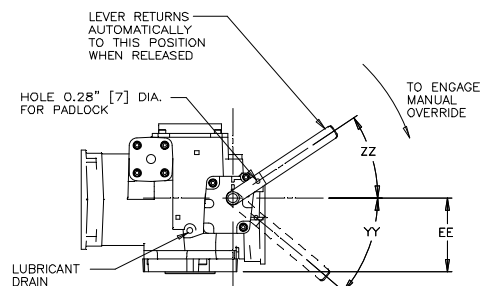
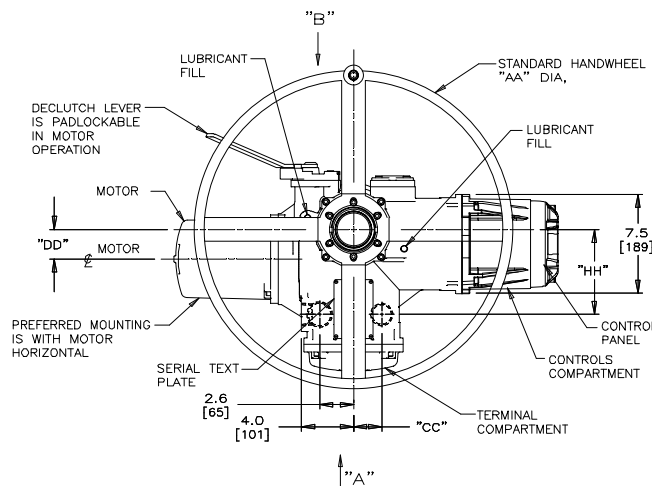
JOB NUMBER		FLOWSERVE 5114 WOODALL ROAD LYNCHBURG, VA 24506-1318			
DRW:	CKR				
DATE:	2-28-20	DRAWING TITLE MX-20 STANDARD UNIT			
CKD:	RHD				
DATE:	2-28-20				
APV:	RHD	SIZE B			
DATE:	3-10-20				
ALL DIMENSIONS ARE REFERENCE UNLESS OTHERWISE SPECIFIED		DRAWING NUMBER 03-653-0006		REV.	
3rd ANGLE		SCALE	NONE	UNITS	SHEET 1 OF 1

D

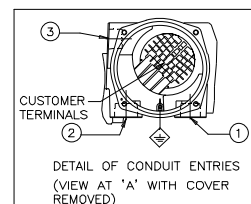
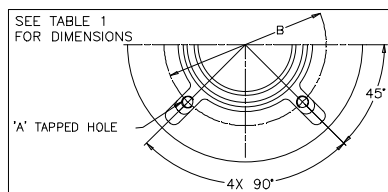
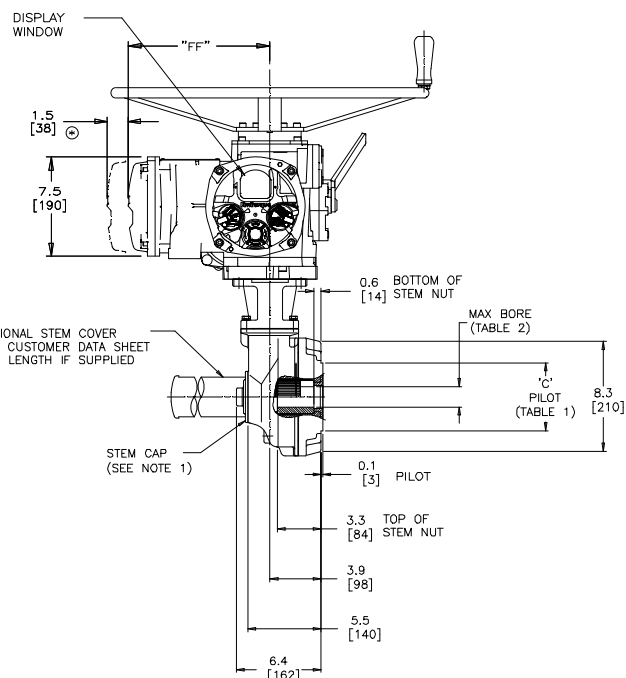
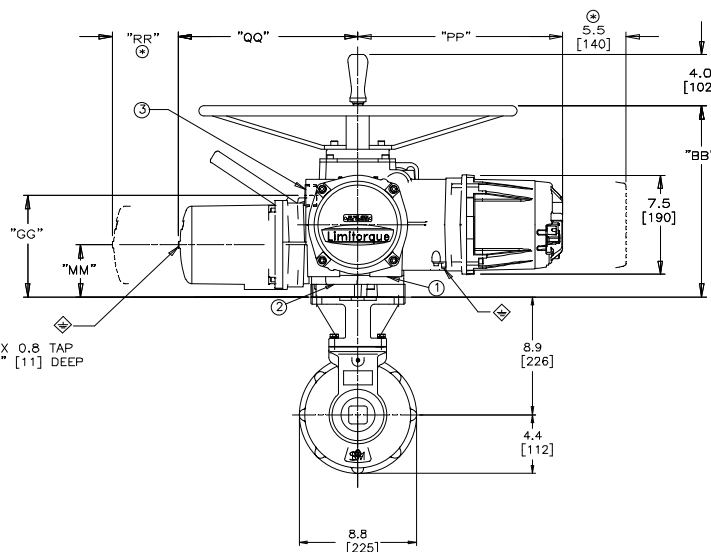
C

B

A



DETAIL OF DECLUTCH LEVER
(VIEW AT "B")



DETAIL OF CONDUIT ENTRIES
(VIEW AT 'A' WITH COVER REMOVED)

ACTUATOR OUTLINE DIMENSIONS (INCHES)

UNIT SIZE	AA	BB	CC	DD	EE	FF	GG	HH	MM	PP	QQ		RR		YY	ZZ
											1-PH	3-PH	1-PH	3-PH		
MX-05	12"	11.4	2.3	1.4	4.1	8.7	6.6	4.4	2.8	14.8	11.5	9.1	4.0	5.0	40"	38"
MX-10	18"	13.4	2.4	1.8	4.6	9.2	6.8	4.9	3.1	15.3	14.7	11.7			40"	38"
MX-20	24"	14.4	2.1	2.2	5.6	10.7	7.7	6.4	3.9	16.0	N/A	13.5			37"	33.5"
MX-40	24"	15.0	2.1	2.7	5.9	11.1	7.8	6.9	4.1	16.3	N/A	13.8			37"	33.5"

ACTUATOR OUTLINE DIMENSIONS (METRIC)

UNIT SIZE	AA	BB	CC	DD	EE	FF	GG	HH	MM	PP	QQ		RR		YY	ZZ
											1-PH	3-PH	1-PH	3-PH		
MX-05	305	289	58	37	105	222	168	113	71	376	292	232	102	127	40"	38"
MX-10	457	340	60	46	117	233	173	125	78	389	374	297			40"	38"
MX-20	610	366	54	57	141	272	196	163	100	406	N/A	344			37"	33.5"
MX-40	610	381	54	68	150	283	198	174	105	414	N/A	350			37"	33.5"

CONDUIT ENTRIES

POSITION	①	②	③
STANDARD	1 1/4" NPT	1 1/2" NPT	1 1/4" NPT
OPTIONS	M32	M40 M32	M32
	PG21	PG29	PG21

- ◆ GROUND/EARTHING LUG
⊙ ALLOWANCE FOR COVER REMOVAL

TABLE 2

MAX BORE	MAX KEYWAY (SQUARE)	MAX KEYWAY (RECTANGLE)	MAX THREAD, STEM
Ø1.750	3/8 X 3/8	—	Ø2.17 [Ø55]
Ø1.750	—	1/2 X 3/8	
[Ø45]	—	[14 X 9]	

TABLE 1

MOUNTING BASE	A	B	C
MSS (INCH) FA-14	(4) 5/8-11 UNC TAP X .94 DP	Ø5.51	Ø3.75
ISO (mm) F-14	(4) M16 TAP X 24 DP	Ø140	Ø100

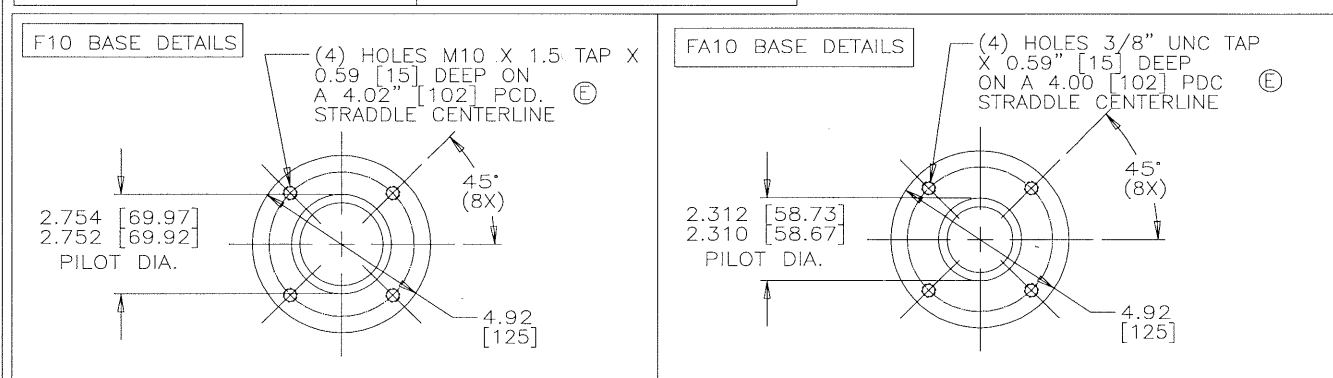
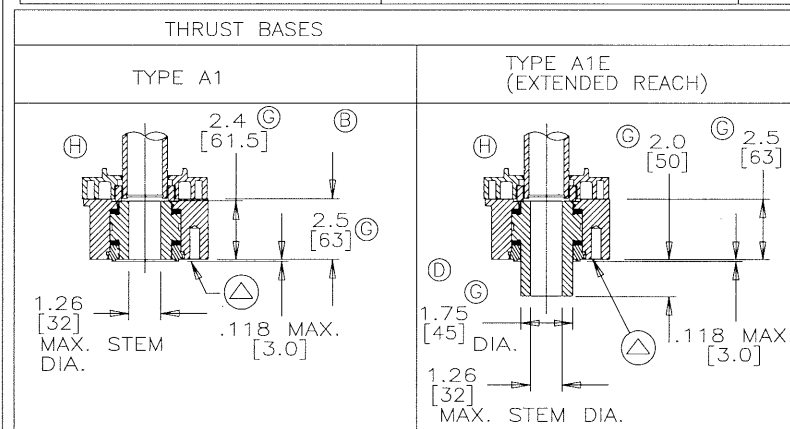
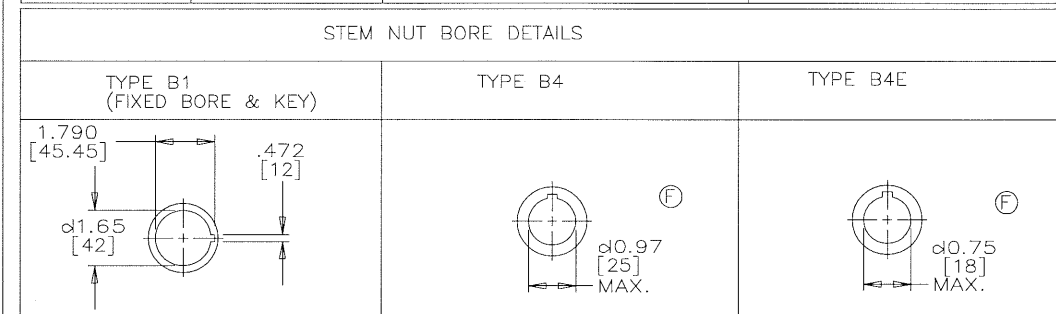
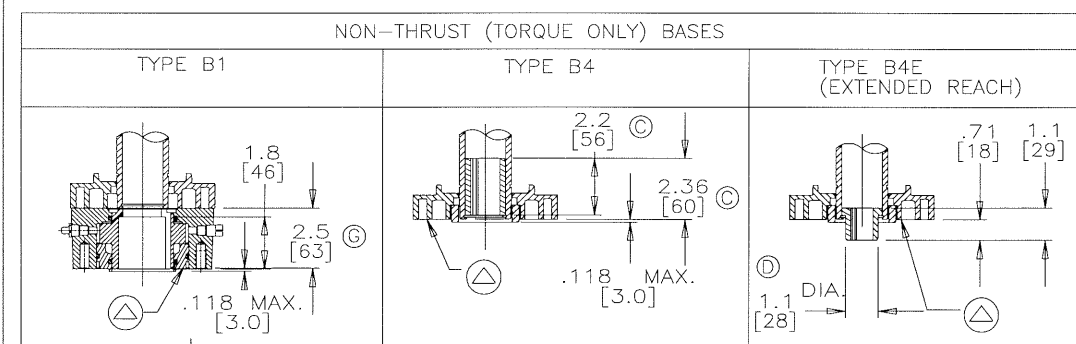
NOTES:

1. REMOVE STEM CAP TO ACCESS PF (BSPP) 3"x11 TPI THREADS
2. SEE CUSTOMER DATA SHEET TO DETERMINE BASE SUPPLIED.

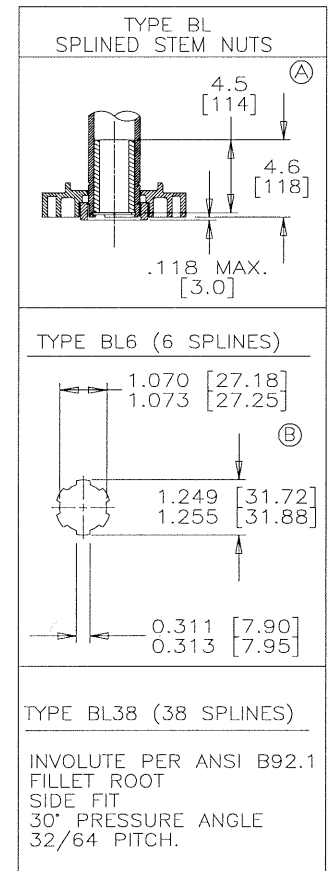
JOB NUMBER	5114 WOODALL ROAD, LYNCHBURG, VA USA 24506-1318		
DRW: KPR	DRAWING TITLE MX-05-40 STANDARD UNITS WITH: A) TOP MOUNTED HANDWHEEL B) V1 BEVEL GEAR OPERATOR C) MSS & ISO MOUNTING BASES D) 1 & 3-PHASE MOTORS		
DATE: 3-4-20			
CKD: SVA			
DATE: 3-4-20			
APV: MWR			
DATE: 3-12-20			
ALL DIMENSIONS ARE REFERENCE UNLESS OTHERWISE SPECIFIED	SIZE C	DRAWING NUMBER 03-654-0094	REV.
3rd ANGLE	SCALE NONE	UNITS	SHEET 1 OF 1

REV	CHG NO	DATE	REV D	CHK D	APV D
1					

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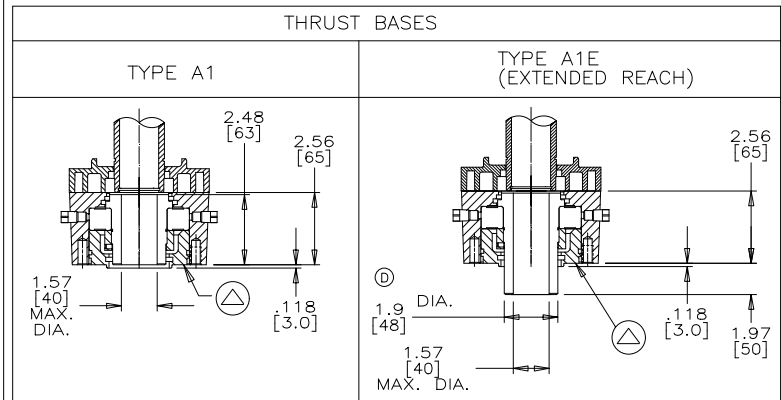
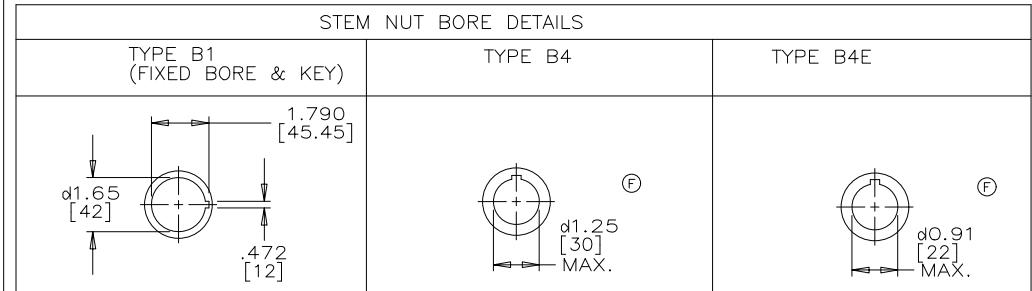
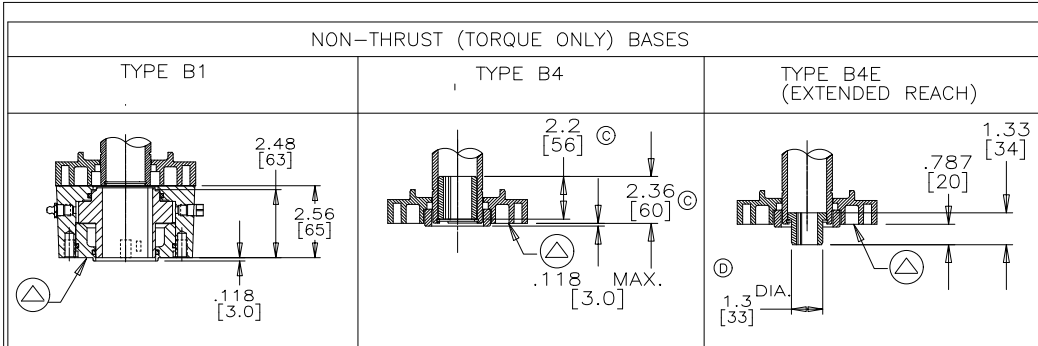
REVISIONS			BY	DATE	APPROV
REV	DESCRIPTION				
A	"4.5 [114] WAS "1.5 [38]; "4.6 [118] WAS "1.6 [42]"		CCR	1-30 1997	DJM
B	REMOVED "NUT LENGTH"; ADD METRIC EQUIVALENT		CCR	2-6 1997	DJM
C	2.2[56] WAS 1.5[38]; 2.36[60] WAS 1.6[42]		CCR	8-12 1997	DJM
D	ADD O.D. FOR EXTENDED STEM NUTS		CCR	9-9 1997	DJM
E	ADD TAP DEPTH: (4) WAS (8)		CCR	12-15 1997	DJM
F	.97 [25] WAS 1.00 [25.4]; [18] WAS [19]		CCR	8-13 1999	DJM
G	2.5[63] WAS 2.1[54]; 2.4[61.5] WAS 2.0[52.5] 1.75[45] WAS 1.50[38]; 2.0[50] WAS 1.57[40]		DKA	3-10 2004	DMP
H	UPDATED STRUCTURE CROSS SECTION		DKA	4-26 2004	JSF



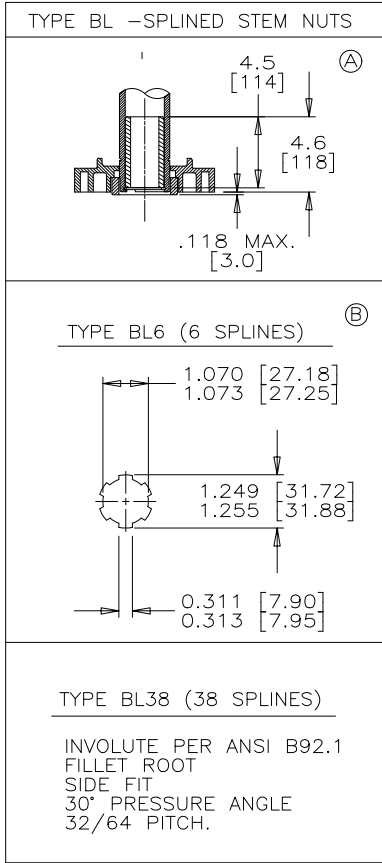
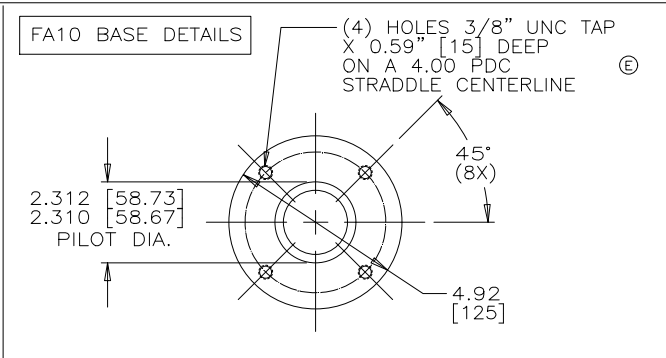
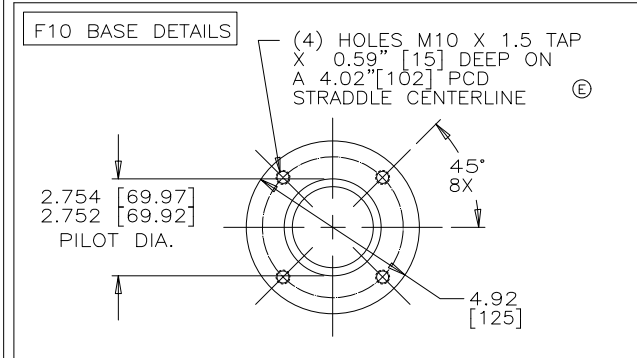
DRAWN CCR		DATE 4-19-96		LIMITORQUE CORPORATION 5114 WOODALL ROAD, LYNCHBURG, VIRGINIA 24506-1318	
CHECKED DJM		DATE 12-6-96			
APPROVED DJM		DATE 12-6-96		TITLE MX-05 BASE DETAILS	
SIZE A		DWG NO 03-615-0002-1			
SCALE NONE		ORDER NO N/A			

[] = MILLIMETER VALUE
UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN INCHES

REVISIONS		BY	DATE	APPVD
REV	DESCRIPTION			
A	"4.5 [114] WAS "1.5 [38]; "4.6 [118] WAS "1.6 [42]"	CCR	1-30 1997	DJM
B	ADD METRIC EQUIVALENT	CCR	2-6 1997	DJM
C	2.2[56] WAS 1.8[48]; 2.36[60] WAS 2.0[52]	CCR	8-12 1997	DJM
D	ADD O.D. FOR EXTENDED STEM NUTS	CCR	9-9 1997	DJM
E	ADD TAP DEPTH; (4) WAS (8)	CCR	12-15 1997	DJM
F	[30] WAS [32]; 0.91 WAS 0.875	CCR	8-13 1999	



△ MOUNTING FACE FOR ATTACHING TO VALVE FLANGE

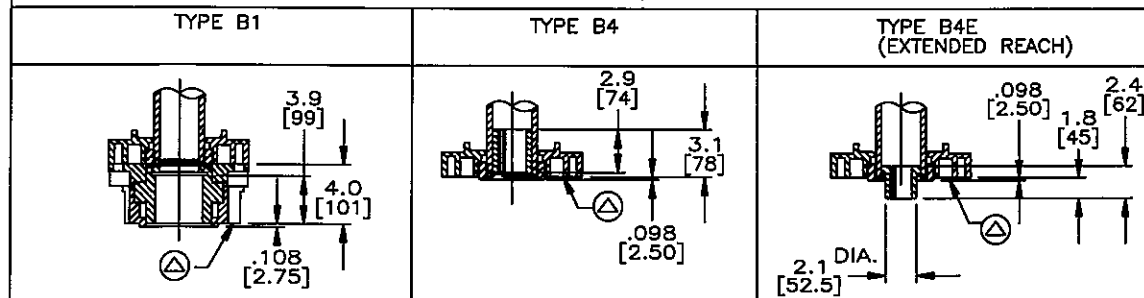


DRAWN CCR		DATE 4-19-96		LIMITORQUE CORPORATION 5114 WOODALL ROAD, LYNCHBURG, VIRGINIA 24506-1318		
CHECKED DJM		DATE 12-6-96				
APPROVED DJM		DATE 12-6-96		TITLE MX-10 BASE DETAILS		
SIZE A		DWG NO 03-615-0005-1		REV F		
SCALE NONE		ORDER NO N/A				

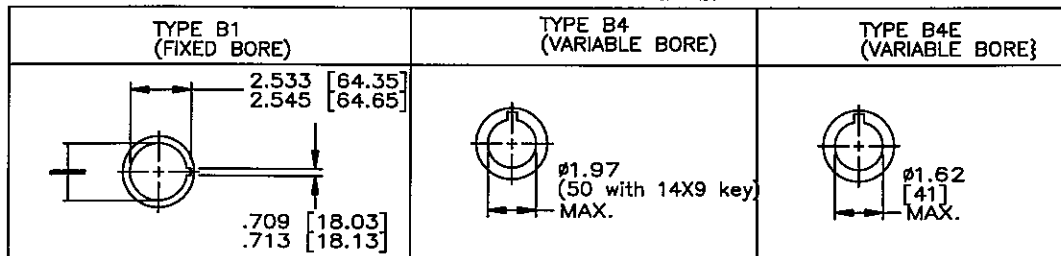
[] = MILLIMETER VALUE
UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN INCHES

REV		REVISIONS	BY	DATE	APPVD
H	CHANGED DIMENSION 4.5[115] TO 4.3[108]		TSC	7-1 2014	DC

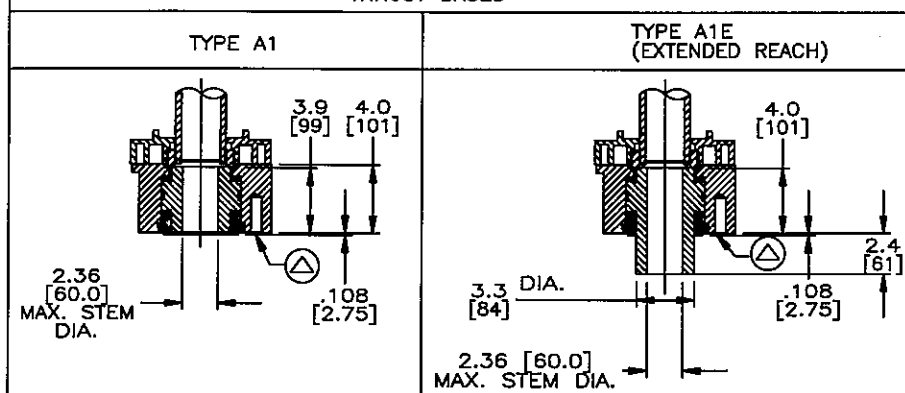
NON-THRUST TORQUE ONLY



STEM NUT BORE DETAILS

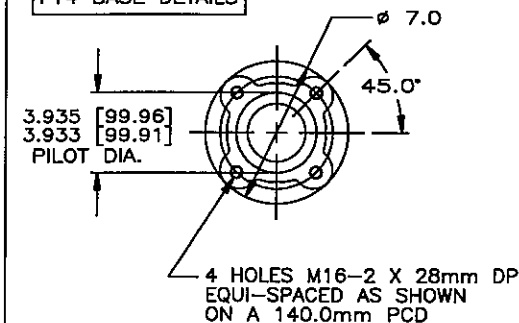


THRUST BASES

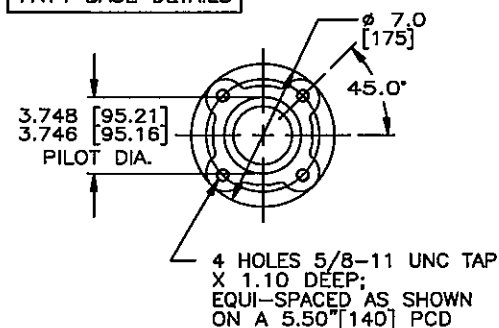


⊙ MOUNTING FACE FOR ATTACHING TO VALVE FLANGE

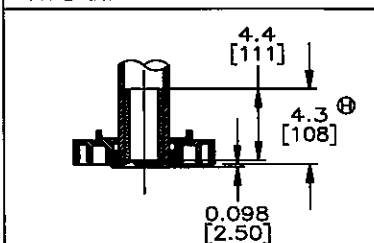
F14 BASE DETAILS



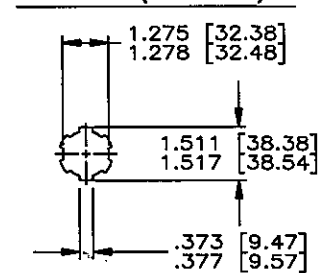
FA14 BASE DETAILS



TYPE BL - SPLINED STEM NUTS



TYPE BL6 (6 SPLINES)

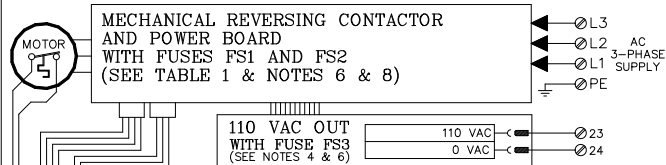


TYPE BL36 (36 SPLINES)

INVOLUTE PER ANSI B92.1
FILLET ROOT
SIDE FIT
30° PRESSURE ANGLE
32/64 PITCH

DATE	4-19-96	FLOWERVE - LIMITORQUE		
CCR		TITLE		
CHECKED	DATE	MX-20 BASE DETAILS		
DJM	12-6-96			
APPROVED	DATE			
DJM	12-6-96			
SIZE	A	DWG NO	03-615-0008-1	REV H
SCALE	NONE	ORDER NO	N/A	

1 = MILLIMETER VALUE
UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN INCHES



NOTES: (SEE INSTALLATION & OPERATION MANUAL VAIOM000071 FOR DETAILS)
CIRCUIT SHOWN WITH VALVE IN FULLY CLOSED POSITION AND WITH POWER OFF.

- REMOTE INPUTS**
SIGNAL THRESHOLD - MINIMUM "ON" 19.2V AC/DC
MAXIMUM "OFF" 5.0V AC/DC
MAX LOAD - 10 mA / 110V AC

REQUIRED CONTROL SIGNAL DURATION = 350ms MIN.
INPUTS 0, 1, & 2 ARE FIELD CONFIGURABLE FOR CLOSE/OPEN/BOTH INHIBIT, USER INPUT, OR ESD.
INPUT 1 MAY ALSO BE CONFIGURED FOR PARTIAL STROKE TEST.
INPUT 2 MAY ALSO BE CONFIGURED FOR CSE REMOTE SELECTION INDICATION.
DEFAULT INPUT CONFIGURATION
INPUT 0 - ESD,
INPUT 1 - OPEN INHIBIT,
INPUT 2 - CLOSE INHIBIT.

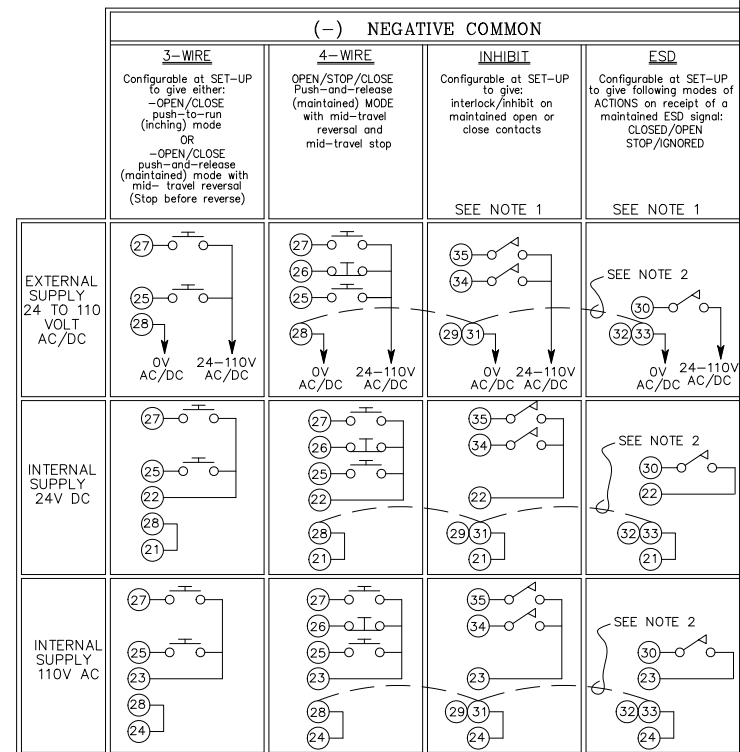
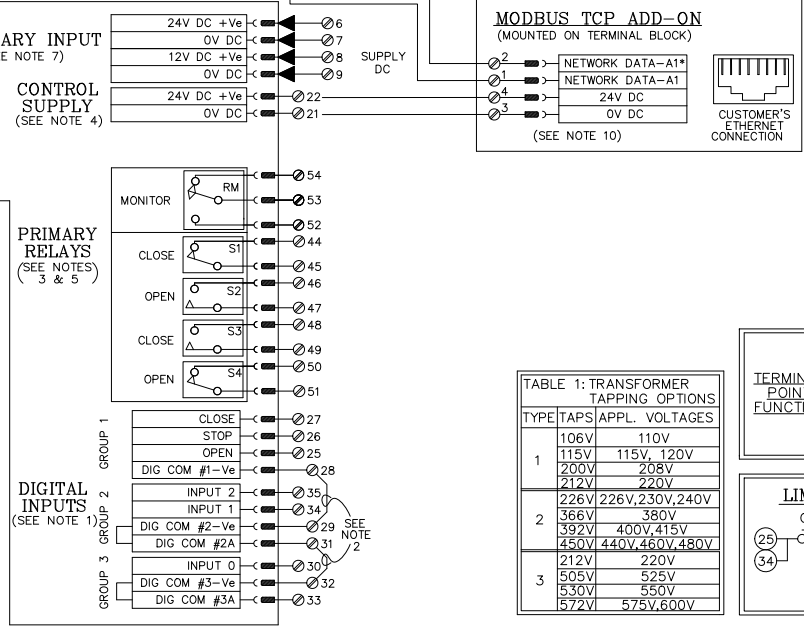
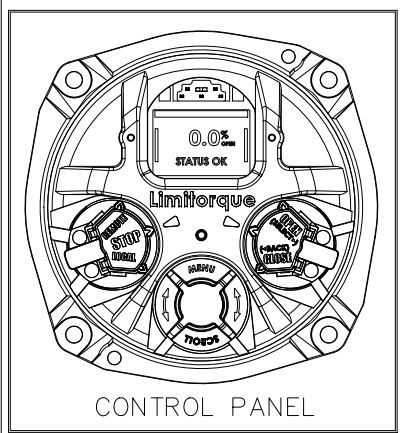
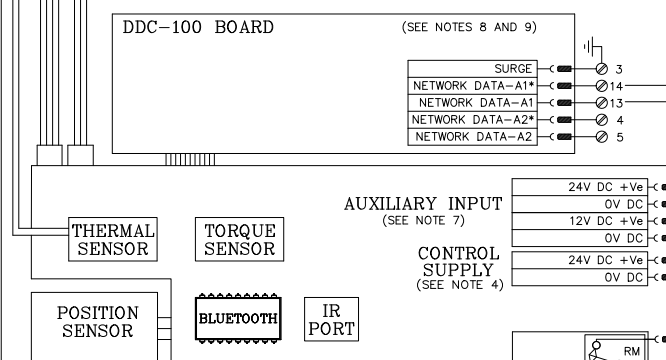
- AUXILIARY INPUTS**
BACK-UP UPS POWER MAY BE CONNECTED: 24VDC TO TERMINALS 6 & 7, MAX. CURRENT DRAW 1 AMP; 12VDC TO TERMINALS 8 & 9, MAX. CURRENT DRAW 2 AMPS. THIS POWERS ALL CONTROLS AND OPTION BOARDS FOR LOCAL/REMOTE INDICATION AND CONFIGURATION. THE REVERSING CONTACTOR WILL NOT BE POWERED. CUSTOMER MUST SUPPLY FUSES AS REQUIRED BY LOCAL ELECTRICAL CODES.
- COMMAND PRIORITY**
PLEASE SEE INSTALLATION AND OPERATION MANUAL VAIOM000071 FOR NETWORK COMMAND PRIORITIES.
- DDC-100**
JP1 AND JP2 MUST BE IN POSITION "A". PREFERRED NETWORK GROUND IS LOCATED ADJACENT TO THE TERMINAL BLOCK. SEE MANUAL FOR DETAILS. THE 6-PIN CONNECTOR WITH BLUE WIRES SHOULD BE CONNECTED TO J2 OF THE DDC-100 BOARD.
- TCP/IP**
REFER TO LMENIM2329-02 DDC-100 FIELD UNIT WITH ETHERNET USER INSTRUCTIONS FOR MODBUS TCP/IP PROGRAMMING INFORMATION.

- REMOTE INPUT JUMPERS**
JUMPERS CAN BE USER WIRED TO CONNECT DIG COMMONS #1, 2 & 3 (AS NEEDED). THE 3 REMOTE WIRING GROUPS ARE ELECTRICALLY ISOLATED, INTERNALLY. ONLY A SINGLE POWER SOURCE MAY BE USED TO POWER ANY ONE GROUP BUT DIFFERENT SOURCES MAY POWER DIFFERENT GROUPS. IF DIFFERENT SOURCES ARE USED, ENSURE THAT THE POLARITIES ARE CORRECT.
- DEFAULT [S] SETTINGS**
THE DEFAULT OPERATING CONFIGURATION FOR THE "S" OUTPUTS ARE SHOWN IN THE TABLE BELOW. THE CONTACT STATES SHOWN IN THE SCHEMATIC REPRESENT A FULLY CLOSED VALVE. SEE OPERATION MANUAL FOR ALTERNATE CONFIGURATIONS.

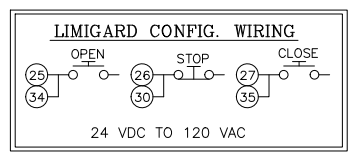
OUTPUT SWITCH CONTACT DEVELOPMENT			
OUTPUT SWITCH	VALVE POSITION		FUNCTION
	FULL CLOSE	FULL OPEN	
S1	—	—	CLOSE LIMIT
S3	—	—	
S2	—	—	OPEN LIMIT
S4	—	—	

— OPEN CONTACT
— CLOSED CONTACT

- MAXIMUM EXTERNAL LOAD**
TERMINALS 21 AND 22 (24 VDC)
- 4W MAX. EXT. LOAD
TERMINALS 23 AND 24 (117VAC)
- 15VA MAX. EXT. LOAD
- CONTACT RATINGS**
S1-S4 - 5 AMPS @ 250 VAC OR 30VDC (RESISTIVE)
THE TOTAL COMBINED CURRENT THROUGH ALL FOUR STATUS CONTACTS (S1-S4) MUST NOT EXCEED 8A
MONITOR - 5 AMPS @ 250 VAC OR 30 VDC (RESISTIVE)
- FUSES**
FS1 (PRIMARY) - 600V AC, 1A, 100ka INT RATING, FAST ACTING 10.3 x 38.1mm TUBE, UL LISTED, CSA CERTIFIED
FS2 (PRIMARY) - 0.2A, 250V, TIME DELAY, 5 x 20mm, GLASS TUBE
FS3 (SECONDARY) - 0.2A, 250V, TIME DELAY, 5 x 20mm, GLASS TUBE

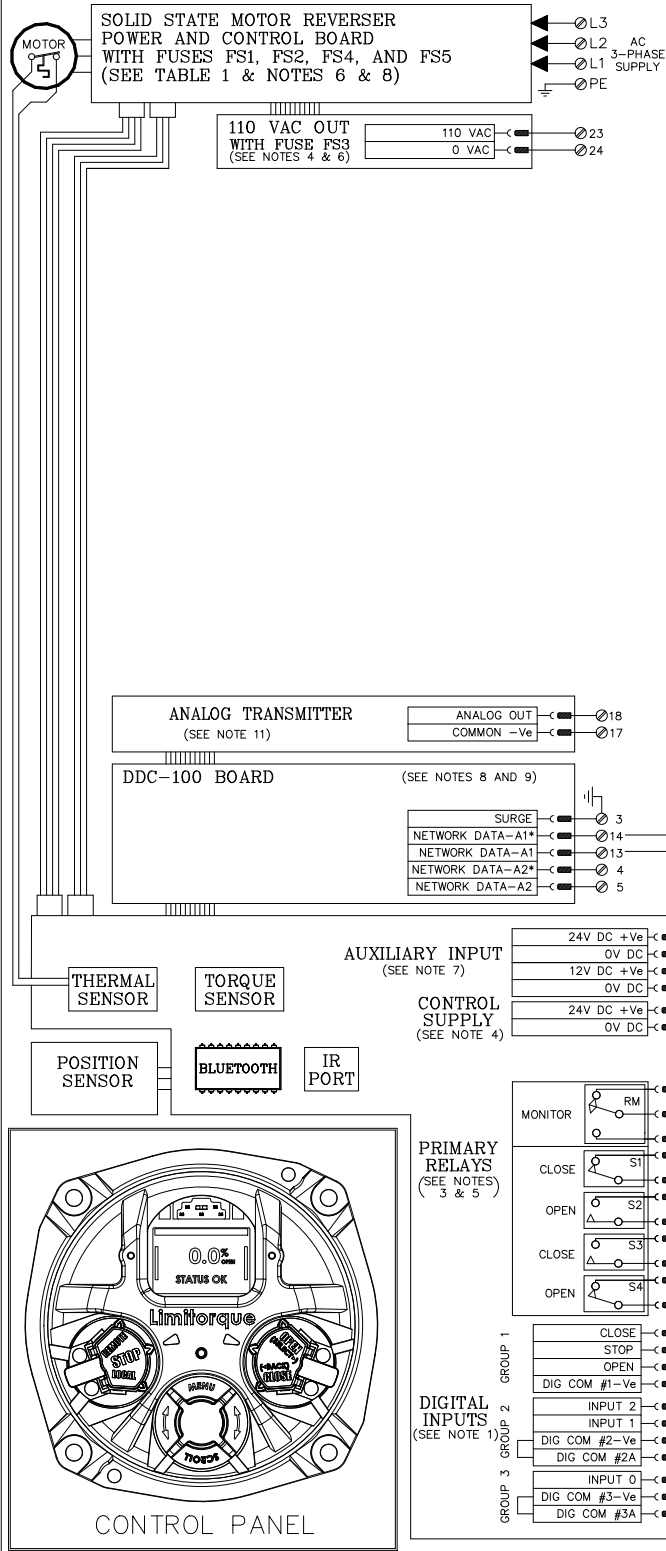


REMOTE WIRING CONNECTIONS									
TERMINAL POINT	CLOSE (27)	+24V DC (22)	0V DC (21)	OPEN INHIBIT (34)	DIG COM #1 (28)	STOP (26)	+110V AC (23)	0V AC (24)	CLOSE INHIBIT (35)
FUNCTION									



DRAWN	DATE	LIMITORQUE Actuation Systems	
NWN	01/16/20	5114 WOODALL ROAD, LYNCHBURG, VIRGINIA 24506-1318	
CHECKED	DATE	TITLE	
JPS	01/17/20	MXb WIRING DIAGRAM	
APPROVED	DATE	DDC With Modbus TCP Add-On 110VAC/15VA	
EMO	01/20/20	No Analog	
		Mechanical Contactor	
		R4 (4) 5A, (1) 5A monitor relay	
		3Wire/4Wire Negative Common	
UNIT TYPE	MXb	DWG NO	WD-MXb-041100-1101
		REV	00B
		SHEET	1 OF 1

DRAWN	DATE	 <i>Limiterorque Actuation Systems</i> 5114 BISSALL ROAD, LYNCHBURG, VIRGINIA 24090-1318	
NWN	01/16/20		
CHECKED	DATE	TITLE Mxb WIRING DIAGRAM DDC With Modbus TCP Add-On 110Vac/15VA Analog Transmitter Bluetooth Mechanical Contactor R4 (4) 5A, (1) 5A monitor relay 3Wire/4Wire Negative Common	
JPS	01/17/20		
APPROVED	DATE		
EMO	01/20/20		
UNIT TYPE	Mxb	DWG NO WD-Mxb-041110-1101	REV 00B SHEET 1 of 1



NOTES: (SEE INSTALLATION & OPERATION MANUAL VAIOM000071 FOR DETAILS)
CIRCUIT SHOWN WITH VALVE IN FULLY CLOSED POSITION AND WITH POWER OFF.

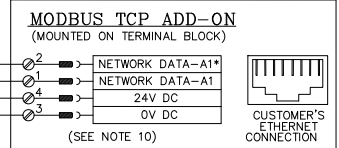
- REMOTE INPUTS**
SIGNAL THRESHOLD - MINIMUM "ON" 19.2V AC/DC
MAXIMUM "OFF" 5.0V AC/DC
MAX LOAD - 10 mA / 110V AC

REQUIRED CONTROL SIGNAL DURATION = 350ms MIN.
INPUTS 0, 1, & 2 ARE FIELD CONFIGURABLE FOR CLOSE/OPEN/BOTH INHIBIT, USER INPUT, OR ESD.
INPUT 1 MAY ALSO BE CONFIGURED FOR PARTIAL STROKE TEST.
INPUT 2 MAY ALSO BE CONFIGURED FOR CSE REMOTE SELECTION INDICATION.
DEFAULT INPUT CONFIGURATION:
INPUT 0 - ESD,
INPUT 1 - OPEN INHIBIT,
INPUT 2 - CLOSE INHIBIT.
- REMOTE INPUT JUMPERS**
JUMPERS CAN BE USER WIRED TO CONNECT DIG COMMONS #1, 2 & 3 (AS NEEDED). THE 3 REMOTE WIRING GROUPS ARE ELECTRICALLY ISOLATED, INTERNALLY. ONLY A SINGLE POWER SOURCE MAY BE USED TO POWER ANY ONE GROUP BUT DIFFERENT SOURCES MAY POWER DIFFERENT GROUPS. IF DIFFERENT SOURCES ARE USED, ENSURE THAT THE POLARITIES ARE CORRECT.
- DEFAULT [S] SETTINGS**
THE DEFAULT OPERATING CONFIGURATION FOR THE "S" OUTPUTS ARE SHOWN IN THE TABLE BELOW. THE CONTACT STATES SHOWN IN THE SCHEMATIC REPRESENT A FULLY CLOSED VALVE.
SEE OPERATION MANUAL FOR ALTERNATE CONFIGURATIONS.

OUTPUT SWITCH CONTACT DEVELOPMENT			
OUTPUT SWITCH	VALVE POSITION		FUNCTION
	FULL CLOSE	FULL OPEN	
S1	—	—	CLOSE LIMIT
S3	—	—	
S2	—	—	OPEN LIMIT
S4	—	—	

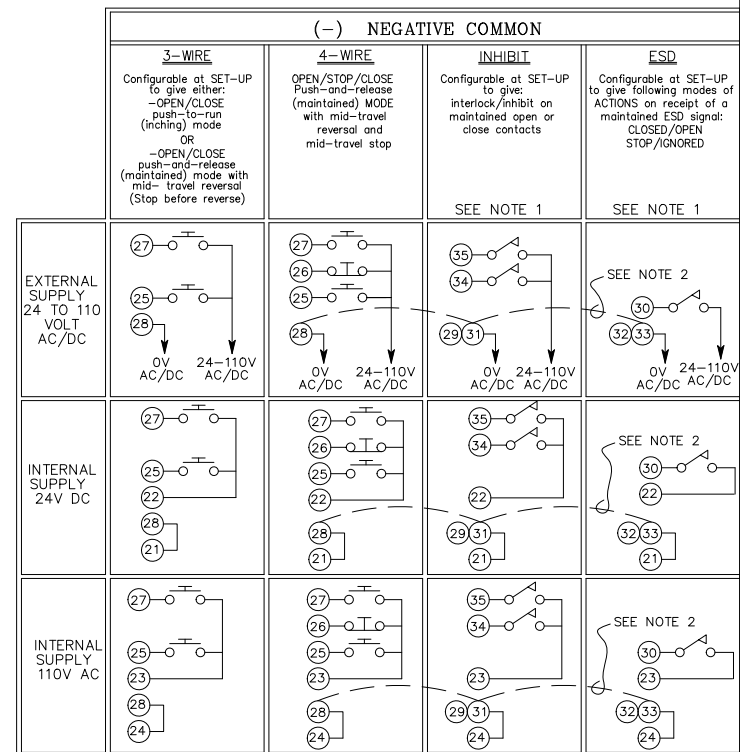
— OPEN CONTACT
— CLOSED CONTACT

- MAXIMUM EXTERNAL LOAD**
TERMINALS 21 AND 22 (24 VDC)
- 4W MAX. EXT. LOAD
TERMINALS 23 AND 24 (117VAC)
- 15VA MAX. EXT. LOAD
- CONTACT RATINGS**
S1-S4 - 5 AMPS @ 250 VAC OR 30VDC (RESISTIVE)
THE TOTAL COMBINED CURRENT THROUGH ALL FOUR STATUS CONTACTS (S1-S4) MUST NOT EXCEED 8A
MONITOR - 5 AMPS @ 250 VAC OR 30 VDC (RESISTIVE)
- FUSES**
FS1 (PRIMARY) - 600V AC, 1A, 100ka INT. RATING, FAST ACTING 10.3 x 38.1mm TUBE, UL LISTED, CSA CERTIFIED
FS2 (PRIMARY) - 0.2A, 250V, TIME DELAY, 5 x 20mm, GLASS TUBE
FS4 FOR SSMR PROTECTION IF MOTOR CIRCUIT FAULT.
FS5 FOR WIRING/MOTOR OVERLOAD PROTECTION.
NOT FOR SHORT CIRCUIT PROTECTION OF THE ACTUATOR WHICH MUST BE PROVIDED BY OTHERS.

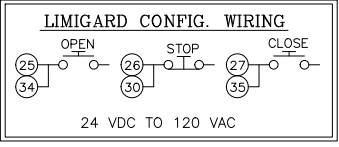


- AUXILIARY INPUTS**
BACK-UP UPS POWER MAY BE CONNECTED: 24VDC TO TERMINALS 6 & 7, MAX. CURRENT DRAW 1 AMP; 12VDC TO TERMINALS 8 & 9, MAX. CURRENT DRAW 2 AMPS. THIS POWERS ALL CONTROLS AND OPTION BOARDS FOR LOCAL/REMOTE INDICATION AND CONFIGURATION. THE REVERSING CONTACTOR WILL NOT BE POWERED. CUSTOMER MUST SUPPLY FUSES AS REQUIRED BY LOCAL ELECTRICAL CODES.
- COMMAND PRIORITY**
PLEASE SEE INSTALLATION AND OPERATION MANUAL VAIOM000071 FOR NETWORK COMMAND PRIORITIES.
- DDC-100**
JP1 AND JP2 MUST BE IN POSITION "A". PREFERRED NETWORK GROUND IS LOCATED ADJACENT TO THE TERMINAL BLOCK. SEE MANUAL FOR DETAILS. THE 6-PIN CONNECTOR WITH BLUE WIRES SHOULD BE CONNECTED TO J2 OF THE DDC-100 BOARD.
- TCP/IP**
REFER TO LMENIM2329-02 DDC-100 FIELD UNIT WITH ETHERNET USER INSTRUCTIONS FOR MODBUS TCP/IP PROGRAMMING INFORMATION.

- ANALOG EXTERNAL LOAD**
4-20 mA OUTPUT SIGNAL - 600 OHMS MAX.
0-10 VDC OUTPUT SIGNAL - 1000 OHMS MAX.
SET JUMPERS JP1 AND JP2 TO POSITION 1-2 ON ANALOG BOARD.
THE 3-PIN CONNECTOR WITH BLUE WIRES SHOULD BE CONNECTED TO J3 ON THE ANALOG BOARD.



REMOTE WIRING CONNECTIONS									
TERMINAL POINT	CLOSE (27)	+24V DC (22)	0V DC (21)	OPEN INHIBIT (34)	DIG COM #1 (28)	STOP (26)	+110V AC (23)	0V AC (24)	CLOSE INHIBIT (35)
FUNCTION	OPEN (25)	EXTERNAL SUPPLY		ESD (30)	DIG COM #2(A) (29/31)				



DRAWN DATE		LIMITORQUE Actuation Systems	
NWN	01/16/20	5114 WOODALL ROAD, LYNCHBURG, VIRGINIA 24506-1318	
CHECKED DATE		TITLE	
JPS	01/17/20	MXb WIRING DIAGRAM	
APPROVED DATE		DDC With Modbus TCP Add-On 110Vac/15VA	
EMO	01/20/20	Analog Transmitter	
UNIT TYPE		Solid State Relay	
MXb		R4 (4) 5A, (1) 5A monitor relay	
DWG NO		3Wire/4Wire Negative Common	
WD-Mxb-141110-1101			
REV		SHEET	
00B		1 OF 1	

TABLE 1: TRANSFORMER TAPPING OPTIONS		
TYPE	TAPS	APPL. VOLTAGES
1	106V	110V
	115V	115V, 120V
	200V	208V
	212V	220V
2	226V	226V, 230V, 240V
	366V	380V
	392V	400V, 415V
	450V	440V, 460V, 480V
3	212V	220V
	505V	525V
	530V	550V
	572V	575V, 600V